MATH1013 & MATH1014: Calculus I & II

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MATH1013 Calculus I

1 Functions

1.1 Functions and Their Graphs

1.1.1 The Notion of a Function

Remark 1.1 — Functions as Machines

A function can be understood as a "machine": given an input, the machine processes it according to some rules and gives a definite output. For example, the machine $x \mapsto x^2$ (e.g. $0 \mapsto 0$, $-5 \mapsto 25$, $\sqrt{2} \mapsto 2$, $-\pi \mapsto \pi^2$) is represented by the formula $f(x) = x^2$, where f is the name of the machine, x an input and $f(x)$ the corresponding output. Characteristics of the machine: what kinds of inputs are allowable; what kinds of outputs are expected; what kinds of outputs are actually achievable.

Definition 1.2. Let A and B be sets. A *function* (or *map*, *mapping*, *transformation*) f *from* A *to* B, denoted by

$$f\colon A \to B,$$

is a relation that connects the elements of A and the elements of B. Given each element $x \in A$, this relation assigns to it a **unique** element $y \in B$. In case y is assigned to x by f, we say "f sends x to y" and write

$$y = f(x).$$

- The set A is called the *domain* of f and the set B is called the *codomain* of f.
- The *range* (or *image*) of $f\colon A \to B$ is the set that contains precisely all the elements in B that are actually assigned to some element in A:

$$\{y \in B : y = f(x) \text{ for some } x \in A\} \qquad (\text{or informally, } \{f(x) : x \in A\}).$$

In particular, the range of f is a subset of the codomain of f.

Remark 1.3 — Machine Analogy Revisited

In the machine analogy, each input can only be processed to give one unique output.
- The domain is the set of all "allowable inputs";
- the codomain describes what kind of objects the "outputs" are;
- the range is the set of all "achievable outputs".

Remark 1.4 — Well-Definedness

A function may be illustrated by an arrow diagram from the domain to the codomain; the elements of the codomain that are hit form the range. Consider three relations $g, h, f\colon A \to B$ with $A = \{1, 2, 3, 4\}$:
- g: nothing is assigned to the element 4 in the domain, i.e. $g(4)$ is undefined. Then g is **not** a function.
- h: both 4 and 6 are assigned to the element 3 in the domain, i.e. $h(3) = 4$ and $h(3) = 6$. Then h is **not** a function either.
- f: although the same element 2 in the codomain is assigned to both 1 and 2 in the domain ($f(1) = f(2) = 2$), exactly one element in the codomain is assigned to each element in the domain. So f **is** a function. An element of the codomain not assigned to anything (e.g. 8) is simply not in the range.

Instead of saying "f is a function" and "g and h are not functions", some authors equivalently say "f is *well-defined*" and "g and h are not well-defined".

1.1.2 Examples of Functions

Example 1.5

The area A of a circular disk is a function of the radius r of the disk: $A\colon (0,+\infty) \to \mathbb{R}$ defined by

$$A(r) = \pi r^2.$$

It is perfectly fine to input a non-positive number into the formula $A(r) = \pi r^2$, but it does not make sense to have a disk with zero or negative radius; so instead of $\mathbb{R}$, we take $(0,+\infty)$ to be the domain of A.

Example 1.6

Suppose you are taking MATH1013. The letter grade g you finally obtain is a function of the total score x you get:

$$g\colon [0,100] \to \{\text{A+, A, A}-, \text{B+, B, B}-, \text{C+, C, C}-, \text{D, F}\},$$

whose defining formula $g(x)$ is currently unknown. (Of course $g(0) = \text{F}$ and $g(100) = \text{A+}$.)

Example 1.7

Let A be the set of all people in the world, and B the set of all women in the world.

- The assignment $f\colon A \to B$, $f(x) =$ the biological mother of x, is a (well-defined) function because everyone has one and only one biological mother. The range of f is the set of all mothers in the world.
- The assignment $g\colon B \to A$, $g(x) =$ the child of x, is **not** a function because a woman may have no child or more than one child.

Example 1.8

Let A be the set of all statements. Define $v\colon A \to \{0,1\}$ by

$$v(P) = \begin{cases} 1 & \text{if } P \text{ is true} \\ 0 & \text{if } P \text{ is false} \end{cases}.$$

For each pair of statements P and Q:

- $v(\sim P) = 1 - v(P)$,
- $v(P \wedge Q) = v(P)\, v(Q)$,
- $v(P \vee Q) = 1 - (1 - v(P))(1 - v(Q)) = v(P) + v(Q) - v(P)v(Q)$, etc.

(Exercise: express $v(P \Rightarrow Q)$ and $v(P \Leftrightarrow Q)$ in terms of $v(P)$ and $v(Q)$.)

Example 1.9

Let S be the set of all finite sets. Define $n\colon S \to \mathbb{N} \cup \{0\}$ by $n(A) =$ the number of elements in A. Then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

for each pair of finite sets A and B.

Example 1.10

Consider the assignment $f\colon \mathbb{Q} \to \mathbb{Z}$ defined by

$$f\left(\frac{m}{n}\right) = m \qquad \text{for each } m \in \mathbb{Z} \text{ and } n \in \mathbb{Z} \setminus \{0\}.$$

Then f is **not** a function, because a rational number has more than one representa-

tion, e.g.

$$\frac{1}{2} = \frac{2}{4} = \frac{-1}{-2} = \cdots,$$

so the same input $\frac{1}{2} = \frac{2}{4}$ yields two different outputs $f(\frac{1}{2}) = 1$ but $f(\frac{2}{4}) = 2$. On the other hand, the assignment $g\colon \mathbb{Q} \to \mathbb{Z}$ defined by

$$g\Big(\frac{m}{n}\Big) = m \qquad \text{if } m \in \mathbb{Z},\ n \in \mathbb{N} \text{ and } m, n \text{ are relatively prime}$$

is a (well-defined) function.

1.1.3 Domain, Codomain and Range

In MATH1013 and MATH1014 we usually deal with *real-valued functions in a real variable*, so domains, codomains and ranges are usually subsets of $\mathbb{R}$. ("Real-valued": codomain $\subseteq \mathbb{R}$; "in a real variable": domain $\subseteq \mathbb{R}$.)

Example 1.11

Let $f\colon \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Although the codomain $\mathbb{R}$ consists of all real numbers:

- each real input x can only give a non-negative output x^2, and
- each non-negative output y is indeed achievable by some real input, namely $\sqrt{y}$ and $-\sqrt{y}$, because $f(\sqrt{y}) = y$ and $f(-\sqrt{y}) = y$.

So the range of f is $[0, +\infty)$ (which is indeed a subset of the codomain).

Example 1.12

Find the range of $f\colon (2, 3] \to \mathbb{R}$ defined by $f(x) = \dfrac{12}{x}$.

Solution.

- Given each $x \in (2, 3]$, we have $2 < x \leq 3$, which implies

  $$\frac{12}{3} \leq \frac{12}{x} < \frac{12}{2},$$

  i.e. $4 \leq f(x) < 6$, so $f(x) \in [4, 6)$. This shows that the range of f is $\subseteq [4, 6)$.
- Conversely, for each $y \in [4, 6)$, the equation $y = f(x)$ has a solution $x = \dfrac{12}{y}$; and since

  $$\frac{12}{6} < \frac{12}{y} \leq \frac{12}{4},$$

  such a solution x indeed belongs to $(2, 3]$. This shows that $[4, 6) \subseteq$ the range of f.

Therefore the range of f is the interval $[4, 6)$ (a subset of the codomain $\mathbb{R}$).

Remark 1.13 — Natural Domain; Finding Ranges

In the previous two examples the domains were specified. If we are given just a defining formula $f(x)$ without a specified domain, we usually take the domain to be its *natural domain*: the largest possible subset of $\mathbb{R}$ on which the formula $f(x)$ is "valid" (in MATH1013, "valid" means "well-defined as a real number").

(i) Useful arithmetic facts for finding natural domains:
- any real number divided by zero is undefined;
- the product of two negative numbers is positive;
- the nth root of a positive number is always positive;

– the nth root of a negative number is $\begin{cases}\text{negative} & \text{if } n \text{ is odd} \\ \text{not a real number} & \text{if } n \text{ is even}\end{cases}$.

(ii) To find the range of f, we try to answer: "for what values of y does the equation $y = f(x)$ have a solution in the domain of f?" There is no general method; it depends very much on the defining formula.

Example 1.14

Find the (natural) domain and the range of each of the following functions:

$$\text{(a) } f(x) = \sqrt{x+2} \qquad \text{(b) } g(x) = \sqrt{x^2+5} \qquad \text{(c) } h(x) = \frac{x^2+5}{x+2}$$

Solution.

(a) $\sqrt{x+2}$ is well-defined iff $x + 2 \geq 0$, i.e. $x \geq -2$; so the natural domain of f is $[-2, +\infty)$. For the range, the question is: "for what values of y does $y = \sqrt{x+2}$ have a solution in $[-2, +\infty)$?" The only arithmetic constraint is $y \geq 0$; conversely, for each $y \geq 0$,

$$y = \sqrt{x+2} \;\Rightarrow\; y^2 = x + 2 \;\Rightarrow\; x = y^2 - 2,$$

and this solution indeed belongs to $[-2, +\infty)$. Therefore the range of f is $[0, +\infty)$.

(b) $\sqrt{x^2+5}$ is well-defined iff $x^2 + 5 \geq 0$, which holds for every real x; so the natural domain of g is $\mathbb{R}$. For the range: the only constraint is $y \geq \sqrt{5}$; conversely, for each $y \geq \sqrt{5}$,

$$y = \sqrt{x^2+5} \;\Rightarrow\; y^2 = x^2 + 5 \;\Rightarrow\; x = \sqrt{y^2-5} \text{ or } x = -\sqrt{y^2-5},$$

and these solutions are indeed real numbers. Therefore the range of g is $[\sqrt{5}, +\infty)$.

(c) $\dfrac{x^2+5}{x+2}$ is well-defined iff $x + 2 \neq 0$, i.e. $x \neq -2$; so the natural domain of h is $\mathbb{R} \setminus \{-2\}$ (or $(-\infty, -2) \cup (-2, +\infty)$). For the range: the equation $y = \dfrac{x^2+5}{x+2}$ rewrites as $(x + 2)y = x^2 + 5$, i.e.

$$x^2 - yx + (5 - 2y) = 0.$$

This quadratic equation has solutions in $\mathbb{R} \setminus \{-2\}$ iff its discriminant is non-negative:

$$(-y)^2 - 4(1)(5 - 2y) \geq 0 \;\Leftrightarrow\; y^2 + 8y - 20 \geq 0 \;\Leftrightarrow\; (y - 2)(y + 10) \geq 0.$$

This holds iff $y \leq -10$ or $y \geq 2$, so the range of h is $(-\infty, -10] \cup [2, +\infty)$.

1.1.4 Graphs of Functions

Definition 1.15. Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The *graph* of f is the set of points in the coordinate plane

$$\{(x, y) : x \in A \text{ and } y = f(x)\} \qquad \text{(or informally, } \{(x, f(x)) : x \in A\}).$$

We also say that the graph of f is described by the equation $y = f(x)$.

Remark 1.16 — Facts about Graphs

(i) (*Vertical line test*) Each vertical line in the coordinate plane intersects the graph of f at one point **at most**.

(ii) The domain of f consists of all those real numbers a such that the vertical line $x = a$ intersects the graph of f at exactly one point.

(iii) The range of f consists of all those real numbers b such that the horizontal line $y = b$ intersects the graph of f at one point **at least**.

Example 1.17

Determine whether each given set of points is the graph of a function; if yes, identify the domain and the range.

Solution. (The three given graphs, drawn through the labeled points, are described verbally.)

- A curve through $(-3, 1)$, $(4, 3)$ with a break at the vertical line $x = 0$: it is the graph of a function, whose domain is $\mathbb{R} \setminus \{0\}$ and range is $\mathbb{R}$.
- A curve through $(-4, 2)$, $(3, 3)$, $(-3, -3)$, $(5, -2)$ with the endpoint at $x = 5$ excluded: it is the graph of a function, whose domain is $[-4, 5)$ and range is $[-3, 3]$.
- A curve through $(1, 2)$, $(3, 5)$, $(2, -1)$, $(-2, -4)$ in which some vertical line meets the curve more than once: **not** the graph of a function.

1.2 Polynomials

The first and perhaps easiest kind of functions are polynomials. Their natural domain is $\mathbb{R}$.

Definition 1.18. Let n be a non-negative integer. A *polynomial* (in a variable x) is an algebraic expression of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_0, a_1, \ldots, a_n$ are numbers.

- For each $k \in \{0, 1, \ldots, n\}$, the number a_k is called the *coefficient* of x^k in f.
- The *degree* of f is $\deg f = n$, if the leading coefficient a_n is non-zero.
- A polynomial of degree 1 (resp. 2, 3) is called a *linear* (resp. *quadratic*, *cubic*) polynomial.

A polynomial f can be regarded as a function $f\colon \mathbb{R} \to \mathbb{R}$ if all its coefficients are real numbers. A *real root* of a polynomial f is a real number a such that $f(a) = 0$.

Recall from high school: slope, inclination angle and intercepts of a linear polynomial $f(x) = mx + c$; slopes of parallel and perpendicular lines; axis of symmetry, intercepts, vertex coordinates and extremum values of a quadratic polynomial $f(x) = ax^2 + bx + c$; ranges of polynomials of odd degree and even degree.

Example 1.19

Determine the signs of a, b and c in the given graphs of $f(x) = ax^2 + bx + c$ (i.e. whether each is positive, negative or zero).

Solution. Recall that from the graph of $f(x) = ax^2 + bx + c$:

- a is positive if the parabola opens upward, negative if it opens downward;
- $b = -2ah$ where h is the x-coordinate of the vertex;
- c is the y-intercept of the parabola.

According to the given graphs:

(a) $a > 0,\ b < 0,\ c < 0$ (b) $a < 0,\ b < 0,\ c < 0$ (c) $a > 0,\ b > 0,\ c > 0$.

Example 1.20

Let $f\colon \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x^2 - 8x + 6$. Find the natural domains and the

ranges of $g(x) = \sqrt{f(x)}$ and $h(x) = \dfrac{1}{f(x)}$.

Solution. First rewrite f in various forms:

$$f(x) = 2x^2 - 8x + 6 = 2(x-1)(x-3) = 2(x-2)^2 - 2,$$

so the graph of f is a parabola which opens upward, intersects the x-axis at 1 and at 3, and has minimum value -2 at $x = 2$.

(a) $\sqrt{f(x)}$ is well-defined iff $f(x) \geq 0$, i.e. $2(x-1)(x-3) \geq 0$, which holds iff $x \leq 1$ or $x \geq 3$. So g has natural domain $(-\infty, 1] \cup [3, +\infty)$. From the graph of f, on the domain of g, f takes every value in $[0, +\infty)$. Since the square root of every non-negative number is non-negative, and each non-negative number is the square root of a non-negative number, the range of $g = \sqrt{f}$ is also $[0, +\infty)$.

(b) $\dfrac{1}{f(x)}$ is well-defined iff $f(x) \neq 0$, i.e. $2(x-1)(x-3) \neq 0$, which holds iff $x \neq 1$ and $x \neq 3$. So h has natural domain $\mathbb{R} \setminus \{1, 3\} = (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$. Noting that the minimum value of f is -2, on the domain of h, f takes every value in $[-2, 0) \cup (0, +\infty)$. So the range of $h = \dfrac{1}{f}$ is $(-\infty, -\dfrac{1}{2}] \cup (0, +\infty)$.

Theorem 1.21 — Number of Real Roots of a Polynomial

Theorem. Let f be a polynomial of degree n. Then f has at most n real roots.

Proof. If f had $n+1$ real roots $c_0, c_1, c_2, \ldots, c_n$, then f could be factorized as

$$f(x) = (x - c_0)(x - c_1)(x - c_2) \cdots (x - c_n)\, g(x)$$

where g is some non-zero polynomial. Thus $\deg f = (n+1) + \deg g \geq n+1$, a contradiction. ■

Example 1.22

Since $f(x) = x^3 + 3x - 1$ is a polynomial of degree 3, it has at most 3 real roots.

1.3 Operations on Functions and Graphs

We build new functions from old ones that we already know, e.g. polynomials.

1.3.1 Arithmetic Operations and Composition

Definition 1.23. Let f and g be two real-valued functions. The *sum* $f + g$, *difference* $f - g$, *product* fg and *quotient* $\dfrac{f}{g}$ are defined by

- $(f+g)(x) := f(x) + g(x)$ for every x,
- $(f-g)(x) := f(x) - g(x)$ for every x,
- $(fg)(x) := f(x)g(x)$ for every x,
- $\left(\dfrac{f}{g}\right)(x) := \dfrac{f(x)}{g(x)}$ for every x which satisfies $g(x) \neq 0$.

(Think: how are the domains of these new functions related to those of f and g?)

Applying the quotient operation to polynomials, we obtain a new kind of functions. Because dividing by zero is "illegal", the natural domain of a rational function is the set of all real numbers except the roots of the polynomial in the denominator.

Definition 1.24. A *rational function* is a quotient of polynomials, i.e. a function f of the form

$$f(x) = \frac{p(x)}{q(x)}$$

where p and q are polynomials and q is not the zero polynomial.

Apart from the four usual arithmetic operations, there is a "fifth arithmetic operation" on functions.

Definition 1.25. Let f and g be two functions. The *composition* $f \circ g$ is the function defined by

$$(f \circ g)(x) := f(g(x)) \qquad \text{for every } x.$$

(Think: how is the domain of $f \circ g$ related to those of f and g?)

Example 1.26

Let $f\colon \mathbb{R} \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$ and $g(x) = x + 1$. What are $f \circ g$ and $g \circ f$? Are they the same?

Solution. For every $x \in \mathbb{R}$,

$$(f \circ g)(x) = f(g(x)) = f(x+1) = (x+1)^2 = x^2 + 2x + 1,$$

and

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1.$$

It is clear that $f \circ g \neq g \circ f$.

Example 1.27

Consider the rational functions $f(x) = \dfrac{1}{x}$ and $g(x) = \dfrac{x+1}{x-2}$, defined on their natural domains respectively. Find $f \circ f$, $f \circ g$ and $g \circ f$.

Solution. (The task includes finding not only the formulas but also the domains.) First note that the domain of f is $\mathbb{R} \setminus \{0\}$, and the domain of g is $\mathbb{R} \setminus \{2\}$.

- For each $x \in \mathbb{R} \setminus \{0\}$,

$$(f \circ f)(x) = f(f(x)) = f\left(\frac{1}{x}\right) = \frac{1}{\frac{1}{x}} = x.$$

  Therefore $f \circ f\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}$ is given by $(f \circ f)(x) = x$.
- For each $x \in \mathbb{R} \setminus \{2\}$,

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{x+1}{x-2}\right) = \frac{1}{\frac{x+1}{x-2}} = \frac{x-2}{x+1} \qquad \text{except when } x = -1.$$

  Therefore $f \circ g\colon \mathbb{R} \setminus \{-1, 2\} \to \mathbb{R}$ is given by $(f \circ g)(x) = \dfrac{x-2}{x+1}$.
- For each $x \in \mathbb{R} \setminus \{0\}$,

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{1}{x}\right) = \frac{\frac{1}{x} + 1}{\frac{1}{x} - 2} = \frac{1+x}{1-2x} \qquad \text{except when } x = \frac{1}{2}.$$

  Therefore $g \circ f\colon \mathbb{R} \setminus \{0, \frac{1}{2}\} \to \mathbb{R}$ is given by $(g \circ f)(x) = \dfrac{1+x}{1-2x}$.

Remark 1.28 — Hidden Requirements on Domains of Compositions

In Example 1.27 we note the following important issue:

- Although the formula of $f \circ f$ can be written as $(f \circ f)(x) = x$, its domain is $\mathbb{R} \setminus \{0\}$ instead of the "natural domain" $\mathbb{R}$. There is a hidden requirement that $x \neq 0$.
- Although the formula of $f \circ g$ can be written as $(f \circ g)(x) = \dfrac{x-2}{x+1}$, its domain is $\mathbb{R} \setminus \{-1, 2\}$ instead of the "natural domain" $\mathbb{R} \setminus \{-1\}$. There is a hidden requirement that $x \neq 2$.
- Although the formula of $g \circ f$ can be written as $(g \circ f)(x) = \dfrac{1+x}{1-2x}$, its domain is $\mathbb{R} \setminus \{0, \frac{1}{2}\}$ instead of the "natural domain" $\mathbb{R} \setminus \{\frac{1}{2}\}$. There is a hidden requirement that $x \neq 0$.

1.3.2 Piecewise-Defined Functions; the Absolute Value Function

We may define a function by giving different rules of assignment to different parts of the domain; such functions are called *piecewise-defined functions*.

Example 1.29

Let $f\colon \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1-x & \text{if } x \leq -1 \\ x^2 & \text{if } x > -1 \end{cases}.$$

Evaluate $f(-2)$, $f(-1)$ and $f(0)$, and sketch the graph of f.

Solution.

- Since $-2 \leq -1$, we have $f(-2) = 1 - (-2) = 3$.
- Since $-1 \leq -1$, we have $f(-1) = 1 - (-1) = 2$.
- Since $0 > -1$, we have $f(0) = 0^2 = 0$.

The graph consists of the line $y = 1 - x$ on $(-\infty, -1]$ (with a solid dot at $(-1, 2)$) joined to the parabola $y = x^2$ on $(-1, +\infty)$ (with a hollow dot at $(-1, 1)$). (A solid dot denotes an endpoint that belongs to the graph; a hollow dot denotes one that does not.)

A frequently used piecewise-defined function is the absolute value function.

Definition 1.30. The *absolute value function* is the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}.$$

It is commonly denoted as $f(x) = |x|$. (Its graph is V-shaped with vertex at the origin.)

Corollary 1.31. Let x be a real number. Then

$$\sqrt{x^2} = |x|.$$

In particular, we must have $|x| \geq 0$. (Notice how $\sqrt{x^2}$ and $(\sqrt{x})^2$ are different from each other.)

Theorem 1.32 — Properties of the Absolute Value

Theorem. Let a and b be real numbers. Then

(i) $|ab| = |a||b|$ (ii) $|a^2| = |a|^2 = a^2$ (iii) $|-a| = |a|$ (iv) $\left|\dfrac{a}{b}\right| = \dfrac{|a|}{|b|}$ if $b \neq 0$.

Remark 1.33 — Removing the Absolute Value Sign

To remove the absolute value sign from an expression $|g(x)|$, first check whether $g(x)$

inside is negative or not:
- if $g(x) < 0$, write an extra negative sign after removing the absolute value sign: $|g(x)| = -g(x)$;
- if $g(x) \geq 0$, do nothing after removing the absolute value sign: $|g(x)| = g(x)$.

Example 1.34

Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = |x+1| + |x-2|$. Rewrite the formula for f into piecewise form.

Solution. $|x+1|$ and $|x-2|$ change their forms near -1 and near 2, so we divide $\mathbb{R}$ into the three intervals $(-\infty, -1)$, $[-1, 2)$, $[2, +\infty)$.
- If $x \in (-\infty, -1)$, then $x+1 < 0$ and $x-2 < 0$, so $|x+1| + |x-2| = -(x+1) - (x-2) = -2x+1$.
- If $x \in [-1, 2)$, then $x+1 \geq 0$ but $x-2 < 0$, so $|x+1| + |x-2| = (x+1) - (x-2) = 3$.
- If $x \in [2, +\infty)$, then $x+1 \geq 0$ and $x-2 \geq 0$, so $|x+1| + |x-2| = (x+1) + (x-2) = 2x-1$.

In summary,

$$f(x) = \begin{cases} -2x+1 & \text{if } x \in (-\infty, -1) \\ 3 & \text{if } x \in [-1, 2) \\ 2x-1 & \text{if } x \in [2, +\infty) \end{cases}.$$

Example 1.35

For each inequality, find all real numbers x which satisfy it:

$$\text{(a) } |2x-5| > 7 \qquad \text{(b) } |x-2| + |2x+1| \leq 4$$

Solution.

(a) If $2x-5 \geq 0$, i.e. $x \geq \frac{5}{2}$, the inequality becomes $2x-5 > 7$, so $x > 6$. Combining $x \geq \dfrac{5}{2}$ and $x > 6$, every $x > 6$ is a solution.

If $2x-5 < 0$, i.e. $x < \frac{5}{2}$, the inequality becomes $-(2x-5) > 7$, so $x < -1$. Combining $x < \dfrac{5}{2}$ and $x < -1$, every $x < -1$ is also a solution.

Therefore the solutions consist of all $x \in (-\infty, -1) \cup (6, +\infty)$.

(b) If $x \in (-\infty, -\frac{1}{2})$, then both $x-2 < 0$ and $2x+1 < 0$, and the inequality becomes

$$-(x-2) - (2x+1) \leq 4,$$

so $x \geq -1$. Combining with $x < -\frac{1}{2}$, every $x \in [-1, -\frac{1}{2})$ is a solution.

If $x \in [-\frac{1}{2}, 2)$, then $x-2 < 0$ but $2x+1 \geq 0$, and the inequality becomes

$$-(x-2) + (2x+1) \leq 4,$$

so $x \leq 1$. Combining with $-\frac{1}{2} \leq x < 2$, every $x \in [-\frac{1}{2}, 1]$ is a solution.

If $x \in [2, +\infty)$, then both $x-2 \geq 0$ and $2x+1 \geq 0$, and the inequality becomes

$$(x-2) + (2x+1) \leq 4,$$

so $x \leq \dfrac{5}{3}$. But no x satisfies both $x \geq 2$ and $x \leq \dfrac{5}{3}$, so this case gives no solutions.

Therefore the solutions consist of all $x \in [-1, 1]$.

Corollary 1.36 — Equations and Inequalities Involving Absolute Values

Corollary. Let y be a real number and c a non-negative real number.

(i) $|y| < c$ if and only if $-c < y < c$.
(ii) $|y| = c$ if and only if either $y = c$ or $y = -c$.
(iii) $|y| > c$ if and only if either $y > c$ or $y < -c$.

If c were negative, then $|y| < c$ is always impossible, $|y| = c$ is always impossible, and $|y| > c$ is always true.

Remark 1.37 — Absolute Value as Distance

Let $a, b \in \mathbb{R}$. The quantity $|a - b|$ represents the *distance* between a and b:

$$|a - b| = \begin{cases} b - a & \text{if } a < b \\ a - b & \text{if } a \geq b \end{cases}.$$

Now suppose $a \in \mathbb{R}$ is fixed and $\delta > 0$ is fixed. The inequality

$$|x - a| < \delta$$

represents all real numbers x whose distance from a is less than δ; i.e. its solutions are $x \in (a - \delta, a + \delta)$, the open interval centered at a with radius δ. (δ is the Greek letter "d", standing for "distance" here.)

Theorem 1.38 — Triangle Inequality

Theorem. Let a and b be real numbers. Then
(i) $-|a| \leq a \leq |a|$,
(ii) (Triangle inequality) $|a + b| \leq |a| + |b|$.

Proof. The proof of (i) is easy and is omitted. Given real numbers a and b, (i) implies

$$-|a| \leq a \leq |a| \qquad \text{and} \qquad -|b| \leq b \leq |b|;$$

thus

$$-(|a| + |b|) \leq a + b \leq |a| + |b|,$$

and so $|a + b| \leq |a| + |b|$ by Corollary 1.36 (i). Therefore (ii) follows. ■

Example 1.38′

Let x be a real number and $\delta > 0$. Prove each of the following statements.
(a) If $|x - 2| < 1$, then $|x + 2| < 5$.
(b) If $|x - 1| < \delta$, then $|x^2 - 6x + 5| < \delta(\delta + 4)$.

Solution.
(a) If $|x - 2| < 1$, then $-1 < x - 2 < 1$, i.e. $3 < x + 2 < 5$. Thus $|x + 2| < 5$.
Alternatively: if $|x - 2| < 1$, then by the triangle inequality,

$$|x + 2| = |(x - 2) + 4| \leq |x - 2| + 4 < 1 + 4 = 5.$$

(b) If $|x - 1| < \delta$, then by the triangle inequality,

$$|x - 5| = |(x - 1) + (-4)| \leq |x - 1| + |-4| < \delta + 4.$$

Therefore

$$|x^2 - 6x + 5| = |(x - 1)(x - 5)| = |x - 1| \cdot |x - 5| < \delta(\delta + 4).$$

Example 1.38″

Find an example of a positive number δ such that the following statement is true: "if $|x - 2| < \delta$, then $|x| > 1$."

Solution. Note that $|x - 2| < \delta$ iff $-\delta < x - 2 < \delta$, i.e.

$$2 - \delta < x < 2 + \delta.$$

To obtain the conclusion $|x| > 1$, it is enough to require $x > 1$. Choosing $2 - \delta = 1$, we see that $\delta = 1$ is an appropriate example. (Not the only one; every $\delta \in (0, 1]$ works.) Two ways to prove the statement for $\delta = 1$:

- If $|x - 2| < 1$, then $-1 < x - 2 < 1$, i.e. $1 < x < 3$. Therefore $|x| > 1$.
- Alternatively, if $|x - 2| < 1$, then by the triangle inequality,

$$2 = |x - (x - 2)| \leq |x| + |x - 2| < |x| + 1,$$

and so $|x| > 2 - 1 = 1$.

1.3.3 The Floor Function

Definition 1.39. The *floor function* (or *greatest integer function*) is the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = n \qquad \text{if } n \leq x < n + 1,$$

for each integer n. In other words, for each real number x, $f(x)$ is the greatest integer that is smaller than or equal to x. (In particular, when $x \geq 0$, $f(x)$ is the "integer part" of x.) This function is commonly denoted $f(x) = \lfloor x \rfloor$. (Its graph is a staircase: on each interval $[n, n + 1)$ it is the constant n, with a solid dot at (n, n) and a hollow dot at $(n + 1, n)$.)

Example 1.40

Some values of the floor function:

- $\lfloor 1 \rfloor = 1$, $\lfloor 2 \rfloor = 2$, $\lfloor 3 \rfloor = 3$, ... In general, $\lfloor n \rfloor = n$ if $n \in \mathbb{Z}$.
- $\lfloor 1.5 \rfloor = 1$, $\lfloor 2.75 \rfloor = 2$.
- $\lfloor \sqrt{2} \rfloor = 1$, $\lfloor \pi \rfloor = 3$.
- $\lfloor -1.5 \rfloor = -2$, $\lfloor -2.75 \rfloor = -3$.

1.3.4 Transformations of Graphs

Apart from operations on functions, we can apply transformations on the graph of a function $\{(x, y) : y = f(x)\}$ to obtain graphs of related functions; they correspond to actions on the variables x and y.

Theorem 1.41 — Six Basic Transformations

Theorem. Let $k > 0$, $c > 1$, and the graph of a function f be given. Then the graph of each of the following functions is obtained by the corresponding action on the graph of f:

- $y = f(x) - k$: shift the graph of $y = f(x)$ **downward** by k units
- $y = f(x + k)$: shift the graph of $y = f(x)$ to the **left** by k units
- $y = \dfrac{1}{c} f(x)$: **compress** the graph of $y = f(x)$ **vertically** by a factor of c
- $y = f(cx)$: **compress** the graph of $y = f(x)$ **horizontally** by a factor of c
- $y = -f(x)$: **reflect** the graph of $y = f(x)$ across the x-axis
- $y = f(-x)$: **reflect** the graph of $y = f(x)$ across the y-axis

These basic transformations can be further composed with each other to produce more complicated transformations.

Remark 1.42 — How to Remember Theorem 1.41

Let $k > 0$ and $c > 1$. One can think of

- replacing y by $y + k$ as shifting the **coordinate plane** (NOT the graph) upward by k units,

- replacing x by $x + k$ as shifting the coordinate plane to the right by k units,
- replacing y by cy as stretching the coordinate plane vertically by a factor of c, and so on.

Example 1.43

Given the graph of $y = |x|$, sketch the graph of $y = 4-|3x+3|$ on the same coordinate plane.
Solution. Start with the graph of $y = |x|$. A horizontal compression by a factor of 3 gives the graph of $y = |3x|$; a left translation of 1 unit then gives the graph of

$$y = |3(x + 1)| = |3x + 3|$$

(a V-shape with vertex at $(-1, 0)$). Next, a reflection across the x-axis gives the graph of $y = -|3x + 3|$ (an upside-down V with vertex at $(-1, 0)$), and an upward translation of 4 units gives the graph of

$$y = 4 - |3x + 3|,$$

an upside-down V with vertex at $(-1, 4)$, which is the graph we want.

1.4 Properties of Functions

1.4.1 Odd, Even and Periodic Functions

Definition 1.44. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function.

(i) f is an *odd function* if $f(-x) = -f(x)$ for every $x \in \mathbb{R}$;
(ii) f is an *even function* if $f(-x) = f(x)$ for every $x \in \mathbb{R}$;
(iii) f is a *periodic function* if there exists $c > 0$ such that $f(x + c) = f(x)$ for every $x \in \mathbb{R}$. This positive number c is called a *period* of f. If there is a smallest such positive number c, it is called the *fundamental period* of f.

Remark 1.45 — Symmetries of Graphs

The graph of an odd function is rotationally symmetric about the origin by half a revolution. The graph of an even function is symmetric across the y-axis. The graph of a periodic function is translationally symmetric along the x-axis.

Example 1.46

The simplest examples of odd and even functions are the monomials $f(x) = x^n$, where n is a non-negative integer: f is odd if n is odd, and even if n is even.

Example 1.47

Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x|x|$ and $g(x) = x^2|x|$. Determine whether each is odd, and whether each is even.
Solution. For every $x \in \mathbb{R}$,

$$f(-x) = (-x)|-x| = -x|x| = -f(x) \qquad \text{and} \qquad g(-x) = (-x)^2|-x| = x^2|x| = g(x).$$

So f is an odd function and g is an even function.

Remark 1.48

Odd functions and even functions are **not** opposite concepts (unlike the same pair of terminology for integers). Some functions are both odd and even, while some are neither odd nor even. (Can you construct examples for these situations?)

Example 1.49

Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function and suppose there exists a positive constant a such that

$$f(x-y) = f(x)f(y) - f(a-x)f(a+y)$$

for every pair of real numbers x and y. If $f(0) = 1$, (a) find the value of $f(a)$, and (b) hence show that f is even.

Solution. The given equality holds for every $x, y \in \mathbb{R}$, so we may replace x, y by any real numbers we like.

(a) To find $f(a)$, put $x = y = 0$:

$$f(0-0) = f(0)f(0) - f(a-0)f(a+0) \;\Rightarrow\; 1 = 1^2 - [f(a)]^2 \;\Rightarrow\; f(a) = 0.$$

(b) To prove f is even, we show $f(-y) = f(y)$ for every real y. Putting $x = 0$ into the given equality, for every $y \in \mathbb{R}$,

$$f(-y) = f(0-y) = f(0)f(y) - f(a-0)f(a+y) = 1 \cdot f(y) - 0 \cdot f(a+y) = f(y).$$

Therefore f is an even function. ■

Example 1.50

The cosine and sine functions $\cos, \sin\colon \mathbb{R} \to \mathbb{R}$ are periodic:

$$\cos(x+2\pi) = \cos x \qquad \text{and} \qquad \sin(x+2\pi) = \sin x \qquad \text{for every } x \in \mathbb{R},$$

so 2π is a period of these two functions. It turns out that 2π is also their fundamental period.

Example 1.51

Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = x - \lfloor x \rfloor$. For each $x \in \mathbb{R}$, there exists an integer n such that $n \le x < n+1$, so

$$\lfloor x \rfloor = n \qquad \text{and} \qquad \lfloor x+1 \rfloor = n+1.$$

Thus we always have

$$f(x+1) = (x+1) - \lfloor x+1 \rfloor = (x+1) - (n+1) = x - n = x - \lfloor x \rfloor = f(x),$$

which shows that f is periodic and 1 is a period of f. (Exercise: prove that 1 is the fundamental period of f.)

Example 1.52 (Even Periodic Extension)

Let f be defined on $[0,1]$ by $f(x) = x$ for every $x \in [0,1]$ (graph: the line segment from $(0,0)$ to $(1,1)$). Then we can extend f so that $f\colon \mathbb{R} \to \mathbb{R}$ becomes an **even periodic** function with period 2. The graph of the extension is a triangular wave: it rises from 0 at even integers to 1 at odd integers and falls back, repeating with period 2.

Example 1.53 (Odd Periodic Extension)

Let f be defined on $[0,2]$ by

$$f(x) = \begin{cases} x & \text{if } x \in [0,1] \\ 2-x & \text{if } x \in [1,2] \end{cases}$$

(graph: a tent rising from $(0,0)$ to the peak $(1,1)$ and falling to $(2,0)$). Then we can extend f so that $f\colon \mathbb{R} \to \mathbb{R}$ becomes an **odd periodic** function with period 4. The graph of the extension is a triangular wave alternating between peaks of height 1 (at $x = 1+4k$) and troughs of depth -1 (at $x = -1+4k$), passing through 0 at every

even integer.

1.4.2 Monotonicity

Definition 1.54. Let A be a non-empty subset of $\mathbb{R}$. A function $f\colon A \to \mathbb{R}$ is said to be

$$\left\{\begin{array}{l}\text{(monotonic) increasing}\\ \text{(monotonic) decreasing}\\ \text{strictly increasing}\\ \text{strictly decreasing}\end{array}\right\} \text{ on } A \quad \text{if} \quad \left\{\begin{array}{l}f(a) \geq f(b)\\ f(a) \leq f(b)\\ f(a) > f(b)\\ f(a) < f(b)\end{array}\right\} \quad \text{for every pair } a > b \text{ in } A.$$

Example 1.55

Let $f\colon \mathbb{R} \to \mathbb{R}$ be the monomial $f(x) = x^2$. Show that f is strictly increasing on $[0, +\infty)$.

Solution. Proof. For all non-negative real numbers a and b with $a > b$, we have

$$a^2 - b^2 = (a-b)(a+b) > 0$$

as both factors are positive. Therefore $f(a) = a^2 > b^2 = f(b)$, which shows that f is strictly increasing on $[0, +\infty)$. (Or simply: $a^2 > ab \geq b^2$ since $a > b \geq 0$.) ■

Example 1.56

Let $f\colon \mathbb{R} \to \mathbb{R}$ be the monomial $f(x) = x^3$. Show that f is strictly increasing on $\mathbb{R}$.

Solution. Proof. For all real numbers a and b with $a > b$, consider $a^3 - b^3$ under three cases:

- If $a > b \geq 0$, then

$$a^3 - b^3 = \underbrace{(a-b)}_{>0}\big(\underbrace{a^2}_{>0} + \underbrace{ab}_{\geq 0} + \underbrace{b^2}_{\geq 0}\big) > 0.$$

- If $a \geq 0 > b$, then $a^3 \geq 0$ and $b^3 < 0$, so obviously $a^3 - b^3 > 0$.
- If $0 > a > b$, then

$$a^3 - b^3 = \underbrace{(a-b)}_{>0}\big(\underbrace{a^2}_{>0} + \underbrace{ab}_{>0} + \underbrace{b^2}_{>0}\big) > 0.$$

Therefore in each case $f(a) = a^3 > b^3 = f(b)$, which shows that f is strictly increasing on $\mathbb{R}$. ■

Alternative proof. For all real numbers a and b with $a > b$,

$$\begin{aligned} a^3 - b^3 = (a-b)(a^2+ab+b^2) &= (a-b)\left(\left(a^2 + ab + \frac{1}{4}b^2\right) + \frac{3}{4}b^2\right) \\ &= (a-b)\left(\left(a + \frac{1}{2}b\right)^2 + \left(\frac{\sqrt{3}}{2}b\right)^2\right). \end{aligned}$$

The first factor $a - b$ is obviously positive. The second factor is also positive (why is it non-zero?). Therefore $f(a) = a^3 > b^3 = f(b)$, which shows that f is strictly increasing on $\mathbb{R}$. ■

Example 1.57

In general, let n be a positive integer and $f\colon \mathbb{R} \to \mathbb{R}$ the monomial $f(x) = x^n$.

- If n is odd, then f is strictly increasing on $\mathbb{R}$.

- If n is even, then f is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$.

(Exercise: prove this by mimicking Examples 1.55–1.56.)

Example 1.58

Let $f\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be $f(x) = \dfrac{1}{x}$. For all positive real numbers a and b with $a > b$,

$$\frac{1}{a} - \frac{1}{b} = \frac{\overbrace{b-a}^{<0}}{\underbrace{a}_{>0}\underbrace{b}_{>0}} < 0.$$

Therefore $f(a) = \dfrac{1}{a} < \dfrac{1}{b} = f(b)$, which shows that f is strictly decreasing on $(0, +\infty)$. Similarly, f is strictly decreasing on $(-\infty, 0)$ as well.

Think

Does Example 1.58 imply that $f(x) = \dfrac{1}{x}$ is strictly decreasing on $\mathbb{R} \setminus \{0\}$? **No!** We have $1 > -1$ and yet $f(1) > f(-1)$.

Example 1.59

Let $f\colon \mathbb{R} \to \mathbb{R}$ be the absolute value function $f(x) = |x|$. Then f is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$.

Example 1.60

Let $f\colon \mathbb{R} \to \mathbb{R}$ be the floor function $f(x) = \lfloor x \rfloor$. Then f is monotonic increasing on $\mathbb{R}$.

1.4.3 Boundedness

Definition 1.61. Let A be a set and $f\colon A \to \mathbb{R}$ a function.

(i) If there exists a real number M such that $f(x) \leq M$ for every $x \in A$, then f is *bounded from above* on A, and M is called an *upper bound* of f on A.

(ii) If there exists a real number m such that $f(x) \geq m$ for every $x \in A$, then f is *bounded from below* on A, and m is called a *lower bound* of f on A.

(iii) If f is both bounded from above and bounded from below on A, then f is *bounded* on A.

Example 1.62

Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$. Then f is not bounded from above, but is bounded from below by 0. Note that f is also bounded from below by every negative number. We say 0 is the *greatest lower bound* of f.

Example 1.63

Let $f\colon \mathbb{R} \to \mathbb{R}$ be

$$f(x) = \frac{1}{x^2 + 2x + 2} = \frac{1}{(x+1)^2 + 1}.$$

Since $(x+1)^2 + 1 \geq 1$ for every $x \in \mathbb{R}$, it follows that

$$0 < f(x) \leq 1 \qquad \text{for every } x \in \mathbb{R}.$$

So f is bounded from above by 1 and bounded from below by 0; in other words, f is a bounded function. Here 1 is the *least upper bound* of f and 0 is the *greatest lower bound* of f.

Example 1.64

Let $a < b$ be real numbers and $f\colon [a,b] \to \mathbb{R}$ an increasing function. Then

$$f(a) \leq f(x) \leq f(b) \qquad \text{for every } x \in [a,b],$$

so f is bounded on $[a,b]$. Similarly, a decreasing function on $[a,b]$ is also bounded on $[a,b]$.

1.4.4 Injectivity, Surjectivity and Bijectivity

Definition 1.65. Let A and B be sets and $f\colon A \to B$ a function.

(i) f is *one-to-one* (or *injective*) if for every $a, b \in A$, $a \neq b$ implies $f(a) \neq f(b)$, i.e. different elements in A take different values in B. (Equivalently: for every $a, b \in A$, $f(a) = f(b)$ implies $a = b$.)

(ii) f is *onto* (or *surjective*) if for each y in the codomain B, there exists (at least one) x in the domain A such that $f(x) = y$. (Equivalently: the range of f equals the whole codomain of f.)

(iii) f is a *one-to-one correspondence* (or *bijective*) if it is both one-to-one and onto.

Example 1.66

Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = 2x + 1$.

- For every $a, b \in \mathbb{R}$, if we assume $f(a) = f(b)$, then

  $$2a + 1 = 2b + 1 \;\Rightarrow\; 2a = 2b \;\Rightarrow\; a = b.$$

  This shows that f is one-to-one.

- For each $y \in \mathbb{R}$, consider the real number $x = \dfrac{y-1}{2}$. Then

  $$f(x) = f\left(\frac{y-1}{2}\right) = 2\left(\frac{y-1}{2}\right) + 1 = (y-1) + 1 = y.$$

  This shows that f is onto.

Therefore f is bijective.

Remark 1.67 — Horizontal Line Test

A function f is one-to-one if and only if every horizontal line in the coordinate plane intersects the graph of f at no more than one point.

Example 1.68

Every strictly increasing function is one-to-one. Every strictly decreasing function is also one-to-one.

Example 1.69

For each function, determine whether it is one-to-one and whether it is onto:

(a) $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

(b) $g\colon [0,+\infty) \to \mathbb{R}$, $g(x) = x^2$
(c) $h\colon [0,+\infty) \to [0,+\infty)$, $h(x) = x^2$

Solution. (To disprove a "for-all" statement we just need a counter-example.)

(a) Since 1 and -1 are two different real numbers with $f(1) = 1 = f(-1)$, f is not one-to-one. Moreover, for the real number -2, there is no real x such that $f(x) = x^2 = -2$ (since $x^2 \geq 0$); so f is not onto either.

(b) Suppose a, b are non-negative reals with $g(a) = g(b)$. Then $a^2 = b^2$; taking positive square roots on both sides gives $a = b$, so g is one-to-one. (Why does this reasoning fail for f in (a)?) There is still no real x with $g(x) = -2$, so the same proof as in (a) shows g is not onto.

(c) The same proof as in (b) shows h is one-to-one. This time, for each non-negative real y, consider $x = \sqrt{y}$. Then

$$h(x) = h(\sqrt{y}) = (\sqrt{y})^2 = y,$$

which shows h is onto. (Why does this reasoning fail for f and g in (a) and (b)?) Therefore h is bijective.

Remark 1.70

From Example 1.69 we observe:

- Whether a function is one-to-one depends very much on its **domain**. If a function is not one-to-one, it is sometimes possible to make it one-to-one by shrinking its domain.
- Whether a function is onto depends on its **codomain**. If a function is not onto, we can make it onto by replacing its codomain by just its range.

1.4.5 Inverse Functions

One-to-one functions are important because they have inverses. An inverse can be understood as a "reverse machine" which undoes everything the given function does.

Theorem 1.71 — Existence and Uniqueness of the Inverse

Theorem. Let A and B be sets, let $f\colon A \to B$ be a one-to-one function and let $C \subseteq B$ be the range of f. Then there exists a unique function $g\colon C \to A$ such that

$$(f \circ g)(y) = y \quad \text{for every } y \in C \qquad \text{and} \qquad (g \circ f)(x) = x \quad \text{for every } x \in A.$$

Proof. We define $g\colon C \to A$ as follows. For each $y \in C$, since C is the range of f, there exists $x \in A$ such that $y = f(x)$; there is only one such x because f is one-to-one. (If there were no such x or more than one such x, we could not define a function this way.) So we can define

$$g(y) := x.$$

Now we check that this g indeed satisfies

$$(f \circ g)(y) = f(g(y)) = f(x) = y \quad \text{for every } y \in C,$$
$$(g \circ f)(x) = g(f(x)) = g(y) = x \quad \text{for every } x \in A.$$

To prove uniqueness, let $h\colon C \to A$ be a function such that $(f \circ h)(y) = y$ for every $y \in C$. Then $h(y) = (g \circ f)(h(y)) = g((f \circ h)(y)) = g(y)$ for every $y \in C$, so $g = h$. ■

Definition 1.72. Let f be a one-to-one function. The function g constructed as in Theorem 1.71 is called the *inverse* of f, denoted f^{-1}.

Warning

f^{-1} does **NOT** mean $\dfrac{1}{f}$.

Example 1.66 (continued)

Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = 2x + 1$. We have seen that f is one-to-one and onto (i.e. its range is the whole $\mathbb{R}$). Let $g\colon \mathbb{R} \to \mathbb{R}$ be $g(x) = \dfrac{x-1}{2}$. Then for every $x \in \mathbb{R}$,

$$(f \circ g)(x) = f\left(\frac{x-1}{2}\right) = 2\left(\frac{x-1}{2}\right) + 1 = x,$$

$$(g \circ f)(x) = g(2x+1) = \frac{(2x+1)-1}{2} = x.$$

So g is the inverse of f, i.e. $f^{-1}\colon \mathbb{R} \to \mathbb{R}$ is given by $f^{-1}(x) = \dfrac{x-1}{2}$.

Remark 1.73. Given a one-to-one function f, Theorem 1.71 implies that

- the domain of f^{-1} is (defined to be) the range of f, and
- the range of f^{-1} is the domain of f.

Corollary 1.74. Let A and B be sets. If $f\colon A \to B$ is bijective, then f has an inverse $f^{-1}\colon B \to A$. Moreover, (i) f^{-1} is also bijective, and (ii) $(f^{-1})^{-1} = f$.

Remark 1.75 — How to Find an Inverse

To find the inverse of a given one-to-one function f:

(i) For each y in the range of f, write $y = f(x)$.

(ii) Find a solution x (in the domain of f) to this equation in terms of y. This is $f^{-1}(y)$.

(iii) Rename the input as x instead of y.

Example 1.76

Let $f\colon (-\infty, 0] \to \mathbb{R}$ be defined by $f(x) = x^2$. Find the inverse of f and sketch the graphs of f and of f^{-1} on the same coordinate plane.

Solution. The range of f is $[0, +\infty)$. Writing $y = f(x)$ with $y \in [0, +\infty)$, we solve for x in the domain $(-\infty, 0]$ of f:

$$y = x^2 \;\Rightarrow\; x = -\sqrt{y}.$$

Therefore $f^{-1}(y) = -\sqrt{y}$. We usually prefer the symbol x (instead of y) for the input, so the inverse $f^{-1}\colon [0, +\infty) \to \mathbb{R}$ is given by

$$f^{-1}(x) = -\sqrt{x}.$$

The graph of f is the left half of the parabola $y = x^2$; the graph of f^{-1} is its reflection across the line $y = x$, i.e. the lower half of the sideways parabola $x = y^2$.

Remark 1.77. Suppose the graph of a one-to-one function f is given. Then the graph of f^{-1} is obtained by reflecting the graph of f across the line with slope 1 that passes through the origin (i.e. the line $y = x$).

Theorem 1.78 — Inverse of a Composition

Theorem. Let f and g be one-to-one functions. Then $f \circ g$ is also one-to-one and

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

Proof. To show $f \circ g$ is one-to-one, assume $(f \circ g)(a) = (f \circ g)(b)$; we want to show $a = b$. Now $f(g(a)) = f(g(b))$. Since f is one-to-one, $g(a) = g(b)$. Since g is one-to-one, $a = b$ as required. Therefore $f \circ g$ is one-to-one.
For the second statement, note that for every x in the range of $f \circ g$,

$$(f \circ g)\big((f \circ g)^{-1}(x)\big) = x,$$

i.e. $f\big(g\big((f \circ g)^{-1}(x)\big)\big) = x$. Now f has an inverse f^{-1}; applying f^{-1} on both sides,

$$g\big((f \circ g)^{-1}(x)\big) = f^{-1}(x).$$

Similarly g has an inverse g^{-1}; applying g^{-1} on both sides,

$$(f \circ g)^{-1}(x) = g^{-1}\big(f^{-1}(x)\big) = (g^{-1} \circ f^{-1})(x).$$

Therefore $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. ■

1.5 Elementary Functions

Apart from polynomials and rational functions introduced in the previous sections, there are many other types of elementary functions useful in daily life.

1.5.1 Power Functions

Definition 1.79. Let a be a real number. Then the function $f(x) = x^a$ is called a *power function*. (For example, $f(x) = x^2$, $f(x) = x^{\frac{1}{2}} = \sqrt{x}$, $f(x) = x^{-1} = \dfrac{1}{x}$.)

Some important special cases of power functions include

- monomials (when a is a non-negative integer),
- the nth root function (when $a = \dfrac{1}{n}$ and n is a positive integer), and
- the reciprocal (when $a = -1$).

(Graphs of $y = x^2$, $y = x$, $y = x^{\frac{1}{2}}$, $y = x^{-1}$ and $y = x^{-2}$ were sketched: when $a > 0$ the graph passes through $(0, 0)$ and $(1, 1)$; when $a < 0$ the graph passes through $(1, 1)$ and has the coordinate axes as asymptotes. Think: what are the domain and the range of a power function? They depend on the exponent a.)

1.5.2 Exponential Functions

Definition 1.80. Let $a > 0$ and $a \neq 1$. The function $f(x) = a^x$ is called the *exponential function with base a*. (When $a = 1$ this is just a boring constant function.)

Comparing the graphs of $y = a^x$ for various bases a (e.g. $2, 3, 5$ versus $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}$), we observe:

- each graph passes through the point $(0, 1)$, and the function values are always positive, so the range is $(0, +\infty)$;
- if $a > 1$, the graph is strictly increasing, with the negative x-axis as a horizontal asymptote;
- if $0 < a < 1$, the graph is strictly decreasing, with the positive x-axis as a horizontal asymptote.

Lemma 1.81 — Exponential Functions are One-to-One

Lemma. For each $a > 0$ and $a \neq 1$, the exponential function $f(x) = a^x$ is a one-to-one function with domain $\mathbb{R}$ and range $(0, +\infty)$.

Remark 1.82 — Power Functions vs. Exponential Functions

Do not mix up the power functions and the exponential functions! The input x in a power function x^a is located at the **base**, while in an exponential function a^x it is located at the **exponent**.

Definition 1.83. In calculus we will often encounter a special base $e \approx 2.71828$, which is an irrational number. The particular exponential function $f(x) = e^x$ with base e is called the *natural exponential function* and is sometimes denoted also as $f(x) = \exp x$.

Example 1.84

Find the natural domain and the range of the function

$$f(x) = \frac{1}{\sqrt{1 - e^x}}.$$

Solution. The expression $\dfrac{1}{\sqrt{1 - e^x}}$ is well-defined if and only if $1 - e^x > 0$, which holds if and only if $x < 0$; so the natural domain of f is $(-\infty, 0)$. To find the range of f, we need to answer the question "for what values of y does the equation $y = \dfrac{1}{\sqrt{1 - e^x}}$ have a solution in $(-\infty, 0)$?". Clearly we need $y > 0$. Also, rearranging the equation we get

$$e^x = 1 - \frac{1}{y^2},$$

which implies that $1 - \dfrac{1}{y^2} > 0$, i.e. $y > 1$ or $y < -1$. Combining these two requirements we have $y > 1$, so the range of f is $(1, +\infty)$.

1.5.3 Logarithmic Functions

Definition 1.85. Let $a > 0$ and $a \neq 1$. The inverse of the exponential function with base a is called the *logarithmic function with base a* and is denoted by $f(x) = \log_a x$.

As expected, we can obtain the graph of the logarithmic function $y = \log_a x$ by reflecting the graph of the corresponding exponential function $y = a^x$ across the line $y = x$. From the graphs for various bases a (e.g. $2, 3, 5$ versus $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}$), we observe:

- each graph passes through the point $(1, 0)$ and is defined for positive numbers only, so the domain is $(0, +\infty)$;
- the y-axis is a vertical asymptote in each case;
- if $a > 1$, the graph is strictly increasing; if $0 < a < 1$, it is strictly decreasing.

Lemma 1.86 — Logarithmic Functions are One-to-One

Lemma. For each $a > 0$ and $a \neq 1$, the logarithmic function $f(x) = \log_a x$ is a one-to-one function with domain $(0, +\infty)$ and range $\mathbb{R}$.

Definition 1.87. The logarithmic function $f(x) = \log_e x$ with the special base $e \approx 2.71828$ is called the *natural logarithmic function* and is commonly denoted by $f(x) = \ln x$.

Since there is an inverse relation between the exponential and logarithmic functions (with the same base), we obtain the following relations from Theorem 1.71.

Proposition 1.88 — Cancellation Relations between a^x and $\log_a x$

Proposition. Let a be a positive number with $a \neq 1$. Then
(i) $a^{\log_a x} = x$ and in particular $e^{\ln x} = x$ for every $x > 0$,
(ii) $\log_a a^x = x$ and in particular $\ln e^x = x$ for every $x \in \mathbb{R}$.

Remark 1.89 — Laws of Indices and Laws of Logarithms

Do remember the laws of indices and the laws of logarithms in your computations.

Laws of indices	*Laws of logarithms*
$a^0 = 1$	$\log_a 1 = 0$
$a^1 = a$	$\log_a a = 1$
$a^{x+y} = a^x a^y$	$\log_a x + \log_a y = \log_a xy$
$a^{x-y} = \dfrac{a^x}{a^y}$	$\log_a x - \log_a y = \log_a \dfrac{x}{y}$
$a^{xy} = (a^x)^y$	$\log_a x^y = y \log_a x$
$a^x b^x = (ab)^x$	$\log_a b = \dfrac{\log_c b}{\log_c a}$
$\dfrac{a^x}{b^x} = \left(\dfrac{a}{b}\right)^x$	

These formulas hold whenever both sides are well-defined.

Example 1.90

Find all the real numbers x which satisfy the equation

$$\log_{2x}(2x^2 + 2x + 12) = 2.$$

Solution. Our first aim is to equate the bases of logarithm on both sides, so we write

$$2 = 2\log_{2x} 2x = \log_{2x}(2x)^2.$$

Then the given equation becomes

$$\begin{aligned} \log_{2x}(2x^2 + 2x + 12) = \log_{2x}(2x)^2 &\Rightarrow 2x^2 + 2x + 12 = (2x)^2 \\ &\Rightarrow -2x^2 + 2x + 12 = 0 \\ &\Rightarrow -2(x-3)(x+2) = 0. \end{aligned}$$

So $x = 3$ or $x = -2$. However, since $2x$ was the base of a logarithm, we must require $2x > 0$. Therefore $x = -2$ is not a solution to the equation, and the only valid solution is $x = 3$.

1.5.4 Trigonometric Functions

Definition 1.91. The *cosine function* $f(x) = \cos x$ and the *sine function* $g(x) = \sin x$ are defined on $\mathbb{R}$ as follows. For each real number x, let P be the point of intersection of the unit circle centered at the origin and the ray emanating from the origin making an angle of x radians from the positive horizontal axis measured counterclockwise. Then $\cos x$ is defined to be the first coordinate of P and $\sin x$ is defined to be the second coordinate of P (i.e. $P = (\cos x, \sin x)$).

Remark 1.92 — Radians, not Degrees

Always use "radians" and never use "degrees" as the unit of the size of an angle in this course. This will help avoid troubles in calculus computations. Bearing in mind that the unit circle has circumference 2π, we can convert between the degree measure

and the radian measure of an angle using the formulas

$$x^\circ = \frac{2\pi}{360^\circ}\, x^\circ = \frac{\pi x}{180} \qquad \text{and} \qquad x = \frac{360^\circ}{2\pi}\, x = \left(\frac{180x}{\pi}\right)^\circ.$$

Definition 1.93. The *tangent function* $f(x) = \tan x$ is defined to be the quotient of the sine by the cosine, i.e. $\tan x := \dfrac{\sin x}{\cos x}$. We also define the *cotangent function*, the *secant function* and the *cosecant function* by

$$\cot x := \frac{\cos x}{\sin x}, \qquad \sec x := \frac{1}{\cos x} \qquad \text{and} \qquad \csc x := \frac{1}{\sin x}$$

respectively.

Remark 1.94 — Graphs and Facts about sin, cos and tan

The following are the graphs of the three basic trigonometric functions and some important facts. (Try to sketch the graphs of cot, sec and csc on your own.)

- $y = \sin x$: a smooth wave through the origin oscillating between -1 and 1, peaking at $x = \dfrac{\pi}{2}$. Domain: $\mathbb{R}$. Range: $[-1, 1]$. Fundamental period: 2π. Parity: odd.
- $y = \cos x$: the same wave, but taking the peak value 1 at the origin. Domain: $\mathbb{R}$. Range: $[-1, 1]$. Fundamental period: 2π. Parity: even.
- $y = \tan x$: strictly increasing branches separated by vertical asymptotes at $x = (n + \dfrac{1}{2})\pi$, $n \in \mathbb{Z}$. Domain: $\mathbb{R} \setminus \{(n + \dfrac{1}{2})\pi : n \in \mathbb{Z}\}$. Range: $\mathbb{R}$. Fundamental period: π. Parity: odd.

Special values:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

(Mnemonic for the sin row: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; the cos row is the same sequence in reverse order.)

Remark 1.95 — A Trigonometric Formula Collection

The following are some important formulas in trigonometry. Make yourself familiar with all these results, which you should have learnt from high school already.

Right-angled triangle ratios. Writing o, a and h for the lengths of the opposite side, the adjacent side and the hypotenuse of a right-angled triangle with angle x:

$$\cos x = \frac{a}{h}, \qquad \sin x = \frac{o}{h}, \qquad \tan x = \frac{o}{a}, \qquad \cot x = \frac{a}{o}, \qquad \sec x = \frac{h}{a}, \qquad \csc x = \frac{h}{o}.$$

Pythagorean identities.

$$\cos^2 x + \sin^2 x = 1;$$

dividing both sides by $\cos^2 x$ or by $\sin^2 x$ respectively gives

$$1 + \tan^2 x = \sec^2 x, \qquad \cot^2 x + 1 = \csc^2 x.$$

Phase-shift identities. Use the A-S-T-C diagram ("All Students Take Calculus": in the four quadrants, all / sine / tangent / cosine are positive respectively). If the angle is shifted by an odd multiple of $\dfrac{\pi}{2}$, switch the "co-"; otherwise don't. For example:

$$\sin(\pi - x) = \sin x, \qquad \sec(-x) = \sec x, \qquad \cos\left(\tfrac{3\pi}{2} - x\right) = -\sin x,$$
$$\csc\left(\tfrac{5\pi}{2} - x\right) = \sec x, \qquad \tan\left(\tfrac{\pi}{2} + x\right) = -\cot x, \qquad \cot\left(\tfrac{3\pi}{2} + x\right) = -\tan x,$$

and so on.

Addition of angles (a.k.a. compound angle formulae).

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$$
$$\sin 2x = 2 \sin x \cos x,$$
$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Proof of the first formula. If $0 < x < \dfrac{\pi}{2}$ and $0 < y < \dfrac{\pi}{2}$, consider a pair of right-angled triangles, with hypotenuses a and b and acute angles x and y respectively, glued along a common side so as to form a large triangle whose two sides of lengths a and b enclose the angle $\pi - x - y$. On the one hand, the area A of the large combined triangle is

$$A = \frac{1}{2}ab\sin(\pi - x - y) = \frac{1}{2}ab\sin(x + y);$$

on the other hand, it is also the sum of the areas of the two right-angled triangles, which is

$$A = \frac{1}{2}(a\sin x)(b\cos y) + \frac{1}{2}(a\cos x)(b\sin y) = \frac{1}{2}ab(\sin x\cos y + \cos x\sin y).$$

Therefore $\sin(x + y) = \sin x \cos y + \cos x \sin y$. The case for general $x, y \in \mathbb{R}$ then follows from the phase-shift identities. ■

Addition of sinusoids (a.k.a. subsidiary angle form). Any sum of constant multiples of $\sin x$ and $\cos x$ can be written in the form $r\sin(x + \theta)$ or $r\cos(x + \theta)$: if a and b are not both zero, then

$$a\sin x + b\cos x = \sqrt{a^2 + b^2}\left(\sin x \cdot \underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\cos\theta} + \cos x \cdot \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\sin\theta}\right) = r\sin(x + \theta),$$

where $r = \sqrt{a^2 + b^2}$, $\cos\theta = \dfrac{a}{\sqrt{a^2 + b^2}}$ and $\sin\theta = \dfrac{b}{\sqrt{a^2 + b^2}}$.

Sum-to-product / product-to-sum formulae.

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}, \qquad \sin x\cos y = \frac{1}{2}[\sin(x+y)+\sin(x-y)],$$
$$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}, \qquad \cos x\sin y = \frac{1}{2}[\sin(x+y)-\sin(x-y)],$$
$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}, \qquad \cos x\cos y = \frac{1}{2}[\cos(x+y)+\cos(x-y)],$$
$$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}, \qquad \sin x\sin y = \frac{1}{2}[\cos(x-y)-\cos(x+y)].$$

Example 1.96

Prove the following identities:

(a) $\cos x\cos 2x\cos 3x = \dfrac{1}{4}(1+\cos 2x+\cos 4x+\cos 6x)$

(b) $\sin^2\left(x-\dfrac{3\pi}{2}\right)+\cos^2\left(\dfrac{\pi}{2}+x\right)+\tan^2(x-2\pi)=\sec^2 x$

Solution. Proof.

(a) For every real number x, we have

$$\begin{aligned}\cos x\cos 2x\cos 3x &= \frac{1}{2}(\cos 3x+\cos x)\cos 3x\\ &= \frac{1}{2}\cos^2 3x+\frac{1}{2}\cos x\cos 3x\\ &= \frac{1}{2}\left(\frac{1+\cos 6x}{2}\right)+\frac{1}{4}(\cos 4x+\cos 2x)\\ &= \frac{1}{4}+\frac{1}{4}\cos 2x+\frac{1}{4}\cos 4x+\frac{1}{4}\cos 6x\\ &= \frac{1}{4}(1+\cos 2x+\cos 4x+\cos 6x).\end{aligned}$$

(b) Whenever each term is well-defined, we have

$$\begin{aligned}\sin^2\left(x-\frac{3\pi}{2}\right)+\cos^2\left(\frac{\pi}{2}+x\right)+\tan^2(x-2\pi) &= \cos^2 x+\sin^2 x+\tan^2 x\\ &= 1+\tan^2 x\\ &= \sec^2 x.\end{aligned}$$

■

Example 1.97

Express $\cos^4 x$ as a sum of constant multiples of terms of the form $\cos nx$ only.

Solution. For every real number x, we have

$$\begin{aligned}\cos^4 x=(\cos^2 x)^2=\left(\frac{1+\cos 2x}{2}\right)^2 &= \frac{1}{4}(1+2\cos 2x+\cos^2 2x)\\ &= \frac{1}{4}\left(1+2\cos 2x+\frac{1+\cos 4x}{2}\right)\\ &= \frac{3}{8}+\frac{1}{2}\cos 2x+\frac{1}{8}\cos 4x.\end{aligned}$$

Example 1.98

Find all the real numbers x which satisfy the equation

$$\cos x - \sqrt{3}\sin x = 1.$$

Solution. We have

$$\begin{aligned}\cos x - \sqrt{3}\sin x = 1 &\Rightarrow 2\left(\cos x \cos\frac{\pi}{3} - \sin x \sin\frac{\pi}{3}\right) = 1\\ &\Rightarrow \cos\left(x + \frac{\pi}{3}\right) = \frac{1}{2}.\end{aligned}$$

This gives $x + \dfrac{\pi}{3} = 2n\pi + \dfrac{\pi}{3}$ or $x + \dfrac{\pi}{3} = 2n\pi - \dfrac{\pi}{3}$, where n is an integer. Thus the solutions are

$$x = 2n\pi \qquad \text{or} \qquad x = 2n\pi - \frac{2\pi}{3},$$

where n is an integer.

Example 1.99

The exact value of $\sin\dfrac{5\pi}{12}$ is given by

$$\begin{aligned}\sin\frac{5\pi}{12} = \sin\left(\frac{\pi}{4} + \frac{\pi}{6}\right) &= \sin\frac{\pi}{4}\cos\frac{\pi}{6} + \cos\frac{\pi}{4}\sin\frac{\pi}{6}\\ &= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}.\end{aligned}$$

1.5.5 Inverse Trigonometric Functions

The trigonometric functions are not one-to-one functions on their natural domains, so in order to define their inverses, we need to restrict their domains so that they become one-to-one. We want to choose domains that are large enough so that the ranges of these functions are not affected. Conventionally we choose the domains $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ for sin, $[0, \pi]$ for cos, and $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ for tan.

Definition 1.100. The *inverse trigonometric functions* are defined as follows.

- The cosine function is one-to-one when its domain is restricted to $[0, \pi]$, and its inverse is denoted by $\arccos\colon [-1, 1] \to [0, \pi]$.
- The sine function is one-to-one when its domain is restricted to $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$, and its inverse is denoted by $\arcsin\colon [-1, 1] \to \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$.
- The tangent function is one-to-one when its domain is restricted to $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$, and its inverse is denoted by $\arctan\colon \mathbb{R} \to \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$.
- We define the inverse trigonometric functions $\operatorname{arcsec}\colon (-\infty, -1] \cup [1, +\infty) \to \left(\dfrac{\pi}{2}, \pi\right] \cup \left[0, \dfrac{\pi}{2}\right)$, $\operatorname{arccsc}\colon (-\infty, -1] \cup [1, +\infty) \to \left[-\dfrac{\pi}{2}, 0\right) \cup \left(0, \dfrac{\pi}{2}\right]$ and $\operatorname{arccot}\colon \mathbb{R} \to (0, \pi)$ in a similar manner.

Remark 1.101 — The Notation $\sin^{-1}$ is Ambiguous

In some textbooks, the inverse trigonometric functions are denoted as $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ and so on. Although these notations are also used commonly, they may lead to confusions. For instance we have $\sin^2 x \equiv (\sin x)^2$, $\sin^3 x \equiv (\sin x)^3$ but

$$\sin^{-1} x \not\equiv (\sin x)^{-1}$$

because the left-hand side means $\arcsin x$ but the right-hand side means $\csc x$.

The graphs of the inverse trigonometric functions can be obtained by reflecting the chosen portion of the graph of the corresponding trigonometric function across the line $y = x$:

- $y = \arcsin x$: a strictly increasing S-shaped curve through the origin, from $\left(-1, -\frac{\pi}{2}\right)$ to $\left(1, \frac{\pi}{2}\right)$. Domain: $[-1, 1]$. Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
- $y = \arccos x$: a strictly decreasing curve from $(-1, \pi)$ to $(1, 0)$. Domain: $[-1, 1]$. Range: $[0, \pi]$.
- $y = \arctan x$: a strictly increasing S-shaped curve through the origin, with horizontal asymptotes $y = \pm\frac{\pi}{2}$. Domain: $\mathbb{R}$. Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(Try to sketch the graphs of arccot, arcsec and arccsc on your own.)

Proposition 1.102 — Cancellation Relations for Inverse Trigonometric Functions

Proposition. We have the following inverse relations:

(i) $\arcsin(\sin x) = x$ for every $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and $\sin(\arcsin x) = x$ for every $x \in [-1, 1]$.

(ii) $\arccos(\cos x) = x$ for every $x \in [0, \pi]$, and $\cos(\arccos x) = x$ for every $x \in [-1, 1]$.

(iii) $\arctan(\tan x) = x$ for every $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and $\tan(\arctan x) = x$ for every $x \in \mathbb{R}$.

Similar relations also exist for the functions arccot, arcsec and arccsc.

Example 1.103

Without using a calculator, find the value of each of the following:

$$\text{(a) } \arcsin\left(\sin\frac{5\pi}{4}\right) \qquad \text{(b) } \tan(\arcsin x) \qquad \text{(c) } \cos\left(\operatorname{arccsc}\frac{\sqrt{x^2+9}}{3}\right)$$

Solution.

(a) Note that $\sin\frac{5\pi}{4} = \sin\left(\pi - \frac{5\pi}{4}\right) = \sin\left(-\frac{\pi}{4}\right)$ and the number $-\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Thus

$$\arcsin\left(\sin\frac{5\pi}{4}\right) = \arcsin\left[\sin\left(-\frac{\pi}{4}\right)\right] = -\frac{\pi}{4}.$$

(b) For each $x \in [-1, 1]$, let's denote $\theta := \arcsin x$ for simplicity. Then
- $\sin\theta = x$.
- Since $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we have $\cos\theta \geq 0$, so $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - x^2}$.

Therefore

$$\tan(\arcsin x) = \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{x}{\sqrt{1-x^2}}$$

if $x \in (-1, 1)$, and is undefined otherwise.

(c) For each $x \in \mathbb{R}$, let's denote $\theta := \operatorname{arccsc} \dfrac{\sqrt{x^2+9}}{3}$ for simplicity. Since $\theta \in \left[-\dfrac{\pi}{2}, 0\right) \cup \left(0, \dfrac{\pi}{2}\right]$ (in fact we are even sure that $\theta \in \left(0, \dfrac{\pi}{2}\right]$, why?), we must have $\cos\theta \geq 0$. Consequently

$$\cos\left(\operatorname{arccsc}\frac{\sqrt{x^2+9}}{3}\right) = \cos\theta = \sqrt{1-\sin^2\theta}$$
$$= \sqrt{1-\frac{3^2}{x^2+9}} = \sqrt{\frac{x^2}{x^2+9}} = \frac{|x|}{\sqrt{x^2+9}}$$

for every $x \in \mathbb{R}$.

All the functions that we are going to study in this course will basically come from this catalog of elementary functions, as well as all those functions that we can construct from them using the operations and transformations introduced in the previous sections.

2 Limits

Limit is a fundamental concept of calculus. In this chapter we study the limit of a sequence and limits of a function.

2.1 Sequences

Definition. A sequence is an ordered infinite list of objects. If the objects are labeled $a_1, a_2, a_3, \dots$, the sequence $(a_1, a_2, a_3, \dots)$ is denoted $(a_n)_{n\in\mathbb{N}}$, $(a_n)_{n=1}^{+\infty}$, or simply (a_n); n is called the index. A sequence of real numbers (a_n) can be regarded as a function $a : \mathbb{N} \to \mathbb{R}$, $a(n) = a_n$.

Index start

It is alright if the index starts with $n = 0$ or some other integer instead of $n = 1$.

Example (arithmetic and geometric sequences)

Let $a, d, r \in \mathbb{R}$.

- **Arithmetic sequence:** $a_n = a + (n-1)d$ for every $n \in \mathbb{N}$; d is the common difference. Recursive form: $a_1 = a$, $a_{n+1} = a_n + d$. E.g. $a = 1, d = 3$: $(1, 4, 7, 10, 13, \dots)$; $a = 5, d = -2$: $(5, 3, 1, -1, -3, \dots)$.
- **Geometric sequence:** $a_n = ar^{n-1}$ for every $n \in \mathbb{N}$; r is the common ratio. Recursive form: $a_1 = a$, $a_{n+1} = ra_n$. E.g. $a = 1, r = 3$: $(1, 3, 9, 27, 81, \dots)$; $a = 2, r = -\frac{1}{2}$: $(2, -1, \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \frac{1}{32}, \dots)$.

Example (Fibonacci sequence)

Define recursively $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = a_n + a_{n-1}$ for every $n \in \mathbb{N}$. This gives

$$(0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots),$$

the Fibonacci sequence. It can be shown that

$$a_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

for every non-negative integer n.

Remark. A sequence of real numbers (a_n) can be presented visually in two ways:

- Image of the sequence: mark the (first few) terms of (a_n) as points on the real line.
- Graph of the sequence: mark the points of the graph $\{(n, a_n) : n \in \mathbb{N}\}$ in the coordinate plane, regarding (a_n) as a function $a : \mathbb{N} \to \mathbb{R}$.

Example (visual presentation of sequences)

Present visually the sequences (a) $a_n = \sqrt{n}$; (b) $a_n = \dfrac{1}{n}$; (c) $a_n = \dfrac{(-1)^n}{n}$.

Solution. For each sequence, draw both its image on the real line and its graph in the plane.

(a) $a_n = \sqrt{n}$: image points $a_0 = 0, a_1 = 1, a_2 \approx 1.41, \dots$ march right on the real line; graph points $(n, \sqrt{n})$ rise like the square-root curve, unbounded.

(b) $a_n = \dfrac{1}{n}$: image points $1, \frac{1}{2}, \frac{1}{3}, \dots$ cluster towards 0 from the right; graph points $(n, \dfrac{1}{n})$ descend towards the n-axis.

(c) $a_n = \dfrac{(-1)^n}{n}$: image points $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \dots$ alternate sides and cluster towards 0; graph points alternate above and below the n-axis with decreasing magnitude.

Definition (informal definition of limit)**.** Let (a_n) be a sequence of real numbers.

- If a_n gets closer and closer to a unique real number L when n gets larger and larger, then L is called the limit of (a_n), written
$$\lim_{n\to+\infty} a_n = L.$$
- (a_n) converges (to L) if it has a limit L which is a finite real number; otherwise (a_n) diverges.
- In the latter case, (a_n) diverges to $+\infty$ (resp. to $-\infty$) if a_n (resp. $-a_n$) gets arbitrarily large and keeps getting larger as n increases. We write
$$\lim_{n\to+\infty} a_n = +\infty \qquad (\text{resp. } \lim_{n\to+\infty} a_n = -\infty).$$

Example (basic limits of sequences)

- **Constant sequence.** For $a \in \mathbb{R}$, the sequence $a_n = a$, i.e. $(a, a, a, \dots)$, converges to a.
- $\lim\limits_{n\to+\infty} \dfrac{1}{n} = 0$, since $\dfrac{1}{n}$ gets closer and closer to 0.
- $a_n = (-1)^n$, i.e. $(-1, 1, -1, 1, \dots)$, diverges.
- **Geometric** $a_n = r^n$**.** (a_n) diverges to $+\infty$ if $r > 1$; converges to 1 if $r = 1$; converges to 0 if $-1 < r < 1$; diverges (neither to $+\infty$ nor $-\infty$) if $r \le -1$.
- **Power** $a_n = \dfrac{1}{n^p}$**.** (a_n) converges to 0 if $p > 0$; converges to 1 if $p = 0$; diverges to $+\infty$ if $p < 0$.

The following properties of limits of sequences are stated without proof for the time being, since we only have an informal definition of limit.

Uniqueness of limit; arithmetic rules; reciprocal rule

Theorem (uniqueness of limit)**.** If a sequence of real numbers (a_n) converges to both real numbers a and b, then $a = b$.

Theorem (arithmetic rules)**.** Let $a, b \in \mathbb{R}$ and let $(a_n), (b_n)$ be sequences of real

numbers, both convergent, with $\lim_{n\to+\infty} a_n = a$ and $\lim_{n\to+\infty} b_n = b$. Then

(i) $(a_n + b_n)$ and $(a_n - b_n)$ both converge, and $\lim_{n\to+\infty}(a_n \pm b_n) = a \pm b$;

(ii) $(a_n b_n)$ converges, and $\lim_{n\to+\infty} a_n b_n = ab$;

(iii) if $b \neq 0$, then $\left(\frac{a_n}{b_n}\right)$ also converges and $\lim_{n\to+\infty} \frac{a_n}{b_n} = \frac{a}{b}$.

Theorem. If (a_n) diverges to $+\infty$ or to $-\infty$, then $(\frac{1}{a_n})$ converges and

$$\lim_{n\to+\infty} \frac{1}{a_n} = 0.$$

Example (using the arithmetic rules)

Evaluate each limit, if it exists.

$$\text{(a)} \lim_{n\to+\infty} \frac{n^3}{n^4+n^2+1} \qquad \text{(b)} \lim_{n\to+\infty} \left(\sqrt{n+1} - \sqrt{n}\right)$$

$$\text{(c)} \lim_{n\to+\infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{n}{n^2}\right)$$

Solution.

(a) $\dfrac{n^3}{n^4+n^2+1} = \dfrac{\frac{1}{n}}{1+\frac{1}{n^2}+\frac{1}{n^4}}$. Since $\lim_{n\to+\infty} \dfrac{1}{n} = 0$, the arithmetic rules give $\lim_{n\to+\infty} \dfrac{1}{n^2} = \lim_{n\to+\infty} \dfrac{1}{n^4} = 0$ and $\lim_{n\to+\infty} \left(1 + \dfrac{1}{n^2} + \dfrac{1}{n^4}\right) = 1$; hence by the quotient rule

$$\lim_{n\to+\infty} \frac{n^3}{n^4+n^2+1} = \lim_{n\to+\infty} \frac{\frac{1}{n}}{1+\frac{1}{n^2}+\frac{1}{n^4}} = \frac{0}{1} = 0.$$

(b) Rationalize:

$$\sqrt{n+1} - \sqrt{n} = (\sqrt{n+1} - \sqrt{n}) \frac{\sqrt{n+1}+\sqrt{n}}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}.$$

Since $\sqrt{n+1}+\sqrt{n}$ diverges to $+\infty$, the reciprocal rule gives $\lim_{n\to+\infty}(\sqrt{n+1}-\sqrt{n}) = 0$.

(c) Since $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$,

$$\lim_{n\to+\infty} \left(\frac{1}{n^2} + \cdots + \frac{n}{n^2}\right) = \lim_{n\to+\infty} \frac{n(n+1)}{2n^2} = \lim_{n\to+\infty} \left(\frac{1}{2} + \frac{1}{2n}\right) = \frac{1}{2}.$$

Invalid uses of the arithmetic rules

Before applying the arithmetic rules, check that all relevant sequences converge.

- It is wrong to write $\lim_{n\to+\infty} \dfrac{n^3}{n^4+n^2+1} = \dfrac{\lim n^3}{\lim(n^4+n^2+1)}$, since numerator and denominator both diverge to $+\infty$.
- Worse: $\lim(\sqrt{n+1} - \sqrt{n}) = \lim\sqrt{n+1} - \lim\sqrt{n} = (+\infty) - (+\infty) = 0$. Not only do both sequences diverge to $+\infty$, but $(+\infty) - (+\infty)$ is NOT 0.
- The sum rule generalizes to finitely many terms but is invalid for infinitely many

terms: it is wrong to split $\lim(\frac{1}{n^2}+\frac{2}{n^2}+\cdots+\frac{n}{n^2})$ as $\lim\frac{1}{n^2}+\lim\frac{2}{n^2}+\cdots=0+0+\cdots=0$.

How do we justify convergence of a complicated sequence, so that the computational rules become applicable? Two tools: the Squeeze Theorem and the monotone sequence theorem.

Comparison lemma and Squeeze Theorem for sequences

Lemma. Let $(a_n),(b_n)$ be sequences of real numbers such that $a_n \le b_n$ for every sufficiently large n.

(i) If both converge, then $\lim_{n\to+\infty} a_n \le \lim_{n\to+\infty} b_n$.

(ii) If (a_n) diverges to $+\infty$, then (b_n) also diverges to $+\infty$.

(iii) If (b_n) diverges to $-\infty$, then (a_n) also diverges to $-\infty$.

Theorem (Squeeze Theorem)**.** Let $(a_n),(b_n),(c_n)$ be sequences of real numbers such that

$$a_n \le b_n \le c_n$$

for every sufficiently large n. If (a_n) and (c_n) both converge and $\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} c_n$, then (b_n) also converges to the same number.

Proof. Direct application of part (i) of the lemma. ■

Remark. Even with strict inequalities $a_n < b_n$, the best conclusion is still $\lim a_n \le \lim b_n$: the limits may be equal. E.g. $a_n = \frac{1}{n+1} < \frac{1}{n} = b_n$ for every n, yet both converge to 0.

Remark. In the lemma and the Squeeze Theorem the condition is only required "for every sufficiently large n": the limit of a sequence depends only on its tail. Inserting, removing or modifying the first few (even the first 10000) terms does not affect the limit. The same relaxation applies to every theorem whose conclusion is only about the limit.

Example (Squeeze Theorem for sequences)

Evaluate (a) $\lim_{n\to+\infty}\frac{(-1)^n}{n}$; (b) $\lim_{n\to+\infty}\frac{2^n}{n!}$.

Solution.

(a) For every integer $n \ge 1$, $-\frac{1}{n} \le \frac{(-1)^n}{n} \le \frac{1}{n}$. Since $\lim_{n\to+\infty}\left(-\frac{1}{n}\right) = \lim_{n\to+\infty}\frac{1}{n} = 0$, by the Squeeze Theorem $\lim_{n\to+\infty}\frac{(-1)^n}{n} = 0$.

(b) For every integer $n \ge 2$,

$$(n-1)! = (n-1)(n-2)\cdots(2)(1) \ge (2)(2)\cdots(2)(1) = 2^{n-2},$$

so $0 < \frac{2^n}{n!} \le \frac{4}{n}$. Since $\lim_{n\to+\infty}\frac{4}{n} = 0$, by the Squeeze Theorem $\lim_{n\to+\infty}\frac{2^n}{n!} = 0$.

Definition. A sequence of real numbers (a_n) is <u>increasing</u> if $a_{n+1} \ge a_n$ for all n; <u>decreasing</u> if $a_{n+1} \le a_n$ for all n.

Example (monotonicity)

$(n!)$ is increasing; $(\frac{1}{2^n})$ is decreasing; $((-1)^n)$ is neither increasing nor decreasing; the constant sequence (0) is both.

Definition. Let (a_n) be a sequence of real numbers.

(i) (a_n) is bounded from above if there exists $M \in \mathbb{R}$ with $a_n \leq M$ for every n.
(ii) (a_n) is bounded from below if there exists $m \in \mathbb{R}$ with $a_n \geq m$ for every n.
(iii) (a_n) is bounded if it is bounded both from above and from below.

Example (boundedness)

$(\sin n)$ is bounded; $((-1)^n)$ is bounded; $((-1)^n n)$ is bounded neither from above nor from below. Every convergent sequence is bounded, but not every bounded sequence converges: $((-1)^n)$ is bounded yet diverges.

Monotone sequence theorem

Lemma. An increasing sequence of real numbers either converges or diverges to $+\infty$; a decreasing sequence either converges or diverges to $-\infty$.

Theorem (monotone sequence theorem)**.** Let (a_n) be a sequence of real numbers.

(i) If (a_n) is increasing, then (a_n) converges if and only if it is bounded from above.
(ii) If (a_n) is decreasing, then (a_n) converges if and only if it is bounded from below.

Example (recursive sequence; monotone convergence)

Let $a_1 = 2$ and $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$ for every $n \in \mathbb{N}$. Evaluate $\lim_{n\to+\infty} a_n$ if it exists.

Solution.

- Since $a_1 = 2 > 0$, and $a_k > 0$ implies $a_{k+1} = \frac{1}{2}(a_k + \frac{2}{a_k}) > 0$, induction gives $a_n > 0$ for every n; so (a_n) is bounded from below.
- Since $2 - a_1^2 = 2 - 2^2 < 0$ and

$$2 - a_n^2 = 2 - \frac{1}{4}\left(a_{n-1} + \frac{2}{a_{n-1}}\right)^2 = -\frac{1}{4}\left(a_{n-1} - \frac{2}{a_{n-1}}\right)^2 \leq 0$$

for every integer $n \geq 2$, we have

$$a_{n+1} - a_n = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right) - a_n = \frac{1}{2a_n}\left(2 - a_n^2\right) \leq 0$$

for every n (note $a_n > 0$), so (a_n) is decreasing.

Therefore (a_n) converges by the monotone sequence theorem; write $\lim_{n\to+\infty} a_n = L$. Taking limits on both sides of $a_{n+1} = \frac{1}{2}(a_n + \frac{2}{a_n})$ gives

$$L = \frac{1}{2}\left(L + \frac{2}{L}\right),$$

so $L = \sqrt{2}$ (the other solution $L = -\sqrt{2}$ is rejected since $a_n > 0$ for all n). Hence $\lim_{n\to+\infty} a_n = \sqrt{2}$.

An important consequence of the monotone sequence theorem:

Convergence of $(1+\frac{1}{n})^n$; Euler's number

Theorem. The sequence (a_n) defined by

$$a_n = \left(1+\frac{1}{n}\right)^n$$

converges, and its limit belongs to the open interval $(2,3)$.

Proof. We show (a_n) is increasing and bounded from above. For every integer $n \geq 2$, the binomial theorem gives

$$a_n = \left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = 1+1+\sum_{k=2}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!}\,\frac{1}{n^k}$$
$$= 2+\sum_{k=2}^{n} \frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right).$$

Each term in the sum is positive.

(i) As the index increases from n to $n+1$, each term of the last sum increases, and the number of terms also increases. Hence (a_n) is increasing.

(ii) The equality also gives $2 < a_n < 2+\sum_{k=2}^{n} \frac{1}{k!}$ for $n \geq 2$. Since

$$\frac{1}{k!} = \frac{1}{k(k-1)\cdots(2)(1)} \leq \frac{1}{(2)(2)\cdots(2)(1)} = \frac{1}{2^{k-1}}$$

for every integer $k \geq 2$,

$$2 < a_n < 2+\sum_{k=2}^{n} \frac{1}{k!} \leq 2+\sum_{k=2}^{n} \frac{1}{2^{k-1}} = 3-\frac{1}{2^{n-1}} < 3.$$

In particular (a_n) is bounded from above by 3.

Therefore (a_n) converges, with limit in the open interval $(2,3)$. ■

Remark. By the comparison lemma alone we would only conclude the limit lies in $[2,3]$; evaluating more initial terms before estimating improves the bounds, so the limit equals neither 2 nor 3.

Definition. Euler's number $e \in (2,3)$ is defined by

$$e := \lim_{n\to+\infty} \left(1+\frac{1}{n}\right)^n.$$

2.2 Limits of a Function

Similar to limits of sequences, we define limits of a function informally but intuitively. Whenever a is a number and f is a function well-defined near a (but not necessarily at a), we can talk about "the limit of $f(x)$ as x tends to a".

Definition (informal definition of limit)**.** Let $a \in \mathbb{R}$ and let f be well-defined on an open interval containing a, except possibly at a. If $f(x)$ gets closer and closer to a unique real number L when x gets closer and closer to a but $x \neq a$, then L is called the limit of $f(x)$ as x tends to a, written "$f(x) \to L$ as $x \to a$", or

$$\lim_{x\to a} f(x) = L.$$

Such a limit L may or may not exist.

Example (first function limits)

- For any $a \in \mathbb{R}$, obviously $\lim_{x \to a} x = a$.

(a) $\lim_{x \to 2} x^2 = 4$: as $x \to 2$, $x \neq 2$, the number x^2 gets closer and closer to 4.

(b) $\lim_{x \to 2} \dfrac{1}{x-2}$ does not exist: as $x \to 2$, $\dfrac{1}{x-2}$ becomes a very large positive or very large negative number, never settling near any real number.

(c) Let $f(x) = \dfrac{x^2-4}{x-2}$. For every $x \neq 2$,

$$f(x) = \frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2.$$

As $x \to 2$ with $x \neq 2$, $f(x)$ gets closer to 4, so

$$\lim_{x \to 2} \frac{x^2-4}{x-2} = \lim_{x \to 2}(x+2) = 4.$$

(Although $f(2)$ is undefined, we do not care.)

Example (Heaviside step function)

Let $f : \mathbb{R} \to \mathbb{R}$ be the Heaviside step function

$$f(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x \geq 0. \end{cases}$$

Evaluate $\lim_{x \to 0} f(x)$ if it exists.

Solution. No matter how close x gets to 0 with $x \neq 0$, $f(x)$ is sometimes 0 and sometimes 1; it is not getting closer to a unique real number. Hence $\lim_{x \to 0} f(x)$ does not exist.

Remark. It is important to distinguish

$$\lim_{x \to a} f(x) = L \qquad \text{and} \qquad f(a) = L.$$

$\lim_{x \to a} f(x)$ concerns values of f when the input is close to but not equal to a; $f(a)$ is the value at exactly a. When computing the limit we do not care about $f(a)$ at all: it may be well-defined or undefined, but is irrelevant. That is why we may cancel common factors $(x-a)$ in numerator and denominator, as in (c) above.

Uniqueness and sequential limit theorem

Theorem (uniqueness of limit)**.** Let $a \in \mathbb{R}$ and f be a function. If L and M are both limits of $f(x)$ as x tends to a, then $L = M$.

Theorem (sequential limit theorem)**.** Let $a, L \in \mathbb{R}$ and f be a function. The following are equivalent:

(i) $\lim_{x \to a} f(x) = L$.

(ii) For every sequence of real numbers (x_n) converging to a with no term equal to a, we have

$$\lim_{n \to +\infty} f(x_n) = L.$$

Example ($\lim_{x\to 0}\cos(\frac{1}{x})$ does not exist)

Evaluate $\lim_{x\to 0}\cos\frac{1}{x}$ if it exists.

Solution. Consider the sequences

$$x_n = \frac{1}{2n\pi}, \qquad y_n = \frac{1}{(2n-1)\pi},$$

i.e. $(x_n) = (\frac{1}{2\pi}, \frac{1}{4\pi}, \frac{1}{6\pi}, \dots)$ and $(y_n) = (\frac{1}{\pi}, \frac{1}{3\pi}, \frac{1}{5\pi}, \dots)$. Both converge to 0 and no term is 0. However,

$$\cos\frac{1}{x_n} = \cos(2n\pi) = 1, \qquad \cos\frac{1}{y_n} = \cos(2n-1)\pi = -1$$

for every positive integer n, so $(\cos\frac{1}{x_n})$ and $(\cos\frac{1}{y_n})$ converge to different limits. By the sequential limit theorem, $\lim_{x\to 0}\cos\frac{1}{x}$ does not exist.

The trend of $f(x)$ for x near a splits into $x \to a$ "from the left" and "from the right": one-sided limits.

Definition. Let $a \in \mathbb{R}$.

- Let f be well-defined on an open interval with right end-point a. If $f(x)$ gets closer and closer to a unique real number L when x gets closer and closer to a and $x < a$, then L is the left-hand limit of $f(x)$ as x tends to a, written
$$\lim_{x\to a^-} f(x) = L.$$
- If f is well-defined on an open interval with left end-point a, the right-hand limit $\lim_{x\to a^+} f(x)$ is defined similarly (with $x > a$ instead).

Example (one-sided limits)

For the Heaviside function $f(x) = 0$ $(x < 0)$, $f(x) = 1$ $(x \geq 0)$:

$$\lim_{x\to 0^-} f(x) = 0 \qquad \text{and} \qquad \lim_{x\to 0^+} f(x) = 1.$$

For $f(x) = \cos\frac{1}{x}$ $(x \neq 0)$: both $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$ do not exist.

Example (reading one-sided limits from a graph)

The graph of a function f (defined on a neighborhood of $0, 2, 4$) shows: as x approaches 0 from either side the graph tends to height 2 and $f(0) = 2$; as $x \to 2^-$ the graph tends to 4, as $x \to 2^+$ it tends to 1, and $f(2) = 2$; as $x \to 4^-$ it tends to 0, as $x \to 4^+$ it tends to 3, and $f(4) = 3$. Hence

$$\lim_{x\to 0^-} f(x) = 2, \quad \lim_{x\to 0^+} f(x) = 2, \quad f(0) = 2;$$
$$\lim_{x\to 2^-} f(x) = 4, \quad \lim_{x\to 2^+} f(x) = 1, \quad f(2) = 2;$$
$$\lim_{x\to 4^-} f(x) = 0, \quad \lim_{x\to 4^+} f(x) = 3, \quad f(4) = 3.$$

Limit existence criterion

Theorem (limit existence criterion)**.** Let $a \in \mathbb{R}$ and let f be well-defined on an open interval containing a, except possibly at a. Then

$$\lim_{x\to a} f(x) = L \qquad \Longleftrightarrow \qquad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

Example ($\lim_{x\to 0} |x|$)

Let $f(x) = |x| = \begin{cases} -x & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases}$ Then

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (-x) = 0, \qquad \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x = 0,$$

so $\lim_{x\to 0} |x| = 0$.

Remark (piecewise defined functions)**.** Suppose f has different formulas for $x < a$ and for $x > a$. To find $\lim_{x\to a} f(x)$:

(i) Find $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$. If either fails to exist, neither does $\lim_{x\to a} f(x)$.

(ii) If both exist, check whether they are equal. If equal, $\lim_{x\to a} f(x)$ exists and equals the common value; otherwise it does not exist.

Again, existence of the limit is irrelevant to whether $f(a)$ is well-defined.

Example (composition read from a graph)

Using the same graph of f as above, evaluate $\lim_{x\to 0^-} f(f(x+2))$. Does $\lim_{x\to 0} f(f(x+2))$ exist?

Solution.

(i) As $x \to 0^-$, $x+2 \to 2^-$, so from the graph $f(x+2) \to 4^+$; from the graph again, $f(f(x+2)) \to 3$. Hence $\lim_{x\to 0^-} f(f(x+2)) = 3$.

(ii) As $x \to 0^+$, $x+2 \to 2^+$, so $f(x+2) \to 1^-$; then $f(f(x+2)) \to 3$. Hence $\lim_{x\to 0^+} f(f(x+2)) = 3$.

From (i) and (ii), $\lim_{x\to 0} f(f(x+2))$ exists and equals 3.

2.3 Rules and Techniques of Computing Limits

Arithmetic rules for limits of functions

Theorem. Let $a, c \in \mathbb{R}$ and let f, g be functions such that both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist as finite real numbers. Then

(i) $\lim_{x\to a} c = c$;

(ii) $\lim_{x\to a}[f(x) \pm g(x)] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$;

(iii) $\lim_{x\to a} cf(x) = c \lim_{x\to a} f(x)$;

(iv) $\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$;

(v) $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$, provided $\lim_{x\to a} g(x) \neq 0$;

(vi) $\lim_{x\to a}[f(x)]^c = \left[\lim_{x\to a} f(x)\right]^c$, provided both sides are well-defined.

Similar statements hold for one-sided limits.

Example (applying the rules step by step)

Evaluate $\lim_{x\to 1}(3x^2+\sqrt{x})$, indicating which rules are used.

Solution. We have $\lim_{x\to 1} x = 1$. Then

- $\lim_{x\to 1} x^2 = (\lim_{x\to 1} x)^2 = 1^2 = 1$ by (vi), so $\lim_{x\to 1} 3x^2 = 3\lim_{x\to 1} x^2 = 3$ by (iii);
- $\lim_{x\to 1} \sqrt{x} = \lim_{x\to 1} x^{\frac{1}{2}} = (\lim_{x\to 1} x)^{\frac{1}{2}} = 1^{\frac{1}{2}} = 1$ by (vi) again.

Therefore, by (ii),

$$\lim_{x\to 1}(3x^2+\sqrt{x}) = \lim_{x\to 1} 3x^2 + \lim_{x\to 1}\sqrt{x} = 3+1 = 4.$$

Conditions of the arithmetic rules

The rules fail when their conditions are not met. It is incorrect to write

$$\lim_{x\to 2}\frac{1}{x-2} = \frac{\lim 1}{\lim(x-2)} \qquad \text{or} \qquad \lim_{x\to 2}\frac{x^2-4}{x-2} = \frac{\lim(x^2-4)}{\lim(x-2)},$$

because $\lim_{x\to 2}(x-2) = 0$, whereas rule (v) requires $\lim g(x) \neq 0$.

Also, the formulas hold only when the limits on both sides all exist as real numbers. If some limit on one side does not exist, we cannot conclude anything about the other side. E.g. with $f(x) = \dfrac{1}{x}$, $g(x) = -\dfrac{1}{x}$, $h(x) = x$:

- $\lim_{x\to 0} f(x)$ and $\lim_{x\to 0} g(x)$ do not exist, yet $\lim_{x\to 0}[f(x)+g(x)] = \lim_{x\to 0} 0 = 0$ exists; rule (ii) is not applicable.
- $\lim_{x\to 0} f(x)$ does not exist, yet $\lim_{x\to 0}[f(x)h(x)] = \lim_{x\to 0} 1 = 1$ exists; rule (iv) is not applicable.

Example (recovering $\lim f$ from a difference quotient)

Let $a, b \in \mathbb{R}$ and suppose $\lim_{x\to a}\dfrac{f(x)-b}{x-a}$ exists as a finite real number. Find $\lim_{x\to a} f(x)$.

Solution. For every $x \neq a$ in the domain of f, $f(x) = \dfrac{f(x)-b}{x-a}\cdot(x-a)+b$. Writing $\lim_{x\to a}\dfrac{f(x)-b}{x-a} = L$,

$$\lim_{x\to a} f(x) = \lim_{x\to a}\left[\frac{f(x)-b}{x-a}\cdot(x-a)+b\right] = L\cdot\lim_{x\to a}(x-a)+b = L\cdot 0 + b = b,$$

where the splitting is valid since all three limits exist.

Direct substitution for elementary functions

Theorem. Let f be a polynomial, rational function, power function, exponential function, logarithmic function, trigonometric function or inverse trigonometric function. Then for every a in the interior of the domain of f,

$$\lim_{x\to a} f(x) = \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a).$$

Example (valid and invalid substitutions)

Valid (the point 1 is interior to each domain):

$$\lim_{x\to 1}\frac{x^2+3x+1}{4x+1}=\frac{1+3+1}{4+1}=1, \qquad \lim_{x\to 1}\sin x=\sin 1, \qquad \lim_{x\to 1}e^x=e.$$

Each of the following is incorrect (the point is not in the interior of the domain):

$$\lim_{x\to 2}\frac{x^2-4}{x-2}=\frac{2^2-4}{2-2}=\frac{0}{0}\ \times, \qquad \lim_{x\to \frac{\pi}{2}}\tan x=\tan\frac{\pi}{2}\ \times, \qquad \lim_{x\to 0}\ln x=\ln 0\ \times .$$

Example (matching one-sided limits with parameters)

Let $a, b \in \mathbb{R}$ and

$$f(x)=\begin{cases} a\cos x+2 & \text{if } x<\pi,\\ b & \text{if } x=\pi,\\ \dfrac{a^2x^2}{\pi^2} & \text{if } x>\pi. \end{cases}$$

Find all values of a and b such that $\lim_{x\to\pi} f(x)$ exists.

Solution. From the definition of f,

$$\lim_{x\to\pi^-} f(x)=\lim_{x\to\pi^-}(a\cos x+2)=a\cos\pi+2=-a+2, \qquad \lim_{x\to\pi^+} f(x)=\lim_{x\to\pi^+}\frac{a^2x^2}{\pi^2}=a^2.$$

For the limit to exist we need $-a+2=a^2$, i.e.

$$a^2+a-2=0 \quad\Longrightarrow\quad (a+2)(a-1)=0,$$

giving $a=-2$ or $a=1$. On the other hand $b=f(\pi)$ can be any real number, as it is irrelevant to the existence of the limit.

Direct substitution sometimes yields an indeterminate form $\frac{0}{0}$; this happens when the limit point is not in the domain. Factorization and rationalization handle such forms. Useful identities:

$$a^2-b^2=(a-b)(a+b), \qquad a^3-b^3=(a-b)(a^2+ab+b^2),$$
$$a^n-b^n=(a-b)(a^{n-1}+a^{n-2}b+\cdots+ab^{n-2}+b^{n-1}).$$

Example (factorization and rationalization)

Evaluate:

$$\text{(a)}\ \lim_{x\to 2}\frac{\sqrt{2+x}-2}{x-2} \qquad \text{(b)}\ \lim_{x\to 0}\frac{4^x+4^{-x}-2}{4^x-4^{-x}} \qquad \text{(c)}\ \lim_{x\to 1}\frac{\sqrt[3]{x}-1}{x-1}$$

Solution.

(a) (Cancel the factor $(x-2)$.)

$$\lim_{x\to 2}\frac{\sqrt{2+x}-2}{x-2}=\lim_{x\to 2}\frac{\sqrt{2+x}-2}{x-2}\cdot\frac{\sqrt{2+x}+2}{\sqrt{2+x}+2}=\lim_{x\to 2}\frac{(2+x)-2^2}{(x-2)(\sqrt{2+x}+2)}$$
$$=\lim_{x\to 2}\frac{1}{\sqrt{2+x}+2}=\frac{1}{\sqrt{4}+2}=\frac{1}{4}.$$

(b) (Cancel a factor that makes the denominator zero.)

$$\lim_{x\to 0}\frac{4^x+4^{-x}-2}{4^x-4^{-x}}=\lim_{x\to 0}\frac{(2^x)^2+(2^{-x})^2-2(2^x)(2^{-x})}{(2^x)^2-(2^{-x})^2}$$
$$=\lim_{x\to 0}\frac{(2^x-2^{-x})^2}{(2^x-2^{-x})(2^x+2^{-x})}$$
$$=\lim_{x\to 0}\frac{2^x-2^{-x}}{2^x+2^{-x}}=\frac{2^0-2^0}{2^0+2^0}=0.$$

(c) (Cancel the factor $(x-1)$.)

$$\lim_{x\to 1}\frac{\sqrt[3]{x}-1}{x-1}=\lim_{x\to 1}\frac{\sqrt[3]{x}-1}{x-1}\cdot\frac{(\sqrt[3]{x})^2+\sqrt[3]{x}+1}{(\sqrt[3]{x})^2+\sqrt[3]{x}+1}=\lim_{x\to 1}\frac{x-1}{(x-1)\,[(\sqrt[3]{x})^2+\sqrt[3]{x}+1]}$$
$$=\lim_{x\to 1}\frac{1}{(\sqrt[3]{x})^2+\sqrt[3]{x}+1}=\frac{1}{1+1+1}=\frac{1}{3}.$$

Change of variables

Theorem. Let $a, b \in \mathbb{R}$ and let f, g be functions. If
- $\lim_{x\to a} g(x) = b$, and
- $g(x) \neq b$ for any x near a, except possibly at $x = a$,

then one can apply the change of variable $t = g(x)$:

$$\lim_{x\to a} f(g(x)) = \lim_{g(x)\to b} f(g(x)) = \lim_{t\to b} f(t),$$

provided the last limit exists. A similar statement holds for one-sided limits.

Example (cube-root limit by change of variable)

Evaluate $\lim_{x\to 1}\dfrac{\sqrt[3]{x}-1}{x-1}$ using a change of variable.

Solution. Let $t = \sqrt[3]{x}$, so $x = t^3$; then $t \to 1$ as $x \to 1$, and $t \neq 1$ except for $x = 1$. Hence

$$\lim_{x\to 1}\frac{\sqrt[3]{x}-1}{x-1}=\lim_{t\to 1}\frac{t-1}{t^3-1}=\lim_{t\to 1}\frac{t-1}{(t-1)(t^2+t+1)}=\lim_{t\to 1}\frac{1}{t^2+t+1}=\frac{1}{3}.$$

Limit inequalities and Squeeze Theorem

Lemma. Let $a \in \mathbb{R}$ and let f, g be functions such that both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist.

(i) If $f(x) \leq g(x)$ for every x near a (except possibly at a), then $\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x)$.

(ii) If $\lim_{x\to a} f(x) < \lim_{x\to a} g(x)$, then $f(x) < g(x)$ for every x near a (except possibly at a).

Similar statements hold for one-sided limits.

Theorem (Squeeze Theorem)**.** Let $a \in \mathbb{R}$ and let f, g, h be functions. If $g(x) \leq f(x) \leq h(x)$ for every x near a except possibly at a, and if

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x),$$

then $\lim_{x\to a} f(x)$ exists and equals the same limit. A similar statement holds for one-sided limits.

Equality signs in limit inequalities

Think carefully about the presence or absence of equality in the lemma.
(i) If $f(x) < g(x)$ for every x near a, must $\lim f(x) < \lim g(x)$? **No.**
(ii) If $\lim f(x) \le \lim g(x)$, must $f(x) \le g(x)$ near a? **No.**

The Squeeze Theorem is useful when the target function is bounded by two simpler functions with the same limit. The inequalities $-1 \le \sin x \le 1$ and $-1 \le \cos x \le 1$ (for every real x) are often useful.

Example ($x \sin(\frac{1}{x^2})$ by squeeze)

Evaluate $\lim\limits_{x\to 0} x \sin \dfrac{1}{x^2}$ if it exists.

Solution. Since $-1 \le \sin(\dfrac{1}{x^2}) \le 1$ for every $x \neq 0$,

$$-x \le x \sin \frac{1}{x^2} \le x \ (x > 0), \qquad x \le x \sin \frac{1}{x^2} \le -x \ (x < 0).$$

Combining both cases, $-|x| \le x \sin \dfrac{1}{x^2} \le |x|$ for every $x \neq 0$. Since $\lim\limits_{x\to 0}(-|x|) = 0$ and $\lim\limits_{x\to 0} |x| = 0$, the Squeeze Theorem gives

$$\lim_{x\to 0} x \sin \frac{1}{x^2} = 0.$$

The fundamental trigonometric limit

Theorem.

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Lemma. For every $x \in (0, \dfrac{\pi}{2})$, $\sin x \le x \le \tan x$. Consequently, for every real x, $|\sin x| \le |x|$.

Proof of lemma. Suppose $0 < x < \dfrac{\pi}{2}$. In the coordinate plane consider $O = (0,0)$, $P = (\cos x, \sin x)$, $A = (1,0)$, and let B be the intersection of the line OP with the line $x = 1$, so $B = (1, \tan x)$; both P and B lie in the first quadrant. Clearly

$$\text{Area of } \triangle OPA \le \text{Area of sector } OPA \le \text{Area of } \triangle OAB,$$

i.e. $\dfrac{1}{2}\sin x \le \dfrac{1}{2}x \le \dfrac{1}{2}\tan x$; hence $\sin x \le x \le \tan x$. ■

Proof of theorem. First evaluate the right-hand limit. Suppose $0 < x < \dfrac{\pi}{2}$, so $\cos x > 0$. Rearranging the lemma's inequality gives

$$\cos x \le \frac{\sin x}{x} \le 1.$$

Since $\lim\limits_{x\to 0^+} \cos x = \cos 0 = 1$ and $\lim\limits_{x\to 0^+} 1 = 1$, the Squeeze Theorem gives $\lim\limits_{x\to 0^+} \dfrac{\sin x}{x} = 1$. With the change of variables $t = -x$,

$$\lim_{x\to 0^-} \frac{\sin x}{x} = \lim_{t\to 0^+} \frac{\sin(-t)}{-t} = \lim_{t\to 0^+} \frac{\sin t}{t} = 1.$$

Therefore $\lim_{x\to 0}\dfrac{\sin x}{x}$ exists and equals 1. ■

This fact is useful for $\dfrac{0}{0}$ indeterminate forms involving trigonometric functions.

Example (trigonometric limits)

Evaluate:

$$\text{(a) } \lim_{x\to 0}\frac{\tan 5x}{\sin 3x} \qquad \text{(b) } \lim_{x\to 0}\frac{1-\cos x}{x^2}$$

$$\text{(c) } \lim_{x\to \pi}\frac{\sin x}{x-\pi} \qquad \text{(d) } \lim_{x\to a}\frac{\sin x-\sin a}{x-a}\ (a\in\mathbb{R})$$

Solution.

(a)
$$\lim_{x\to 0}\frac{\tan 5x}{\sin 3x}=\lim_{x\to 0}\left(\frac{\sin 5x}{5x}\cdot\frac{3x}{\sin 3x}\cdot\frac{5}{3\cos 5x}\right)=1\cdot 1\cdot\frac{5}{3\cdot 1}=\frac{5}{3},$$
noting that as $x\to 0$ also $5x\to 0$ and $3x\to 0$.

(b) Since $1-\cos x=2\sin^2\dfrac{x}{2}$ and $\dfrac{x}{2}\to 0$ as $x\to 0$,
$$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\lim_{x\to 0}\frac{2\sin^2(\frac{x}{2})}{x^2}=\lim_{x\to 0}\frac{1}{2}\left(\frac{\sin(\frac{x}{2})}{\frac{x}{2}}\right)^2=\frac{1}{2}\cdot 1^2=\frac{1}{2}.$$

(c) With $t=x-\pi$,
$$\lim_{x\to \pi}\frac{\sin x}{x-\pi}=\lim_{t\to 0}\frac{\sin(t+\pi)}{t}=\lim_{t\to 0}\left(-\frac{\sin t}{t}\right)=-1.$$

(d) By the sum-to-product formula $\sin x-\sin a=2\cos\dfrac{x+a}{2}\sin\dfrac{x-a}{2}$,
$$\lim_{x\to a}\frac{\sin x-\sin a}{x-a}=\lim_{x\to a}\frac{\sin\frac{x-a}{2}}{\frac{x-a}{2}}\cdot\lim_{x\to a}\cos\frac{x+a}{2}=1\cdot\cos\frac{a+a}{2}=\cos a,$$
since $\dfrac{x-a}{2}\to 0$ as $x\to a$. This includes (c) as the special case $a=\pi$.

2.4 Infinite Limits and Limits at Infinity

Definition (limit at infinity)**.** Let f be a function.

- Suppose f is well-defined on an open interval unbounded on the right. If $f(x)$ gets closer and closer to a unique real number L when x gets larger and larger, then L is the limit of $f(x)$ as x tends to $+\infty$, written
$$\lim_{x\to+\infty}f(x)=L.$$
- If f is well-defined on an open interval unbounded on the left, $\lim_{x\to-\infty}f(x)$ is defined similarly (with $-x$ getting larger and larger).

Remark. Uniqueness of limit, the sequential limit theorem, the arithmetic rules, change of variables, the limit inequalities and the Squeeze Theorem all hold for limits at infinity.

Definition (infinite limit)**.** Let $a\in\mathbb{R}$ and let f be well-defined on an open interval containing a, except possibly at a.

- The limit of $f(x)$ as $x \to a$ is $+\infty$ if $f(x)$ gets arbitrarily large and keeps getting larger when x gets closer and closer to a but $x \neq a$; written $\lim\limits_{x\to a} f(x) = +\infty$.
- The limit is $-\infty$, written $\lim\limits_{x\to a} f(x) = -\infty$, if $\lim\limits_{x\to a}(-f(x)) = +\infty$.

Remark. One-sided infinite limits are defined similarly with "$x \neq a$" replaced by "$x > a$" or "$x < a$"; the notion also extends to infinite limits at infinity, e.g. $\lim\limits_{x\to+\infty} f(x) = +\infty$. Formulations are similar and omitted.

Example (basic infinite limits and limits at infinity)

From the graphs of elementary functions:

- Reciprocal: $\lim\limits_{x\to0^+} \dfrac{1}{x} = +\infty$, $\lim\limits_{x\to0^-} \dfrac{1}{x} = -\infty$, $\lim\limits_{x\to+\infty} \dfrac{1}{x} = 0$, $\lim\limits_{x\to-\infty} \dfrac{1}{x} = 0$.
- Exponential/logarithmic, for each $a > 1$:

  $$\lim_{x\to+\infty} a^x = +\infty, \quad \lim_{x\to-\infty} a^x = 0, \quad \lim_{x\to+\infty} \log_a x = +\infty, \quad \lim_{x\to0^+} \log_a x = -\infty;$$

  in particular $\lim\limits_{x\to+\infty} e^x = +\infty$, $\lim\limits_{x\to-\infty} e^x = 0$, $\lim\limits_{x\to+\infty} \ln x = +\infty$, $\lim\limits_{x\to0^+} \ln x = -\infty$.
- For each $a \in (0, 1)$:

  $$\lim_{x\to+\infty} a^x = 0, \quad \lim_{x\to-\infty} a^x = +\infty, \quad \lim_{x\to+\infty} \log_a x = -\infty, \quad \lim_{x\to0^+} \log_a x = +\infty.$$

- Tangent/arctangent:

  $$\lim_{x\to(\frac{\pi}{2})^-} \tan x = +\infty, \quad \lim_{x\to(-\frac{\pi}{2})^+} \tan x = -\infty,$$
  $$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \quad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

Remark. Many textbooks regard an infinite limit as a limit that "does not exist"; it is more appropriate to regard these limits as existing but not as real numbers. We emphasize that $+\infty$ and $-\infty$ are not real numbers.

Example ($\dfrac{1}{x^2}$ at 0)

Since $\lim\limits_{x\to0^-} \dfrac{1}{x^2} = +\infty$ and $\lim\limits_{x\to0^+} \dfrac{1}{x^2} = +\infty$, the limit

$$\lim_{x\to0} \frac{1}{x^2} = +\infty$$

exists (but is not a real number).

Example ($\sin x$ at infinity does not exist)

Consider $\lim\limits_{x\to+\infty} \sin x$. Let $x_n = n\pi$ and $y_n = 2n\pi + \dfrac{\pi}{2}$. Both sequences diverge to $+\infty$, none of their terms equals $+\infty$; but

$$\sin x_n = \sin n\pi = 0, \qquad \sin y_n = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1$$

for every n, so $(\sin x_n)$ and $(\sin y_n)$ converge to different limits. By the sequential limit theorem, $\lim\limits_{x\to+\infty} \sin x$ does not exist.

Example (limits at infinity)

Evaluate:

$$\text{(a)}\ \lim_{x\to+\infty}\frac{6x^3-7x+1}{-2x^3+x^2} \qquad \text{(b)}\ \lim_{x\to+\infty}\frac{x^3+1}{1-2x^2}$$

$$\text{(c)}\ \lim_{x\to0^-}\frac{1}{3+e^{\frac{1}{x}}} \qquad \text{(d)}\ \lim_{x\to-\infty}\left(\sqrt{x^2-2x+3}+x\right)$$

Solution.

(a)

$$\lim_{x\to+\infty}\frac{6x^3-7x+1}{-2x^3+x^2}=\lim_{x\to+\infty}\frac{6-\frac{7}{x^2}+\frac{1}{x^3}}{-2+\frac{1}{x}}=\frac{6-0+0}{-2+0}=-3.$$

(b) With $t=\dfrac{1}{x}$ (so $t\to0^+$ as $x\to+\infty$),

$$\lim_{x\to+\infty}\frac{x^3+1}{1-2x^2}=\lim_{x\to+\infty}\frac{x^3\left(1+\frac{1}{x^3}\right)}{x^3\left(\frac{1}{x^3}-\frac{2}{x}\right)}=\lim_{t\to0^+}\frac{1+t^3}{t^3-2t}=\lim_{t\to0^+}\frac{1+t^3}{t(t^2-2)}=-\infty,$$

since the numerator $\to 1$ and the denominator $\to 0^+\cdot(-2)=0^-$.

(c) With $t=\dfrac{1}{x}$, $\lim\limits_{x\to0^-}e^{\frac{1}{x}}=\lim\limits_{t\to-\infty}e^t=0$, so

$$\lim_{x\to0^-}\frac{1}{3+e^{\frac{1}{x}}}=\frac{1}{3+\lim\limits_{x\to0^-}e^{\frac{1}{x}}}=\frac{1}{3+0}=\frac{1}{3}.$$

(d) Rationalize, using $\sqrt{x^2}=-x$ for $x<0$:

$$\lim_{x\to-\infty}\left(\sqrt{x^2-2x+3}+x\right)=\lim_{x\to-\infty}\frac{(x^2-2x+3)-x^2}{\sqrt{x^2-2x+3}-x}=\lim_{x\to-\infty}\frac{-2x+3}{-x\sqrt{1-\frac{2}{x}+\frac{3}{x^2}}-x}$$

$$=\lim_{x\to-\infty}\frac{-2+\frac{3}{x}}{-\sqrt{1-\frac{2}{x}+\frac{3}{x^2}}-1}=\frac{-2+0}{-\sqrt{1-0+0}-1}=1.$$

Never replace part of an expression by its limit

When computing a limit, NEVER replace part of the expression by its limit without handling the rest. Writing

$$\lim_{x\to-\infty}\frac{-2x+3}{-x\sqrt{1-\frac{2}{x}+\frac{3}{x^2}}-x}=\lim_{x\to-\infty}\frac{-2x+3}{-x\sqrt{1-0+0}-x}=\lim_{x\to-\infty}\frac{-2x+3}{-2x}=1$$

is as logically incorrect as writing

$$\lim_{x\to0}\frac{x}{x}=\frac{\lim x}{\lim x}=\lim_{x\to0}\frac{0}{x}=\lim_{x\to0}0=0.$$

(The correct full computation in (d) above keeps all parts until the end.)

Example (Squeeze Theorem at infinity)

Evaluate $\lim\limits_{x\to+\infty}\dfrac{\cos x}{e^x}$ if it exists.

Solution. Since $-1 \le \cos x \le 1$ for every real x,

$$-\frac{1}{e^x} \le \frac{\cos x}{e^x} \le \frac{1}{e^x} \qquad \text{for every real } x.$$

Since $\lim_{x\to+\infty}\left(-\frac{1}{e^x}\right) = 0$ and $\lim_{x\to+\infty}\frac{1}{e^x} = 0$, the Squeeze Theorem gives $\lim_{x\to+\infty}\frac{\cos x}{e^x} = 0$.

Comparing limits at infinity with limits of sequences, a natural question: are $\lim_{x\to+\infty} f(x)$ and $\lim_{n\to+\infty} f(n)$ the same?

From function limits to sequence limits

Theorem. Let $f : [1, +\infty) \to \mathbb{R}$ be a function. If the limit of function $\lim_{x\to+\infty} f(x) = L$ exists (either as a real number or infinite), then the limit of sequence $\lim_{n\to+\infty} f(n)$ also exists and equals the same L.

Proof. The sequence $(n) = (1, 2, 3, 4, \dots)$ diverges to $+\infty$ and none of its terms equals $+\infty$, so the result follows from the sequential limit theorem (direction (i)⇒(ii)). ■

Example. Since $\lim_{x\to+\infty}\frac{\cos x}{e^x} = 0$, we also have the limit of sequence $\lim_{n\to+\infty}\frac{\cos n}{e^n} = 0$.

Remark. Later techniques may make $\lim_{x\to+\infty} f(x)$ much easier to evaluate than $\lim_{n\to+\infty} f(n)$; then this theorem gives $\lim_{n\to+\infty} f(n) = \lim_{x\to+\infty} f(x)$.

Remark (converse fails)**.** Even if $\lim_{n\to+\infty} f(n)$ exists, $\lim_{x\to+\infty} f(x)$ may not. Example: $f(x) = \sin \pi x$. The sequence $(f(n)) = (\sin n\pi)$ is the constant zero sequence, so $\lim_{n\to+\infty} f(n) = 0$; but $\lim_{x\to+\infty} f(x)$ does not exist (argue as with $\sin x$ above).

The limit $(1 + \frac{1}{x})^x \to e$

Theorem.

$$\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = \lim_{x\to-\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Proof. For each $x \ge 1$ let $n = \lfloor x \rfloor$. Then $n \le x < n + 1$, so

$$1 + \frac{1}{n+1} < 1 + \frac{1}{x} \le 1 + \frac{1}{n}.$$

Each component exceeds 1, hence

$$\left(1+\frac{1}{n+1}\right)^n < \left(1+\frac{1}{x}\right)^x < \left(1+\frac{1}{n}\right)^{n+1}.$$

Now n is a positive integer and $n \to +\infty$ as $x \to +\infty$, so

$$\lim_{x\to+\infty}\left(1+\frac{1}{n+1}\right)^n = \lim_{n\to+\infty}\left(1+\frac{1}{n+1}\right)^n = \lim_{n\to+\infty}\frac{\left(1+\frac{1}{n+1}\right)^{n+1}}{1+\frac{1}{n+1}} = \frac{e}{1+0} = e,$$

and

$$\lim_{x\to+\infty}\left(1+\frac{1}{n}\right)^{n+1} = \lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^n\left(1+\frac{1}{n}\right) = e(1+0) = e.$$

By the Squeeze Theorem, $\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e$. For the second limit, substitute $t = -x$:

$$\begin{aligned}\lim_{x\to-\infty}\left(1+\frac{1}{x}\right)^x &= \lim_{t\to+\infty}\left(1-\frac{1}{t}\right)^{-t} = \lim_{t\to+\infty}\left(\frac{t}{t-1}\right)^t \\ &= \lim_{t\to+\infty}\left(1+\frac{1}{t-1}\right)^t = \lim_{t\to+\infty}\left(1+\frac{1}{t-1}\right)^{t-1}\left(1+\frac{1}{t-1}\right) \\ &= e(1+0) = e.\end{aligned}$$

■

Corollary. Applying the change of variable $t = \frac{1}{x}$:

$$\lim_{x\to0}(1+x)^{\frac{1}{x}} = e.$$

Example ($(\frac{x+1}{x-1})^x$)

Evaluate $\lim_{x\to+\infty}\left(\frac{x+1}{x-1}\right)^x$.

Solution. With $t = \frac{x-1}{2}$, we have $x = 2t+1$, so

$$\begin{aligned}\lim_{x\to+\infty}\left(\frac{x+1}{x-1}\right)^x &= \lim_{x\to+\infty}\left(1+\frac{2}{x-1}\right)^x = \lim_{t\to+\infty}\left(1+\frac{1}{t}\right)^{2t+1} \\ &= \left[\lim_{t\to+\infty}\left(1+\frac{1}{t}\right)^t\right]^2 \cdot \lim_{t\to+\infty}\left(1+\frac{1}{t}\right) = e^2\cdot 1 = e^2.\end{aligned}$$

An asymptote of the graph of a function is a line that approximates the trend of the function.

Definition. Let $m, c \in \mathbb{R}$ and f be a function.

(i) If $\lim_{x\to c^-} f(x) = +\infty$ or $-\infty$, or $\lim_{x\to c^+} f(x) = +\infty$ or $-\infty$, then the line $x = c$ is a vertical asymptote of the graph of f.

(ii) If $\lim_{x\to+\infty}[f(x)-(mx+c)] = 0$ or $\lim_{x\to-\infty}[f(x)-(mx+c)] = 0$, then the line $y = mx+c$ is a horizontal asymptote when $m = 0$, or a slant asymptote when $m \neq 0$.

Example (immediate asymptotes)

- $f(x) = \frac{1}{x}$: vertical asymptote $x = 0$; horizontal asymptote $y = 0$.
- $f(x) = e^x$: horizontal asymptote $y = 0$; no vertical or slant asymptotes.
- $f(x) = \ln x$: vertical asymptote $x = 0$; no horizontal or slant asymptotes.
- $f(x) = \tan x$: vertical asymptotes $x = n\pi + \frac{\pi}{2}$ for all integers n; no horizontal or slant asymptotes.
- $f(x) = \arctan x$: two horizontal asymptotes $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$; no vertical or slant asymptotes.

Example (all asymptotes of a rational function)

Let $f(x) = \dfrac{2x^3 - x^2 - 4x - 4}{x^2 - 4}$. Find all asymptotes of the graph of f.

Solution.

(i) **Vertical asymptotes.** Factor $x^2 - 4 = (x-2)(x+2)$. For every $a \neq \pm 2$,

$$\lim_{x \to a} f(x) = \frac{2a^3 - a^2 - 4a - 4}{(a-2)(a+2)},$$

which is finite, so no vertical asymptote there. Check $x = -2$ and $x = 2$.

- As $x \to -2$: the numerator $\to 2(-2)^3 - (-2)^2 - 4(-2) - 4 = -16$, and $(x-2) \to -4$, $(x+2) \to 0^+$ as $x \to -2^+$; hence

$$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} \frac{2x^3 - x^2 - 4x - 4}{(x-2)(x+2)} = +\infty.$$

  So $x = -2$ is a vertical asymptote.
- As $x \to 2$: the numerator $\to 2(2)^3 - (2)^2 - 4(2) - 4 = 0$, so $(x-2)$ is a factor:

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{(x-2)(2x^2 + 3x + 2)}{(x-2)(x+2)} = \lim_{x \to 2} \frac{2x^2 + 3x + 2}{x+2} = \frac{2(4) + 3(2) + 2}{2+2} = 4,$$

  which is finite; $x = 2$ is not a vertical asymptote.

(ii) **Horizontal/slant asymptotes.** Long division gives

$$f(x) = \frac{2x^3 - x^2 - 4x - 4}{x^2 - 4} = 2x - 1 + \frac{4x - 8}{x^2 - 4}.$$

Thus

$$\lim_{x \to \pm\infty} [f(x) - (2x-1)] = \lim_{x \to \pm\infty} \frac{4x - 8}{x^2 - 4} = \lim_{x \to \pm\infty} \frac{\frac{4}{x} - \frac{8}{x^2}}{1 - \frac{4}{x^2}} = \frac{0 - 0}{1 - 0} = 0,$$

so $y = 2x - 1$ is a slant asymptote. There are no horizontal asymptotes.

Finding m and c of an asymptote

Lemma. Let $m, c \in \mathbb{R}$ and f be a function. If $y = mx + c$ is a horizontal or slant asymptote of the graph of f, then either

$$m = \lim_{x \to +\infty} \frac{f(x)}{x} \quad \text{and} \quad c = \lim_{x \to +\infty} [f(x) - mx],$$

$$\text{or} \qquad m = \lim_{x \to -\infty} \frac{f(x)}{x} \quad \text{and} \quad c = \lim_{x \to -\infty} [f(x) - mx].$$

Proof. Left as an exercise. ■

Procedure for finding all asymptotes

(i) Identify the domain of f.

(ii) For each boundary point a of the domain, compute $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. If at least one is infinite, $x = a$ is a vertical asymptote.

(iii) If the domain is unbounded towards $+\infty$, compute $\lim_{x \to +\infty} \dfrac{f(x)}{x}$. If it is a finite real number m, compute $\lim_{x \to +\infty} [f(x) - mx]$; if this is a finite real number c, then $y = mx + c$ is a horizontal or slant asymptote. Do the same towards $-\infty$.

Example (asymptotes of $x\sin(\frac{1}{x})$)

Let $f(x) = x\sin\dfrac{1}{x}$. Find all asymptotes of the graph of f.

Solution. The domain is $(-\infty, 0) \cup (0, +\infty)$.

- The only boundary point is 0. By a Squeeze Theorem argument as before, $\lim\limits_{x\to 0} x\sin\dfrac{1}{x} = 0$. Since neither one-sided limit is infinite, there is no vertical asymptote.
- Next,
$$\lim_{x\to\pm\infty}\frac{f(x)}{x} = \lim_{x\to\pm\infty}\sin\frac{1}{x} = \sin 0 = 0,$$
and with the change of variable $t = \dfrac{1}{x}$,
$$\lim_{x\to\pm\infty}[f(x) - 0x] = \lim_{x\to\pm\infty} x\sin\frac{1}{x} = \lim_{t\to 0^{\pm}}\frac{\sin t}{t} = 1.$$
Hence $y = 0x + 1$, i.e. $y = 1$, is a horizontal asymptote. There are no slant asymptotes.

2.5 Rigorous Definition of Limits

So far almost all proofs were omitted because the definitions were informal: we never made precise "a_n gets closer and closer to L" or "x gets closer and closer to a". What we really mean is that a_n can be made as close to L as we like, if n gets sufficiently large: given any distance $\varepsilon > 0$ (no matter how small), a_n will have distance $< \varepsilon$ from L once n exceeds a certain number N.

Definition (limit of a sequence)**.** Let (a_n) be a sequence of real numbers. A real number L is a limit of (a_n) if for each $\varepsilon > 0$, there exists $N > 0$ such that
$$n \text{ is an integer and } n \geq N \quad\Longrightarrow\quad |a_n - L| < \varepsilon.$$
Pictorially: $\lim\limits_{n\to+\infty} a_n = L$ if, no matter how narrow a horizontal strip around L on the graph is drawn, the points (n, a_n) eventually all fall within the strip.

Remark. $\varepsilon > 0$ controls the distance between a_n and L: "for each $\varepsilon > 0$ (no matter how small)" rigorously says "a_n can be made as close to L as we like". N measures how large the index is: "there exists $N > 0$" and "if $n \geq N$" rigorously say "if n gets sufficiently large". N may depend on ε; usually smaller ε requires larger N.

Similarly, "$f(x)$ gets closer and closer to L when x gets closer and closer to a but $x \neq a$" means $f(x)$ can be made as close to L as we like, if x is sufficiently close to a but $x \neq a$: given any $\varepsilon > 0$, $f(x)$ has distance $< \varepsilon$ from L whenever the distance between x and a is nonzero but less than a certain δ.

Definition (limit of a function)**.** Let $I \subseteq \mathbb{R}$ be an open interval, $a \in I$, and $f : I \setminus \{a\} \to \mathbb{R}$. A real number L is a limit of $f(x)$ as x tends to a if for each $\varepsilon > 0$, there exists $\delta > 0$ such that
$$0 < |x - a| < \delta \quad\Longrightarrow\quad |f(x) - L| < \varepsilon.$$
Pictorially: no matter how narrow the horizontal strip $(L - \varepsilon, L + \varepsilon)$ is, we can draw the vertical strip $(a - \delta, a + \delta)$ narrow enough that the portion of the graph within the vertical strip (except possibly at a) falls inside the horizontal strip.

Remark. ε controls the distance between $f(x)$ and L; δ measures how close x and a are. The phrases "there exists $\delta > 0$" and "if $0 < |x - a| < \delta$" rigorously say "if x gets sufficiently close to a but $x \neq a$". δ may depend on ε; usually smaller ε requires smaller δ.

The rigorous definition gives a concrete way to prove the theorems about limits. First: uniqueness of limit (which came as "free lunch" under the informal definition, but needs proof now).

Uniqueness of limit (rigorous proof)

Theorem (uniqueness of limit)**.** Let $I \subseteq \mathbb{R}$ be an open interval, $a \in I$, and $f : I \setminus \{a\} \to \mathbb{R}$. If real numbers L and M are both limits of $f(x)$ as x tends to a, then $L = M$.

Proof. By definition, for each $\varepsilon > 0$,

- there exists $\delta_1 > 0$ such that $0 < |x - a| < \delta_1 \Rightarrow |f(x) - L| < \dfrac{\varepsilon}{2}$;
- there exists $\delta_2 > 0$ such that $0 < |x - a| < \delta_2 \Rightarrow |f(x) - M| < \dfrac{\varepsilon}{2}$.

Let $x \in I \setminus \{a\}$. If $0 < |x - a| < \min\{\delta_1, \delta_2\}$, then both conditions hold, so by the triangle inequality

$$|L - M| = |L - f(x) + f(x) - M| \leq |L - f(x)| + |f(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus the non-negative number $|L - M|$ is smaller than every positive real number, so $|L - M| = 0$, i.e. $L = M$. ■

Definition. Let $S \subseteq \mathbb{R}$. The maximum (resp. minimum) of S is the largest (resp. smallest) element contained in S:

- $\max S = M$ if $M \in S$ and $M \geq x$ for every $x \in S$;
- $\min S = m$ if $m \in S$ and $m \leq x$ for every $x \in S$.

So $\min\{\delta_1, \delta_2\}$ denotes the smaller of the two numbers.

Example (max and min may fail to exist)

Not every subset of $\mathbb{R}$ has a maximum or minimum:

$$\max\{0,1\} = 1, \quad \max[0,1] = 1, \quad \max(0,1) \text{ does not exist,}$$
$$\min\{0,1\} = 0, \quad \min[0,1] = 0, \quad \min(0,1) \text{ does not exist.}$$

Now we construct rigorous proofs of limits computed intuitively before. The triangle inequality and inequalities like $|\sin x| \leq |x|$ are often useful.

Example (ε-δ proof that $\lim\limits_{x\to 2} x^2 = 4$)

Prove using the rigorous definition that $\lim\limits_{x\to 2} x^2 = 4$.

Solution. Rough work: given arbitrary $\varepsilon > 0$, we need $\delta > 0$ such that $0 < |x-2| < \delta \Rightarrow |x^2 - 4| < \varepsilon$. Regardless of δ, if $0 < |x - 2| < \delta$ then by the triangle inequality

$$|x^2 - 4| = |(x-2)(x+2)| = |x-2|\,|x-2+4| \leq |x-2|(|x-2|+4) < \delta(\delta+4).$$

So choose δ with $\delta(\delta+4) \leq \varepsilon$: it suffices to require $\delta \leq 1$ and $\delta \leq \dfrac{\varepsilon}{5}$ (then $\delta(\delta+4) \leq \dfrac{\varepsilon}{5}(1+4) = \varepsilon$).

Proof: Let $\varepsilon > 0$ be arbitrary. Choose $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\} > 0$. If $0 < |x-2| < \delta$, then

$$|x^2 - 4| = |x-2|\,|x-2+4| \leq |x-2|(|x-2|+4) < \delta(\delta+4) \leq \frac{\varepsilon}{5}(1+4) = \varepsilon,$$

using the triangle inequality in the middle. Hence $\lim_{x\to 2} x^2 = 4$ by definition. ■

Example (ε-δ proof that $\lim_{x\to 1}(x+\frac{1}{x}) = 2$)

Prove using the rigorous definition that $\lim_{x\to 1}\left(x+\frac{1}{x}\right) = 2$.

Solution. Rough work: we need $\delta > 0$ such that $0 < |x-1| < \delta \Rightarrow \left|\left(x+\frac{1}{x}\right) - 2\right| < \varepsilon$. Regardless of δ, if $0 < |x-1| < \delta$ then

$$\left|\left(x+\frac{1}{x}\right) - 2\right| = \left|\frac{x^2-2x+1}{x}\right| = \frac{1}{|x|}|x-1|^2 < \frac{1}{|x|}\delta^2.$$

We need a lower bound for $|x|$: when x is near 1, it cannot be near 0. Triangle inequality:

$$1 = |(x-1)-x| \le |x-1| + |x| \quad \Longrightarrow \quad |x| \ge 1 - |x-1|.$$

Requiring $\delta \le \frac{1}{2}$ gives $|x| \ge 1 - |x-1| > 1 - \frac{1}{2} = \frac{1}{2}$, so the expression is $< 2\delta^2$. Finally require also $\delta \le \sqrt{\frac{\varepsilon}{2}}$ so that $2\delta^2 \le \varepsilon$.

Proof: Let $\varepsilon > 0$ be arbitrary. Choose $\delta = \min\left\{\frac{1}{2}, \sqrt{\frac{\varepsilon}{2}}\right\} > 0$. If $0 < |x-1| < \delta$, then by the triangle inequality

$$|x| \ge 1 - |x-1| > 1 - \delta \ge 1 - \frac{1}{2} = \frac{1}{2},$$

so

$$\left|\left(x+\frac{1}{x}\right) - 2\right| = \frac{1}{|x|}|x-1|^2 < 2|x-1|^2 < 2\delta^2 \le 2\left(\sqrt{\frac{\varepsilon}{2}}\right)^2 = \varepsilon.$$

Hence $\lim_{x\to 1}\left(x+\frac{1}{x}\right) = 2$ by definition. ■

With the rigorous definition one can prove all the other theorems of this chapter: the limit existence criterion, the arithmetic rules, limits of elementary functions, change of variables, the limit inequalities and the Squeeze Theorem. We give one of the easiest proofs and leave the rest as exercises.

Example (rigorous proof of the sum rule)

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ exist as real numbers. Prove that

$$\lim_{x\to a}[f(x)+g(x)] = L + M.$$

Solution. Let $\varepsilon > 0$ be arbitrary. Since the two limits exist,

- there exists $\delta_1 > 0$ such that $0 < |x-a| < \delta_1 \Rightarrow |f(x)-L| < \frac{\varepsilon}{2}$;
- there exists $\delta_2 > 0$ such that $0 < |x-a| < \delta_2 \Rightarrow |g(x)-M| < \frac{\varepsilon}{2}$.

Choose $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x-a| < \delta$, then by the triangle inequality

$$|(f(x)+g(x))-(L+M)| = |(f(x)-L)+(g(x)-M)| \le |f(x)-L|+|g(x)-M| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon.$$

Therefore $\lim_{x\to a}[f(x)+g(x)] = L+M$. ■

Infinite limits and limits at infinity are also rigorously formulated.

Definition (limit at infinity)**.** Let $f : (c, +\infty) \to \mathbb{R}$. A real number L is a limit of $f(x)$ as x tends to $+\infty$ if for each $\varepsilon > 0$, there exists $r \in \mathbb{R}$ such that

$$x > r \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Definition (infinite limit)**.** Let $I \subseteq \mathbb{R}$ be an open interval, $a \in I$, and $f : I \setminus \{a\} \to \mathbb{R}$. The limit of $f(x)$ as $x \to a$ is $+\infty$ if for each $r \in \mathbb{R}$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) > r.$$

2.6 Continuity

Elementary functions allow direct substitution when computing limits at points of their domain; functions with this property are called continuous.

Definition. Let I be an interval, $f : I \to \mathbb{R}$, and $a \in I$.

(i) If a is in the interior of I, f is continuous at a if $\lim_{x\to a} f(x) = f(a)$.

(ii) If a is the left end-point of I, f is continuous at a if $\lim_{x\to a^+} f(x) = f(a)$.

(iii) If a is the right end-point of I, f is continuous at a if $\lim_{x\to a^-} f(x) = f(a)$.

f is discontinuous at a if it is not continuous at a.

Remark. Saying f is continuous at a implicitly requires all three of:

- f is well-defined at a (i.e. a is in the domain of f);
- the limit $\lim_{x\to a} f(x)$ (or one-sided limit at an end-point) exists as a real number;
- $\lim_{x\to a} f(x) = f(a)$.

Intuitively, f is continuous at a if its graph does not "break" and has no "sudden jump" through $(a, f(a))$; discontinuities arise from a hole, a jump, or a missing point.

Definition. f is continuous on an interval if it is continuous at every number in that interval; f is continuous everywhere (or simply continuous) if it is continuous on its whole domain. Intuitively, f is continuous on an interval if its whole graph can be drawn without lifting the pen.

Continuous functions: basic classes and closure properties

Theorem. Polynomials, rational functions, power functions, exponential functions, logarithmic functions, trigonometric functions and inverse trigonometric functions are all continuous.

Theorem. The absolute value function is continuous.

Proof. For every $a \neq 0$, near a the absolute value function is just a monomial (x or $-x$), hence continuous at a by the previous theorem. At 0: $\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0$, $\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} (-x) = 0$, and $|0| = 0$; so it is continuous at 0 also. ■

Theorem. Let f, g be continuous. Then $f+g$, $f-g$ and fg are all continuous, and $\dfrac{f}{g}$ is continuous at every number except those a where $g(a) = 0$.

Example (floor function)

The floor function $f(x) = \lfloor x \rfloor$ is discontinuous at every integer, although the integers all belong to its domain.

Example (continuity with a parameter)

Let

$$f(x) = \begin{cases} a^2x^2 - 2a & \text{if } x \le 1, \\ a\cos \pi x + 1 & \text{if } x > 1. \end{cases}$$

Determine all values of a such that f is continuous everywhere.
Solution. On $(-\infty, 1)$ and $(1, +\infty)$, f is continuous, being sums of polynomials and trigonometric functions. Continuity at 1 requires

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1).$$

Compute:

$$\lim_{x\to 1^-} f(x) = a^2 - 2a, \qquad \lim_{x\to 1^+} f(x) = a\cos \pi + 1 = -a + 1, \qquad f(1) = a^2 - 2a.$$

The requirement becomes $a^2 - 2a = -a + 1$, i.e. $a^2 - a - 1 = 0$, giving

$$a = \frac{1 + \sqrt{5}}{2} \qquad \text{or} \qquad a = \frac{1 - \sqrt{5}}{2}.$$

Example (functional equation and continuity)

Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)f(y)$ for every $x, y \in \mathbb{R}$, and suppose $f(0) \neq 0$.
(a) Evaluate $f(0)$.
(b) If f is continuous at 0, show that f is continuous on $\mathbb{R}$.
Solution.
(a) Putting $x = y = 0$: $f(0) = f(0)f(0)$, i.e. $f(0)[f(0) - 1] = 0$. Since $f(0) \neq 0$, $f(0) = 1$.
(b) Continuity at 0 gives $\lim_{t\to 0} f(t) = f(0) = 1$. For every $a \in \mathbb{R}$,

$$\begin{aligned} \lim_{x\to a} f(x) &= \lim_{x\to a} f((x-a)+a) = \lim_{x\to a} f(x-a)f(a) \\ &= f(a)\lim_{x\to a} f(x-a) = f(a)\lim_{t\to 0} f(t) \\ &= f(a)\cdot 1 = f(a), \end{aligned}$$

so f is continuous at a. Since a is arbitrary, f is continuous on $\mathbb{R}$. ■

It is often convenient to move the limit sign through a function symbol; this swapping is valid if the function is continuous (a consequence of the change-of-variables theorem).

Passing limits through continuous functions

Theorem. Let $a \in \mathbb{R}$ and f, g be functions. If $\lim_{x\to a} g(x) = b$ exists as a real number and f is well-defined near b and continuous at b, then

$$\lim_{x\to a} f(g(x)) = f\Big(\lim_{x\to a} g(x)\Big) = f(b).$$

A similar statement holds for one-sided limits and limits at infinity.
Theorem. Let $b \in \mathbb{R}$, let (a_n) converge to b, and let f be well-defined near b and

continuous at b. Then $(f(a_n))$ converges and

$$\lim_{n\to+\infty} f(a_n) = f\left(\lim_{n\to+\infty} a_n\right) = f(b).$$

Theorem. The composition of two continuous functions is continuous. In particular, if f is continuous, then $|f|$, f^a, e^f, $\ln f$, $\cos f$, $\sin f$, $\arccos f$, $\arcsin f$, etc. are all continuous.

Example (swapping limit and function)

Evaluate $\lim_{x\to\frac{\pi}{2}} \tan(\sin x)$.

Solution. Since $\lim_{x\to\frac{\pi}{2}} \sin x = \sin\frac{\pi}{2} = 1$ and tan is continuous at 1,

$$\lim_{x\to\frac{\pi}{2}} \tan(\sin x) = \tan\left(\lim_{x\to\frac{\pi}{2}} \sin x\right) = \tan 1.$$

Example (continuity plus sequence limit)

Let $f:\mathbb{R}\to\mathbb{R}$ be continuous, with $f\left(\frac{1}{n}\right) = \left(1+\frac{1}{n}\right)^n$ for every $n\in\mathbb{N}$. Find $f(0)$.

Solution. Since f is continuous at 0,

$$f(0) = f\left(\lim_{n\to+\infty}\frac{1}{n}\right) = \lim_{n\to+\infty} f\left(\frac{1}{n}\right) = \lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^n = e.$$

(It is incorrect to deduce $f(x) = (1+x)^{\frac{1}{x}}$ for every $x>0$: the given condition is valid only at the reciprocals of positive integers.)

Example ($x\ln(1+\frac{1}{x})$ and $\frac{a^h-1}{h}$)

Evaluate $\lim_{x\to+\infty} x\ln\left(1+\frac{1}{x}\right)$; hence, for each $a>0$, evaluate $\lim_{h\to0}\frac{a^h-1}{h}$.

Solution.

(i) For every $x>0$, $x\ln\left(1+\frac{1}{x}\right) = \ln\left(1+\frac{1}{x}\right)^x$. Since $\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e$ and ln is continuous at e,

$$\lim_{x\to+\infty} x\ln\left(1+\frac{1}{x}\right) = \lim_{x\to+\infty}\ln\left(1+\frac{1}{x}\right)^x = \ln\left[\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x\right] = \ln e = 1.$$

(ii) Similarly one obtains $\lim_{x\to-\infty} x\ln\left(1+\frac{1}{x}\right) = 1$. Given $a>0$, substitute $h = \log_a\left(1+\frac{1}{x}\right) = \frac{1}{\ln a}\ln\left(1+\frac{1}{x}\right)$ (so $a^h = 1+\frac{1}{x}$, and $h\to0^\pm$ corresponds to $x\to\pm\infty$):

$$\lim_{h\to0^\pm}\frac{a^h-1}{h} = \lim_{x\to\pm\infty}\frac{(1+\frac{1}{x})-1}{\frac{1}{\ln a}\ln(1+\frac{1}{x})} = \frac{\ln a}{\lim_{x\to\pm\infty} x\ln(1+\frac{1}{x})} = \frac{\ln a}{1} = \ln a.$$

Therefore $\lim_{h\to0}\frac{a^h-1}{h} = \ln a$.

Three important properties of continuous functions of a real variable: sign stability, the Intermediate Value Theorem, and the Extreme Value Theorem.

Sign stability

Lemma (sign stability)**.** Let $I \subseteq \mathbb{R}$ be an interval, $a \in I$, and $f : I \to \mathbb{R}$ continuous at a. If $f(a) > 0$, then $f(x) > 0$ for every x near a; more precisely, there exists $\delta > 0$ such that $f(x) > 0$ for every $x \in (a - \delta, a + \delta) \cap I$. (This $\delta > 0$ encodes the mathematical meaning of the word "near".)

The Intermediate Value Theorem follows from sign stability. Using it one can also show the inverse f^{-1} of a one-to-one continuous function $f : [a, b] \to \mathbb{R}$ is continuous (done in MATH2033).

Intermediate Value Theorem

Theorem (Intermediate Value Theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous on the bounded closed interval $[a, b]$. If

$$f(a)f(b) < 0$$

(one of $f(a), f(b)$ is positive and the other negative), then there exists $c \in (a, b)$ such that $f(c) = 0$.

IVT is often used to show the existence of points taking certain function values, e.g. zeros of functions or solutions of equations.

Example (locating roots via IVT)

Let $f(x) = x^3 - 8x + 5$. Since f is a polynomial, it is continuous everywhere. Since

$$f(0) = 5 > 0 \qquad \text{and} \qquad f(1) = 1 - 8 + 5 = -2 < 0,$$

by IVT, f has at least one root in $(0, 1)$.
Show that f has exactly three real roots.
Solution. (Without the cubic formula, existence of three roots can still be shown via IVT.)

- We have $f(0) > 0$ and $f(1) < 0$. Moreover, $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to+\infty} f(x) = +\infty$, so there exist $a < 0$ and $b > 1$ with $f(a) < 0$ and $f(b) > 0$ (or simply note $f(-100) < 0$, $f(100) > 0$). Since f is continuous everywhere, IVT gives at least one root in each of $(a, 0)$, $(0, 1)$, $(1, b)$; so f has at least three distinct real roots.
- Since f has degree 3, it has at most three real roots.

Therefore f has exactly three real roots. ■

Example (Fixed Point Theorem)

Let $f : [0, 1] \to [0, 1]$ be continuous. Show that the equation $f(x) = x$ has at least one solution in $[0, 1]$.
Solution. Consider $g : [0, 1] \to \mathbb{R}$, $g(x) = f(x) - x$; g is continuous since f is. Since $f(0), f(1) \in [0, 1]$,

$$g(0) = f(0) - 0 \geq 0, \qquad g(1) = f(1) - 1 \leq 0.$$

We claim there exists $c \in [0, 1]$ with $g(c) = 0$:
(i) if $g(0) = 0$ or $g(1) = 0$, take $c = 0$ or $c = 1$;
(ii) if $g(0) > 0$ and $g(1) < 0$, the claim follows from IVT.
Then $g(c) = 0$ means $f(c) = c$, i.e. c solves $f(x) = x$. ■

Definition. Let $A \subseteq \mathbb{R}$ be the domain of $f : A \to \mathbb{R}$, and $c \in A$.

- If $f(x) \le f(c)$ for every $x \in A$, f attains global (absolute) maximum at c; $f(c)$ is a global maximum value and $(c, f(c))$ a global maximum point. We write $f(c) = \max\{f(x) : x \in A\}$.
- If $f(x) \ge f(c)$ for every $x \in A$, f attains global (absolute) minimum at c; similarly for the global minimum value/point, $f(c) = \min\{f(x) : x \in A\}$.

Global maxima and minima are collectively global extrema.

Definition. Let $A \subseteq \mathbb{R}$ be the domain of $f : A \to \mathbb{R}$, and $c \in A$.

- If $f(x) \le f(c)$ for every $x \in A$ near c, i.e. there exists $\delta > 0$ with
$$f(x) \le f(c) \quad \text{for every } x \in (c-\delta, c+\delta) \cap A,$$
then f attains local (relative) maximum at c; $f(c)$ is a local maximum value and $(c, f(c))$ a local maximum point.
- Similarly for local (relative) minimum with $f(x) \ge f(c)$.

Local maxima and minima are collectively local extrema. (On a typical graph, local maxima/minima alternate along the curve; one local maximum may also be the global maximum, and one local minimum the global minimum.)

Extreme Value Theorem

Theorem (Extreme Value Theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous on the bounded closed interval $[a, b]$. Then f attains global maximum and global minimum somewhere in $[a, b]$: there exist $c, d \in [a, b]$ such that
$$f(c) \le f(x) \le f(d) \qquad \text{for every } x \in [a, b].$$
(The proof is technical and deferred to MATH2033.)

Remark. Graphically, EVT says the graph of a continuous function on a closed interval has at least one highest point and one lowest point.

- EVT only asserts existence; it does not tell us how to find the extrema (chapter 4 treats this for special classes of functions).
- EVT fails on open intervals: $f(x) = \tan x$ is continuous on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, but its graph has neither highest nor lowest point since $\lim_{x \to (\frac{\pi}{2})^-} \tan x = +\infty$ and $\lim_{x \to (-\frac{\pi}{2})^+} \tan x = -\infty$. It fails even for bounded continuous functions on open intervals.

Remark (Dirichlet function)**.** The Dirichlet function $f : \mathbb{R} \to \mathbb{R}$,
$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational,} \end{cases}$$
is discontinuous everywhere, because there is always a rational number between any two irrational numbers, and an irrational number between any two rational numbers.

2.7 Supplement: Triangle Inequality

In the lecture we have seen the following property of the absolute value function.

Triangle inequality

Theorem. Let a and b be real numbers. Then
$$|a + b| \le |a| + |b|.$$

This inequality is called the triangle inequality. Its name comes from its generalization to "higher dimensions". Using $|x| = \sqrt{x^2}$, it can be rewritten as
$$\sqrt{(a+b)^2} \le \sqrt{a^2} + \sqrt{b^2}.$$

In fact there is a "two-dimensional version" (studied in MATH1014): for all real numbers a_1, a_2, b_1, b_2,

$$\sqrt{(a_1+b_1)^2+(a_2+b_2)^2} \le \sqrt{a_1^2+a_2^2}+\sqrt{b_1^2+b_2^2}.$$

Now consider three points P, Q, R in the coordinate plane with coordinates (x_1, y_1), (x_2, y_2), (x_3, y_3) respectively. Setting

$$a_1 = x_3 - x_1, \quad a_2 = y_3 - y_1, \quad b_1 = x_2 - x_3, \quad b_2 = y_2 - y_3$$

in the two-dimensional version gives

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \le \sqrt{(x_3-x_1)^2+(y_3-y_1)^2}+\sqrt{(x_2-x_3)^2+(y_2-y_3)^2}.$$

This says precisely that

$$PQ \le PR + RQ,$$

i.e. the length of an edge of a triangle is never greater than the sum of the lengths of the two other edges. This explains the name "triangle inequality".

2.8 Supplement: Stolz–Cesàro Theorem

L'Hôpital's rule and limits of sequences. A popular and natural question after learning L'Hôpital's rule:

Question. Can L'Hôpital's rule be applied to compute limits of sequences? E.g., for

$$\lim_{n\to+\infty} \frac{\ln n}{n},$$

can we simply say that because $\lim\limits_{n\to+\infty} \frac{\frac{1}{n}}{1} = 0$ exists, by L'Hôpital's rule

$$\lim_{n\to+\infty} \frac{\ln n}{n} = \lim_{n\to+\infty} \frac{\frac{d}{dn}\ln n}{\frac{d}{dn}n} = \lim_{n\to+\infty} \frac{\frac{1}{n}}{1} = 0\ ?$$

The brief answer: "the result is correct but the reasoning is incorrect". The symbol n takes positive integer values only; it makes no sense to "differentiate with respect to n", because we can only differentiate functions defined on open intervals. This leads to:

Question. What if we relabel n by x and allow x to take values in $(0, +\infty)$? Does L'Hôpital's rule then become applicable?

Good news — the answer is "yes", because of the following theorem (a consequence of the sequential limit theorem).

From function limit to sequence limit

Theorem. Let $f : [1, +\infty) \to \mathbb{R}$ be a function. If the limit of function $\lim\limits_{x\to+\infty} f(x) = L$ exists (either as a real number or infinite), then the limit of sequence $\lim\limits_{n\to+\infty} f(n)$ also exists and equals the same L.

In summary, the correct way of "applying L'Hôpital's rule to compute the limit of a sequence":

Example ($\lim \frac{\ln n}{n}$ via L'Hôpital on functions)

Compute $\lim\limits_{n\to+\infty} \frac{\ln n}{n}$.

Solution. Let $f, g : (0, +\infty) \to \mathbb{R}$ be $f(x) = \ln x$ and $g(x) = x$. Since $\lim\limits_{x\to+\infty} g(x) =$

$+\infty$ and

$$\lim_{x\to+\infty}\frac{f'(x)}{g'(x)}=\lim_{x\to+\infty}\frac{\frac{1}{x}}{1}=0,$$

by L'Hôpital's rule $\lim_{x\to+\infty}\dfrac{f(x)}{g(x)}=0$. Thus by the theorem above (or by the sequential limit theorem),

$$\lim_{n\to+\infty}\frac{\ln n}{n}=\lim_{n\to+\infty}\frac{f(n)}{g(n)}=\lim_{x\to+\infty}\frac{f(x)}{g(x)}=0.$$

Although differentiation with respect to the index n makes no sense, there is an analogous operation on sequences, the (one-step) difference:

$$\Delta a_n := a_{n+1}-a_n.$$

Indeed there is a result about sequences analogous to L'Hôpital's rule.

Stolz–Cesàro theorem, $\frac{0}{0}$ case

Theorem (Stolz–Cesàro, $\frac{0}{0}$ case)**.** Let (a_n) and (b_n) be sequences of real numbers. If

(i) $\lim_{n\to+\infty} a_n=\lim_{n\to+\infty} b_n=0$ and (b_n) is strictly decreasing, and

(ii) $\lim_{n\to+\infty}\dfrac{a_{n+1}-a_n}{b_{n+1}-b_n}$ exists (either as a finite real number or infinity),

then $\lim_{n\to+\infty}\dfrac{a_n}{b_n}$ also exists and

$$\lim_{n\to+\infty}\frac{a_n}{b_n}=\lim_{n\to+\infty}\frac{a_{n+1}-a_n}{b_{n+1}-b_n}.$$

Stolz–Cesàro theorem, */infinity case

Theorem (Stolz–Cesàro, $\frac{*}{\infty}$ case)**.** Let (a_n) and (b_n) be sequences of real numbers. If

(i) (b_n) is (strictly increasing and diverges to $+\infty$) or (strictly decreasing and diverges to $-\infty$), and

(ii) $\lim_{n\to+\infty}\dfrac{a_{n+1}-a_n}{b_{n+1}-b_n}$ exists (either as a finite real number or infinity),

then $\lim_{n\to+\infty}\dfrac{a_n}{b_n}$ also exists and

$$\lim_{n\to+\infty}\frac{a_n}{b_n}=\lim_{n\to+\infty}\frac{a_{n+1}-a_n}{b_{n+1}-b_n}.$$

Remark. The above results are stated without proof. You are encouraged to prove them by mimicking the ε-δ proofs of L'Hôpital's rule (Supplementary Note D); note that the proofs use "ε-N" instead of "ε-δ", since we are dealing with limits of sequences.

Example ($\lim\frac{\ln n}{n}$ via Stolz–Cesàro)

Compute $\lim_{n\to+\infty}\dfrac{\ln n}{n}$ again, this time using the Stolz–Cesàro Theorem.

Solution. The denominator sequence (n) is strictly increasing and diverges to $+\infty$.

Since

$$\lim_{n\to+\infty}\frac{\ln(n+1)-\ln n}{(n+1)-n}=\lim_{n\to+\infty}\frac{\ln\left(1+\frac{1}{n}\right)}{1}=\frac{\ln(1+0)}{1}=0,$$

by the Stolz–Cesàro Theorem ($\frac{*}{\infty}$ case) we conclude again that

$$\lim_{n\to+\infty}\frac{\ln n}{n}=0.$$

3 Derivatives

3.1 Derivatives and Differentiability

Remark 3.1 — Slope of a Secant Line

Let $I\subseteq\mathbb{R}$ be an interval, let $a,b\in I$ be a distinct pair of numbers and let $f:I\to\mathbb{R}$. Geometrically, the quantity

$$\frac{f(b)-f(a)}{b-a}$$

represents the slope of the secant line joining the two points $(a,f(a))$ and $(b,f(b))$ on the graph of f. As $b\to a$, this secant line approaches a line which touches the graph of f at the single point $(a,f(a))$ — a tangent line. (Picture: a curve $y=f(x)$ with a secant through $(a,f(a))$ and $(b,f(b))$ rotating into the tangent at $(a,f(a))$ as $b\to a$.)

Definition 3.2. Let f be a function.

- Let a be a real number. The **derivative of f at a** is defined by the limit

$$f'(a):=\lim_{b\to a}\frac{f(b)-f(a)}{b-a},$$

or equivalently (with the change of variable $h=b-a$)

$$f'(a):=\lim_{h\to0}\frac{f(a+h)-f(a)}{h}.$$

- Replacing the fixed number a by a real variable x, the **derivative of f** is the function f' defined by

$$f'(x):=\lim_{h\to0}\frac{f(x+h)-f(x)}{h}$$

for every x in the domain of f at which the limit exists as a (finite) real number. The domain of f' is therefore a subset of the domain of f.

Definition 3.3. Let f be a function.

- Let a be a real number. f is said to be **differentiable at a** if $f'(a)$ exists, i.e. if the limit

$$\lim_{b\to a}\frac{f(b)-f(a)}{b-a},\qquad\text{or}\qquad\lim_{h\to0}\frac{f(a+h)-f(a)}{h},$$

exists as a finite real number.
- f is said to be **differentiable on an interval** if it is differentiable at every number in that interval.
- The process of finding the derivative of a function is called **differentiation**. To differentiate f means to find its derivative f'.

Remark 3.4 — Geometric Meaning of the Derivative

Let a be a real number and let f be differentiable at a. Then the graph of f is smooth near a. Geometrically, $f'(a)$ represents the **slope of the tangent line** to the graph of f at the point $(a, f(a))$.

Example 3.5 — Derivative of $f(x) = x^2$ from Definition

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Find the derivative of f from definition.
Solution. For every real number x,

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0} \frac{2hx + h^2}{h} = \lim_{h\to 0} (2x + h) = 2x.$$

Therefore f is differentiable everywhere on $\mathbb{R}$, and its derivative $f' : \mathbb{R} \to \mathbb{R}$ is given by $f'(x) = 2x$.

Example 3.6 — Derivative of $f(x) = \frac{1}{x}$ from Definition

Let $f : \mathbb{R}\setminus\{0\} \to \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Find the derivative of f from definition.
Solution. For each non-zero real number x,

$$\begin{aligned}\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h\to 0} \frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \lim_{h\to 0} \frac{x - (x+h)}{hx(x+h)} \\ &= \lim_{h\to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.\end{aligned}$$

Therefore f is differentiable on $\mathbb{R} \setminus \{0\}$, and $f' : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ is given by $f'(x) = -\frac{1}{x^2}$.

Example 3.7 — Derivative of $f(x) = \sqrt{x}$ from Definition

Let $f : [0, +\infty) \to \mathbb{R}$ be defined by $f(x) = \sqrt{x}$. Find the derivative of f from definition.
Solution. For each positive real number x (the computation does not work for $x = 0$),

$$\begin{aligned}\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h\to 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}\right) \\ &= \lim_{h\to 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.\end{aligned}$$

Therefore f is differentiable on $(0, +\infty)$, and $f' : (0, +\infty) \to \mathbb{R}$ is given by $f'(x) = \frac{1}{2\sqrt{x}}$. Note that this time the domain of f' is different from the domain of f.

Example 3.8 — A Functional Equation

Let $f : \mathbb{R} \to \mathbb{R}$ be a non-constant function satisfying

$$f(x+y) = f(x) + f(y) + f(x)f(y) \qquad \text{for every } x, y \in \mathbb{R}.$$

If $f(0) = 0$ and if f is differentiable at 0, show that f is differentiable on $\mathbb{R}$.

Solution. Since f is differentiable at 0 and $f(0) = 0$, the limit

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{f(h)}{h}$$

exists as a real number. Then for every $x \in \mathbb{R}$,

$$\begin{aligned}\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h\to 0} \frac{[f(x) + f(h) + f(x)f(h)] - f(x)}{h} = \lim_{h\to 0} \frac{f(h) + f(x)f(h)}{h} \\ &= \lim_{h\to 0} \frac{(1+f(x))f(h)}{h} = (1+f(x)) \lim_{h\to 0} \frac{f(h)}{h} = (1+f(x))f'(0),\end{aligned}$$

where $1 + f(x)$ is independent of h. This limit exists as a real number, so f is differentiable on $\mathbb{R}$. ■

Remark 3.9 — Average Rate of Change

In applications, write $y = f(x)$ with y depending on the independent variable x. Rename the denominator h as Δx (a small change in x) and the numerator $f(x + \Delta x) - f(x)$ as Δy (the corresponding change in y). Then

$$f'(x) = \lim_{\Delta x\to 0} \frac{\Delta y}{\Delta x}$$

is the limit of the **average rate of change of y with respect to x.**

Remark 3.10 — Notations for the Derivative

- (Lagrange) When we emphasize that the derivative of f is a function, we write f'.
- (Leibniz) When we emphasize that the derivative is the rate of change of a quantity $y = f(x)$ with respect to x, we write $\dfrac{dy}{dx}$. The derivative of f at a is then denoted $\left.\dfrac{dy}{dx}\right|_{x=a}$.
- Without labelling the function: the derivative of $f(x) = x^2 + 3x$ can be written $\dfrac{d}{dx}(x^2 + 3x)$; at a we write $\left.\dfrac{d}{dx}(x^2 + 3x)\right|_{x=a}$.

The differentiability of f at a, i.e. the existence of $f'(a) = \lim\limits_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$, requires that the one-sided limits

$$\lim_{h\to 0^-} \frac{f(a+h) - f(a)}{h} \qquad \text{and} \qquad \lim_{h\to 0^+} \frac{f(a+h) - f(a)}{h}$$

both exist and are equal.

Definition 3.11. Let a be a real number and f be a function. The one-sided limits

$$f'_-(a) := \lim_{h\to 0^-} \frac{f(a+h) - f(a)}{h} \qquad \text{and} \qquad f'_+(a) := \lim_{h\to 0^+} \frac{f(a+h) - f(a)}{h}$$

are respectively called the **left-hand derivative** and the **right-hand derivative** of f at a.

Theorem 3.12 — Differentiability via One-sided Derivatives

Let a be a real number and f be a function. Then f is differentiable at a if and only

if $f'_-(a)$ and $f'_+(a)$ both exist as finite real numbers and

$$f'_-(a) = f'_+(a).$$

Example 3.13 — Differentiability of $f(x) = |x|$

Let $f : \mathbb{R} \to \mathbb{R}$ be the absolute value function $f(x) = |x|$. At which real numbers is f differentiable?

Solution.

- For each $a > 0$:

$$\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h\to 0} \frac{|a+h| - |a|}{h} = \lim_{h\to 0} \frac{(a+h) - a}{h} = 1,$$

  a finite real number, so f is differentiable at each $a > 0$ with $f'(a) = 1$.
- Similarly, for each $a < 0$:

$$\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h\to 0} \frac{-(a+h) + a}{h} = -1,$$

  so f is differentiable at each $a < 0$ with $f'(a) = -1$.
- At 0:

$$f'_-(0) = \lim_{h\to 0^-} \frac{|h| - |0|}{h} = \lim_{h\to 0^-} \frac{-h}{h} = -1, \qquad f'_+(0) = \lim_{h\to 0^+} \frac{|h| - |0|}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

  Since $f'_-(0) \neq f'_+(0)$, f is not differentiable at 0.

Therefore f is differentiable on $(-\infty, 0) \cup (0, +\infty)$, with

$$f'(x) = \begin{cases} -1 & \text{if } x < 0, \\ 1 & \text{if } x > 0. \end{cases}$$

Theorem 3.14 — Differentiability Implies Continuity

Let a be a real number and f be a function. If f is differentiable at a, then f is continuous at a.

Proof. For every x near a and $x \neq a$, we have

$$f(x) = f(a) + [f(x) - f(a)] = f(a) + \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

Since f is differentiable at a, the limit $f'(a) = \lim_{x\to a} \dfrac{f(x) - f(a)}{x - a}$ exists as a finite real number. So

$$\begin{aligned} \lim_{x\to a} f(x) &= f(a) + \lim_{x\to a} \left[\frac{f(x) - f(a)}{x - a} \cdot (x - a) \right] = f(a) + \lim_{x\to a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x\to a} (x - a) \\ &= f(a) + f'(a) \cdot (a - a) = f(a). \end{aligned}$$

Therefore f is continuous at a. ■

Theorem 3.14 is graphically intuitive: if the graph is smooth, it can be drawn without lifting the pen. With this theorem, whenever we are given differentiability, we can automatically use continuity as well.

Example 3.15 — Determining Constants from Differentiability

Let

$$f(x) = \begin{cases} 4x+1 & \text{if } x < 2, \\ a & \text{if } x = 2, \\ bx^2 + c & \text{if } x > 2. \end{cases}$$

If f is differentiable at 2, find the values of a, b and c.

Solution. Since f is differentiable at 2, it is continuous at 2, so

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2).$$

Now

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} (4x+1) = 4(2)+1 = 9,$$
$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} (bx^2+c) = 4b+c, \qquad f(2) = a.$$

These together imply $a = 9 = 4b + c$. Next we use differentiability at 2:

$$f'_-(2) = \lim_{x\to 2^-} \frac{f(x)-f(2)}{x-2} = \lim_{x\to 2^-} \frac{(4x+1)-9}{x-2} = \lim_{x\to 2^-} \frac{4x-8}{x-2} = \lim_{x\to 2^-} 4 = 4,$$

and

$$f'_+(2) = \lim_{x\to 2^+} \frac{(bx^2+c)-(4b+c)}{x-2} = \lim_{x\to 2^+} \frac{b(x^2-4)}{x-2} = \lim_{x\to 2^+} b(x+2) = 4b.$$

So the existence of the limit implies $4b = 4$. Therefore $a = 9$, $b = 1$ and $c = 5$.

Remark 3.16 — Geometry of One-sided Derivatives

Let a be a real number and f be continuous at a with both one-sided derivatives existing. Geometrically, $f'_-(a)$ is the slope of the tangent line to the graph of f from the left of $(a, f(a))$, while $f'_+(a)$ is the slope of the tangent from the right.

Example 3.17 — Sketching a Graph from Conditions

Sketch the graph of a function f satisfying all the following conditions:

(i) The domain of f is $[0, 5]$, and the range of f is $[-2, 2]$.
(ii) f is continuous on $[0, 5]$.
(iii) $f'(x) < 0$ for every $x < 1$.
(iv) $\lim\limits_{x\to 1^+} \dfrac{f(x)+1}{x-1} = 2$.
(v) $f'(3) = 0$.
(vi) $f'(x) < 0$ for every $x > 3$.
(vii) $\lim\limits_{x\to 5^-} \dfrac{f(x)+2}{x-5} = -\infty$.

Solution. Translate each condition into graphical information.

- Conditions (i)–(ii): the graph is one single piece of curve whose leftmost point hits $x = 0$, rightmost point hits $x = 5$, highest point(s) hit $y = 2$ and lowest point(s) hit $y = -2$; it must pass the vertical line test.
- Condition (iii): on $[0, 1]$ the graph is smooth and the slope of its tangent line is negative.
- The limit in (iv) being finite implies $\lim\limits_{x\to 1^+} f(x) = -1$ (cf. Example 2.51). Since f is continuous at 1 by (ii), $f(1) = -1$, i.e. the graph passes through $(1, -1)$. The limit in (iv) is then $f'_+(1) = 2$: the tangent line from the right of $(1, -1)$ has slope 2.

- Condition (v): the graph has a horizontal tangent line near $x = 3$; it says nothing about the value $f(3)$.
- Condition (vi): on $[3, 5]$ the graph is smooth with negative tangent slope.
- Condition (vii) implies $\lim_{x\to 5^-} f(x) \geq -2$ (the limit exists by continuity at 5). Since f is continuous at 5, $f(5) \geq -2$ (if $f(5) = -2$, the limit in (vii) is $f'_-(5)$). The graph passes through a point $(5, a)$ with $a \geq -2$, and the tangent line from the left of this point is vertical.

(The resulting sketch: a curve on $[0, 5]$ inside the strip $-2 \leq y \leq 2$, decreasing smoothly on $[0, 1]$, passing through $(1, -1)$ with right tangent slope 2, having a horizontal tangent at $x = 3$, decreasing again on $[3, 5]$, and ending at $(5, a)$, $a \geq -2$, with a vertical tangent from the left.)

Remark 3.18 — Continuous but NOT Differentiable

The converse to Theorem 3.14 does not hold: there exist functions continuous but not differentiable at a point.

- $f(x) = |x|$ is continuous on $\mathbb{R}$ but non-differentiable at 0 because $f'_+(0) \neq f'_-(0)$ (Example 3.13). The slopes of tangent from the two sides of $(0, 0)$ do not match: the graph has a **corner** at $(0, 0)$.
- $f(x) = x^{\frac{1}{3}}$ is continuous on $\mathbb{R}$ but non-differentiable at 0 because

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{x^{\frac{1}{3}}}{x} = \lim_{x\to 0} \frac{1}{x^{\frac{2}{3}}} = +\infty.$$

  The graph has a **vertical tangent line** at $(0, 0)$.
- The function

$$f(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

  is also continuous on $\mathbb{R}$ but non-differentiable at 0 (prove this on your own). The graph has **no tangent line at all** at $(0, 0)$.

One can construct continuous functions non-differentiable at a discrete set of points; to the extreme, there exist functions that are everywhere continuous but nowhere differentiable.

Remark 3.19 — Logical Relations

The following implications are important:

$$\underbrace{f \text{ is differentiable at } a}_{\substack{f'_-(a)=f'_+(a) \\ \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \text{ exists and is finite}}} \implies \underbrace{f \text{ is continuous at } a}_{\lim_{x\to a} f(x)=f(a)} \implies \underbrace{f \text{ is well-defined at } a}_{\substack{f(a) \text{ exists} \\ \lim_{x\to a} f(x) \text{ exists}}}$$

3.2 Rules of Differentiation

Our next aim is to establish rules of differentiation so that we can obtain a library of formulas of derivatives of elementary functions.

Theorem 3.20 — Linearity of Differentiation

Let f and g be differentiable functions and c be a real number. Then

$$(f + g)' = f' + g', \qquad (f - g)' = f' - g', \qquad (cf)' = cf'.$$

Proof. We only prove the first equality. Recall that $f+g$ is defined by $(f+g)(x) = f(x)+g(x)$, so for every x in the domain of $f+g$,

$$\begin{aligned}\lim_{h\to 0}\frac{(f+g)(x+h)-(f+g)(x)}{h} &= \lim_{h\to 0}\frac{[f(x+h)+g(x+h)]-[f(x)+g(x)]}{h}\\ &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}+\lim_{h\to 0}\frac{g(x+h)-g(x)}{h}\\ &= f'(x)+g'(x).\end{aligned}$$

Therefore $f+g$ is differentiable and $(f+g)' = f'+g'$. ■

Theorem 3.21 — Product Rule and Quotient Rule

Let f and g be differentiable functions. Then

$$(fg)' = f'g+fg' \qquad \text{and} \qquad \left(\frac{f}{g}\right)' = \frac{f'g-fg'}{g^2}.$$

Note that in the second equality, the domain of $(\frac{f}{g})'$ excludes all those x with $g(x) = 0$.

Proof. To prove the product rule, recall that fg is defined by $(fg)(x) = f(x)g(x)$, so for every x in the domain of fg,

$$\begin{aligned}&\lim_{h\to 0}\frac{(fg)(x+h)-(fg)(x)}{h} = \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h}\\ &= \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x+h)+f(x)g(x+h)-f(x)g(x)}{h}\\ &= \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}g(x+h)+f(x)\frac{g(x+h)-g(x)}{h}\right]\\ &= \left[\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}\right]\left[\lim_{h\to 0}g(x+h)\right]+f(x)\left[\lim_{h\to 0}\frac{g(x+h)-g(x)}{h}\right]\\ &= f'(x)g(x)+f(x)g'(x),\end{aligned}$$

where $\lim_{h\to 0} g(x+h) = g(x)$ follows from the continuity of g (Theorem 3.14). Therefore fg is differentiable and $(fg)' = f'g+fg'$.

To prove the quotient rule, observe that $f = \frac{f}{g}\cdot g$. Applying the product rule,

$$f' = \left(\frac{f}{g}\right)' g+\frac{f}{g}g',$$

and rearranging gives

$$\left(\frac{f}{g}\right)' = \frac{f'g-fg'}{g^2}.$$

■

Example 3.22 — Quotient Rule

Find the derivative of $f(x) = \dfrac{1}{x\sqrt{x}}$.

Solution. Note that $f(x) = \frac{\frac{1}{x}}{\sqrt{x}}$ for every $x > 0$. From Examples 3.6 and 3.7, the derivative of $\frac{1}{x}$ is $-\frac{1}{x^2}$ and the derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$. By the quotient rule, for every $x > 0$,

$$f'(x) = \frac{\left(-\frac{1}{x^2}\right)(\sqrt{x}) - \left(\frac{1}{x}\right)\left(\frac{1}{2\sqrt{x}}\right)}{(\sqrt{x})^2} = \frac{-x^{-\frac{3}{2}} - \frac{1}{2}x^{-\frac{3}{2}}}{x} = -\frac{3}{2}x^{-\frac{5}{2}}.$$

Example 3.23 — Evaluating a Derivative at a Point

Let f and g be such that

$$f(1) = -1, \qquad g(1) = 2, \qquad f'(1) = 0, \qquad g'(1) = 4.$$

Compute $\left.\frac{d}{dx}\left(\frac{f(x)}{g(x)} + \frac{g(x)}{f(x)}\right)\right|_{x=1}$.

Solution. (Put $x = 1$ after differentiating the function $\frac{f}{g} + \frac{g}{f}$.)

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)} + \frac{g(x)}{f(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} + \frac{g'(x)f(x) - g(x)f'(x)}{[f(x)]^2}.$$

At $x = 1$,

$$\left.\frac{d}{dx}\left(\frac{f(x)}{g(x)} + \frac{g(x)}{f(x)}\right)\right|_{x=1} = \frac{(0)(2) - (-1)(4)}{2^2} + \frac{(4)(-1) - (2)(0)}{(-1)^2} = -3.$$

Theorem 3.24 — Derivative of a Constant

Let c be a real number. The derivative of the constant function $f(x) = c$ is the constant function $f'(x) = 0$. In Leibniz's notation, $\frac{d}{dx}c = 0$.

Proof. Given $f(x) = c$ for every real number x,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h} = \lim_{h\to 0}\frac{c - c}{h} = \lim_{h\to 0} 0 = 0$$

for every real number x. ■

Theorem 3.24 is a special case of the following theorem.

Theorem 3.25 — Power Rule for Integer Powers

Let n be an integer. Then

$$\frac{d}{dx}x^n = nx^{n-1}.$$

(In Lagrange's notation: the derivative of $f(x) = x^n$ is $f'(x) = nx^{n-1}$.)

Proof. The case $n = 0$ is Theorem 3.24. First let n be a positive integer. Then

$$\begin{aligned}
\frac{d}{dx}x^n &= \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} \\
&= \lim_{h\to 0}\frac{\left[x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-1}xh^{n-1} + h^n\right] - x^n}{h} \\
&= \lim_{h\to 0}\frac{\binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-1}xh^{n-1} + h^n}{h} \\
&= \lim_{h\to 0}\left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1}\right] \\
&= nx^{n-1} + 0 + \cdots + 0 = nx^{n-1}.
\end{aligned}$$

Next, when n is a negative integer, say $n = -k$ with k a positive integer, by the quotient rule,

$$\frac{d}{dx}x^n = \frac{d}{dx}\frac{1}{x^k} = \frac{\left(\frac{d}{dx}1\right)(x^k) - (1)\left(\frac{d}{dx}x^k\right)}{(x^k)^2} = \frac{0\cdot x^k - 1\cdot kx^{k-1}}{x^{2k}} = -kx^{-k-1} = nx^{n-1}$$

as well. ■

In fact Theorem 3.25 holds even when n is any real number; we will prove this later (Theorem 3.45).

Example 3.26 — Applying the Basic Rules

Evaluate the derivatives of the following functions.

(a) $f(x) = (x^2 - 3x + 1)(x + 2) + x^{1013}$ (b) $g(t) = \dfrac{2t^2 + 5t + 3}{t + 2}$

Solution. (a) Apply the product rule to the first term. For every $x \in \mathbb{R}$,

$$\begin{aligned}
f'(x) &= \left\{\left[\frac{d}{dx}(x^2 - 3x + 1)\right](x + 2) + (x^2 - 3x + 1)\left[\frac{d}{dx}(x + 2)\right]\right\} + \frac{d}{dx}x^{1013} \\
&= [(2x - 3)(x + 2) + (x^2 - 3x + 1)(1)] + 1013x^{1012} = 1013x^{1012} + 3x^2 - 2x - 5.
\end{aligned}$$

(b) Apply the quotient rule. For every $t \in \mathbb{R} \setminus \{-2\}$,

$$\begin{aligned}
g'(t) &= \frac{\left[\frac{d}{dt}(2t^2 + 5t + 3)\right](t + 2) - (2t^2 + 5t + 3)\left[\frac{d}{dt}(t + 2)\right]}{(t + 2)^2} \\
&= \frac{(4t + 5)(t + 2) - (2t^2 + 5t + 3)(1)}{(t + 2)^2} = \frac{2t^2 + 8t + 7}{(t + 2)^2}.
\end{aligned}$$

Theorem 3.27 — Derivatives of Trigonometric Functions

$$\begin{aligned}
&\frac{d}{dx}\sin x = \cos x, && \frac{d}{dx}\tan x = \sec^2 x, && \frac{d}{dx}\sec x = \sec x\tan x, \\
&\frac{d}{dx}\cos x = -\sin x, && \frac{d}{dx}\cot x = -\csc^2 x, && \frac{d}{dx}\csc x = -\csc x\cot x.
\end{aligned}$$

Proof. To prove $\dfrac{d}{dx}\sin x = \cos x$:

$$\frac{d}{dx}\sin x = \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h} = \lim_{h\to 0}\frac{2\cos\left(x+\frac{h}{2}\right)\sin\frac{h}{2}}{h}$$
$$= \lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)\cdot\lim_{h\to 0}\frac{\sin\frac{h}{2}}{\frac{h}{2}} = \cos x\cdot 1 = \cos x.$$

The proof of $\dfrac{d}{dx}\cos x = -\sin x$ is similar and is left as an exercise.

To prove $\dfrac{d}{dx}\tan x = \sec^2 x$, recall $\tan x = \dfrac{\sin x}{\cos x}$ and apply the quotient rule:

$$\frac{d}{dx}\tan x = \frac{d}{dx}\frac{\sin x}{\cos x} = \frac{\left(\frac{d}{dx}\sin x\right)(\cos x)-(\sin x)\left(\frac{d}{dx}\cos x\right)}{(\cos x)^2}$$
$$= \frac{(\cos x)(\cos x)-(\sin x)(-\sin x)}{(\cos x)^2} = \frac{\cos^2 x+\sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

The proofs of the three remaining formulas are similar and are left as exercises. ■

Example 3.28 — Trigonometric Quotient

Find the derivative of $h(\theta) = \dfrac{\sec\theta\tan\theta}{\theta^3}$.

Solution.

$$h'(\theta) = \frac{\left[\frac{d}{d\theta}(\sec\theta\tan\theta)\right](\theta^3)-(\sec\theta\tan\theta)\left(\frac{d}{d\theta}\theta^3\right)}{(\theta^3)^2}$$
$$= \frac{[(\sec\theta\tan\theta)(\tan\theta)+(\sec\theta)(\sec^2\theta)](\theta^3)-(\sec\theta\tan\theta)(3\theta^2)}{\theta^6}$$
$$= \frac{\theta\sec\theta\tan^2\theta+\theta\sec^3\theta-3\sec\theta\tan\theta}{\theta^4}.$$

Theorem 3.29 — Derivatives of Exponential Functions

Let $a>0$. Then

$$\frac{d}{dx}a^x = a^x\ln a.$$

Proof. Let $a>0$ and $f(x)=a^x$. Then for every real number x,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{a^{x+h}-a^x}{h} = a^x\lim_{h\to 0}\frac{a^h-1}{h} = a^x\ln a,$$

where $\lim\limits_{h\to 0}\dfrac{a^h-1}{h} = \ln a$ was computed in Example 2.111. ■

Corollary 3.30 — Derivative of the Natural Exponential Function

$$\frac{d}{dx}e^x = e^x.$$

Remark 3.31 — Why "Natural"?

The derivative of an exponential function $f(x) = a^x$ is proportional to itself (with $\ln a$ the proportionality constant). The base e is specially chosen to make this constant equal to 1, i.e.

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1.$$

In other words, the slope of the tangent to the graph of $y = e^x$ at the point $(0, 1)$ is $f'(0) = 1$. This is why it is called "natural".

Example 3.32 — Exponential Quotient

Find the derivative of $f(x) = \dfrac{e^x}{\sqrt{x}}$.

Solution. For every $x > 0$,

$$f'(x) = \frac{\left(\dfrac{d}{dx}e^x\right)(\sqrt{x}) - (e^x)\left(\dfrac{d}{dx}\sqrt{x}\right)}{(\sqrt{x})^2} = \frac{(e^x)(\sqrt{x}) - (e^x)\left(\dfrac{1}{2\sqrt{x}}\right)}{x} = \left(1 - \frac{1}{2x}\right)\frac{e^x}{\sqrt{x}}.$$

3.3 Chain Rule

After the rules for the sum, difference, product and quotient, we now study the rule for the composition of functions.

Theorem 3.33 — Chain Rule

Let x be a real number and let f and g be functions. If g is differentiable at x and f is differentiable at $g(x)$, then $f \circ g$ is also differentiable at x and its derivative is given by

$$(f \circ g)'(x) = f'(g(x))\, g'(x).$$

If we write $u = g(x)$ and $y = f(u) = (f \circ g)(x)$, then in Leibniz's notation this becomes

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

The following sloppy proof is enough to illustrate why the rule should hold; the rigorous proof is recorded in the supplement at the end of this chapter.

A sloppy proof. Recalling that $f \circ g$ is defined by $(f \circ g)(x) = f(g(x))$,

$$\begin{aligned}(f \circ g)'(x) &= \lim_{h\to 0} \frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} = \lim_{h\to 0} \frac{f(g(x+h)) - f(g(x))}{h} \\ &= \lim_{h\to 0} \frac{f(g(x) + g(x+h) - g(x)) - f(g(x))}{g(x+h) - g(x)} \cdot \frac{g(x+h) - g(x)}{h} \\ &= \lim_{h\to 0} \frac{f(g(x) + g(x+h) - g(x)) - f(g(x))}{g(x+h) - g(x)} \cdot \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}.\end{aligned}$$

With the change of variable $t = g(x+h) - g(x)$, we have $t \to 0$ as $h \to 0$ since g is continuous. Therefore

$$(f \circ g)'(x) = \lim_{t\to 0} \frac{f(g(x) + t) - f(g(x))}{t} \cdot \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} = f'(g(x))\, g'(x).$$

■

Think. Why is this proof sloppy? Where is the possible flaw? (See the supplement at the end of this chapter.)

Example 3.34 — Chain Rule Computations

Find the derivative of each of the following functions.

(a) $f(x) = \sin(x^2)$ (b) $g(x) = \sin^2 x = (\sin x)^2$ (c) $h(x) = \tan\dfrac{x^2+3}{x-5}$ (d) $p(t) = \cos(5^{t^2})$ (e) $q(s) = \left(1-\sqrt{s^2+1}\right)^3$

Solution. (Either Lagrange's or Leibniz's notation can be used to present the solution.)

(a) Writing $v(u) = \sin u$ and $u(x) = x^2$, we have $f(x) = v(u(x))$. So

$$f'(x) = v'(u)u'(x) = (\cos u)(2x) = 2x\cos(x^2).$$

Equivalently, in Leibniz's notation: let $y = \sin u$ where $u = x^2$; then $\dfrac{dy}{dx} = \dfrac{dy}{du}\cdot\dfrac{du}{dx} = (\cos u)(2x) = 2x\cos(x^2)$.

(b) Writing $v(u) = u^2$ and $u(x) = \sin x$, we have $g(x) = v(u(x))$. So

$$g'(x) = v'(u)u'(x) = (2u)(\cos x) = 2\sin x\cos x.$$

(c) $h(x)$ is the composition of $v(u) = \tan u$ and $u(x) = \dfrac{x^2+3}{x-5}$. So

$$h'(x) = v'(u)u'(x) = (\sec^2 u)\cdot\frac{(2x)(x-5)-(x^2+3)(1)}{(x-5)^2} = \frac{x^2-10x-3}{(x-5)^2}\sec^2\frac{x^2+3}{x-5}.$$

(d) $p(t)$ is the composition of three elementary functions $w(v) = \cos v$, $v(u) = 5^u$ and $u(t) = t^2$. So

$$p'(t) = w'(v)v'(u)u'(t) = (-\sin v)(5^u\ln 5)(2t) = \left[-\sin(5^{t^2})\right](5^{t^2}\ln 5)(2t)$$
$$= (-2t\ln 5)\,5^{t^2}\sin(5^{t^2}).$$

(e) q is also a composition of three elementary functions. Let $y = q(s) = \left(1-\sqrt{s^2+1}\right)^3$. Write $y = u^3$ where $u = 1-\sqrt{v}$ and $v = s^2+1$. So

$$q'(s) = \frac{dy}{ds} = \frac{dy}{du}\cdot\frac{du}{dv}\cdot\frac{dv}{ds} = (3u^2)\left(-\frac{1}{2\sqrt{v}}\right)(2s)$$
$$= 3\left(1-\sqrt{s^2+1}\right)^2\left(-\frac{1}{2\sqrt{s^2+1}}\right)(2s)$$
$$= \frac{-3s\left(1-\sqrt{s^2+1}\right)^2}{\sqrt{s^2+1}}.$$

Example 3.35 — Derivatives from a Table of Values

The following table shows some values of f, g, f' and g'.

t	-1	1	2
$f(t)$	1	-2	-1
$g(t)$	-5	-1	-2
$f'(t)$	-3	0	2
$g'(t)$	3	-4	0

Using the table, evaluate:

(a) $\dfrac{d}{dt}\left[t^2 g(f(t))\right]\Big|_{t=2}$ (b) $\dfrac{d}{dt}\left[\dfrac{f(g(t))}{[f(t)]^2}\right]\Big|_{t=1}$

Solution. (a) By the product rule and the chain rule,

$$\frac{d}{dt}\left[t^2 g(f(t))\right] = (2t)g(f(t)) + (t^2)\left[g'(f(t))f'(t)\right].$$

At $t = 2$ (plug in the value of t after evaluating the derivative),

$$\begin{aligned}\frac{d}{dt}\left[t^2 g(f(t))\right]\Big|_{t=2} &= 2(2)g(f(2)) + (2^2)g'(f(2))f'(2)\\ &= 4g(-1) + 4g'(-1)f'(2) = 4(-5) + 4(3)(2) = 4.\end{aligned}$$

(b) By the quotient rule and the chain rule,

$$\frac{d}{dt}\frac{f(g(t))}{[f(t)]^2} = \frac{[f'(g(t))g'(t)][f(t)]^2 - [f(g(t))][2f(t)f'(t)]}{[f(t)]^4}.$$

At $t = 1$,

$$\begin{aligned}\frac{d}{dt}\frac{f(g(t))}{[f(t)]^2}\Big|_{t=1} &= \frac{[f'(g(1))g'(1)][f(1)]^2 - [f(g(1))][2f(1)f'(1)]}{[f(1)]^4}\\ &= \frac{[f'(-1)\cdot(-4)](-2)^2 - [f(-1)][2(-2)(0)]}{(-2)^4}\\ &= \frac{[(-3)\cdot(-4)](-2)^2 - (1)[2(-2)(0)]}{(-2)^4} = 3.\end{aligned}$$

Remark 3.36 — Only the Given Values Are Known

When we are only given some particular function values as in Example 3.35, we cannot assume the differentiability, continuity, or even well-definedness of the functions f, g, f' and g' at any other numbers not mentioned in the table.

As a consequence of the chain rule, we obtain two more techniques of differentiation: **implicit differentiation** and **logarithmic differentiation**. The first enables us to differentiate the inverse of a one-to-one differentiable function, leading to the derivative formulas of the remaining elementary functions — the logarithmic functions and the inverse trigonometric functions.

Remark 3.37 — Implicit Relations

Sometimes the relationship between two quantities x and y cannot be expressed explicitly as $y = f(x)$; instead they are related implicitly via an equation $F(x, y) = 0$. For example, the unit circle $x^2 + y^2 - 1 = 0$: here y cannot be expressed as a function of x because the circle fails the vertical line test. However, we can still "pretend that y is a function of x" and find $\dfrac{dy}{dx}$ in terms of x and y using the chain rule. This technique is called **implicit differentiation**.

- Although the circle fails the vertical line test, y can in fact be written as two functions of x: $y = \sqrt{1-x^2}$ or $y = -\sqrt{1-x^2}$, corresponding to the upper and lower halves of the circle. So apart from the two points $(1, 0)$ and $(-1, 0)$, it is alright to locally regard y as one single function of x.

The method will be fully justified in more advanced courses.

Example 3.38 — Implicit Differentiation

Find $\dfrac{dy}{dx}$ if x and y are related implicitly by

$$x^2 + y^2 + x\cos y = 1.$$

Solution. Differentiating both sides with respect to x,

$$2x + \frac{d}{dx}y^2 + \left[(1)(\cos y) + (x)\frac{d}{dx}\cos y\right] = 0.$$

Applying the chain rule,

$$\begin{aligned} 2x + 2y\frac{dy}{dx} + \left[(1)(\cos y) + (x)(-\sin y)\frac{dy}{dx}\right] &= 0, \\ (2y - x\sin y)\frac{dy}{dx} &= -2x - \cos y, \\ \frac{dy}{dx} &= \frac{2x + \cos y}{x\sin y - 2y}. \end{aligned}$$

Example 3.39 — Tangent Line to an Implicit Curve

Find the equation of the tangent line to the curve $3x^2y + y^3 = 4$ at the point $(1, 1)$.
Solution. Differentiating both sides with respect to x,

$$\left(6xy + 3x^2\frac{dy}{dx}\right) + 3y^2\frac{dy}{dx} = 0,$$

i.e. $\dfrac{dy}{dx} = -\dfrac{2xy}{x^2 + y^2}$. Thus the slope of the tangent line at $(1, 1)$ is

$$\left.\frac{dy}{dx}\right|_{(1,1)} = -\frac{2(1)(1)}{1^2 + 1^2} = -1.$$

So the tangent line at $(1, 1)$ is $y - 1 = -1(x - 1)$, i.e. $x + y - 2 = 0$.

The chain rule or implicit differentiation enables us to find the derivative of the inverse of a one-to-one differentiable function.

Theorem 3.40 — Derivative of the Inverse

Let f be a one-to-one differentiable function with inverse f^{-1}, and let y be in the range of f. If $f'(f^{-1}(y)) \neq 0$ and if f^{-1} is differentiable at y, then

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Denoting $y = f(x)$ and $x = f^{-1}(y)$, the formula reads $(f^{-1})'(y) = \dfrac{1}{f'(x)}$, or in Leibniz's notation,

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

Proof. For every y in the range of f, we have $f(f^{-1}(y)) = y$. Differentiating both sides with respect to y and applying the chain rule,

$$f'(f^{-1}(y)) \cdot (f^{-1})'(y) = 1.$$

If $f'(f^{-1}(y)) \neq 0$, then $(f^{-1})'(y) = \dfrac{1}{f'(f^{-1}(y))}$. ■

(Geometrically: the graphs $y = f(x)$ and $y = f^{-1}(x)$ are mirror images across the line $y = x$; if the slope of the tangent to the graph of f at x is m, then the slope of the tangent to the graph of f^{-1} at y is $\dfrac{1}{m}$.)

Remark 3.41 — What Theorem 3.40 Does NOT Say

Theorem 3.40 is just a computation formula of $(f^{-1})'(y)$ for the case when we already know that f^{-1} exists and is differentiable at y. It does not help to justify that f^{-1} is differentiable. Sufficient conditions for the differentiability of f^{-1} will be discussed in more advanced courses.

Example 3.42 — Derivative of an Inverse Without Finding It

Consider $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3 + x + 1$. It is given that f is one-to-one and f^{-1} is differentiable at 3. Compute $(f^{-1})'(3)$.

Solution. (It is too difficult to first find f^{-1} and then differentiate; use Theorem 3.40.) For every $x \in \mathbb{R}$, $f'(x) = 3x^2 + 1$. Notice that

$$f(1) = 1^3 + 1 + 1 = 3.$$

Since f is one-to-one, $f^{-1}(3) = 1$. Thus by Theorem 3.40,

$$(f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(1)} = \frac{1}{3(1)^2 + 1} = \frac{1}{4}.$$

Theorem 3.43 — Derivatives of Logarithms

Let $a \neq 1$ be a positive number. Then, as functions defined on $(0, +\infty)$,

$$\frac{d}{dx} \ln x = \frac{1}{x} \qquad \text{and} \qquad \frac{d}{dx} \log_a x = \frac{1}{x \ln a}.$$

Proof. Let $g(x) = \ln x$ for $x > 0$ and $f(x) = e^x$. Since g is the inverse of f, by Theorem 3.40,

$$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{e^{g(x)}} = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

for every $x > 0$. The second formula follows from the first since $\log_a x = \dfrac{\ln x}{\ln a}$ for every $x > 0$. ■

Remark 3.44 — Derivative of $\ln|x|$

The formulas in Theorem 3.43 are valid on $(0, +\infty)$ only. On $(-\infty, 0)$, by the chain rule,

$$\frac{d}{dx} \ln(-x) = \frac{1}{x}.$$

Combining both intervals,

$$\frac{d}{dx} \ln|x| = \frac{1}{x}, \qquad \text{valid on } (-\infty, 0) \cup (0, +\infty).$$

We mentioned that Theorem 3.25 holds even when the exponent is any real number. Now we can prove this using the chain rule.

Theorem 3.45 — Power Rule (General)

Let r be a real number. Then

$$\frac{d}{dx}x^r = rx^{r-1}.$$

Proof. Note that $x^r = e^{r\ln x}$ for every $x > 0$. By the chain rule,

$$\frac{d}{dx}x^r = \frac{d}{dx}e^{r\ln x} = e^{r\ln x}\cdot\frac{r}{x} = x^r\cdot\frac{r}{x} = rx^{r-1}.$$

■

Theorem 3.46 — Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\arctan x = \frac{1}{1+x^2}, \quad \frac{d}{dx}\operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}},$$
$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\operatorname{arccot} x = -\frac{1}{1+x^2}, \quad \frac{d}{dx}\operatorname{arccsc} x = -\frac{1}{|x|\sqrt{x^2-1}}.$$

(Think before the proof: why does each pair differ just by a minus sign?)

Proof. For every $x \in (-1,1)$, let $y = \arcsin x$. Then $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $x = \sin y$, so

$$\frac{dx}{dy} = \cos y.$$

Therefore, since $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$ implies $\cos y > 0$,

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$

The proofs of the remaining formulas are similar and are left as exercises. ■

Example 3.47 — Inverse Trigonometric Compositions

Find the derivative of each of the following functions.

(a) $f(x) = \arccos(\frac{1}{x})$ (b) $g(t) = \sin[\arctan(\ln t)]$

Solution. (a) For every $x \in (-\infty,-1)\cup(1,+\infty)$,

$$f'(x) = -\frac{1}{\sqrt{1-(\frac{1}{x})^2}}\cdot\left(-\frac{1}{x^2}\right) = \frac{1}{x^2\sqrt{1-\frac{1}{x^2}}} = \frac{1}{\sqrt{x^4-x^2}}.$$

(b) For every $t > 0$,

$$g'(t) = \cos[\arctan(\ln t)]\cdot\frac{1}{1+(\ln t)^2}\cdot\frac{1}{t} = \frac{\cos[\arctan(\ln t)]}{t[1+(\ln t)^2]}.$$

To differentiate functions involving exponents or indices, e.g. complicated exponential and algebraic functions, the method of **logarithmic differentiation** is sometimes useful.

Corollary 3.48 — Logarithmic Differentiation

Let f be a differentiable function. Then for every x at which $f(x) \neq 0$,

$$\frac{d}{dx} \ln |f(x)| = \frac{f'(x)}{f(x)}.$$

Proof. Theorem 3.43 and the chain rule. ■

Example 3.49 — Logarithmic Differentiation I

Find the derivative of

$$f(x) = \frac{x^{\frac{1}{3}}\sqrt{x^2+1}}{(3x+2)^3}.$$

Solution. (This can be done by the product rule and chain rule, but logarithmic differentiation saves a lot of effort.) For every $x \in \mathbb{R} \setminus \{0, -\frac{2}{3}\}$, consider

$$\ln |f(x)| = \ln \left| \frac{x^{\frac{1}{3}}\sqrt{x^2+1}}{(3x+2)^3} \right| = \ln \frac{|x|^{\frac{1}{3}}\sqrt{x^2+1}}{|3x+2|^3} = \ln |x|^{\frac{1}{3}} + \ln \sqrt{x^2+1} - \ln |3x+2|^3$$
$$= \frac{1}{3} \ln |x| + \frac{1}{2} \ln(x^2+1) - 3 \ln |3x+2|.$$

Differentiating both sides with respect to x,

$$\frac{f'(x)}{f(x)} = \frac{1}{3}\left(\frac{1}{x}\right) + \frac{1}{2}\left(\frac{2x}{x^2+1}\right) - 3\left(\frac{3}{3x+2}\right) = \frac{1}{3x} + \frac{x}{x^2+1} - \frac{9}{3x+2}.$$

Therefore

$$f'(x) = f(x)\left(\frac{1}{3x} + \frac{x}{x^2+1} - \frac{9}{3x+2}\right) = \frac{x^{\frac{1}{3}}\sqrt{x^2+1}}{(3x+2)^3}\left(\frac{1}{3x} + \frac{x}{x^2+1} - \frac{9}{3x+2}\right).$$

Example 3.50 — Logarithmic Differentiation II: x^x

Find the derivative of $f : (0, +\infty) \to \mathbb{R}$ defined by $f(x) = x^x$.
Solution. For each $x > 0$, taking natural logarithm on both sides,

$$\ln f(x) = x \ln x.$$

Differentiating both sides with respect to x,

$$\frac{f'(x)}{f(x)} = (1)(\ln x) + (x)\left(\frac{1}{x}\right) = \ln x + 1.$$

Therefore

$$f'(x) = f(x)(\ln x + 1) = x^x(\ln x + 1).$$

Remark 3.51 — Summary: Rules and Formulas of Differentiation

Rules of differentiation (equalities hold assuming all relevant derivatives exist):

- (Linearity I) $(cf)'(x) = cf'(x)$ for every $c \in \mathbb{R}$ $\qquad \frac{d}{dx}(cu) = c\frac{du}{dx}$
- (Linearity II) $(f \pm g)'(x) = f'(x) \pm g'(x)$ $\qquad \frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$
- (Product rule) $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$ $\qquad \frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$

- (Quotient rule) $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$ $\qquad \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\frac{du}{dx}v - u\frac{dv}{dx}}{v^2}$
- (Chain rule) $(f \circ g)'(x) = f'(g(x))g'(x)$ $\qquad \frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx}$
- (Inverse) $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ $\qquad \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$

Derivatives of elementary functions (valid on open intervals):

- Polynomials and powers: $\frac{d}{dx}c = 0$ $(c \in \mathbb{R})$; $\quad \frac{d}{dx}x^r = rx^{r-1}$ $(r \in \mathbb{R})$.
- Trigonometric: $\frac{d}{dx}\sin x = \cos x$, $\frac{d}{dx}\cos x = -\sin x$, $\frac{d}{dx}\tan x = \sec^2 x$, $\frac{d}{dx}\cot x = -\csc^2 x$, $\frac{d}{dx}\sec x = \sec x \tan x$, $\frac{d}{dx}\csc x = -\csc x \cot x$.
- Exponential and logarithmic: $\frac{d}{dx}e^x = e^x$; $\quad \frac{d}{dx}a^x = a^x \ln a$ $(a > 0)$; $\quad \frac{d}{dx}\ln x = \frac{1}{x}$; $\frac{d}{dx}\log_a x = \frac{1}{x \ln a}$ $(a \in (0, +\infty) \setminus \{1\})$.
- Inverse trigonometric: $\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$, $\frac{d}{dx}\arccos x = \frac{-1}{\sqrt{1-x^2}}$, $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$, $\frac{d}{dx}\operatorname{arccot} x = \frac{-1}{1+x^2}$, $\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}}$, $\frac{d}{dx}\operatorname{arccsc} x = \frac{-1}{|x|\sqrt{x^2-1}}$.

Remark 3.52 — Formulas Are Valid on Open Intervals Only

The differentiation formulas in Remark 3.51 are valid on open intervals only.

- For example, if $f : [0,1] \to \mathbb{R}$ is defined by $f(x) = x^2$ for $x \in [0,1]$, then the "power rule" only says $f'(x) = 2x$ for $x \in (0,1)$. f is not differentiable at the endpoints 0 and 1, because $f'_-(0)$ and $f'_+(1)$ do not exist as f is not defined for $x < 0$ or $x > 1$.

As a result, when f is piecewise defined, be careful when computing f':

- On each open interval where f is elementary, evaluate f' using the differentiation formulas.
- At those points where the definitions of f on the two sides are different, go back to the basics and apply the definition of derivative.

Example 3.53 — Derivative of a Piecewise Function

Find the derivative of $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 - \cos x & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases}$$

Solution. Consider three cases. (We cannot deduce $f'(x) = 1$ for $x \geq 0$ directly from "$f(x) = x$ for $x \geq 0$"; differentiation formulas are valid on open intervals only.)

(i) For $x < 0$ (by the differentiation formulas): $f'(x) = \frac{d}{dx}(1 - \cos x) = \sin x$.

(ii) For $x > 0$ (by the power rule): $f'(x) = \frac{d}{dx}x = 1$.

(iii) For $x = 0$, apply the definition of derivative:

$$\lim_{x\to 0^+} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0^+} \frac{x-0}{x-0} = \lim_{x\to 0^+} 1 = 1,$$

$$\lim_{x\to 0^-} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0^-} \frac{(1-\cos x)-0}{x-0} = \lim_{x\to 0^-} \frac{2\sin^2(\frac{x}{2})}{x}$$
$$= \lim_{x\to 0^-} \frac{\sin(\frac{x}{2})}{\frac{x}{2}} \cdot \lim_{x\to 0^-} \sin\frac{x}{2} = 1 \cdot 0 = 0.$$

Therefore $f'(0)$ does not exist.

In summary, the derivative $f' : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ is given by

$$f'(x) = \begin{cases} \sin x & \text{if } x < 0, \\ 1 & \text{if } x > 0. \end{cases}$$

Remark 3.54 — $\lim f'(x)$ vs. $f'(a)$: A Wrong Argument

In Example 3.53 it is tempting to say "for $x = 0$,

$$\lim_{x\to 0^-} f'(x) = \lim_{x\to 0^-} \sin x = 0 \qquad \text{and} \qquad \lim_{x\to 0^+} f'(x) = \lim_{x\to 0^+} 1 = 1$$

which are different, so $f'(0)$ does not exist". Although this argument leads to the correct conclusion, its reasoning is incorrect: what it really implies is that "$\lim_{x\to 0} f'(x)$ does not exist", instead of that "$f'(0)$ does not exist". The two quantities $\lim_{x\to 0} f'(x)$ and $f'(0)$ are different in general; they are the same only when f' is continuous at 0.

Example 3.55 — A Differentiable Function with Discontinuous Derivative

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Show that its derivative f' is well-defined at 0 but discontinuous at 0.

Solution.

- First,
$$\lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0} \frac{x^2\sin(\frac{1}{x})-0}{x-0} = \lim_{x\to 0} x\sin\frac{1}{x} = 0$$
by the Squeeze Theorem, so $f'(0) = 0$ exists as a real number.
- On the other hand, for every $x \neq 0$, by the chain rule,
$$f'(x) = (2x)\left(\sin\frac{1}{x}\right) + (x^2)\left(-\frac{1}{x^2}\cos\frac{1}{x}\right) = 2x\sin\frac{1}{x} - \cos\frac{1}{x}.$$
Thus $\lim_{x\to 0} f'(x)$ does not exist, because otherwise
$$\lim_{x\to 0} \cos\frac{1}{x} = \underbrace{2\lim_{x\to 0} x\sin\frac{1}{x}}_{=0 \text{ by Squeeze}} - \underbrace{\lim_{x\to 0} f'(x)}_{\text{exists}}$$
would exist, contradicting Example 2.40. In particular f' is discontinuous at 0. ■

Remark 3.56. If f is differentiable and its derivative f' is also continuous, then we say that f is **continuously differentiable**, or f is **of class** C^1.

Example 3.57 — A Trap: Two Formulas, One Function

Find the derivative of $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^x & \text{if } x \neq 0, \\ \cos x & \text{if } x = 0. \end{cases}$$

Solution. It is incorrect to say that

$$f'(x) = \begin{cases} e^x & \text{if } x \neq 0, \\ -\sin x & \text{if } x = 0. \end{cases} \qquad \text{(wrong!)}$$

In fact, noting that $f(0) = \cos 0 = 1 = e^0$, we see that

$$f(x) = e^x \qquad \text{for every } x \in \mathbb{R}.$$

Therefore $f'(x) = e^x$ for every $x \in \mathbb{R}$ also.

3.4 Higher Derivatives

Differentiation can be done more than once.

Definition 3.58. Let f be a function. Then

- the derivative of its derivative f', denoted f'', is called the **second derivative** of f;
- the derivative of f'', denoted f''' or $f^{(3)}$, is called the **third derivative** of f;
- in a similar way, for each positive integer n, the n**th derivative** of f is the derivative of the $(n-1)$th derivative of f, and is denoted $f^{(n)}$.

Remark 3.59 — Notations for Higher Derivatives

- (Lagrange) When we emphasize that the nth derivative of f is a function, we write $f^{(n)}$.
- (Leibniz) Presenting a function as $y = f(x)$, we write the nth derivative as $\dfrac{d^n y}{dx^n}$; its value at a is $\left.\dfrac{d^n y}{dx^n}\right|_{x=a}$.
- We may also write the nth derivative as $\dfrac{d^n}{dx^n} f(x)$ with the defining formula replacing $f(x)$, e.g. $\dfrac{d^n}{dx^n}\cos(x^2+3x)$; its value at a is $\left.\dfrac{d^n}{dx^n}\cos(x^2+3x)\right|_{x=a}$.

Example 3.60 — Second Derivative via Chain and Product Rules

Find the second derivative of $f(x) = \cos(e^x)$.

Solution. By the chain rule,

$$f'(x) = (-\sin(e^x))(e^x) = -e^x \sin(e^x).$$

Then by the product rule and the chain rule,

$$f''(x) = (-e^x)(\sin(e^x)) + (-e^x)[(\cos(e^x))(e^x)] = -e^x \sin(e^x) - e^{2x}\cos(e^x).$$

Example 3.61 — Second Derivative of an Implicit Relation

If x and y are related implicitly by $x^2 + y^2 = 9$, show that

$$y^3 \frac{d^2 y}{dx^2} + 9 = 0.$$

Solution. Differentiating both sides of $x^2 + y^2 = 9$ with respect to x gives $2x + 2y\dfrac{dy}{dx} = 0$, so $x + y\dfrac{dy}{dx} = 0$. (Why don't we write $\dfrac{dy}{dx} = -\dfrac{x}{y}$ before moving on? Keeping the mixed form makes the next differentiation cleaner.) Differentiating with respect to x once more,

$$1 + \left(\frac{dy}{dx}\right)\left(\frac{dy}{dx}\right) + (y)\left(\frac{d^2y}{dx^2}\right) = 0.$$

Multiplying both sides by y^2 gives $y^2 + \left(y\dfrac{dy}{dx}\right)^2 + y^3\dfrac{d^2y}{dx^2} = 0$. Since $y\dfrac{dy}{dx} = -x$,

$$y^2 + x^2 + y^3\frac{d^2y}{dx^2} = 0.$$

Finally, since $x^2 + y^2 = 9$, we obtain the desired equation $y^3\dfrac{d^2y}{dx^2} + 9 = 0$. ■

Example 3.62 — nth Derivatives

For each non-negative integer n, find the nth derivative of each of the following.
(a) $f(x) = x^m$, where m is a non-negative integer (b) $g(x) = \cos x$
Solution. (a) Observe that

$$f'(x) = mx^{m-1}, \qquad f''(x) = m(m-1)x^{m-2}, \qquad f'''(x) = m(m-1)(m-2)x^{m-3},$$

and so on. According to the pattern, we guess

$$f^{(n)}(x) = m(m-1)(m-2)\cdots(m-n+1)x^{m-n}.$$

Using the factorial notation,

$$f^{(n)}(x) = \begin{cases} \dfrac{m!}{(m-n)!}x^{m-n} & \text{if } 0 \le n \le m, \\ 0 & \text{if } n > m. \end{cases}$$

The verification by mathematical induction is left as an exercise.
(b) Observe that

$$g'(x) = -\sin x, \qquad g''(x) = -\cos x, \qquad g'''(x) = \sin x, \qquad g^{(4)}(x) = \cos x,$$
$$g^{(5)}(x) = -\sin x, \qquad g^{(6)}(x) = -\cos x,$$

and so on. According to the pattern, we guess

$$g^{(n)}(x) = \begin{cases} -\sin x & \text{if } n = 4k+1, \\ -\cos x & \text{if } n = 4k+2, \\ \sin x & \text{if } n = 4k+3, \\ \cos x & \text{if } n = 4k, \end{cases} \quad \text{for some non-negative integer } k,$$

which can be written compactly as

$$g^{(n)}(x) = \cos\left(x + \frac{n\pi}{2}\right).$$

The verification by mathematical induction is left as an exercise.

Remark 3.63 — High Derivatives of Polynomials Vanish

Example 3.62(a) shows that if f is a polynomial of degree m, then $f^{(n)} = 0$ for every integer $n > m$.

Let a be a real number and f a function. Even if $f'(a)$ exists (i.e. f is differentiable

at a), it is not always the case that $f''(a)$ also exists.

Definition 3.64. Let a be a real number, n a positive integer and f a function.

- f is said to be **n-times differentiable at a** if $f^{(n)}(a)$ exists as a finite real number (for $n = 2$, f is said to be **twice differentiable at a**).
- f is said to be **n-times differentiable on an interval** if it is n-times differentiable at every number in that interval.

Example 3.65 — $f(x) = x|x|$

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x|x|$.
(a) Find $f'(x)$ for every $x \in \mathbb{R}$. (b) Is $f' : \mathbb{R} \to \mathbb{R}$ continuous at 0? (c) Is f twice differentiable at 0?
Solution. (a) For $x > 0$: $f(x) = x(x) = x^2$, so $f'(x) = 2x$. For $x < 0$: $f(x) = x(-x) = -x^2$, so $f'(x) = -2x$. For $x = 0$:

$$f'_+(0) = \lim_{x\to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{x|x| - 0}{x - 0} = \lim_{x\to 0^+} \frac{x^2 - 0}{x - 0} = \lim_{x\to 0^+} x = 0,$$

$$f'_-(0) = \lim_{x\to 0^-} \frac{x|x| - 0}{x - 0} = \lim_{x\to 0^-} \frac{-x^2 - 0}{x - 0} = \lim_{x\to 0^-} (-x) = 0,$$

so $f'(0)$ exists and $f'(0) = 0$ also. In summary, $f' : \mathbb{R} \to \mathbb{R}$ is given by

$$f'(x) = \begin{cases} -2x & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ 2x & \text{if } x > 0. \end{cases}$$

(b) Since $f'(0) = 0$ and

$$\lim_{x\to 0^-} f'(x) = \lim_{x\to 0^-} (-2x) = 0 \qquad \text{and} \qquad \lim_{x\to 0^+} f'(x) = \lim_{x\to 0^+} (2x) = 0,$$

it follows that $\lim_{x\to 0} f'(x)$ exists and equals $f'(0)$. Therefore f' is continuous at 0.
(c) The issue is whether $f''(0)$ exists. Since

$$(f')'_+(0) = \lim_{x\to 0^+} \frac{f'(x) - f'(0)}{x - 0} = \lim_{x\to 0^+} \frac{2x - 0}{x - 0} = \lim_{x\to 0^+} 2 = 2,$$

$$(f')'_-(0) = \lim_{x\to 0^-} \frac{f'(x) - f'(0)}{x - 0} = \lim_{x\to 0^-} \frac{-2x - 0}{x - 0} = \lim_{x\to 0^-} (-2) = -2,$$

it follows that $f''(0)$ does not exist. In other words, f is not twice differentiable at 0.

Example 3.66 — $f(x) = x^3 \sin(\frac{1}{x})$

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^3 \sin(\frac{1}{x}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(a) Find $f'(x)$ for every $x \in \mathbb{R}$. (b) Is $f' : \mathbb{R} \to \mathbb{R}$ continuous at 0? (c) Is f twice differentiable at 0?
Solution. (a) For $x \neq 0$, by the differentiation formulas and the chain rule,

$$f'(x) = (3x^2)\left(\sin\frac{1}{x}\right) + (x^3)\left(-\frac{1}{x^2}\cos\frac{1}{x}\right) = 3x^2 \sin\frac{1}{x} - x\cos\frac{1}{x}.$$

For $x = 0$:

$$\lim_{x\to 0}\frac{f(x)-f(0)}{x-0} = \lim_{x\to 0}\frac{x^3\sin(\frac{1}{x})-0}{x-0} = \lim_{x\to 0} x^2\sin\frac{1}{x}.$$

Since $-1 \le \sin(\frac{1}{x}) \le 1$ for every $x \neq 0$, we have $-x^2 \le x^2\sin\frac{1}{x} \le x^2$ for every $x \neq 0$. Since $\lim_{x\to 0}(-x^2) = 0 = \lim_{x\to 0} x^2$, by the Squeeze Theorem $\lim_{x\to 0} x^2\sin\frac{1}{x} = 0$. Therefore $f'(0)$ exists and $f'(0) = 0$. In summary,

$$f'(x) = \begin{cases} 3x^2\sin(\frac{1}{x}) - x\cos(\frac{1}{x}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(b) Since $f'(0) = 0$ and

$$\lim_{x\to 0} f'(x) = \lim_{x\to 0}\left(3x^2\sin\frac{1}{x} - x\cos\frac{1}{x}\right) = \underbrace{3\lim_{x\to 0}\left(x^2\sin\frac{1}{x}\right)}_{=0 \text{ by Squeeze Thm}} - \underbrace{\lim_{x\to 0}\left(x\cos\frac{1}{x}\right)}_{=0 \text{ by Squeeze Thm}} = 0,$$

it follows that $\lim_{x\to 0} f'(x)$ exists and equals $f'(0)$. Therefore f' is continuous at 0.

(c) The issue is whether $f''(0)$ exists. Since

$$\lim_{x\to 0}\frac{f'(x)-f'(0)}{x-0} = \lim_{x\to 0}\frac{[3x^2\sin(\frac{1}{x}) - x\cos(\frac{1}{x})]-0}{x-0} = \lim_{x\to 0}\left(3x\sin\frac{1}{x} - \cos\frac{1}{x}\right)$$

does not exist (otherwise $\lim_{x\to 0}\cos\frac{1}{x} = 3\lim_{x\to 0} x\sin\frac{1}{x} - \lim_{x\to 0}\left(3x\sin\frac{1}{x} - \cos\frac{1}{x}\right)$ would exist, contradicting Example 2.40), it follows that $f''(0)$ does not exist. In other words, f is not twice differentiable at 0.

The sum, difference and constant multiple rules generalize easily to higher derivatives.

Theorem 3.67 — Linearity of Higher Derivatives

Let n be a positive integer, let f and g be n-times differentiable functions, and let c be a real number. Then

$$(f+g)^{(n)} = f^{(n)} + g^{(n)}, \qquad (f-g)^{(n)} = f^{(n)} - g^{(n)}, \qquad (cf)^{(n)} = cf^{(n)}.$$

Proof. It follows immediately from Theorem 3.20. ■

The following is the product rule for higher derivatives. The formula looks similar to the Binomial Theorem, and the proof is similar too.

Theorem 3.68 — Leibniz's Rule

Let n be a positive integer and let f and g be n-times differentiable functions. Then

$$(fg)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(n-k)}g^{(k)} = f^{(n)}g + \binom{n}{1}f^{(n-1)}g' + \cdots + \binom{n}{n-1}f'g^{(n-1)} + fg^{(n)}.$$

Proof. We induce on n. The equality holds for $n = 1$ as it is just the ordinary product rule. Assuming that the equality holds for some positive integer $n = m$, we check that

$$\begin{aligned}
(fg)^{(m+1)} &= \big((fg)^{(m)}\big)' \\
&= (f^{(m)}g)' + \binom{m}{1}(f^{(m-1)}g')' + \binom{m}{2}(f^{(m-2)}g'')' + \cdots \\
&\quad + \binom{m}{m-1}(f'g^{(m-1)})' + (fg^{(m)})' \\
&= (f^{(m+1)}g + f^{(m)}g') + \binom{m}{1}(f^{(m)}g' + f^{(m-1)}g'') + \binom{m}{2}(f^{(m-1)}g'' + f^{(m-2)}g''') \\
&\quad + \cdots + \binom{m}{m-1}(f''g^{(m-1)} + f'g^{(m)}) + (f'g^{(m)} + fg^{(m+1)}) \\
&= f^{(m+1)}g + \left(1 + \binom{m}{1}\right) f^{(m)}g' + \left(\binom{m}{1} + \binom{m}{2}\right) f^{(m-1)}g'' + \cdots \\
&\quad + \left(\binom{m}{m-1} + 1\right) f'g^{(m)} + fg^{(m+1)} \\
&= f^{(m+1)}g + \binom{m+1}{1} f^{(m)}g' + \binom{m+1}{2} f^{(m-1)}g'' + \cdots \\
&\quad + \binom{m+1}{m} f'g^{(m)} + fg^{(m+1)},
\end{aligned}$$

so the equality holds for $n = m + 1$ as well. ■

Example 3.69 — Applying Leibniz's Rule

Find the 11th derivative of $f(x) = x^2 \cos x$.

Solution. According to Leibniz's rule (terms involving $\dfrac{d^3}{dx^3}x^2$ or higher vanish),

$$\begin{aligned}
f^{(11)}(x) &= \left(\frac{d^{11}}{dx^{11}}\cos x\right)(x^2) + 11\left(\frac{d^{10}}{dx^{10}}\cos x\right)\left(\frac{d}{dx}x^2\right) \\
&\quad + \frac{11\cdot 10}{2}\left(\frac{d^9}{dx^9}\cos x\right)\left(\frac{d^2}{dx^2}x^2\right) \\
&= (\sin x)(x^2) + 11(-\cos x)(2x) + \frac{11\cdot 10}{2}(-\sin x)(2) \\
&= (x^2 - 110)\sin x - 22x\cos x.
\end{aligned}$$

Example 3.70 — Legendre-type Identity

Let n be a non-negative integer and let $f : \mathbb{R} \to \mathbb{R}$ be the polynomial defined by

$$f(x) = (1 - x^2)^n.$$

(a) Show that $(1 - x^2)f'(x) + 2nxf(x) = 0$ for every $x \in \mathbb{R}$.
(b) Hence show that

$$(1 - x^2)f^{(n+2)}(x) - 2xf^{(n+1)}(x) + n(n + 1)f^{(n)}(x) = 0$$

for every $x \in \mathbb{R}$.

Solution. (a) The derivative of f is

$$f'(x) = n(1 - x^2)^{n-1}(-2x) = -2nx(1 - x^2)^{n-1}$$

for every $x \in \mathbb{R}$, so

$$(1-x^2)f'(x) + 2nxf(x) = (1-x^2)\left[-2nx(1-x^2)^{n-1}\right] + 2nx(1-x^2)^n = 0$$

for every $x \in \mathbb{R}$. ■

(b) Differentiate both sides of the result from (a) for $(n+1)$ times. According to Leibniz's rule,

$$\begin{aligned}&\Big[(1-x^2)f^{(n+2)}(x) + (n+1)(-2x)f^{(n+1)}(x) \\ &\qquad + \frac{(n+1)(n)}{2}(-2)f^{(n)}(x)\Big] \\ &\qquad + \left[(2nx)f^{(n+1)}(x) + (n+1)(2n)f^{(n)}(x)\right] = 0\end{aligned}$$

for every $x \in \mathbb{R}$, so

$$\begin{aligned}&(1-x^2)f^{(n+2)}(x) + \left[(n+1)(-2x) + (2nx)\right] f^{(n+1)}(x) \\ &+ \left[\frac{(n+1)(n)}{2}(-2) + (n+1)(2n)\right] f^{(n)}(x) = 0,\end{aligned}$$

i.e.

$$(1-x^2)f^{(n+2)}(x) - 2xf^{(n+1)}(x) + n(n+1)f^{(n)}(x) = 0$$

for every $x \in \mathbb{R}$. ■

3.5 Linear Approximation and Newton's Method

Remark 3.71 — The Approximation Behind Differentiability

Recall that f is differentiable at a if the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. Thus, if f is differentiable at a and if x is very close to a, we have the approximation $f'(a) \approx \dfrac{f(x) - f(a)}{x - a}$, i.e.

$$f(x) \approx f(a) + f'(a)(x - a).$$

Equivalently, if h is very close to 0, then $f(a + h) \approx f(a) + hf'(a)$.

Definition 3.72. Let a be a real number and f differentiable at a. The linear polynomial L defined by

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linear approximation** (or the **linearization**) of f at a.

Remark 3.73 — Geometry of the Linearization

The graph of L is a line which is just the tangent line to the graph of f at a. (Picture: the curve $y = f(x)$ together with its tangent line $y = L(x)$ at the point $(a, f(a))$; near a the line stays very close to the curve.)

Example 3.74 — Finding a Linear Approximation

Find the linear approximation of $f(x) = \arctan x$ at 1. (Finding the linear approximation of f at 1 is similar to finding the equation of the tangent line to the graph of f at $(1, f(1))$.)

Solution. f is differentiable with derivative $f'(x) = \dfrac{1}{1+x^2}$, so the linear approximation of f at 1 is

$$L(x) = f(1) + f'(1)(x-1) = \arctan 1 + \left(\frac{1}{1+1^2}\right)(x-1) = \frac{1}{2}x + \frac{\pi-2}{4}.$$

Example 3.75 — Numerical Estimates via Linear Approximation

Find an approximate value of each of the following using linear approximation:
(a) $\sqrt[3]{1.009}$ (b) $\cos\dfrac{49\pi}{100}$

Solution. (a) Consider the differentiable function $f(x) = \sqrt[3]{x}$. Then $f'(x) = \dfrac{1}{3}x^{-\frac{2}{3}}$, and

$$f(1.009) \approx f(1) + f'(1)(0.009) \quad \Longrightarrow \quad \sqrt[3]{1.009} \approx \sqrt[3]{1} + \frac{1}{3}(1)^{-\frac{2}{3}}(0.009) = 1.003.$$

(b) Consider the differentiable function $f(x) = \cos x$. Then $f'(x) = -\sin x$, and

$$f\left(\frac{49\pi}{100}\right) \approx f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(\frac{49\pi}{100} - \frac{\pi}{2}\right)$$

gives

$$\cos\frac{49\pi}{100} \approx \cos\frac{\pi}{2} + \left(-\sin\frac{\pi}{2}\right)\left(-\frac{\pi}{100}\right) = 0 + (-1)\left(-\frac{\pi}{100}\right) = \frac{\pi}{100}.$$

Remark 3.76 — Differentials

Denote a small change in x by Δx and the corresponding change in $y = f(x)$ by Δy; then $\Delta y \approx f'(x)\Delta x$. This motivates us to sloppily interpret the symbols dy and dx in the derivative by writing

$$dy = f'(x)\,dx.$$

These symbols dy and dx are called **differentials**; their precise meanings are well out of the scope of this course. (In fact the above equality is rigorous under the precise definitions of differentials.)

Example 3.77 — Computing Differentials

For each of the following, evaluate the differential dy in terms of dx. (There is no big difference between evaluating the differential and finding the derivative.)
(a) $y = x^3 - 3\sqrt{x}$ (b) $y = 3\csc(1 - 2\sqrt{x})$ (c) $y = \ln\dfrac{x+1}{\sqrt{x-1}}$

Solution.

(a) $dy = \left(3x^2 - \dfrac{3}{2\sqrt{x}}\right) dx.$

(b) $$dy = \left[-3\csc(1-2\sqrt{x})\cot(1-2\sqrt{x})\left(-\frac{1}{\sqrt{x}}\right)\right] dx$$
$$= \frac{3}{\sqrt{x}}\csc(1-2\sqrt{x})\cot(1-2\sqrt{x})\, dx.$$

(c) Note that $y = \ln\dfrac{x+1}{\sqrt{x-1}} = \ln(x+1) - \dfrac{1}{2}\ln(x-1)$. So

$$dy = \left[\frac{1}{x+1} - \frac{1}{2(x-1)}\right] dx = \frac{x-3}{2(x+1)(x-1)}\, dx.$$

Newton's method is a procedure of finding an approximate solution to an equation like $\cos x = x$, which is very hard, if not impossible, to solve analytically. It is an application of the idea of linear approximation.

Remark 3.78 — The Idea of Newton's Method

Let f be a function and suppose we want to solve $f(x) = 0$. Start with an initial guess x_0. If $f(x_0) = 0$ we are done — but usually not. Using the linear approximation of f at x_0,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0),$$

we look for x with $f(x_0) + f'(x_0)(x - x_0) \approx 0$, i.e. $x \approx x_0 - \dfrac{f(x_0)}{f'(x_0)}$, provided $f'(x_0) \neq 0$. Denote this improved guess by

$$x_1 := x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Graphically, x_1 is obtained from x_0 by drawing the tangent to the graph of f at $(x_0, f(x_0))$ and finding where this tangent meets the x-axis. If $f(x_1) \neq 0$, further improve the guess using the linear approximation at x_1:

$$x_2 := x_1 - \frac{f(x_1)}{f'(x_1)},$$

provided $f'(x_1) \neq 0$. Repeat as many times as needed until the approximation is accurate enough. (Picture: tangents at $x_0, x_1, x_2, \ldots$ successively meeting the x-axis at $x_1, x_2, x_3, \ldots$, converging to the true root.)

Theorem 3.79 — Convergent Newton Iterates Land on a Root

Let f be a differentiable function such that f' is continuous and never zero in some open interval, and let $(x_0, x_1, x_2, \ldots)$ be a sequence of numbers in the open interval satisfying

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If $\lim\limits_{n\to+\infty} x_n = r$ exists as a real number in the open interval, then

$$f(r) = 0,$$

i.e. r is a root of f.

Proof. Taking limits on both sides of $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$,

$$\lim_{n\to+\infty} x_{n+1} = \lim_{n\to+\infty} \left[x_n - \frac{f(x_n)}{f'(x_n)} \right].$$

Since f and f' are both continuous and $f'\left(\lim\limits_{n\to+\infty} x_n\right) = f'(r) \neq 0$,

$$\lim_{n\to+\infty} x_{n+1} = \lim_{n\to+\infty} x_n - \frac{f\left(\lim\limits_{n\to+\infty} x_n\right)}{f'\left(\lim\limits_{n\to+\infty} x_n\right)}, \qquad \text{i.e.} \qquad r = r - \frac{f(r)}{f'(r)}.$$

Therefore $f(r) = 0$. ■

Remark 3.80 — Newton's Method

Let f be a continuously differentiable function (i.e. f' is continuous). To find an approximated solution to $f(x) = 0$:

(i) First give an initial trial x_0 which is close to the required solution.

(ii) Compute the iterations $x_1, x_2, \ldots$ by successively applying the iteration formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

for each non-negative integer n, provided $f'(x_n) \neq 0$.

(iii) Stop when the approximation is accurate enough, i.e. when x_{n+1} and x_n are close enough.

Example 3.81 — Newton's Method in Two Steps

Use Newton's method with the initial trial $x_0 = 0$ to estimate the real solution to

$$x^3 + 3x + 1 = 0$$

in two steps.

Solution. Let $f(x) = x^3 + 3x + 1$. Then $f'(x) = 3x^2 + 3$. Newton's iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 + 3x_n + 1}{3x_n^2 + 3} = \frac{2x_n^3 - 1}{3x_n^2 + 3}.$$

With $x_0 = 0$:

$$x_1 = \frac{2x_0^3 - 1}{3x_0^2 + 3} = \frac{2(0)^3 - 1}{3(0)^2 + 3} = -\frac{1}{3} \qquad \text{and} \qquad x_2 = \frac{2x_1^3 - 1}{3x_1^2 + 3} = \frac{2(-\frac{1}{3})^3 - 1}{3(-\frac{1}{3})^2 + 3} = -\frac{29}{90}.$$

So the real solution is approximately equal to $-\dfrac{29}{90}$.

Example 3.82 — Approximating e by Newton's Method

Find the approximate value of e corrected to 4 decimal places by solving $\ln x = 1$ using Newton's method.

Solution. Let $f(x) = \ln x - 1$. Then $f'(x) = \dfrac{1}{x}$, and Newton's iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\ln x_n - 1}{\frac{1}{x_n}} = 2x_n - x_n \ln x_n.$$

With the initial guess $x_0 = 3$:

$$x_1 = 2x_0 - x_0 \ln x_0 = 2(3) - 3\ln 3 \approx 2.704163,$$
$$x_2 = 2x_1 - x_1 \ln x_1 \approx 2(2.704163) - 2.704163 \ln 2.704163 \approx 2.718245,$$
$$x_3 = 2x_2 - x_2 \ln x_2 \approx 2(2.718245) - 2.718245 \ln 2.718245 \approx 2.718282,$$
$$x_4 = 2x_3 - x_3 \ln x_3 \approx 2(2.718282) - 2.718282 \ln 2.718282 \approx 2.718282.$$

Corrected to 4 decimal places, x_3 and x_4 both give $e \approx 2.7183$. Therefore we stop the iteration with the approximate solution $e \approx 2.7183$.

Example 3.83 — Dependence on the Initial Guess

Find a solution to

$$xe^{-x} + 1 = 0$$

up to 4 significant digits by Newton's method, using the initial guesses 0 and 2 respectively.

Solution. Let $f(x) = xe^{-x} + 1$. Then $f'(x) = e^{-x} - xe^{-x} = (1-x)e^{-x}$. Newton's iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n e^{-x_n} + 1}{(1-x_n)e^{-x_n}} = \frac{x_n^2 + e^{x_n}}{x_n - 1}.$$

With $x_0 = 0$:

$$x_1 = -1, \qquad x_2 \approx -0.68394, \qquad x_3 \approx -0.57745,$$
$$x_4 \approx -0.56723, \qquad x_5 \approx -0.56714, \qquad x_6 \approx -0.56714.$$

So a solution to $xe^{-x} + 1 = 0$ is approximately -0.5671. On the other hand, starting with $x_0 = 2$,

$$x_1 \approx 11.389 \qquad \text{and} \qquad x_2 \approx 8516.1,$$

so the iteration sequence blows up, and we cannot find a solution using the initial guess $x_0 = 2$.

Remark 3.84 — Different Initial Trials May Differ

There may be more than one solution to an equation $f(x) = 0$, so different initial trials x_0 may yield different solutions.

Summary of Chapter 3

- Definition of the derivative of a function; definition of differentiability at a point and on an interval.
- Geometric interpretation of the derivative: slopes of tangent lines; various notations of the derivative; one-sided derivatives.
- Differentiability implies continuity; non-differentiability: discontinuity, corner point, vertical tangent line.
- Rules of differentiation: sum and difference, product rule and quotient rule, chain rule.
- Derivatives of elementary functions: polynomials and power functions, trigonometric functions, exponential functions, logarithmic functions, inverse trigonometric functions.
- Consequences of the chain rule: implicit differentiation, derivative of the inverse, logarithmic differentiation.
- Higher derivatives; Leibniz's rule.
- Rules and formulas of differentiation are valid on open intervals only; differentiability of piecewise defined functions; continuity and differentiability of the derivative and higher-order derivatives.
- Linear approximation of a differentiable function; estimating function values via linear approximation; Newton's method for approximate solutions of equations.

3.6 Supplement: Proof of the Chain Rule

Recall the statement of the chain rule.

Theorem 3.33 (Chain Rule)

Let x be a real number and let f and g be functions. If g is differentiable at x and f is differentiable at $g(x)$, then $f \circ g$ is also differentiable at x and its derivative is given by

$$(f \circ g)'(x) = f'(g(x))\, g'(x).$$

If we write $u = g(x)$ and $y = f(u) = (f \circ g)(x)$, then in Leibniz's notation this formula becomes

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

The following proof is sloppy but intuitive.

A sloppy proof. Recalling that the function $f \circ g$ is defined by $(f \circ g)(x) = f(g(x))$, we have

$$\begin{aligned}
(f \circ g)'(x) &= \lim_{h \to 0} \frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} \\
&= \lim_{h \to 0} \frac{f(g(x+h)) - f(g(x))}{h} \\
&= \lim_{h \to 0} \frac{f(g(x) + g(x+h) - g(x)) - f(g(x))}{g(x+h) - g(x)} \cdot \frac{g(x+h) - g(x)}{h} \\
&= \lim_{h \to 0} \frac{f(g(x) + g(x+h) - g(x)) - f(g(x))}{g(x+h) - g(x)} \cdot \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}.
\end{aligned}$$

With the change of variables $t = g(x+h) - g(x)$, we have $t \to 0$ as $h \to 0$ since g is continuous. Therefore

$$(f \circ g)'(x) = \lim_{t \to 0} \frac{f(g(x) + t) - f(g(x))}{t} \cdot \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = f'(g(x))\, g'(x).$$

■

This proof is sloppy because when we apply the change of variables $t = g(x+h) - g(x)$ to evaluate the limit, we have to require that (cf. Theorem 2.56)

$$g(x+h) - g(x) \neq 0 \quad \text{for every } h \text{ near } 0, \text{ except at } h = 0.$$

This requirement may not be always satisfied by the differentiable function g in the chain rule, as this function g can be something as strange as

$$g(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

(the requirement is violated for $x = 0$), or g may simply be a constant function (the requirement is violated for all x). To alleviate this problem, we have the following more rigorous proof instead.

Proof of Theorem 3.33. With the number x fixed, we let p be the function

$$p(t) := \begin{cases} \dfrac{f(t) - f(g(x))}{t - g(x)} & \text{if } t \neq g(x), \\ f'(g(x)) & \text{if } t = g(x). \end{cases}$$

This function p is continuous at the number $g(x)$, because

$$\lim_{t\to g(x)} p(t) = \lim_{t\to g(x)} \frac{f(t)-f(g(x))}{t-g(x)} = f'(g(x)) = p(g(x)).$$

Now the construction of p ensures that we always have the equality

$$f(t) - f(g(x)) = p(t)\,(t-g(x))$$

for every t, no matter $t = g(x)$ or $t \neq g(x)$. Thus applying this equality with $t = g(x+h)$, we have

$$\begin{aligned}
(f\circ g)'(x) &= \lim_{h\to 0} \frac{f(g(x+h))-f(g(x))}{h} \\
&= \lim_{h\to 0} \frac{p(g(x+h))\,(g(x+h)-g(x))}{h} \\
&= \lim_{h\to 0} p(g(x+h)) \cdot \lim_{h\to 0} \frac{g(x+h)-g(x)}{h} && \text{because } \lim_{h\to 0} p(g(x+h)) = p(g(x)), \text{ since } g \text{ is continuous at } x \text{ and } p \text{ is continuous at } g(x). \\
&= p(g(x))\,g'(x) \\
&= f'(g(x))\,g'(x).
\end{aligned}$$

■

3.7 Supplement: Differentiability and Existence of Derivative

In Definition 3.3, we defined the "differentiability" of a function to be the same as "existence of derivative".

Definition 3.3 (recalled). Let f be a function and let a be a real number. f is said to be **differentiable at** a if its derivative at a exists, i.e. if the limit

$$\lim_{x\to a} \frac{f(x)-f(a)}{x-a}, \qquad \text{or} \qquad \lim_{h\to 0} \frac{f(a+h)-f(a)}{h},$$

exists as a finite real number.

This is actually an oversimplification of the genuine notion of "differentiability", and is true for functions with one variable only. In fact, a better way to define the notion of "differentiability" is as follows.

Definition. Let f be a function and let a be a real number. f is said to be **differentiable at** a if there exists a number m such that

$$\lim_{x\to a} \frac{|f(x)-f(a)-m(x-a)|}{|x-a|} = 0.$$

In this new definition, the expression

$$L(x) := f(a) + m(x-a)$$

appearing in the numerator is actually the **linear approximation of f at a** (see Definition 3.72). Thus, a function f being "differentiable at a" actually means that when x gets closer and closer to a, the function value $f(x)$ gets closer and closer to $L(x)$ even more quickly. In other words, the linear approximation of f at a is "well approximating the value $f(x)$".

Let us see why the two definitions above are equivalent for one-variable functions.

Theorem — Equivalence of the Two Definitions

Let f be a function and let a be a real number. Then the limit

$$\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$$

exists as a finite real number if and only if there exists a number m such that

$$\lim_{x\to a}\frac{|f(x)-f(a)-m(x-a)|}{|x-a|}=0.$$

Proof. ($\Rightarrow$) Suppose that the limit $\lim\limits_{x\to a}\dfrac{f(x)-f(a)}{x-a}$ exists as a finite real number. Then we let m be this limit. It follows that

$$\lim_{x\to a}\frac{f(x)-f(a)-m(x-a)}{x-a}=\underbrace{\lim_{x\to a}\frac{f(x)-f(a)}{x-a}}_{=m}-m\cdot\lim_{x\to a}\frac{x-a}{x-a}=m-m=0,$$

and so

$$\lim_{x\to a}\frac{|f(x)-f(a)-m(x-a)|}{|x-a|}=\left|\lim_{x\to a}\frac{f(x)-f(a)-m(x-a)}{x-a}\right|=0.$$

($\Leftarrow$) Suppose that there exists a number m such that

$$\lim_{x\to a}\frac{|f(x)-f(a)-m(x-a)|}{|x-a|}=0.$$

Then we also have

$$\lim_{x\to a}\frac{f(x)-f(a)-m(x-a)}{x-a}=0.$$

Now for every x near a but $x\neq a$, we have

$$\frac{f(x)-f(a)}{x-a}=\frac{f(x)-f(a)-m(x-a)}{x-a}+m;$$

so the limit

$$\lim_{x\to a}\frac{f(x)-f(a)}{x-a}=\lim_{x\to a}\left(\frac{f(x)-f(a)-m(x-a)}{x-a}+m\right)=0+m=m$$

exists as a finite real number. ■

In this course MATH1013, let us be content with the fact that "differentiability" and "existence of derivatives" are equivalent for functions in one variable. However, when we study functions in several variables in the future (e.g. MATH2023), we will see that the notions of "differentiability" and "existence of (partial) derivatives" are actually different. To give a preview, the following is the definition of differentiability for a function in two variables.

Definition. Let f be a function in two variables, and let a and b be real numbers so that (a,b) is a point in the coordinate plane. f is said to be **differentiable at the point** (a,b) if there exist two numbers m_1 and m_2 such that

$$\lim_{(x,y)\to(a,b)}\frac{|f(x,y)-f(a,b)-m_1(x-a)-m_2(y-b)|}{\sqrt{(x-a)^2+(y-b)^2}}=0.$$

4 Mean Value Theorem

The Mean Value Theorem is a fundamental theorem about derivatives. It has numerous important consequences and is the main reason why differential calculus is useful. Before introducing the Mean Value Theorem in §4.2, we first look at how the global / local extrema of a function are related to the derivative.

4.1 Extrema and the Derivative

Definition. Let f be a function and c a number in the interior of the domain of f. If either

$$f'(c) = 0 \qquad \text{or} \qquad f'(c) \text{ is undefined,}$$

then c is called a critical number of f, and the point $(c, f(c))$ on the graph of f is called a critical point of f.

(Description of figure: the graph $y = f(x)$ over an interval with marked points $x_1 < \cdots < x_9$. Every local extremum occurs at one of: a point with horizontal tangent ($f' = 0$, e.g. x_2, x_3, x_6, x_8), a point with no tangent (sharp corner, f' undefined, e.g. x_4), or an end-point of a closed interval in the domain (e.g. x_1, x_9).)

Example 4.2

Find all the critical numbers of

$$f(x) = \begin{cases} x^3 - 3x, & x < 0, \\ \sin x, & x \geq 0. \end{cases}$$

Solution. We look for real numbers at which f' is either zero or undefined.

(i) For $x < 0$, $f'(x) = 3x^2 - 3$, so $f'(x) = 0$ at $x = -1$.

(ii) For $x > 0$, $f'(x) = \cos x$, so $f'(x) = 0$ at $x = \left(n - \frac{1}{2}\right)\pi$ for each $n \in \mathbb{N}$.

(iii) At $x = 0$,

$$f'_-(0) = \lim_{h \to 0^-} \frac{(h^3 - 3h) - \sin 0}{h} = \lim_{h \to 0^-} (h^2 - 3) = -3,$$

$$f'_+(0) = \lim_{h \to 0^+} \frac{\sin h - \sin 0}{h} = \lim_{h \to 0^+} \frac{\sin h}{h} = 1.$$

So $f'(0)$ does not exist.

Therefore the critical numbers of f are -1, 0, and every $\left(n - \frac{1}{2}\right)\pi$ with $n \in \mathbb{N}$.

Fermat's Theorem

Let $f : (a, b) \to \mathbb{R}$ be a function and let $c \in (a, b)$. If f attains a local maximum or a local minimum at c, then c is a critical number of f.

Proof. Suppose f attains a local maximum at c, i.e. $f(c + h) \leq f(c)$ for every h near 0. The issue is to show that if f is differentiable at c, then $f'(c) = 0$.

(i) We have

$$\lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h}.$$

(ii) If $h > 0$ and h is small, then $\dfrac{f(c+h) - f(c)}{h} \leq 0$. So $\lim\limits_{h \to 0^+} \dfrac{f(c+h) - f(c)}{h} \leq 0$ by Lemma 2.57 (i).

(iii) If $h < 0$ and h is small, then $\dfrac{f(c+h) - f(c)}{h} \geq 0$. So $\lim\limits_{h \to 0^-} \dfrac{f(c+h) - f(c)}{h} \geq 0$ by Lemma 2.57 (i).

Combining (i), (ii) and (iii),

$$0 \leq \lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h} = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = \lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h} \leq 0,$$

and so $f'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = 0$. If f attains a local minimum at c, the proof follows similarly. ■

Remark 4.4

Fermat's Theorem confirms that the initial observation reflects the truth. Let $f : A \to \mathbb{R}$ be a function. To find all local maxima and minima of f, the only possible candidates are:

- numbers x in the interior of A with $f'(x) = 0$ (critical numbers);
- numbers x in the interior of A with $f'(x)$ undefined (critical numbers);
- the end-points of the closed intervals that A consists of.

Remark 4.5

The converse of Fermat's Theorem is false. Once the three kinds of numbers above are identified as the only possible candidates for local extrema, it does NOT mean each candidate is indeed a local maximum or minimum: it may be a local maximum, a local minimum, or neither. Example: in the graph of f, the numbers x_5 and x_7 are critical numbers but each is neither a local maximum nor a local minimum. (Description of figure: graph of f where x_5 ($f'(x_5) = 0$) and x_7 ($f'(x_7)$ undefined) are critical points but neither local max nor local min, while x_2, x_4, x_8 are local maxima and x_3, x_6 are local minima.)

Remark 4.6

To classify each candidate as a local maximum / local minimum / neither, more machinery is needed (later in §4.4). However, if f is continuous on a bounded closed interval, then by the Extreme Value Theorem (Theorem 2.119) f attains global maximum and global minimum. In this special case the global extrema are found by comparing the values of f at all the candidates above.

Example 4.7

Let $f : [1, 4] \to \mathbb{R}$, $f(x) = \frac{2}{x} + \ln x$. Find the global extrema of f. **Solution.** The derivative is

$$f'(x) = -\frac{2}{x^2} + \frac{1}{x} = \frac{x-2}{x^2} \qquad (1 < x < 4),$$

so $f'(x) = 0$ only if $x = 2$. The only critical number of f is 2. (Why is 0 not a critical number of f?)

Since f is continuous on the bounded closed interval $[1, 4]$, compare the values at the critical number 2 and at the end-points 1 and 4:

$$f(1) = 2, \qquad f(2) = 1 + \ln 2 \approx 1.69315, \qquad f(4) = \frac{1}{2} + 2\ln 2 \approx 1.88629.$$

Thus on $[1, 4]$, f has global maximum value 2 (attained at 1) and global minimum value $1 + \ln 2$ (attained at 2).

Example 4.8

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^{\frac{2}{3}}(5-x)$. Find the global extrema of f on $[-1, 3]$. **Solution.** Write $f(x) = 5x^{\frac{2}{3}} - x^{\frac{5}{3}}$ for every $x \in \mathbb{R}$, so

$$f'(x) = 5\left(\tfrac{2}{3}x^{-\frac{1}{3}}\right) - \tfrac{5}{3}x^{\frac{2}{3}} = \frac{5(2-x)}{3x^{\frac{1}{3}}} \qquad (x \neq 0).$$

Thus $f'(x) = 0$ only if $x = 2$, and $f'(0)$ does not exist. The only critical numbers of f are 0 and 2.
Since f is continuous on $\mathbb{R}$, compare the values at the critical numbers $0, 2$ and at the end-points $-1, 3$:

$$f(-1) = 6, \qquad f(0) = 0, \qquad f(2) = 3 \cdot 2^{\frac{2}{3}}, \qquad f(3) = 2 \cdot 3^{\frac{2}{3}}.$$

Thus on $[-1, 3]$, f has global maximum value 6 (attained at -1) and global minimum value 0 (attained at 0).

Remark 4.9

The classification of global extrema by direct comparison of values is valid only when f is continuous on a bounded closed interval. Otherwise it fails because global extrema may not exist (Remark 2.120).

The following relates the trend of a function to its derivative. Recall the notions of increasing/decreasing and strictly increasing/strictly decreasing functions (Definition 1.54).

Theorem 4.10

Let f be continuous on $[a, b]$ and differentiable on (a, b).
(i) If f is increasing on $[a, b]$, then $f'(x) \geq 0$ for every $x \in (a, b)$.
(ii) If f is decreasing on $[a, b]$, then $f'(x) \leq 0$ for every $x \in (a, b)$.

Proof. We prove (i); (ii) is similar. Let $x \in (a, b)$ and h a small real number with $x + h \in (a, b)$. If f is increasing on $[a, b]$, then whether $h > 0$ or $h < 0$ we always have

$$\frac{f(x+h) - f(x)}{h} \geq 0.$$

Therefore $f'(x) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h} \geq 0$ by Lemma 2.57 (i). ■

Remark 4.11

In Theorem 4.10, even if f is strictly increasing on $[a, b]$, the best we can conclude is still $f'(x) \geq 0$ for every $x \in (a, b)$ (NOT $f'(x) > 0$). Indeed, even if $\dfrac{f(x+h) - f(x)}{h} > 0$ for every h near 0, Lemma 2.57 (i) only gives $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h} \geq 0$, not > 0.
Example: $f(x) = x^3$ is strictly increasing on $\mathbb{R}$, while $f'(x) = 3x^2$ is only non-negative on $\mathbb{R}$ and $f'(0) = 0$.

Theorem 4.10 says the derivative of an increasing function is non-negative. The natural question is whether the converse holds, i.e. whether a function with non-negative derivative must be increasing. The answer is "yes", and it is a consequence of the Mean Value Theorem below.

4.2 Mean Value Theorem

The spirit of the Mean Value Theorem is that if we know some values of a differentiable function, then we can deduce some information about its derivative.

Mean Value Theorem (Theorem 4.12)

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Remark 4.13

Geometrically, the Mean Value Theorem says that if the graph of f is smooth, then there always exists c between a and b such that the tangent line to the graph at c is parallel to the line joining the points $(a, f(a))$ and $(b, f(b))$ on the graph. (Description of figure: graph of f on $[a, b]$ with a chord from $(a, f(a))$ to $(b, f(b))$ and a parallel tangent line touching at an interior point c.)

Remark 4.14

- The Mean Value Theorem asserts the existence of at least one such number c; there can be more than one.
- It does not tell us how to find the number(s) c.

To prove the Mean Value Theorem we first need the following special case.

Rolle's Theorem (Theorem 4.15)

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then f has a critical number in (a, b), i.e. there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

(Description of figure: graph of f on $[a, b]$ with $f(a) = f(b)$ and an interior point c where the tangent is horizontal.)

Proof. Since $f(a) = f(b)$, consider two cases:

(i) If $f(x) = f(a)$ for all $x \in [a, b]$, then f is constant and $f' = 0$ everywhere.

(ii) Suppose for some x we have $f(x) \neq f(a)$; say $f(x) < f(a)$ WLOG. Since f is continuous on the closed interval $[a, b]$, it attains a global minimum on $[a, b]$ by the Extreme Value Theorem. By the assumption, this global minimum is attained neither at a nor at b, so it is attained at some $c \in (a, b)$. In particular f also attains a local minimum at c, so $f'(c) = 0$ by Fermat's Theorem.

■

Example 4.16

How many real roots does $f(x) = x^5 + 3x^3 + 2x + 5$ have? **Solution.**

- f is continuous (polynomial). Since $f(-1) = -1 < 0$ and $f(0) = 5 > 0$, f has a root $a \in (-1, 0)$ by the Intermediate Value Theorem.
- If f has any other real root $b \neq a$, then $f(a) = f(b) = 0$, so by Rolle's Theorem there exists a real number c between a and b with $f'(c) = 0$. But this is impossible because $f'(x) = 5x^4 + 9x^2 + 2 > 0$ for every real x.

Therefore f has exactly one real root, lying in $(-1, 0)$. ■

Example 4.17

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If there exists $c \in (a, b)$ with $f(c) > f(a)$ and $f(c) > f(b)$, show that there exists $d \in (a, b)$ with $f'(d) = 0$. **Solution.** (Motivation: we aim to use Rolle's Theorem, so we seek two distinct points at which f takes the same value. Description of figure: points a, c, b with $f(c)$ above the horizontal level m, and m above $f(a), f(b)$.)
Let $k = \max\{f(a), f(b)\}$. Since $f(c) > f(a)$ and $f(c) > f(b)$, we have $f(c) > k$. Let m satisfy $f(c) > m > k$. Then

(i) Since $f(c) > m$ and $f(a) < m$, by the Intermediate Value Theorem there exists $d_1 \in (a, c)$ with $f(d_1) = m$.

(ii) Since $f(c) > m$ and $f(b) < m$, by the Intermediate Value Theorem there exists $d_2 \in (c, b)$ with $f(d_2) = m$.

Now $d_1 < d_2$ and $f(d_1) = f(d_2)$, so by Rolle's Theorem there exists $d \in (d_1, d_2)$ with $f'(d) = 0$. ■

Proof of the Mean Value Theorem

Let $g : [a, b] \to \mathbb{R}$ be

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

(Description of figure: graph of f with a point x and the vertical gap $g(x)$ between the graph and the chord.) Then g is continuous on $[a, b]$, differentiable on (a, b) with

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

and $g(a) = 0 = g(b)$. So by Rolle's Theorem there exists $c \in (a, b)$ with $g'(c) = 0$, i.e.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

■

Remark 4.18

The Mean Value Theorem resembles the Intermediate Value Theorem only in that both assert existence of some number satisfying conditions. Major differences:

- IVT applies to continuous functions; MVT applies to differentiable functions.
- IVT asserts existence of c at which f attains a certain value $f(c)$; MVT asserts existence of c at which f has a certain derivative $f'(c)$.

Example 4.19

Show that for any real numbers a and b, $|\sin b - \sin a| \leq |b - a|$. **Solution.** Consider $f(x) = \sin x$, differentiable on $\mathbb{R}$ with $f'(x) = \cos x$. Given a, b:

(i) If $a = b$, both sides are 0, so the inequality holds trivially.

(ii) If $a \neq b$, applying the Mean Value Theorem to f gives $\dfrac{\sin b - \sin a}{b - a} = \cos c$ for some c between a and b. Since $|\cos c| \leq 1$, $\left|\dfrac{\sin b - \sin a}{b - a}\right| = |\cos c| \leq 1$, so $|\sin b - \sin a| \leq |b - a|$.

■

In the rest of this chapter we investigate various consequences of Rolle's Theorem and the Mean Value Theorem:

- "arbitrary constant" in finding antiderivatives;
- a function with positive (negative) derivative is increasing (decreasing);
- L'Hôpital's rule for evaluating limits;
- classification of local extrema by the first or second derivative test;
- concavity of a function using the second derivative.

It is clear that the derivative of a constant function is zero, i.e. if f is constant on $[a,b]$ then $f'(x)=0$ on (a,b) (cf. Theorem 3.24). The next theorem is the converse.

Theorem 4.20

Let f be continuous on $[a,b]$ and differentiable on (a,b). If $f'(x)=0$ for every $x\in(a,b)$, then f is constant on $[a,b]$. (Description: f constant on $[a,b] \iff f'=0$ on (a,b).)

Proof. Let $x<y$ be arbitrary in $[a,b]$. By the Mean Value Theorem there exists $c\in(x,y)$ with $f'(c)=\dfrac{f(y)-f(x)}{y-x}$. But $f'(c)=0$ by assumption, so $f(y)=f(x)$. Since $f(y)=f(x)$ for every $x,y\in[a,b]$ with $x<y$, f is constant on $[a,b]$. ■

Corollary 4.21

Let f and g be differentiable functions on (a,b) such that $f'=g'$. Then $f-g$ is a constant function on (a,b).

Remark 4.22

Corollary 4.21 explains why an arbitrary constant C is added when computing antiderivatives, e.g. $\displaystyle\int \cos x\,dx=\sin x+C$. Antiderivatives are discussed at the end of this chapter and in MATH1014.

Example 4.23

Let $f:\mathbb{R}\to\mathbb{R}$ be differentiable with $f'(x)=f(x)$ for every $x\in\mathbb{R}$. Show that there is a real constant C with $f(x)=Ce^x$ for every $x\in\mathbb{R}$. **Solution.** Consider $f(x)e^{-x}$. Its derivative is

$$\frac{d}{dx}\left(f(x)e^{-x}\right)=f'(x)e^{-x}+f(x)(-e^{-x})=(f'(x)-f(x))e^{-x}=0$$

for every $x\in\mathbb{R}$. By Theorem 4.20, $f(x)e^{-x}$ is constant on $\mathbb{R}$: there is a real constant C with $f(x)e^{-x}=C$, i.e. $f(x)=Ce^x$ for every $x\in\mathbb{R}$. ■

Remark 4.24

The equation $f'(x)=f(x)$ is a differential equation (in the unknown f). Its general solution is $f(x)=Ce^x$, where C is an arbitrary constant.

Example 4.25

Let $f,g:\mathbb{R}\to\mathbb{R}$ be differentiable satisfying: $f'(x)=g(x)$ for every x; $g'(x)=-f(x)$ for every x; $f(0)=0$ and $g(0)=1$. By considering

$$h(x)=(f(x)-\sin x)^2+(g(x)-\cos x)^2,$$

show that $f(x)=\sin x$ and $g(x)=\cos x$ for every $x\in\mathbb{R}$. **Solution.** For every $x\in$

$\mathbb{R}$,

$$\begin{aligned} h'(x) &= 2(f(x)-\sin x)(f'(x)-\cos x)+2(g(x)-\cos x)(g'(x)+\sin x) \\ &= 2(f(x)-\sin x)(g(x)-\cos x)+2(g(x)-\cos x)(-f(x)+\sin x)=0. \end{aligned}$$

By Theorem 4.20, there is a constant C with $h(x)=C$. Since $f(0)=0$ and $g(0)=1$, $h(0)=(f(0)-\sin 0)^2+(g(0)-\cos 0)^2=0$, so $C=0$ and $h(x)=0$ for every $x\in\mathbb{R}$:

$$(f(x)-\sin x)^2+(g(x)-\cos x)^2=0 \qquad \text{for every } x\in\mathbb{R}.$$

Both terms are non-negative, so $f(x)-\sin x=0$ and $g(x)-\cos x=0$ for every x, i.e. $f(x)=\sin x$ and $g(x)=\cos x$. ■

Next is the converse of Theorem 4.10.

Theorem 4.26

Let f be continuous on $[a,b]$ and differentiable on (a,b).
(i) If $f'(x)\geq 0$ for every $x\in(a,b)$, then f is (monotonic) increasing on $[a,b]$.
(ii) If $f'(x)\leq 0$ for every $x\in(a,b)$, then f is (monotonic) decreasing on $[a,b]$.

Proof. We prove (i); (ii) is similar. Let $x<y$ be arbitrary in $[a,b]$. By the Mean Value Theorem there exists $c\in(x,y)$ with $f'(c)=\dfrac{f(y)-f(x)}{y-x}$. But $f'(c)\geq 0$ by assumption, so $f(y)\geq f(x)$. Since $f(y)\geq f(x)$ for every $x,y\in[a,b]$ with $x<y$, f is increasing on $[a,b]$. ■

Theorem 4.27

Let f be continuous on $[a,b]$ and differentiable on (a,b).
- If $f'(x)>0$ for every $x\in(a,b)$, then f is strictly increasing on $[a,b]$.
- If $f'(x)<0$ for every $x\in(a,b)$, then f is strictly decreasing on $[a,b]$.
- $f'(x)\geq 0$ on (a,b) if and only if f is (monotonic) increasing on $[a,b]$.
- $f'(x)\leq 0$ on (a,b) if and only if f is (monotonic) decreasing on $[a,b]$.
- $f'(x)=0$ on (a,b) if and only if f is constant on $[a,b]$.

(Description: three graphs illustrating f increasing on $[a,b] \iff f'\geq 0$ on (a,b); f decreasing on $[a,b] \iff f'\leq 0$ on (a,b); f constant on $[a,b] \iff f'=0$ on (a,b).)

Thanks to this relationship many inequalities about functions can be proved using derivatives.

Example 4.28

(a) Show that $e^x>1+x$ for every $x>0$. Hence show further that $e^x>1+x+\dfrac{x^2}{2}$ for every $x>0$.

(b) Using (a), evaluate $\lim\limits_{x\to+\infty}\dfrac{x}{e^x}$ and $\lim\limits_{x\to+\infty}\dfrac{\ln x}{x}$.

(c) Using (b), evaluate $\lim\limits_{n\to+\infty}\sqrt[n]{n}$. **Solution.**

(a) Let $f(x)=e^x-1-x$. Then $f'(x)=e^x-1>0$ for every $x>0$, so f is strictly increasing on $[0,+\infty)$. Thus for every $x>0$, $f(x)>f(0)=0$, so $e^x-1-x>0$, i.e. $e^x>1+x$.

Next let $g(x)=e^x-1-x-\dfrac{x^2}{2}$. Then $g'(x)=e^x-1-x=f(x)>0$ for every $x>0$, so g is strictly increasing on $[0,+\infty)$. Thus for every $x>0$, $g(x)>g(0)=$

0, i.e. $e^x > 1 + x + \dfrac{x^2}{2}$.

(b) For every $x > 0$, by (a) $e^x > \dfrac{x^2}{2}$, hence $0 < \dfrac{x}{e^x} < \dfrac{2}{x}$. Since $\lim\limits_{x\to+\infty} \dfrac{2}{x} = 0$, by the Squeeze Theorem $\lim\limits_{x\to+\infty} \dfrac{x}{e^x} = 0$. With $t = e^x$ we also get $\lim\limits_{t\to+\infty} \dfrac{\ln t}{t} = 0$, hence $\lim\limits_{x\to+\infty} \dfrac{\ln x}{x} = 0$.

(c) Note that $\sqrt[n]{n} = e^{\frac{\ln n}{n}}$ for every positive integer n. From (b), $\lim\limits_{n\to+\infty} \dfrac{\ln n}{n} = 0$ as a limit of a sequence (cf. Theorem 2.76). Since the exponential function is continuous,

$$\lim_{n\to+\infty} \sqrt[n]{n} = \lim_{n\to+\infty} e^{\frac{\ln n}{n}} = e^{\lim_{n\to+\infty} \frac{\ln n}{n}} = e^0 = 1$$

by Theorem 2.108.

Example 4.29

Let $a < b$ be real numbers. (a) Let $g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$, differentiable on (a, b), with $g(a) = g(b) = 0$ and $g' : (a, b) \to \mathbb{R}$ strictly decreasing on (a, b). Show that there exists $c \in (a, b)$ such that g is strictly increasing on $[a, c]$ and strictly decreasing on $[c, b]$. Hence show that $g(x) \geq 0$ for every $x \in [a, b]$.
(b) Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$, twice differentiable on (a, b), with $f''(x) > 0$ for every $x \in (a, b)$. Using (a), show that

$$f(x) \leq \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b) \qquad \text{for every } x \in [a, b].$$

Solution.

(a) Apply Rolle's Theorem to g (continuous on $[a, b]$, differentiable on (a, b), $g(a) = g(b)$): there exists $c \in (a, b)$ with $g'(c) = 0$. Since g' (not g) is strictly decreasing on (a, b):
- $g'(x) > g'(c) = 0$ for every $x \in (a, c)$, so g is strictly increasing on $[a, c]$ by Theorem 4.27;
- $g'(x) < g'(c) = 0$ for every $x \in (c, b)$, so g is strictly decreasing on $[c, b]$ by Theorem 4.27.

It remains to show $g(x) \geq 0$ for every $x \in [a, b]$. Given $x \in [a, b]$:
- if $x = a$ or $x = b$, then $g(x) = 0$ by assumption;
- if $x \in (a, c]$, then $g(x) > g(a) = 0$ since g is strictly increasing on $[a, c]$;
- if $x \in [c, b)$, then $g(x) > g(b) = 0$ since g is strictly decreasing on $[c, b]$.

Therefore $g(x) \geq 0$ always.

(b) Let $h : [a, b] \to \mathbb{R}$ be

$$h(x) = \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b) - f(x).$$

Since f is continuous on $[a, b]$ and twice differentiable on (a, b), so is h. Now

$$h(a) = \frac{b-a}{b-a} f(a) + \frac{a-a}{b-a} f(b) - f(a) = 0$$

and

$$h(b) = \frac{b-b}{b-a} f(a) + \frac{b-a}{b-a} f(b) - f(b) = 0,$$

and $h'(x) = \dfrac{f(b)-f(a)}{b-a} - f'(x)$ is strictly decreasing on (a,b) since $h''(x) = -f''(x) < 0$ for every $x \in (a,b)$. So h satisfies all requirements for g in (a). Applying (a) to h gives $h(x) \geq 0$ for every $x \in [a,b]$, i.e.

$$\frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) - f(x) \geq 0,$$

which is $f(x) \leq \dfrac{b-x}{b-a}f(a) + \dfrac{x-a}{b-a}f(b)$.

■

Remark 4.30

Geometrically, Example 4.29(b) means that if $f'' > 0$ on (a,b), then the graph of f on $[a,b]$ lies below the line segment joining $(a, f(a))$ and $(b, f(b))$. More on the graphical meaning of f'' is in §4.4. (Description of figure: graph of f below the chord $y = \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b)$.)

Example 4.31

(a) Show that $x \ln x - x + 1 \geq 0$ for every $x > 0$.

(b) Let $a > 0$ and let $g : (0, +\infty) \to \mathbb{R}$, $g(x) = \dfrac{a^x - 1}{x}$. Using (a), show that g is (monotonic) increasing on $(0, +\infty)$.

(c) Let $p > q > 0$ be real numbers, n a positive integer, and $a_1, \ldots, a_n > 0$ with $\sum_{k=1}^{n} a_k^q = n$. Using (b), show that $\sum_{k=1}^{n} a_k^p \geq n$. **Solution.**

(a) Let $f(x) = x \ln x - x + 1$ on $(0, +\infty)$. For every $x > 0$,

$$f'(x) = (1)(\ln x) + (x)\left(\tfrac{1}{x}\right) - 1 = \ln x.$$

Since $f'(x) < 0$ on $(0,1)$ and $f'(x) > 0$ on $(1, +\infty)$, f is strictly decreasing on $(0,1]$ and strictly increasing on $[1, +\infty)$. Thus f attains its global minimum at 1, i.e. $f(x) \geq f(1)$ for every $x > 0$, which is $x \ln x - x + 1 \geq 0$.

(b) For every $x > 0$,

$$g'(x) = \frac{(a^x \ln a)(x) - (a^x - 1)(1)}{x^2} = \frac{a^x \ln(a^x) - a^x + 1}{x^2}.$$

Since $a^x > 0$, applying (a) to the number a^x gives $a^x \ln(a^x) - a^x + 1 \geq 0$. Hence $g'(x) \geq 0$ for every $x > 0$, so g is increasing on $(0, +\infty)$.

(c) For each $k \in \{1, \ldots, n\}$, replace the positive number a in (b) by a_k. Since g is increasing on $(0, +\infty)$ and $p > q > 0$, $g(p) \geq g(q)$, i.e.

$$\frac{a_k^p - 1}{p} \geq \frac{a_k^q - 1}{q}.$$

Summing from $k = 1$ to $k = n$:

$$\sum_{k=1}^{n} \frac{a_k^p - 1}{p} \geq \sum_{k=1}^{n} \frac{a_k^q - 1}{q}.$$

Since $\sum_{k=1}^{n} a_k^q = n$,

$$\frac{\left(\sum_{k=1}^{n} a_k^p\right) - \left(\sum_{k=1}^{n} 1\right)}{p} \geq \frac{\left(\sum_{k=1}^{n} a_k^q\right) - \left(\sum_{k=1}^{n} 1\right)}{q} = \frac{n-n}{q} = 0,$$

and so $\sum_{k=1}^{n} a_k^p \geq p \cdot 0 + \sum_{k=1}^{n} 1 = n.$

■

4.3 L'Hôpital's Rule

Differentiation can be applied to compute limits. In Chapter 2, factorization and rationalization handled limits involving the indeterminate forms

$$\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad 0 \cdot \infty, \qquad \infty - \infty, \qquad 1^{\infty}, \qquad 0^0, \qquad \infty^0$$

upon direct substitution. The following theorem provides another effective way to handle such forms.

L'Hôpital's Rule, $\frac{0}{0}$ case (Theorem 4.32)

Let f and g be functions differentiable near a real number a, except possibly at a. Also suppose $g'(x) \neq 0$ for every x near a, except possibly at a. If

(i) $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, and

(ii) $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists (finite or infinite),

then $\lim_{x\to a} \frac{f(x)}{g(x)}$ also exists and

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

Remark 4.33

L'Hôpital's rule also holds for one-sided limits and limits at infinity, and for the "$\frac{*}{\infty}$ case", i.e. when condition (i) is modified as

$$\text{(i)'} \quad \left(\lim_{x\to a^+} g(x) = +\infty \text{ or } -\infty\right) \text{ and } \left(\lim_{x\to a^-} g(x) = +\infty \text{ or } -\infty\right).$$

Example 4.34

Evaluate: (a) $\lim_{x\to 0} \frac{3^x - 2^x}{x}$; (b) $\lim_{x\to 0^+} \frac{\ln \sin x}{\ln \tan x}$. **Solution.**

(a) As $x \to 0$, numerator and denominator both tend to 0, so

$$\lim_{x\to 0} \frac{3^x - 2^x}{x} = \lim_{x\to 0} \frac{3^x \ln 3 - 2^x \ln 2}{1} = 3^0 \ln 3 - 2^0 \ln 2 = \ln 3 - \ln 2$$

(L'Hôpital's rule).

(b) As $x \to 0^+$, $\ln \tan x \to -\infty$, so

$$\lim_{x\to 0^+} \frac{\ln \sin x}{\ln \tan x} = \lim_{x\to 0^+} \frac{\frac{1}{\sin x}\cos x}{\frac{1}{\tan x}\sec^2 x} = \lim_{x\to 0^+} \cos^2 x = \cos^2 0 = 1$$

(L'Hôpital's rule).

Example 4.35

Evaluate $\lim_{x\to+\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$. **Solution.** This is an indeterminate form $\frac{\infty}{\infty}$, but applying L'Hôpital's rule gives

$$\frac{f(x)}{g(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \Rightarrow \frac{f'(x)}{g'(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \Rightarrow \frac{f''(x)}{g''(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \Rightarrow \cdots$$

None of these limits is simpler. So L'Hôpital's rule does not help. Using Chapter 2 methods,

$$\lim_{x\to+\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x\to+\infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{1-0}{1+0} = 1.$$

To prove L'Hôpital's rule we need the next lemma. It reduces to the Mean Value Theorem when g is the identity function.

Cauchy's Mean Value Theorem (Lemma 4.36)

Let f and g be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$g'(c)[f(b) - f(a)] = f'(c)[g(b) - g(a)].$$

Proof. Let h be defined on $[a, b]$ by

$$h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)].$$

Then h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = f(a)g(b) - g(a)f(b) = h(b)$. By Rolle's Theorem there exists $c \in (a, b)$ with $h'(c) = 0$, i.e. $g'(c)[f(b) - f(a)] = f'(c)[g(b) - g(a)]$. ■

Proof of Theorem 4.32 ($\frac{0}{0}$ case, a finite)

Since $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, we may (re-)define $f(a) = g(a) = 0$ to make f and g continuous at a also. (This does not affect validity: the conclusion concerns only limits at a.)

First show $\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}$. By assumption let $x > a$ be so close to a that $g'(t) \neq 0$ for every $t \in (a, x)$. (This forces $g(x) \neq 0$, because otherwise Rolle's Theorem would give $g'(t) = 0$ for some $t \in (a, x)$.) Now f and g are continuous on $[a, x]$ and differentiable on (a, x), so by Cauchy's Mean Value Theorem there exists $c \in (a, x)$ with

$$g'(c)[f(x) - f(a)] = f'(c)[g(x) - g(a)].$$

Since $g(x) \neq 0$ and $g'(c) \neq 0$, we have $\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}$. As $x > a$ was arbitrary, taking

$x \to a^+$,

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}.$$

(As $x \to a^+$ we also have $c \to a^+$, because c lies between a and x.) Similarly $\lim_{x\to a^-} \frac{f(x)}{g(x)} = \lim_{x\to a^-} \frac{f'(x)}{g'(x)}$, and therefore $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$. ■

Each of the other five indeterminate forms can be reduced to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by simple techniques, such as clearing denominators of fractions and taking logarithms.

Example 4.37

Evaluate: (a) $\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{\ln x}\right)$; (b) $\lim_{x\to 0^+} x(\ln x)^n$ (n a fixed non-negative integer); (c) $\lim_{x\to 0^+}(1-\cos x)^{\frac{1}{\ln x}}$. **Solution.**

(a)

$$\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{\ln x}\right) = \lim_{x\to 1} \frac{x\ln x - (x-1)}{(x-1)\ln x} = \lim_{x\to 1} \frac{\ln x + x(\frac{1}{x}) - 1}{\ln x + (x-1)(\frac{1}{x})}$$

(L'Hôpital's rule),

$$= \lim_{x\to 1} \frac{\ln x}{\ln x + 1 - \frac{1}{x}} = \lim_{x\to 1} \frac{\frac{1}{x}}{\frac{1}{x} + \frac{1}{x^2}} = \frac{1}{2}$$

(L'Hôpital's rule).

(b)

$$\lim_{x\to 0^+} x(\ln x)^n = \lim_{x\to 0^+} \frac{(\ln x)^n}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{n(\ln x)^{n-1}(\frac{1}{x})}{-\frac{1}{x^2}} = -n \lim_{x\to 0^+} \frac{(\ln x)^{n-1}}{\frac{1}{x}}$$

(L'Hôpital's rule)

$$= -n \lim_{x\to 0^+} \frac{(n-1)(\ln x)^{n-2}(\frac{1}{x})}{-\frac{1}{x^2}} = (-1)^2 n(n-1) \lim_{x\to 0^+} \frac{(\ln x)^{n-2}}{\frac{1}{x}} = \cdots$$

(apply L'Hôpital's rule for $(n-2)$ more times)

$$= (-1)^n n! \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}} = (-1)^n n! \lim_{x\to 0^+} x = 0.$$

(c) Note $(1-\cos x)^{\frac{1}{\ln x}} = e^{\frac{\ln(1-\cos x)}{\ln x}}$ for every $x \in (0, 2\pi)$. First evaluate

$$\lim_{x\to 0^+} \frac{\ln(1-\cos x)}{\ln x} = \lim_{x\to 0^+} \frac{\frac{1}{1-\cos x}\sin x}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{x\sin x}{1-\cos x}$$

(L'Hôpital's rule)

$$= \lim_{x\to 0^+} \frac{\sin x + x\cos x}{\sin x} = \lim_{x\to 0^+} \left(1 + \frac{x}{\sin x}\cos x\right) = 1 + 1\cos 0 = 2$$

(L'Hôpital's rule). Since the exponential function is continuous,

$$\lim_{x\to 0^+}(1-\cos x)^{\frac{1}{\ln x}} = \lim_{x\to 0^+} e^{\frac{\ln(1-\cos x)}{\ln x}} = e^{\lim_{x\to 0^+}\frac{\ln(1-\cos x)}{\ln x}} = e^2.$$

In Example 4.28 we found $\lim_{x\to+\infty} \frac{\ln x}{x} = 0$ and $\lim_{x\to+\infty} \frac{x}{e^x} = 0$. As $x \to +\infty$, the three

functions $\ln x$, x and e^x all tend to $+\infty$; the limits mean that e^x "grows faster than" x, and $\ln x$ "grows slower than" x. L'Hôpital's rule extends this observation.

Example 4.38

Let p be a positive real number. Show that $\lim\limits_{x\to+\infty} \dfrac{\ln x}{x^p} = 0$ and $\lim\limits_{x\to+\infty} \dfrac{x^p}{e^x} = 0$.

Solution. Both numerators and denominators tend to $+\infty$.

- For the first limit, apply L'Hôpital's rule once:

$$\lim_{x\to+\infty} \frac{\ln x}{x^p} = \lim_{x\to+\infty} \frac{\frac{1}{x}}{px^{p-1}} = \frac{1}{p} \lim_{x\to+\infty} \frac{1}{x^p} = \frac{1}{p} \cdot 0 = 0.$$

- For the second limit, let n be a positive integer with $n > p$. Applying L'Hôpital's rule n times,

$$\begin{aligned}\lim_{x\to+\infty} \frac{x^p}{e^x} &= \lim_{x\to+\infty} \frac{px^{p-1}}{e^x} = \lim_{x\to+\infty} \frac{p(p-1)x^{p-2}}{e^x} = \cdots \\ &= \lim_{x\to+\infty} \frac{p(p-1)\cdots(p-n+1)x^{p-n}}{e^x} \\ &= p(p-1)\cdots(p-n+1) \lim_{x\to+\infty} \frac{1}{x^{n-p}e^x} = 0.\end{aligned}$$

■

This shows that as $x \to +\infty$, e^x grows faster than every power function x^p $(p > 0)$, and $\ln x$ grows slower than every x^p $(p > 0)$.

Remark 4.39 (Growth rates of functions as $x \to +\infty$)

For $p, q, r > 0$ and $a > 1$,

$$(\ln x)^q \ll x^p \ll x^p(\ln x)^r \ll a^x \ll x^x \qquad \text{as } x \to +\infty.$$

Remark 4.40 (Growth rates of sequences)

For $p, q, r > 0$ and $a > 1$,

$$(\ln n)^q \ll n^p \ll n^p(\ln n)^r \ll a^n \ll n! \ll n^n \qquad \text{as } n \to +\infty.$$

Here "$f(x) \ll g(x)$ as $x \to +\infty$" means "$g(x)$ grows faster than $f(x)$ as $x \to +\infty$", and "$a_n \ll b_n$ as $n \to +\infty$" means "b_n grows faster than a_n"; precisely, $\lim\limits_{x\to+\infty} \dfrac{f(x)}{g(x)} = 0$ and $\lim\limits_{n\to+\infty} \dfrac{a_n}{b_n} = 0$.

Remark 4.41

Be careful about when L'Hôpital's rule can be applied. It cannot be applied in any of the following situations.

- If $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)}$ is not of the indeterminate form $\dfrac{0}{0}$ or $\dfrac{*}{\infty}$.
- If, no matter how close to a, there is always some $x \neq a$ with $g'(x) = 0$.
- If $\lim\limits_{x\to a} \dfrac{f'(x)}{g'(x)}$ does not exist.

Example 4.42

Evaluate $\lim_{x\to+\infty} \dfrac{x+\sin x}{x}$. **Solution.** Let $f(x) = x + \sin x$ and $g(x) = x$. Applying L'Hôpital's rule,

$$\lim_{x\to+\infty} \frac{f'(x)}{g'(x)} = \lim_{x\to+\infty} \frac{1+\cos x}{1}$$

does not exist, so L'Hôpital's rule cannot be applied here. (Do NOT conclude that $\lim_{x\to+\infty} \dfrac{x+\sin x}{x}$ "does not exist".) In fact,

$$\lim_{x\to+\infty} \frac{x+\sin x}{x} = \lim_{x\to+\infty} \left(1 + \frac{\sin x}{x}\right) = 1 + 0 = 1,$$

where $\lim_{x\to+\infty} \dfrac{\sin x}{x} = 0$ by a Squeeze Theorem argument.

Example 4.43

Let $f(x) = 2x - \sin 2x$ and $g(x) = (2x - \sin 2x)e^{\cos x}$. Clearly $\lim_{x\to+\infty} \dfrac{f(x)}{g(x)} = \lim_{x\to+\infty} \dfrac{1}{e^{\cos x}}$ does not exist (if it existed and equaled L, then $L \geq e^{-1} > 0$ and $\lim_{x\to+\infty} \cos x = -\ln L$ would exist, a contradiction). The given limit is an $\dfrac{\infty}{\infty}$ form; applying L'Hôpital's rule,

$$\begin{aligned}\lim_{x\to+\infty} \frac{f'(x)}{g'(x)} &= \lim_{x\to+\infty} \frac{2 - 2\cos 2x}{(2 - 2\cos 2x)e^{\cos x} + (2x - \sin 2x)(e^{\cos x})(-\sin x)} \\ &= \lim_{x\to+\infty} \frac{4e^{-\cos x}\sin x}{4\sin x + \sin 2x - 2x} = 0\end{aligned}$$

(using $2 - 2\cos 2x = 4\sin^2 x$ and the Squeeze Theorem, since the numerator is bounded while the denominator tends to $-\infty$). L'Hôpital's rule does not apply because $g'(n\pi) = 0$ for every $n \in \mathbb{Z}$, i.e. the roots of g' can be arbitrarily close to $+\infty$.

4.4 Graph Sketching

This section investigates how differentiation helps understand the shape of the graph of a function: on what intervals f is increasing/decreasing, positions of local maxima and minima, how the graph "bends", whether there are asymptotes, etc.

Recall that the local extrema of $f : A \to \mathbb{R}$ must come from: numbers $x \in A$ with $f'(x) = 0$ (critical numbers); numbers $x \in A$ with $f'(x)$ undefined (critical numbers); and the end-points of the closed intervals that A consists of. We now classify these candidates.

First Derivative Test (Corollary 4.44)

Let f be a function and c a critical number of f at which f is continuous; suppose f is differentiable near c, except possibly at c.

- If for all x near c we have $f'(x) \geq 0$ for $x < c$ and $f'(x) \leq 0$ for $x > c$ (sign of $f'(x)$ changes from $+$ to $-$ as x increases through c), then f attains a local maximum at c.
- If for all x near c we have $f'(x) \leq 0$ for $x < c$ and $f'(x) \geq 0$ for $x > c$ (sign of $f'(x)$ changes from $-$ to $+$), then f attains a local minimum at c.

- If for all x near c the sign of $f'(x)$ stays the same for $x < c$ and $x > c$, then f does not attain any local extremum at c.

(Description: three sketches — increasing then decreasing through c: local maximum; decreasing then increasing: local minimum; keeps increasing through c: no local extremum.)

Proof. In the first case there exist $a < c < b$ with $f'(x) \geq 0$ for every $x \in (a, c)$ and $f'(x) \leq 0$ for every $x \in (c, b)$. Thus f is increasing on $[a, c]$ and decreasing on $[c, b]$ by Theorem 4.27, so $f(c) \geq f(x)$ for every $x \in [a, b]$ whether $x < c$ or $x > c$; i.e. f attains a local maximum at c. The other two cases are similar. ■

Remark 4.45

If none of the above happens, the test is inconclusive: f may or may not attain a local extremum at c.

Example 4.46

Let $f(x) = x^3 - 3x^2 + 1$. Find and classify all local extrema of f. **Solution.** The domain is $\mathbb{R}$ and $f'(x) = 3x^2 - 6x = 3x(x-2)$ for every $x \in \mathbb{R}$. We have $f'(x) = 0$ iff $x = 0$ or $x = 2$, so the critical numbers are $0, 2$. By the first derivative test, check the sign of f' on three intervals:

(i) For $x < 0$, $f'(x) > 0$: f strictly increasing on $(-\infty, 0]$.
(ii) For $0 < x < 2$, $f'(x) < 0$: f strictly decreasing on $[0, 2]$.
(iii) For $x > 2$, $f'(x) > 0$: f strictly increasing on $[2, +\infty)$.

x	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, +\infty)$
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$	$\nearrow$	max.	$\searrow$	min.	$\nearrow$

So f has a local maximum point $(0, f(0)) = (0, 1)$ and a local minimum point $(2, f(2)) = (2, -3)$.

Example 4.47

Let $f(x) = (6x^2 - x^3)^{\frac{1}{3}}$. Find and classify all local extremum points of f. **Solution.** The domain is $\mathbb{R}$ and

$$f'(x) = \frac{1}{3}(6x^2 - x^3)^{-\frac{2}{3}}(12x - 3x^2) = \frac{12x - 3x^2}{3x^{\frac{4}{3}}(6-x)^{\frac{2}{3}}} = \frac{4 - x}{x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}.$$

Now $f'(x) = 0$ iff $x = 4$, and f' is undefined at 0 and 6. So the critical numbers are $0, 4, 6$. By the first derivative test, check the sign of f' on four intervals:

(i) For $x < 0$, $f'(x) < 0$: f strictly decreasing on $(-\infty, 0]$.
(ii) For $0 < x < 4$, $f'(x) > 0$: f strictly increasing on $[0, 4]$.
(iii) For $4 < x < 6$, $f'(x) < 0$: f strictly decreasing on $[4, 6]$.
(iv) For $x > 6$, $f'(x) < 0$: f strictly decreasing on $[6, +\infty)$.

x	$x < 0$	$x = 0$	$0 < x < 4$	$x = 4$	$4 < x < 6$	$x = 6$	$x > 6$
$f'(x)$	$-$	undef.	$+$	0	$-$	undef.	$-$
$f(x)$	$\searrow$	min.	$\nearrow$	max.	$\searrow$	?	$\searrow$

So f has a local maximum point $(4, 2\sqrt[3]{4})$ and a local minimum point $(0, 0)$.

Second Derivative Test (Corollary 4.48)

Let f be a function and c a critical number of f of the type $f'(c) = 0$, and suppose f is twice differentiable at c.

- If $f''(c) < 0$, then f attains a local maximum at c.
- If $f''(c) > 0$, then f attains a local minimum at c.
- If $f''(c) = 0$, the test is inconclusive: f may or may not attain a local extremum at c.

Proof. Suppose $f''(c) < 0$. Since $f'(c) = 0$,

$$0 > f''(c) = \lim_{x \to c} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \to c} \frac{f'(x)}{x - c},$$

so $\dfrac{f'(x)}{x - c} < 0$ for every $x \neq c$ near c. This implies that for x near c, $f'(x) > 0$ if $x < c$ and $f'(x) < 0$ if $x > c$. Therefore f attains a local maximum at c by the first derivative test. The case $f''(c) > 0$ is similar. ■

Example 4.46 (revisited, second derivative test)

Let $f(x) = x^3 - 3x^2 + 1$. Find and classify all local extrema of f. **Solution.** The domain is $\mathbb{R}$. For every real x, $f'(x) = 3x^2 - 6x = 3x(x - 2)$ and $f''(x) = 6x - 6$. Now $f'(x) = 0$ only if $x = 0$ or $x = 2$, so the critical numbers are $0, 2$. Applying the second derivative test,

$$f''(0) = 6(0) - 6 = -6 < 0, \qquad f''(2) = 6(2) - 6 = 6 > 0.$$

So f has a local maximum point $(0, 1)$ and a local minimum point $(2, -3)$.

Remark 4.49

The second derivative test is sometimes convenient if f'' is easy to compute. However, it is weaker than the first derivative test: the first derivative test can classify every critical number the second derivative test can, but some critical numbers classified by the first derivative test cannot be classified by the second derivative test. E.g. the critical number 0 of $f(x) = x^4$.

To sketch a graph accurately we also need the "concavity" of f, encoded in f''.

Definition. Let f be a function defined on $[a, b]$. We say that

- the graph of f is concave upward on (a, b) if for every $a \leq x < y < z \leq b$,

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(y)}{z - y};$$

- the graph of f is concave downward on (a, b) if for every $a \leq x < y < z \leq b$,

$$\frac{f(y) - f(x)}{y - x} \geq \frac{f(z) - f(y)}{z - y}.$$

Corollary 4.51

Let f be differentiable on an open interval (a, b).
- If f' is increasing on (a, b), then the graph of f is concave upward on (a, b).
- If f' is decreasing on (a, b), then the graph of f is concave downward on (a, b).

Proof. This is a direct consequence of the Mean Value Theorem. Given $a \leq x < y < z \leq b$, by the Mean Value Theorem there exist $c \in (x, y)$ and $d \in (y, z)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(c) \qquad \text{and} \qquad \frac{f(z) - f(y)}{z - y} = f'(d).$$

If f' is increasing on (a,b), then $f'(c) \le f'(d)$, which gives $\dfrac{f(y)-f(x)}{y-x} \le \dfrac{f(z)-f(y)}{z-y}$; so the graph of f is concave upward on (a,b). ■

Corollary 4.52

Let f be twice differentiable on an open interval (a,b).
- If $f'' > 0$ on (a,b), then the graph of f is concave upward on (a,b).
- If $f'' < 0$ on (a,b), then the graph of f is concave downward on (a,b).

Proof. If $f'' > 0$ on (a,b), then f' is increasing on (a,b) by Theorem 4.27, and so the graph of f is concave upward on (a,b) by Corollary 4.51. ■

Remark 4.53

Let f be defined on $[a,b]$.
- If the graph of f is concave upward (resp. downward) on (a,b), then for each distinct $c,d \in [a,b]$, the line segment joining $(c,f(c))$ and $(d,f(d))$ lies above (resp. below) the graph of f (a consequence of the Mean Value Theorem; cf. Example 4.29 and Remark 4.30).
- If f is differentiable on (a,b) and the graph of f is concave upward (resp. downward) on (a,b), then for each $c \in (a,b)$ the tangent line at $(c,f(c))$ lies below (resp. above) the graph of f (cf. Q10 of Problem Set 8).

(Description: left figure — concave curve with chord above/below; right figure — concave curve with tangent below/above.)

Definition. Let c be a real number and f a function continuous at c and differentiable near c, except possibly at c. The point $(c,f(c))$ on the graph of f is a point of inflection (or inflection point) if

(i) the concavity of the graph of f changes at c (from upward to downward or from downward to upward), and
(ii) there is a tangent line to the graph of f at $(c,f(c))$.

Example 4.55

The figure shows the graph of a continuous function $f : [x_1, x_7] \to \mathbb{R}$. (Description: graph where concavity changes at x_2, x_4, x_5 (inflection points, with f'' undefined at x_2, $f''(x_4) = 0$, $f''(x_5) = 0$); at x_3 f'' is undefined but concavity does not change; at x_6 f'' is undefined and there is no tangent.)
- The graph is concave upward on (x_2,x_3), (x_3,x_4) and (x_5,x_6).
- The graph is concave downward on (x_1,x_2), (x_4,x_5) and (x_6,x_7).
- f has points of inflection at x_2, x_4, x_5; at these numbers f'' is either zero or undefined.
- Although $f''(x_3)$ is undefined, f has no point of inflection at x_3 because f'' does not change sign there. Although $f''(x_6)$ is undefined, f has no point of inflection at x_6 since there is no tangent to the graph at x_6.

Example 4.56

Find all points of inflection of $f(x) = x^{\frac{2}{3}}(5-x)$. **Solution.** The domain is $\mathbb{R}$. Since

$f(x) = 5x^{\frac{2}{3}} - x^{\frac{5}{3}}$,

$$f'(x) = \frac{10}{3}x^{-\frac{1}{3}} - \frac{5}{3}x^{\frac{2}{3}} = \frac{5(2-x)}{3x^{\frac{1}{3}}} \qquad (x \neq 0)$$

(factorizing f' helps solve $f'(x) = 0$, but expanding f' facilitates further differentiation), and

$$f''(x) = -\frac{10}{9}x^{-\frac{4}{3}} - \frac{10}{9}x^{-\frac{1}{3}} = -\frac{10(1+x)}{9x^{\frac{4}{3}}} \qquad (x \neq 0).$$

We have $f''(x) = 0$ iff $x = -1$. Now (i) for every $x < -1$, $f''(x) > 0$; (ii) for every $x > -1$ and $x \neq 0$, $f''(x) < 0$.

x	$x < -1$	$x = -1$	$-1 < x < 0$	$x = 0$	$x > 0$
$f''(x)$	$+$	0	$-$	undef.	$-$
$f(x)$	conc. up	infl.	conc. down	cusp	conc. down

So f has a point of inflection $(-1, 6)$.

Remark 4.57 (Summary of first/second derivative concepts)

Critical points of f : $f'(c) = 0$ or $f'(c)$ undefined

f' changes $+ \to -$, or $f''(c) < 0 \Rightarrow$ local maximum point
f' changes $- \to +$, or $f''(c) > 0 \Rightarrow$ local minimum point
f' no change in sign $\Rightarrow$ neither

$f''(c) = 0$ or $f''(c)$ undefined

f'' changes sign and tangent line exists $\Rightarrow$ point of inflection
f'' no change in sign, or no tangent line $\Rightarrow$ not point of inflection

Remark 4.58

Some people prefer not to require the existence of a tangent in the definition of a point of inflection (i.e. requirement (ii) in Definition 4.54). Our definition is more appropriate for investigating the geometry of the graph, while the other is more appropriate when interpreting the function as a physical quantity.

Example 4.59

Sketch the graph of a function f satisfying:
(i) Domain of f is $(0, +\infty)$.
(ii) f continuous on $(0, +\infty)$.
(iii) $\lim_{x \to 0^+} f(x) = -\infty$.
(iv) $f'(x) > 0$ for every $x \in (0, 2)$.
(v) $f'(x) < 0$ for every $x \in (2, +\infty)$.
(vi) The global maximum value of f is 3.
(vii) $f''(x) < 0$ for every $x \in (0, 1)$.
(viii) $f''(x) > 0$ for every $x \in (1, 2) \cup (2, +\infty)$.
(ix) $\lim_{x \to +\infty} (f(x) + x) = 3$.

Solution. Translate the information to graph features.

- (i),(ii): the graph is one single curve with x-coordinate from 0 to $+\infty$, satisfying the vertical line test.
- (iii): vertical asymptote $x = 0$; the curve points downward indefinitely near the right of the asymptote.

- (iv),(v): f strictly increasing on $(0,2]$ and strictly decreasing on $[2,+\infty)$; f attains its global maximum at 2, and (vi) gives $f(2)=3$.
- (vii),(viii): the graph is concave downward on $(0,1)$ and concave upward on $(1,2)$ and on $(2,+\infty)$.
- (ix): slant asymptote $y=-x+3$.

x	0	$(0,1)$	1	$(1,2)$	2	$(2,+\infty)$
$f'(x)$	undef.	$+$	$+$	$+$	undef.	$-$
$f''(x)$	undef.	$-$	0	$+$	undef.	$+$
$f(x)$	$\to-\infty$		infl.		max. $=3$	

(Description of a possible graph: curve rising from $-\infty$ near $x=0$, inflection near $x=1$, peak value 3 at $x=2$, then decreasing toward the slant line $y=-x+3$ as $x\to+\infty$.)

Remark 4.60 (Seven-step graph sketching procedure)

(i) Identify the domain of f and find f' and f''.
(ii) Find all critical numbers of f (where f' is zero or undefined) and identify the intervals where f is increasing/decreasing.
(iii) Find all numbers where f'' is zero or undefined and identify the intervals where the graph is concave upward/concave downward.
(iv) Present the information from (ii),(iii) in a table and hence identify all local extremum points and points of inflection.
(v) Identify all asymptotes. (Can also be done at the beginning.)
(vi) Locate the x- and y-intercepts. (Can also be done at the beginning.)
(vii) Sketch the graph using all the information found.

Example 4.61

Sketch the graph of $f(x)=\dfrac{1}{x+1}-\dfrac{1}{x+3}+5$. **Solution.** The domain is $\mathbb{R}\setminus\{-1,-3\}$. The derivatives are

$$f'(x)=-\frac{1}{(x+1)^2}+\frac{1}{(x+3)^2}=-\frac{4(x+2)}{(x+1)^2(x+3)^2}$$

and

$$f''(x)=\frac{2}{(x+1)^3}-\frac{2}{(x+3)^3}=\frac{4(3x^2+12x+13)}{(x+1)^3(x+3)^3}=\frac{12(x+2)^2+4}{(x+1)^3(x+3)^3}$$

for every $x\neq-1,-3$. We have $f'(x)=0$ only if $x=-2$, and $f''(x)$ is always non-zero.

x	$(-\infty,-3)$	-3	$(-3,-2)$	-2	$(-2,-1)$	-1	$(-1,+\infty)$
$f'(x)$	$+$	undef.	$+$	0	$-$	undef.	$-$
$f''(x)$	$+$	undef.	$-$	$-$	$-$	undef.	$+$
$f(x)$		undef.		max.		undef.	

So f has a local maximum point $(-2,3)$, no local minimum points and no points of inflection.
Asymptotes:
(i) Since $\lim\limits_{x\to-1^+}f(x)=+\infty$, $\lim\limits_{x\to-1^-}f(x)=-\infty$, $\lim\limits_{x\to-3^+}f(x)=-\infty$ and $\lim\limits_{x\to-3^-}f(x)=+\infty$, the lines $x=-1$ and $x=-3$ are vertical asymptotes.
(ii) Since $\lim\limits_{x\to-\infty}f(x)=\lim\limits_{x\to+\infty}f(x)=5$, the line $y=5$ is a horizontal asymptote.

Intercepts: writing $f(x) = \dfrac{1}{x+1} - \dfrac{1}{x+3} + 5 = \dfrac{5x^2+20x+17}{(x+1)(x+3)}$,

(i) $f(x) = 0$ iff $5x^2+20x+17 = 0$, i.e. $x = \dfrac{-20 \pm \sqrt{20^2 - 4(5)(17)}}{2(5)} = -2 \pm \sqrt{\frac{3}{5}}$, so the x-intercepts are $-2 \pm \sqrt{\frac{3}{5}}$.

(ii) The y-intercept is $f(0) = \dfrac{1}{1} - \dfrac{1}{3} + 5 = \dfrac{17}{3}$.

(Description of graph: two branches separated by the vertical asymptotes $x = -3$ and $x = -1$, peak $(-2, 3)$, horizontal asymptote $y = 5$.)

Example 4.62

Sketch the graph of $f(x) = (x^2-3)e^{-x}$. **Solution.** The domain is $\mathbb{R}$. The derivatives are

$$f'(x) = (2x)(e^{-x}) + (x^2-3)(-e^{-x}) = -(x^2-2x-3)e^{-x} = (1+x)(3-x)e^{-x}$$

and

$$f''(x) = -(2x-2)e^{-x} - (x^2-2x-3)(-e^{-x}) = (x^2-4x-1)e^{-x}$$

for every real x. We have $f'(x) = 0$ only if $x = -1$ or $x = 3$, and $f''(x) = 0$ only if $x = 2-\sqrt{5}$ or $x = 2+\sqrt{5}$.

x	$(-\infty,-1)$	-1	$(-1, 2-\sqrt{5})$	$2-\sqrt{5}$	$(2-\sqrt{5}, 3)$	3
$f'(x)$	$-$	0	$+$	$+$	$+$	0
$f''(x)$	$+$	$+$	$+$	0	$-$	$-$
$f(x)$		local min.		infl.		local max.

x	3	$(3, 2+\sqrt{5})$	$2+\sqrt{5}$	$(2+\sqrt{5}, +\infty)$
$f'(x)$	0	$-$	$-$	$-$
$f''(x)$	$-$	$-$	0	$+$
$f(x)$	local max.		infl.	

So f has a local maximum point $(3, 6e^{-3})$, a local minimum point $(-1, -2e)$, and two points of inflection $\left(2-\sqrt{5}, (6-4\sqrt{5})e^{-2+\sqrt{5}}\right)$ and $\left(2+\sqrt{5}, (6+4\sqrt{5})e^{-2-\sqrt{5}}\right)$.
Asymptotes: since f is well-defined on $\mathbb{R}$, there are no vertical asymptotes. On the other hand (recall the procedure in Remark 2.85),

$$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} (x^2-3)e^{-x} = \lim_{x\to+\infty} \frac{x^2-3}{e^x} = \lim_{x\to+\infty} \frac{2x}{e^x} = \lim_{x\to+\infty} \frac{2}{e^x} = 0$$

by L'Hôpital's rule; while

$$\lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \left(x - \frac{3}{x}\right) \underbrace{e^{-x}}_{\to+\infty} = -\infty$$

is infinite. So the line $y = 0$ is a horizontal asymptote and there are no slant asymptotes.
The graph has x-intercepts $\pm\sqrt{3}$ and y-intercept -3. (Description of graph: local min $(-1, -2e)$; inflection at $2-\sqrt{5}$; local max $(3, 6e^{-3})$; inflection at $2+\sqrt{5}$; horizontal asymptote $y = 0$ as $x \to +\infty$; curve from $-\infty$ behaving like xe^{-x}-growth.)

Example 4.47 (revisited, full sketch)

Sketch the graph of $f(x) = (6x^2-x^3)^{\frac{1}{3}}$. **Solution.** The domain is $\mathbb{R}$. The derivatives

are

$$f'(x) = \frac{1}{3}(6x^2 - x^3)^{-\frac{2}{3}}(12x - 3x^2) = \frac{12x - 3x^2}{3x^{\frac{4}{3}}(6-x)^{\frac{2}{3}}} = \frac{4-x}{x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}$$

and

$$f''(x) = \frac{(-1)x^{\frac{1}{3}}(6-x)^{\frac{2}{3}} - (4-x)\left[\frac{1}{3}x^{-\frac{2}{3}}(6-x)^{\frac{2}{3}} + x^{\frac{1}{3}}\frac{2}{3}(6-x)^{-\frac{1}{3}}(-1)\right]}{x^{\frac{2}{3}}(6-x)^{\frac{4}{3}}} = -\frac{8}{x^{\frac{4}{3}}(6-x)^{\frac{5}{3}}}$$

for every $x \neq 0, 6$. We have $f'(x) = 0$ only if $x = 4$, and $f''(x)$ is always non-zero.

x	$(-\infty, 0)$	0	$(0, 4)$	4	$(4, 6)$	6	$(6, +\infty)$
$f'(x)$	$-$	undef.	$+$	0	$-$	undef.	$-$
$f''(x)$	$-$	undef.	$-$	$-$	$-$	undef.	$+$
$f(x)$		min. cusp		max.		infl.	

So f has a local maximum point $(4, 2\sqrt[3]{4})$, a local minimum point $(0, 0)$, and a point of inflection $(6, 0)$.
Asymptotes: since f is well-defined on $\mathbb{R}$, there are no vertical asymptotes. Also,

$$\lim_{x\to\pm\infty} \frac{f(x)}{x} = \lim_{x\to\pm\infty} \frac{(6x^2 - x^3)^{\frac{1}{3}}}{x} = \lim_{x\to\pm\infty} \left(\frac{6}{x} - 1\right)^{\frac{1}{3}} = (0-1)^{\frac{1}{3}} = -1,$$

and

$$\lim_{x\to\pm\infty} [f(x) - (-x)] = \lim_{x\to\pm\infty} \left[(6x^2 - x^3)^{\frac{1}{3}} + x\right] = \lim_{x\to\pm\infty} \frac{(\frac{6}{x} - 1)^{\frac{1}{3}} + 1}{\frac{1}{x}}$$

$$= \lim_{x\to\pm\infty} \frac{\frac{1}{3}(\frac{6}{x} - 1)^{-\frac{2}{3}}(-\frac{6}{x^2})}{-\frac{1}{x^2}} = \lim_{x\to\pm\infty} 2\left(\frac{6}{x} - 1\right)^{-\frac{2}{3}} = 2(0-1)^{-\frac{2}{3}} = 2$$

(L'Hôpital's rule), so the line $y = -x + 2$ is a slant asymptote. The graph has y-intercept 0 and x-intercepts 0 and 6. (Description of graph: cusp minimum $(0, 0)$, local max $(4, 2\sqrt[3]{4})$, inflection $(6, 0)$, slant asymptote $y = -x + 2$.)

Example 4.63

Sketch the graph of $g(x) = \dfrac{|x|}{x+2}$. **Solution.** The domain is $\mathbb{R} \setminus \{-2\} = (-\infty, -2) \cup (-2, +\infty)$. Write g in piecewise form (first removing the absolute value):

$$g(x) = \begin{cases} \dfrac{x}{x+2} = 1 - \dfrac{2}{x+2}, & x \geq 0, \\ \dfrac{-x}{x+2} = \dfrac{2}{x+2} - 1, & x < 0,\ x \neq -2. \end{cases}$$

Method I: analyze the piecewise-defined function directly.

$$g'(x) = \begin{cases} \dfrac{2}{(x+2)^2}, & x > 0, \\ \dfrac{-2}{(x+2)^2}, & x < 0,\ x \neq -2, \end{cases} \qquad g''(x) = \begin{cases} \dfrac{-4}{(x+2)^3}, & x > 0, \\ \dfrac{4}{(x+2)^3}, & x < 0,\ x \neq -2. \end{cases}$$

At $x = 0$, since

$$g'_-(0) = \lim_{x\to 0^-} \frac{\frac{-x}{x+2} - 0}{x} = -\frac{1}{2}, \qquad g'_+(0) = \lim_{x\to 0^+} \frac{\frac{x}{x+2} - 0}{x} = \frac{1}{2},$$

$g'(0)$ does not exist, so $g''(0)$ does not exist either.

x	$x < -2$	$x = -2$	$-2 < x < 0$	$x = 0$	$x > 0$
$g'(x)$	$-$	undef.	$-$	undef.	$+$
$g''(x)$	$-$	undef.	$+$	undef.	$-$
$g(x)$		undef.		min. corner	

So g has a local minimum point $(0, 0)$, no local maximum points and no points of inflection.
Asymptotes:

(i) Since $\lim_{x\to -2^-} g(x) = \lim_{x\to -2^-} \frac{-x}{x+2} = -\infty$ and $\lim_{x\to -2^+} g(x) = \lim_{x\to -2^+} \frac{-x}{x+2} = +\infty$, the line $x = -2$ is a vertical asymptote.

(ii) Since $\lim_{x\to -\infty} g(x) = \lim_{x\to -\infty} \left(\frac{2}{x+2} - 1\right) = -1$ and $\lim_{x\to +\infty} g(x) = \lim_{x\to +\infty} \left(1 - \frac{2}{x+2}\right) = 1$, the lines $y = -1$ and $y = 1$ are horizontal asymptotes.

Clearly the graph cuts the axes only at the origin $(0, 0)$. (Description of graph: V-shaped corner minimum at $(0, 0)$, vertical asymptote $x = -2$, horizontal asymptotes $y = 1$ (right) and $y = -1$ (left).)

Method II: use transformations of graphs. The curve $y = \frac{-x}{x+2} = -1 + \frac{2}{x+2}$ is obtained from $y = \frac{1}{x}$ by a vertical stretching of factor 2, then translation of 2 units left and 1 unit downward. Finally, to obtain the graph of g, reflect the portion of $y = \frac{-x}{x+2}$ in the right half-plane across the x-axis. (Description: intermediate curve $y = \frac{-x}{x+2}$ with asymptotes $x = -2$, $y = -1$, and its reflected right portion giving g.)

4.5 Antiderivatives

Definition. Let f be a function. If F is a differentiable function such that $F' = f$ on an open interval (a, b), then F is called an antiderivative of f on (a, b).

Example 4.65

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos x$. Then $F(x) = \sin x$ is an antiderivative of f on $\mathbb{R}$. The function $G(x) = \sin x + 1$ is also another antiderivative of f on $\mathbb{R}$.

Remark 4.66

Let f be defined on an open interval (a, b) and suppose F is an antiderivative of f on (a, b). Then

- for each constant function C, $F + C$ is also an antiderivative of f on (a, b);
- by Corollary 4.21 (a consequence of the Mean Value Theorem), every antiderivative of f on (a, b) must be of the form $F + C$ where C is a constant function.

Therefore, once one antiderivative is found, adding an arbitrary constant function C gives all antiderivatives on the interval.

Think: What if f is defined on the union of two disjoint open intervals? How do we

find all possible antiderivatives of f if we have found one of them?

Instead of the "naturally expected" notation $f^{(-1)}$ or $\left(\frac{d}{dx}\right)^{-1} f(x)$, we adopt the following notation.

Definition. Let f be a function defined on an open interval. Then the collection of all antiderivatives of f is denoted

$$\int f(x)\,dx.$$

Thus if F is an antiderivative of f, we write $\int f(x)\,dx = F(x) + C$, where C is an arbitrary constant.

Example 4.68

Find the derivative of $f(x) = x\ln x - x$. Hence find all antiderivatives of $g(x) = \ln x$ on its natural domain $(0, +\infty)$. **Solution.** On $(0, +\infty)$,

$$f'(x) = (1)(\ln x) + (x)\left(\tfrac{1}{x}\right) - 1 = \ln x = g(x).$$

Since $f' = g$, the antiderivatives of g on $(0, +\infty)$ are all functions of the form $f + C$, i.e.

$$\int g(x)\,dx = f(x) + C = x\ln x - x + C,$$

where C is an arbitrary constant function.

Theorem 4.69

Let f and g be functions having antiderivatives and c a real number. Then

(i) $\int (f \pm g)(x)\,dx = \int f(x)\,dx \pm \int g(x)\,dx$;

(ii) $\int (cf)(x)\,dx = c\int f(x)\,dx$.

Proof. These follow immediately from the corresponding rules of differentiation, Theorem 3.20. ■

Remark 4.70

There is NO "product rule" or "quotient rule" for antiderivatives. In particular, do not apply INCORRECT formulas such as

$$\int f(x)g(x)\,dx = \left(\int f(x)\,dx\right) g(x) + f(x)\left(\int g(x)\,dx\right) \quad \times$$

or

$$\int f(x)g(x)\,dx = \int f(x)\,dx \int g(x)\,dx, \quad \times$$

which do not hold in general. Methods for antiderivatives of products of two specific functions are given later in MATH1014.

Example 4.71 (product rule of differentiation)

(a) Let f and g be differentiable. By computing the derivative of fg, show that

$$\int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx = f(x)g(x) + C,$$

where C is an arbitrary constant.

(b) Hence evaluate $\int xe^x\,dx$. **Solution.**

(a) The derivative of the product fg is $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$, so fg is an antiderivative of $f'g + fg'$:

$$\int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx = \int [f'(x)g(x) + f(x)g'(x)]\,dx = f(x)g(x) + C.$$

(b) Let $f(x) = x$ and $g(x) = e^x$. Then $f'(x) = 1$ and $g'(x) = e^x$. By (a),

$$\int (1)(e^x)\,dx + \int (x)(e^x)\,dx = (x)(e^x) + C_1,$$

i.e. $\int e^x\,dx + \int xe^x\,dx = xe^x + C_1$. Since $\int e^x\,dx = e^x + C_2$, we get $\int xe^x\,dx = xe^x - e^x + C$, where $C = C_1 - C_2$.

■

Example 4.72 (chain rule of differentiation)

(a) Let f and g be differentiable. By computing the derivative of $f \circ g$, show that

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C,$$

where C is an arbitrary constant.

(b) Hence evaluate $\int 2x\cos(x^2)\,dx$. **Solution.**

(a) The derivative of the composition $f \circ g$ is $(f \circ g)'(x) = f'(g(x))g'(x)$, so $f(g(x))$ is an antiderivative of $f'(g(x))g'(x)$: $\int f'(g(x))g'(x)\,dx = f(g(x)) + C$.

(b) Let $f(x) = \sin x$ and $g(x) = x^2$. Then $f'(x) = \cos x$, $g'(x) = 2x$, so $f'(g(x))g'(x) = 2x\cos(x^2)$. By (a),

$$\int 2x\cos(x^2)\,dx = \int f'(g(x))g'(x)\,dx = f(g(x)) + C = \sin(x^2) + C.$$

■

Remark 4.73 (Antidifferentiation formulas)

From our knowledge of derivatives of elementary functions, in each of the following C is an arbitrary constant.

(i) Polynomials and power functions:

$$\int x^r\,dx = \frac{1}{r+1}x^{r+1} + C\ (r \neq -1); \qquad \int \frac{1}{x}\,dx = \ln|x| + C.$$

(ii) Exponential functions:

$$\int e^x\,dx = e^x + C; \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C\ \ (a \in (0,1) \cup (1,+\infty)).$$

(iii) Trigonometric functions:

$$\int \cos x\,dx = \sin x + C; \qquad \int \sin x\,dx = -\cos x + C;$$
$$\int \sec^2 x\,dx = \tan x + C; \qquad \int \csc^2 x\,dx = -\cot x + C;$$
$$\int \sec x \tan x\,dx = \sec x + C; \qquad \int \csc x \cot x\,dx = -\csc x + C.$$

(iv) Algebraic functions related to inverse trigonometric functions:

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C; \quad \int \frac{1}{1+x^2}\,dx = \arctan x + C;$$
$$\int \frac{1}{x\sqrt{x^2-1}}\,dx = \operatorname{arcsec}|x| + C.$$

If the domain of a function consists of two or more disjoint open intervals, the arbitrary constant in its antiderivative can be chosen differently on different intervals. E.g. $\int \frac{1}{x}\,dx = \ln|x| + C$ means that $f(x) = \frac{1}{x}$, defined on $(-\infty, 0) \cup (0, +\infty)$, has all its antiderivatives F given by

$$F(x) = \begin{cases} \ln x + C_1, & x > 0, \\ \ln(-x) + C_2, & x < 0, \end{cases}$$

where C_1, C_2 are arbitrary constants.

Example 4.74

Evaluate: (a) $\int e^{t+3}\,dt$; (b) $\int \frac{(u-1)(u+2)}{\sqrt{u}}\,du$; (c) $\int \frac{2x^2}{1+x^2}\,dx$; (d) $\int \tan^2\theta\,d\theta$. **Solution.**

(a)

$$\int e^{t+3}\,dt = \int e^t e^3\,dt = e^3 \int e^t\,dt = e^3 e^t + C = e^{t+3} + C.$$

(b) It is hard to find antiderivatives of products, so expand into a sum.

$$\int \frac{(u-1)(u+2)}{\sqrt{u}}\,du = \int \frac{u^2+u-2}{\sqrt{u}}\,du = \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}} - 2u^{-\frac{1}{2}}\right) du$$
$$= \int u^{\frac{3}{2}} du + \int u^{\frac{1}{2}} du - 2\int u^{-\frac{1}{2}} du = \frac{1}{\frac{5}{2}} u^{\frac{5}{2}} + \frac{1}{\frac{3}{2}} u^{\frac{3}{2}} - 2\left(\frac{1}{\frac{1}{2}} u^{\frac{1}{2}}\right) + C$$
$$= \frac{2}{5}u^{\frac{5}{2}} + \frac{2}{3}u^{\frac{3}{2}} - 4u^{\frac{1}{2}} + C.$$

(c) Long division gives $\frac{2x^2}{1+x^2} = 2 - \frac{2}{1+x^2}$. Thus

$$\int \frac{2x^2}{1+x^2}\,dx = \int \left(2 - \frac{2}{1+x^2}\right) dx = 2\int dx - 2\int \frac{1}{1+x^2}\,dx = 2x - 2\arctan x + C.$$

(d)

$$\int \tan^2\theta\,d\theta = \int (\sec^2\theta - 1)\,d\theta = \int \sec^2\theta\,d\theta - \int d\theta = \tan\theta - \theta + C.$$

Example 4.75

The domains of the following functions are both $\mathbb{R}$. For each, find all antiderivatives on $\mathbb{R}$.

(a) $f(x) = |x|$; (b) $g(x) = \begin{cases} -1 & x < 0, \\ 1 & x \geq 0. \end{cases}$ **Solution.**

(a) Suppose F is an antiderivative of f on $\mathbb{R}$. Recalling

$$f(x) = |x| = \begin{cases} -x, & x < 0, \\ x, & x \geq 0, \end{cases}$$

(antidifferentiation formulas are valid on open intervals, just like differentiation formulas), we consider two open intervals:

(i) For $x < 0$, $F(x) = -\frac{1}{2}x^2 + C_1$ for some constant C_1.

(ii) For $x > 0$, $F(x) = \frac{1}{2}x^2 + C_2$ for some constant C_2.

Since F is an antiderivative it is differentiable, hence continuous at 0. Therefore $\lim\limits_{x\to 0^-} F(x) = \lim\limits_{x\to 0^+} F(x) = F(0)$, i.e.

$$\lim_{x\to 0^-} \left(-\tfrac{1}{2}x^2 + C_1\right) = \lim_{x\to 0^+} \left(\tfrac{1}{2}x^2 + C_2\right) = F(0),$$

and so $C_1 = C_2 = F(0)$. Replacing C_1 and C_2 by a single constant C_0,

$$F(x) = \begin{cases} -\dfrac{x^2}{2} + C_0, & x < 0, \\ C_0, & x = 0, \\ \dfrac{x^2}{2} + C_0, & x > 0, \end{cases}$$

i.e. $F(x) = \frac{1}{2}x|x| + C_0$. Finally check that at 0,

$$F'_-(0) = \lim_{x\to 0^-} \frac{F(x) - F(0)}{x} = \lim_{x\to 0^-} \frac{(-\frac{x^2}{2} + C_0) - C_0}{x} = \lim_{x\to 0^-} -\tfrac{1}{2}x = 0,$$

$$F'_+(0) = \lim_{x\to 0^+} \frac{F(x) - F(0)}{x} = \lim_{x\to 0^+} \frac{(\frac{x^2}{2} + C_0) - C_0}{x} = \lim_{x\to 0^+} \tfrac{1}{2}x = 0,$$

so $F'(0) = 0 = f(0)$ indeed. Therefore

$$\int |x|\,dx = F(x) + C = \tfrac{1}{2}x|x| + C,$$

where C is an arbitrary constant. (The constant C_0 can be absorbed into C.)

(b) Suppose G is an antiderivative of g on $\mathbb{R}$. Then

(i) For $x < 0$, $G(x) = -x + C_1$ for some constant C_1.

(ii) For $x > 0$, $G(x) = x + C_2$ for some constant C_2.

Since G is differentiable, it is continuous at 0. Therefore $\lim\limits_{x\to 0^-} G(x) = \lim\limits_{x\to 0^+} G(x) = G(0)$, giving $C_1 = C_2 = G(0)$. So

$$G(x) = \begin{cases} -x + C_0, & x < 0, \\ C_0, & x = 0, \\ x + C_0, & x > 0, \end{cases}$$

i.e. $G(x) = |x| + C_0$ for some constant C_0. This time

$$G'_-(0) = \lim_{x\to 0^-} \frac{G(x) - G(0)}{x} = \lim_{x\to 0^-} \frac{(-x + C_0) - C_0}{x} = -1,$$

but

$$G'_+(0) = \lim_{x\to 0^+} \frac{G(x)-G(0)}{x} = \lim_{x\to 0^+} \frac{(x+C_0)-C_0}{x} = 1,$$

so G is not differentiable at 0, a contradiction. Therefore g has no antiderivatives on $\mathbb{R}$.

Think: What about $h(x) = \begin{cases} -1 & x<0, \\ 1 & x>0 \end{cases}$? Does h have an antiderivative on its domain? [Yes. The domain of h is $(-\infty,0)\cup(0,+\infty)$, and the antiderivatives of h on its domain are all functions

$$H(x) = \begin{cases} -x+C_1, & x<0, \\ x+C_2, & x>0, \end{cases}$$

where C_1 and C_2 are arbitrary constants.]

Remark 4.76

Example 4.75 shows that some functions have no antiderivatives. In MATH1014 we will prove that every continuous function has an antiderivative. Nevertheless, a function having an antiderivative may not be continuous. As an example (cf. Example 3.54), consider

$$f(x) = \begin{cases} 2x\sin\dfrac{1}{x} - \cos\dfrac{1}{x}, & x\neq 0, \\ 0, & x=0, \end{cases}$$

whose domain is $\mathbb{R}$. Although f is discontinuous at 0, it has an antiderivative on $\mathbb{R}$ given by

$$F(x) = \begin{cases} x^2\sin\dfrac{1}{x}, & x\neq 0, \\ 0, & x=0. \end{cases}$$

Remark 4.77

Using antidifferentiation we can solve a differential equation of the form $\dfrac{dy}{dx} = f(x)$. To "solve" such an equation means to find a possible candidate for the unknown function y; any antiderivative of f works.

Example 4.78

Solve the differential equations: (a) $\dfrac{dy}{d\theta} = \dfrac{\sin\theta}{\cos^2\theta}$; (b) $\dfrac{du}{dx} = 2^{x-2}$. **Solution.**

(a) Since $\dfrac{dy}{d\theta} = \dfrac{\sin\theta}{\cos^2\theta} = \sec\theta\tan\theta$, taking antiderivatives on both sides,

$$y = \int \sec\theta\tan\theta\, d\theta = \sec\theta + C,$$

where C is an arbitrary constant.

(b) Since $\dfrac{du}{dx} = 2^{x-2} = \dfrac{2^x}{4}$, taking antiderivatives on both sides,

$$u = \int \frac{2^x}{4}\, dx = \frac{1}{4}\int 2^x\, dx = \frac{1}{4}\cdot\frac{2^x}{\ln 2} + C = \frac{2^{x-2}}{\ln 2} + C.$$

Remark 4.79

A general solution of the differential equation $\dfrac{dy}{dx} = f(x)$ must involve an arbitrary constant. If we further require that the solution y takes a chosen value at a number

a, e.g. $y(a) = b$, then the arbitrary constant is determined, so that on each open interval containing a there is only one solution satisfying this additional condition. The condition $y(a) = b$ is called an initial condition; a differential equation together with an initial condition is called an initial value problem.

Example 4.80

Suppose that at each point (x, y) on a curve, the slope of the tangent line is $x + \frac{1}{x^2}$. If the curve cuts the x-axis at $x = 2$, find the equation of the curve. **Solution.** The slope of the tangent is the derivative of the function the curve represents, i.e.

$$\frac{dy}{dx} = x + \frac{1}{x^2}.$$

This is equivalent to the initial value problem $\begin{cases} \frac{dy}{dx} = x + \frac{1}{x^2}, \\ y(2) = 0. \end{cases}$ Taking antiderivatives,

$$y = \frac{1}{2}x^2 - \frac{1}{x} + C,$$

where C is some constant. Since the curve passes through $(2, 0)$,

$$0 = \frac{1}{2}(2^2) - \frac{1}{2} + C,$$

so $C = -\frac{3}{2}$. Therefore the equation of the curve is

$$y = \frac{1}{2}x^2 - \frac{1}{x} - \frac{3}{2}.$$

Example 4.81

It is given that at each point (x, y) on a curve, $\frac{d^2y}{dx^2} = \sin x + e^x$. If the curve passes through the point $(0, -1)$ and the slope of the tangent line at this point is 3, find the equation of the curve. **Solution.** Taking antiderivatives twice on both sides of $\frac{d^2y}{dx^2} = \sin x + e^x$,

$$\frac{dy}{dx} = -\cos x + e^x + C$$

and

$$y = -\sin x + e^x + Cx + D,$$

where C and D are some constants. This is equivalent to the initial value problem $\begin{cases} y''(x) = \sin x + e^x, \\ y(0) = -1, \\ y'(0) = 3. \end{cases}$ Since the curve passes through $(0, -1)$ at which the slope of the tangent is 3,

$$\begin{cases} -\cos 0 + e^0 + C = 3, \\ -\sin 0 + e^0 + C(0) + D = -1. \end{cases}$$

Solving gives $C = 3$ and $D = -2$, so the equation of the curve is

$$y = -\sin x + e^x + 3x - 2.$$

4.6 Supplement: Proof of L'Hôpital's Rule

In the lecture we have seen a proof of L'Hôpital's rule for the $\frac{0}{0}$ case. The following gives a proof for the $\frac{*}{\infty}$ case. For simplicity, only one-sided limits are proved, as the two-sided version then follows easily. The proof uses the ε-δ language.

L'Hôpital's Rule, $\frac{*}{\infty}$ case

Let f and g be functions which are differentiable near a real number a, except possibly at a. Also suppose that $g'(x) \neq 0$ for every x near a, except at a. If

(i) $\lim\limits_{x\to a^+} g(x) = +\infty$ or $-\infty$, and

(ii) $\lim\limits_{x\to a^+} \dfrac{f'(x)}{g'(x)}$ exists (either as a finite real number or infinity),

then $\lim\limits_{x\to a^+} \dfrac{f(x)}{g(x)}$ also exists and

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}.$$

Proof. Suppose the one-sided limit $\lim\limits_{x\to a^+} \dfrac{f'(x)}{g'(x)} = L$ is finite. Then for each $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{if } t \in (a, a+\delta), \text{ then } g'(t) \neq 0 \text{ and } L - \frac{\varepsilon}{2} < \frac{f'(t)}{g'(t)} < L + \frac{\varepsilon}{2}.$$

Let $x \in (a, a+\delta)$ be arbitrary. Since $\lim\limits_{x\to a^+} g(x) = \pm\infty$, by choosing a smaller δ if necessary, we may assume that $g(x) \neq 0$. Now f and g are continuous on $[x, a+\delta]$ and differentiable on $(x, a+\delta)$, so applying Cauchy's Mean Value Theorem there exists $c \in (x, a+\delta)$ such that

$$g'(c)[f(a+\delta) - f(x)] = f'(c)[g(a+\delta) - g(x)].$$

Since $g(a+\delta) - g(x)$ and $g'(c) \neq 0$, we have

$$\frac{f'(c)}{g'(c)} = \frac{f(a+\delta) - f(x)}{g(a+\delta) - g(x)}.$$

We aim to relate $\dfrac{f'(c)}{g'(c)}$ with $\dfrac{f(x)}{g(x)}$. Using $g(x) \neq 0$ and rearranging,

$$\frac{f'(c)}{g'(c)}\left(1 - \frac{g(a+\delta)}{g(x)}\right) = \frac{f(x)}{g(x)} - \frac{f(a+\delta)}{g(x)},$$

and so

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)} - \frac{1}{g(x)}\left(f(a+\delta) - \frac{f'(c)}{g'(c)}g(a+\delta)\right).$$

Denote the second term on the right-hand side by

$$r(x) := \frac{1}{g(x)}\left(f(a+\delta) - \frac{f'(c)}{g'(c)}g(a+\delta)\right).$$

Then $\dfrac{f'(c)}{g'(c)} = \dfrac{f(x)}{g(x)} - r(x)$. But since $c \in (a, a+\delta)$ in particular, we have $L - \dfrac{\varepsilon}{2} < \dfrac{f'(c)}{g'(c)} < L + \dfrac{\varepsilon}{2}$ in the first place; thus

$$L - \frac{\varepsilon}{2} + r(x) < \frac{f(x)}{g(x)} < L + \frac{\varepsilon}{2} + r(x) \qquad \text{for every } x \in (a, a+\delta).$$

Now note that

$$\lim_{x\to a^+} r(x) = \lim_{x\to a^+} \frac{1}{g(x)} \underbrace{\left(f(a+\delta) - \frac{f'(c)}{g'(c)} g(a+\delta) \right)}_{\text{fixed and bounded}} = 0,$$

because $\dfrac{1}{g(x)} \to 0$ (as $g(x) \to \pm\infty$) while the second factor is (componentwise) fixed and bounded. So there exists $\delta_1 \in (0, \delta)$ such that if $x \in (a, a+\delta_1)$ then $|r(x)| < \dfrac{\varepsilon}{2}$. Therefore

$$\text{if } x \in (a, a+\delta_1), \text{ then } L - \varepsilon < \frac{f(x)}{g(x)} < L + \varepsilon,$$

which precisely means $\lim\limits_{x\to a^+} \dfrac{f(x)}{g(x)} = L$, as required. The proof works in a similar way when $L = +\infty$ or $-\infty$, where instead of an arbitrarily small $\varepsilon > 0$ we use an arbitrarily large $R > 0$. ■

L'Hôpital's Rule for Limits at Infinity

Let f and g be functions which are differentiable on an interval $(b, +\infty)$. Also suppose that $g'(x) \neq 0$ for every $x > b$. If

(i) either ($\lim\limits_{x\to+\infty} f(x) = \lim\limits_{x\to+\infty} g(x) = 0$) or ($\lim\limits_{x\to+\infty} g(x) = +\infty$ or $-\infty$), and

(ii) $\lim\limits_{x\to+\infty} \dfrac{f'(x)}{g'(x)}$ exists (either as a finite real number or infinity),

then $\lim\limits_{x\to+\infty} \dfrac{f(x)}{g(x)}$ also exists and

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = \lim_{x\to+\infty} \frac{f'(x)}{g'(x)}.$$

It is obvious that the case for $x \to -\infty$ works similarly.

Proof. Suppose that $\lim\limits_{x\to+\infty} \dfrac{f'(x)}{g'(x)} = L$, regardless of whether L is finite or infinite. With the change of variable $t = \dfrac{1}{x}$, we have

$$L = \lim_{t\to0^+} \frac{f'(\frac{1}{t})}{g'(\frac{1}{t})} = \lim_{t\to0^+} \frac{f'(\frac{1}{t}) \cdot \left(-\frac{1}{t^2}\right)}{g'(\frac{1}{t}) \cdot \left(-\frac{1}{t^2}\right)} = \lim_{t\to0^+} \frac{\frac{d}{dt} f(\frac{1}{t})}{\frac{d}{dt} g(\frac{1}{t})}.$$

Now since $\lim_{t\to 0^+} f(\frac{1}{t}) = \lim_{x\to+\infty} f(x)$ and $\lim_{t\to 0^+} g(\frac{1}{t}) = \lim_{x\to+\infty} g(x)$, by either the $\frac{0}{0}$ or the $\frac{*}{\infty}$ version of L'Hôpital's rule for $t \to 0^+$, we have

$$\lim_{t\to 0^+} \frac{f(\frac{1}{t})}{g(\frac{1}{t})} = \lim_{t\to 0^+} \frac{\frac{d}{dt} f(\frac{1}{t})}{\frac{d}{dt} g(\frac{1}{t})} = L$$

and therefore

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = \lim_{t\to 0^+} \frac{f(\frac{1}{t})}{g(\frac{1}{t})} = L$$

as required. ■

MATH1014 Calculus II

5 Integrals

The main goal of this chapter is to study areas of plane regions bounded by curves. Starting with the simple case of a non-negative continuous function f, we study the area of the region bounded between the graph of f and the x-axis on a bounded closed interval $[a, b]$.

5.1 Area and Integrals

Definition. Let $a < b$ be real numbers and let $f : [a, b] \to [0, +\infty)$ be a continuous function.

- The integral of f over $[a, b]$, denoted by
$$\int_a^b f(x)\,dx,$$
is the area of the region in the coordinate plane bounded by the graph of f, the x-axis, the vertical line $x = a$ and the vertical line $x = b$. (Figure: region under $y = f(x)$ between $x = a$ and $x = b$, with area labeled $\int_a^b f(x)\,dx$.)
- The process of finding the integral of a function is called integration. To integrate f means to compute the area of the above region.
- The function f is called the integrand, and the numbers a and b are called the lower limit and the upper limit of the integral respectively.

Remark — Notation of an integral

(i) The symbol dx is called a differential; its precise meaning is out of the scope of this course. Here dx just indicates that x is the independent variable of the integrand f.

(ii) The integral $\int_a^b f(x)\,dx$ is just a number, and its value does not depend on the symbol x:
$$\int_a^b f(x)\,dx = \int_a^b f(u)\,du = \int_a^b f(t)\,dt.$$
We say that x is a dummy variable. To avoid confusion, the dummy variable should not also appear in the limits or outside the integral:
$$\int_a^x f(x)\,dx, \qquad \int_a^b g(a)\,da, \qquad \frac{1}{t}\int_a^b h(t)\,dt \quad \text{are not allowed.}$$
Note: the x appearing in the antiderivative $\int f(x)\,dx$ (chapter 4) is NOT a dummy variable!

Example — Constant function

Let $a < b$ and let $c > 0$. If f is the constant function $f(x) = c$, then
$$\int_a^b f(x)\,dx = c(b-a), \qquad \text{or simply} \qquad \int_a^b c\,dx = c(b-a),$$

as it represents the area of a rectangle with base width $(b-a)$ and height c. (Figure: horizontal line $y = c$ over $[a, b]$.)

Example — Integrals from geometry

Evaluate (a) $\displaystyle\int_0^3 \sqrt{9-x^2}\,dx$; (b) $\displaystyle\int_{-4}^3 |x|\,dx$.

Solution. (a) The integral is the area of a quarter of a circular disk of radius 3 centered at the origin (the graph $y = \sqrt{9-x^2}$ over $[0, 3]$). Hence

$$\int_0^3 \sqrt{9-x^2}\,dx = \frac{1}{4}\cdot\pi(3)^2 = \frac{9\pi}{4}.$$

(b) The graph $y = |x|$ together with the x-axis over $[-4, 3]$ forms two right-angled triangles (through $(-4, 4)$ and $(3, 3)$). Hence

$$\int_{-4}^3 |x|\,dx = \frac{1}{2}(4)(4) + \frac{1}{2}(3)(3) = \frac{25}{2}.$$

We have computed areas of many geometric shapes since elementary school, but what does "area" mean exactly?

Remark — Area as a function

Mathematically, "area" can be understood as a function

$$A : \mathcal{M} \to \mathbb{R},$$

where $\mathcal{M}$ is a collection of certain "nice" ("measurable") subsets of the coordinate plane, containing all familiar shapes and many more. The function A should satisfy intuitive axioms, for example:

- If a rectangle R has length ℓ and width k, then $R \in \mathcal{M}$ and $A(R) = \ell k$.
- If ℓ is a line segment, then $\ell \in \mathcal{M}$ and $A(\ell) = 0$.
- If $X \in \mathcal{M}$ is cut into two disjoint pieces $X_1, X_2 \in \mathcal{M}$, then $A(X) = A(X_1) + A(X_2)$.
- If $X, Y \in \mathcal{M}$ are congruent, then $A(X) = A(Y)$.
- If $X, Y \in \mathcal{M}$ and $X \subseteq Y$, then $A(X) \leq A(Y)$.

These intuitive properties immediately lead to the following theorems. When we later extend integrals to broader classes of functions, we expect such extensions to keep these theorems true.

Theorem (Area property)

Let $a < b < c$ be real numbers and f be a non-negative continuous function.

(i) $\displaystyle\int_a^a f(x)\,dx = 0.$

(ii) $\displaystyle\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx.$

Remark — Reversed limits

The area property suggests that if $a > b$, then we should define

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

This way, the area property remains true for all real numbers a, b, c in whatever order.

Theorem (Integral inequality)

Let $a < b$ be real numbers and let $f, g : [a, b] \to [0, +\infty)$ be non-negative continuous functions.

(i) If $f(x) \le g(x)$ for every $x \in [a, b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

(ii) In particular, if $m \le f(x) \le M$ for every $x \in [a, b]$, then

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

(Figure: region under $y = f(x)$ squeezed between the horizontal lines $y = m$ and $y = M$.)

Example — Bounding a harmonic sum

Let $n \ge 2$ be an integer. Show that

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} \le \int_1^n \frac{1}{x}\,dx \le 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1}.$$

Solution. The function $f : (0, +\infty) \to (0, +\infty)$ defined by $f(x) = \dfrac{1}{x}$ is decreasing on $(0, +\infty)$, so for each $k \in \{1, 2, 3, \ldots, n-1\}$,

$$\frac{1}{k+1} \le \frac{1}{x} \le \frac{1}{k} \qquad \text{for every } x \in [k, k+1].$$

(Figure: the curve $y = \dfrac{1}{x}$ with rectangles of heights $\dfrac{1}{k}$ and $\dfrac{1}{k+1}$ over $[k, k+1]$.) Since f is non-negative and continuous, the integral inequality gives

$$\frac{1}{k+1} \le \int_k^{k+1} \frac{1}{x}\,dx \le \frac{1}{k}.$$

On the other hand, by the area property and induction,

$$\int_1^n \frac{1}{x}\,dx = \int_1^2 \frac{1}{x}\,dx + \int_2^3 \frac{1}{x}\,dx + \int_3^4 \frac{1}{x}\,dx + \cdots + \int_{n-1}^n \frac{1}{x}\,dx.$$

Summing the inequality from $k = 1$ to $k = n - 1$,

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \le \int_1^n \frac{1}{x}\,dx \le 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1},$$

as desired. ■

Example — Estimating $\displaystyle\int_0^{\frac{\pi}{2}} \sin x\,dx$

Let $f, g, h : [0, \dfrac{\pi}{2}] \to \mathbb{R}$ be the continuous functions

$$f(x) = \frac{2}{\pi}x, \qquad g(x) = \sin x, \qquad h(x) = 1.$$

The graph of f is the line segment joining $(0, 0)$ and $(\dfrac{\pi}{2}, 1)$; the graph of h is a horizontal line segment.

Solution. Obviously $g(x) = \sin x \leq 1 = h(x)$ for all $x \in [0, \frac{\pi}{2}]$. Also, since

$$g''(x) = -\sin x < 0 \qquad \text{for every } x \in (0, \frac{\pi}{2}),$$

the graph of g is concave downward on $(0, \frac{\pi}{2})$, so the line segment joining $(0, 0)$ and $(\frac{\pi}{2}, 1)$ (the graph of f) lies below the graph of g. Combining,

$$f(x) \leq g(x) \leq h(x) \qquad \text{for every } x \in [0, \frac{\pi}{2}].$$

Thus by the integral inequality,

$$\int_0^{\frac{\pi}{2}} f(x)\,dx \leq \int_0^{\frac{\pi}{2}} g(x)\,dx \leq \int_0^{\frac{\pi}{2}} h(x)\,dx.$$

The region under f on $[0, \frac{\pi}{2}]$ is a right-angled triangle of area

$$\int_0^{\frac{\pi}{2}} f(x)\,dx = \frac{1}{2}\left(\frac{\pi}{2}\right)(1) = \frac{\pi}{4},$$

and the region under h is a rectangle of area

$$\int_0^{\frac{\pi}{2}} h(x)\,dx = \left(\frac{\pi}{2}\right)(1) = \frac{\pi}{2}.$$

Therefore

$$\frac{\pi}{4} \leq \int_0^{\frac{\pi}{2}} \sin x\,dx \leq \frac{\pi}{2}.$$

Think: the upper bound $\frac{\pi}{2}$ given by $h(x) = 1$ seems too "rough". Can we choose another (linear) function h to get a better upper bound, whose integral is still computable by simple geometry?

Example — Integral of a function and its inverse

Let $0 < a < b$ and let $f : [a, b] \to [0, +\infty)$ be a continuous strictly increasing function, so that f has a continuous inverse $f^{-1} : [f(a), f(b)] \to [a, b]$. Show that

$$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = bf(b) - af(a).$$

Solution. The graph of f^{-1} is the mirror reflection of the graph of f across the line $y = x$. (Figure: on the left, region A under $y = f(x)$ over $[a, b]$ together with region B beside it fills the rectangle $[0, b] \times [0, f(b)]$ minus the rectangle $[0, a] \times [0, f(a)]$; on the right, the same regions seen for $y = f^{-1}(x)$ over $[f(a), f(b)]$.) Therefore the integrals

$$A = \int_a^b f(x)\,dx \qquad \text{and} \qquad B = \int_{f(a)}^{f(b)} f^{-1}(x)\,dx$$

represent the areas of the two indicated regions. By additivity of area, $bf(b) = A + B + af(a)$, so

$$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = bf(b) - af(a),$$

as desired. ■

5.2 Riemann Sums and Integrals

Given any non-negative continuous function $f : [a, b] \to [0, +\infty)$, our goal now is to actually compute $\displaystyle\int_a^b f(x)\,dx$. One way is to estimate the area by approximating the region with thin rectangles and summing their areas. We begin by choosing the base widths of these rectangles.

Definition. Let $a < b$ be real numbers, so that $[a, b]$ is a bounded closed interval.

- A partition of $[a, b]$ is a (finite) set of numbers $P = \{x_0, x_1, \ldots, x_n\}$ such that
$$a = x_0 < x_1 < \cdots < x_n = b.$$
It divides $[a, b]$ into n subintervals $[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n]$.
- The mesh size of a partition P, denoted by $\|P\|$, is the width of the largest subinterval:
$$\|P\| = \max\{x_k - x_{k-1} : k \in \{1, 2, \ldots, n\}\}.$$
- A partition is regular if all subintervals have equal width. A regular partition of $[a, b]$ into n subintervals is given by
$$P = \left\{a,\ a + \tfrac{1}{n}(b-a),\ a + \tfrac{2}{n}(b-a),\ \ldots,\ a + \tfrac{n-1}{n}(b-a),\ a + \tfrac{n}{n}(b-a)\right\},$$
or equivalently, writing $\Delta x := \dfrac{b-a}{n}$,
$$P = \{a,\ a + \Delta x,\ a + 2\Delta x,\ \ldots,\ a + (n-1)\Delta x,\ a + n\Delta x\}, \qquad a + n\Delta x = b.$$

After choosing the base widths, we focus on the heights. If the rectangles are just tall enough to contain the graph, we get an overestimate; if they lie just underneath the graph, we get an underestimate. (Figures: upper Darboux sum — rectangles above the graph; lower Darboux sum — rectangles below the graph.)

Definition. Let $a < b$, let $f : [a, b] \to \mathbb{R}$ be a continuous function and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$.

- The upper Darboux sum of f with respect to P is
$$U(P; f) := \sum_{k=1}^{n} \underbrace{\max\{f(x) : x \in [x_{k-1}, x_k]\}}_{\text{height}} \underbrace{(x_k - x_{k-1})}_{\text{width}};$$
- the lower Darboux sum of f with respect to P is
$$L(P; f) := \sum_{k=1}^{n} \underbrace{\min\{f(x) : x \in [x_{k-1}, x_k]\}}_{\text{height}} \underbrace{(x_k - x_{k-1})}_{\text{width}}.$$

Note that if $f : [a, b] \to [0, +\infty)$ is non-negative and continuous, then for every partition P of $[a, b]$,
$$L(P; f) \le \int_a^b f(x)\,dx \le U(P; f).$$

Example — Darboux sums of $f(x) = x$ on $[0, 1]$

Let $f : [0, 1] \to \mathbb{R}$ be $f(x) = x$. For each positive integer n, let P_n be the regular

partition of $[0,1]$ into n subintervals. Then

$$\begin{aligned} U(P_n;f) &= \frac{1}{n}\cdot\frac{1}{n} + \frac{2}{n}\cdot\frac{1}{n} + \frac{3}{n}\cdot\frac{1}{n} + \cdots + \frac{n}{n}\cdot\frac{1}{n} \\ &= \frac{1+2+3+\cdots+n}{n^2} = \frac{(n+1)n}{2n^2} = \frac{1}{2}\left(1+\frac{1}{n}\right), \end{aligned}$$

and

$$\begin{aligned} L(P_n;f) &= 0\cdot\frac{1}{n} + \frac{1}{n}\cdot\frac{1}{n} + \frac{2}{n}\cdot\frac{1}{n} + \cdots + \frac{n-1}{n}\cdot\frac{1}{n} \\ &= \frac{0+1+2+\cdots+(n-1)}{n^2} = \frac{n(n-1)}{2n^2} = \frac{1}{2}\left(1-\frac{1}{n}\right). \end{aligned}$$

As n increases, the width of each rectangle decreases, the "gaps" between the actual region and the rectangles are filled up, and the two sums give better and better approximations. (Figures: $U(P_6;f)$, $L(P_6;f)$, $U(P_{12};f)$, $L(P_{12};f)$.)
In fact, since

$$L(P_n;f) \le \int_0^1 x\,dx \le U(P_n;f) \qquad \text{for every } n\in\mathbb{N}$$

and $\lim_{n\to+\infty} L(P_n;f) = \lim_{n\to+\infty} U(P_n;f) = \frac{1}{2}$, by the Squeeze Theorem (for sequences),

$$\int_0^1 x\,dx = \frac{1}{2},$$

agreeing with the area of a right triangle.

Example — Darboux sums of $f(x)=x^2$ on $[0,1]$

Let $f:[0,1]\to\mathbb{R}$ be $f(x)=x^2$, and let P_n be the regular partition of $[0,1]$ into n subintervals. Then

$$\begin{aligned} U(P_n;f) &= \left(\frac{1}{n}\right)^2\frac{1}{n} + \left(\frac{2}{n}\right)^2\frac{1}{n} + \cdots + \left(\frac{n}{n}\right)^2\frac{1}{n} \\ &= \frac{1^2+2^2+3^2+\cdots+n^2}{n^3} = \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right), \end{aligned}$$

and

$$\begin{aligned} L(P_n;f) &= \frac{1}{n}(0)^2 + \left(\frac{1}{n}\right)^2\frac{1}{n} + \cdots + \left(\frac{n-1}{n}\right)^2\frac{1}{n} \\ &= \frac{0^2+1^2+2^2+\cdots+(n-1)^2}{n^3} = \frac{(n-1)n(2n-1)}{6n^3} = \frac{1}{6}\left(1-\frac{1}{n}\right)\left(2-\frac{1}{n}\right). \end{aligned}$$

Since $\lim_{n\to+\infty} L(P_n;f) = \lim_{n\to+\infty} U(P_n;f) = \frac{1}{3}$, the same reasoning gives

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$

Remark — On upper and lower Darboux sums

- We need not assume f is non-negative; the upper and lower Darboux sums are still well-defined even if f takes negative values.

- Continuity of f on $[a,b]$ is assumed just to ensure that the maxima and minima all exist (Extreme Value Theorem). Without continuity, these extrema may not exist, but the sums can still be defined by replacing "maximum" with "supremum" and "minimum" with "infimum"; these ideas are properly introduced in MATH2033.

As suggested by the examples above, one expects the upper and lower Darboux sums of a nice function f to approach the actual area under the graph of f as the partition gets finer. This motivates the following definition (where f need not be continuous or non-negative).

Definition. Let $a < b$ be real numbers and let $f : [a,b] \to \mathbb{R}$ be a function. If the limits

$$\lim_{\|P\|\to 0} U(P;f) = \lim_{\|P\|\to 0} L(P;f) = I$$

both exist and equal the same finite real number, then we say that f is Darboux integrable on $[a,b]$. The (Darboux) integral of f over $[a,b]$ is then defined to be this common limit,

$$\int_a^b f(x)\,dx = I.$$

Remark — Meaning of the limit $\|P\| \to 0$

The limits here differ from those of chapter 2. Recall that $\lim_{x\to 0} f(x) = L$ means: for each $\varepsilon > 0$ there exists $\delta > 0$ such that every real number x with $0 < |x| < \delta$ satisfies $|f(x) - L| < \varepsilon$. Here,

$$\lim_{\|P\|\to 0} U(P;f) = I \iff \begin{array}{l}\text{for each } \varepsilon > 0, \text{ there exists } \delta > 0 \text{ such that every}\\ \text{partition } P \text{ of } [a,b] \text{ with } \|P\| < \delta \text{ satisfies}\\ |U(P;f) - I| < \varepsilon.\end{array}$$

Example — Integrability of x and x^2 on $[0,1]$

From the previous examples, the functions $f(x) = x$ and $g(x) = x^2$ are both Darboux integrable on $[0,1]$. In fact, every continuous function on a bounded closed interval $[a,b]$ is Darboux integrable on $[a,b]$. Moreover, if f is continuous and non-negative on $[a,b]$, the meaning of $\int_a^b f(x)\,dx$ here agrees with the area definition (so this definition generalizes the earlier one), as a consequence of the Squeeze Theorem.

Corollary (Net area)

Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be a continuous function. Then $\int_a^b f(x)\,dx$ equals the "net area" under the graph of f on $[a,b]$:

$$\int_a^b f(x)\,dx = (\text{area above the } x\text{-axis and below the graph of } f) - (\text{area below the } x\text{-axis and above the graph of } f).$$

(Figure: a curve crossing the x-axis, with regions A_1, A_2, A_3, A_4 alternately above and below the axis, and $\int_a^b f(x)\,dx = A_1 - A_2 + A_3 - A_4$.)

Proof. It suffices to show that if f is also non-negative, then

$$\int_a^b -f(x)\,dx = -\int_a^b f(x)\,dx.$$

For any partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, if f is non-negative then

$$\max\{-f(x) : x \in [x_{k-1}, x_k]\} = -\min\{f(x) : x \in [x_{k-1}, x_k]\}$$

for every $k \in \{1, 2, \ldots, n\}$. Thus $U(P; -f) = -L(P; f)$ for every partition P, and the result follows upon taking limits. ■

The following two corollaries follow directly from symmetry. Recall: f is odd if $f(-x) = -f(x)$ for every x; f is even if $f(-x) = f(x)$ for every x; f has period T if $f(x + T) = f(x)$ for every x.

Corollary (Odd and even functions)

Let $a \geq 0$ and let f be Darboux integrable on $[-a, a]$.

(i) If f is odd, then $\displaystyle\int_{-a}^{a} f(x)\,dx = 0$.

(ii) If f is even, then $\displaystyle\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

(Figures: for an even function the two halves contribute $A + A$; for an odd function they contribute $A - A$.)

Corollary (Periodic functions)

Let $f : \mathbb{R} \to \mathbb{R}$ be periodic with period $T > 0$, and let $a, b \in \mathbb{R}$.

(i) Suppose $a < b$. Then f is Darboux integrable on $[a, b]$ if and only if it is Darboux integrable on $[a + T, b + T]$; and

$$\int_a^b f(x)\,dx = \int_{a+T}^{b+T} f(x)\,dx.$$

(ii) If f is Darboux integrable on $[a, a + T]$, then f is Darboux integrable on $[b, b + T]$ and

$$\int_a^{a+T} f(x)\,dx = \int_b^{b+T} f(x)\,dx.$$

(Figures: shifting the interval of integration by one period does not change the integral; the integral over any full period is the same.)

Example — Using symmetry and periodicity

- $\displaystyle\int_{-\pi}^{\pi} \sin x\,dx = 0$ since the integrand is odd;
- $\displaystyle\int_{-1}^{1} x^2\,dx = 2\int_0^1 x^2\,dx = \frac{2}{3}$ since the integrand is even (using $\displaystyle\int_0^1 x^2\,dx = \frac{1}{3}$);
- $\displaystyle\int_{\pi}^{3\pi} \sin x\,dx = \int_{-\pi}^{\pi} \sin x\,dx = 0$ since the integrand has period 2π.

Instead of always taking the maximum or minimum function values as heights, we may choose the function value at any favorite point of each subinterval. This gives the notion of a Riemann sum.

Definition. Let $a < b$ and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. If we equip

P with a choice of sample points

$$\omega_1 \in [x_0, x_1],\ \omega_2 \in [x_1, x_2],\ \ldots,\ \omega_n \in [x_{n-1}, x_n],$$

then P becomes a tagged partition of $[a, b]$.

Definition. Let $a < b$, let $f : [a, b] \to \mathbb{R}$ be a function and let $P = \{x_0, x_1, \ldots, x_n\}$ be a tagged partition of $[a, b]$ with sample points $\omega_k \in [x_{k-1}, x_k]$. The Riemann sum of f (with respect to the tagged partition P) is

$$S(P; f) := \sum_{k=1}^{n} \underbrace{f(\omega_k)}_{\text{height}} \underbrace{(x_k - x_{k-1})}_{\text{width}}.$$

Remark — Left, mid-point and right Riemann sums

When the sample points are chosen systematically:

- the left Riemann sum: choose $\omega_k = x_{k-1}$, giving $\displaystyle\sum_{k=1}^{n} f(x_{k-1})(x_k - x_{k-1})$;
- the mid-point Riemann sum: choose $\omega_k = \dfrac{x_{k-1} + x_k}{2}$, giving $\displaystyle\sum_{k=1}^{n} f\left(\frac{x_{k-1} + x_k}{2}\right)(x_k - x_{k-1})$;
- the right Riemann sum: choose $\omega_k = x_k$, giving $\displaystyle\sum_{k=1}^{n} f(x_k)(x_k - x_{k-1})$.

(Figures: three diagrams of $y = f(x)$ with rectangles whose heights are taken at left endpoints, mid-points, and right endpoints respectively.)

Example — Estimating an area by Riemann sums

Estimate the area of the region bounded between the curve $y = x^3$ and the x-axis from $x = 1$ to $x = 4$ using respectively (a) the mid-point Riemann sum with three subintervals of equal width, and (b) the right Riemann sum with six subintervals of equal width.

Solution. Let $f(x) = x^3$. Since f is non-negative on $[1, 4]$, the area can be approximated by Riemann sums of f on $[1, 4]$.

(a) The width of each subinterval is $\dfrac{4-1}{3} = 1$, so we use the regular partition $\{1, 2, 3, 4\}$. The mid-point Riemann sum is

$$(1.5)^3(1) + (2.5)^3(1) + (3.5)^3(1) = 61.875,$$

so the area is approximately 61.875 square units.

(b) The width of each subinterval is $\dfrac{4-1}{6} = 0.5$, so we use the regular partition $\{1, 1.5, 2, 2.5, 3, 3.5, 4\}$. The right Riemann sum is

$$(1.5)^3(0.5) + (2)^3(0.5) + (2.5)^3(0.5) + (3)^3(0.5) + (3.5)^3(0.5) + (4)^3(0.5) = 80.4375,$$

so the area is approximately 80.4375 square units.

Definition. Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a function. If the limit

$$\lim_{\|P\| \to 0} S(P; f) = I$$

exists as the same finite real number (for every choice of sample points within each tagged partition P of $[a, b]$), then we say that f is Riemann integrable on $[a, b]$. The (Riemann)

integral of f over $[a,b]$ is then this common limit,

$$\int_a^b f(x)\,dx = I.$$

Remark — Origin of the integral notation

Spelling out the definition,

$$\int_a^b f(x)\,dx := \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(\omega_k)(x_k - x_{k-1}).$$

As one passes to the limit: the summation symbol $\sum$ (Greek letter S) is replaced by $\int$ (an "elongated S"); $f(\omega_k)$ is replaced by $f(x)$; and $x_k - x_{k-1}$ (or Δx_k) is replaced by the symbol dx.

According to the following theorem, "Darboux integrable" and "Riemann integrable" are equivalent notions, and the two integrals coincide. From now on we simply say "integrable" and "integral".

Theorem (Darboux $\iff$ Riemann)

Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be a function. Then f is Darboux integrable on $[a,b]$ if and only if f is Riemann integrable on $[a,b]$.

Proof. We need not assume f continuous, but then the Darboux sums involve supremum and infimum, whose properties are out of scope. We only prove the easy case when f is further assumed continuous on $[a,b]$.

($\Rightarrow$) For every tagged partition P of $[a,b]$ with sample points $\{\omega_1, \omega_2, \ldots, \omega_n\}$, by definition of minimum and maximum,

$$\min\{f(x) : x \in [x_{k-1}, x_k]\} \le f(\omega_k) \le \max\{f(x) : x \in [x_{k-1}, x_k]\}$$

for every $k \in \{1, 2, \ldots, n\}$. Multiplying by $(x_k - x_{k-1})$ and summing from $k = 1$ to $k = n$,

$$L(P; f) \le S(P; f) \le U(P; f)$$

for every tagged partition P. If f is Darboux integrable, both $L(P; f)$ and $U(P; f)$ converge to a common limit as $\|P\| \to 0$; so by the Squeeze Theorem, $S(P; f)$ also converges to the same limit. Hence f is Riemann integrable and the two integrals are equal.

($\Leftarrow$) Since f is continuous on $[a,b]$, for every partition P and each k, f attains its maximum on $[x_{k-1}, x_k]$ by the Extreme Value Theorem, i.e. there exists $\omega_k \in [x_{k-1}, x_k]$ with

$$f(\omega_k) = \max\{f(x) : x \in [x_{k-1}, x_k]\}.$$

Thus the upper Darboux sum $U(P; f)$ is just a special Riemann sum with a specific choice of sample points; the same holds for $L(P; f)$. If f is Riemann integrable, every Riemann sum $S(P; f)$ converges to a common limit as $\|P\| \to 0$; in particular both $U(P; f)$ and $L(P; f)$ converge to the same limit, i.e. f is Darboux integrable and the two integrals are equal. ∎

It is usually very difficult to prove Riemann integrability directly from the definition. Luckily the following theorem gives an easy-to-check sufficient condition (its proof is out of scope). Recall: f is bounded on $[a,b]$ if there exist $m, M \in \mathbb{R}$ such that $m \le f(x) \le M$ for every $x \in [a,b]$.

Theorem (Sufficient condition for integrability)

Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a function. If
- f is bounded on $[a, b]$, and
- f is piecewise continuous on $[a, b]$ (i.e. continuous on $[a, b]$ except perhaps at finitely many numbers),

then f is integrable on $[a, b]$. In particular, every continuous function on $[a, b]$ is integrable.

On the other hand, the following is a useful necessary condition.

Theorem (Integrable ⇒ bounded)

Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a function. If f is integrable on $[a, b]$, then f is bounded on $[a, b]$.

Proof. Suppose f is unbounded on $[a, b]$. Then for each partition P of $[a, b]$, no matter how small $\|P\|$ is, there must exist a subinterval $[x_{k-1}, x_k]$ on which f is also unbounded. Thus we can always choose a sample point $\omega_k \in [x_{k-1}, x_k]$ so that $f(\omega_k)(x_k - x_{k-1})$ is as large as we want, and the limit of $S(P; f)$ cannot be a finite real number. ■

Example — An unbounded function is not integrable

The function $f : [0, 1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \dfrac{1}{\sqrt{x}} & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0 \end{cases}$$

is unbounded on $[0, 1]$, so it is not integrable on $[0, 1]$. (Figure: the graph $y = \dfrac{1}{\sqrt{x}}$ blowing up near 0.) However, we will see in chapter 6 that f still has a finite area under its graph on $[0, 1]$, in the sense of an improper integral.

Remark — Computing the integral via convenient Riemann sums

If we already know that f is integrable on $[a, b]$, then in computing the limit we may use Riemann sums that are easy to handle. For example, using right Riemann sums with respect to a regular partition of $[a, b]$ into n subintervals, $\|P\| = \dfrac{b-a}{n} \to 0$ as $n \to +\infty$, so

$$\int_a^b f(x)\,dx = \lim_{n\to+\infty} \sum_{k=1}^{n} f\left(a + k \cdot \frac{b-a}{n}\right) \cdot \frac{b-a}{n};$$

on $[a, b] = [0, 1]$ in particular, this reduces to

$$\int_0^1 f(x)\,dx = \lim_{n\to+\infty} \sum_{k=1}^{n} f\left(\frac{k}{n}\right) \cdot \frac{1}{n}.$$

Example — Writing limits of sequences as integrals

Express each limit as an integral:

(a) $\displaystyle\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{1}{n} \ln\left(1 + \frac{k}{n}\right)$

(b) $\lim_{n\to+\infty} \sum_{k=1}^{n} \dfrac{n^2+k^2}{(n+k)^3}$

(c) $\lim_{n\to+\infty} \left(\dfrac{1}{n} + \dfrac{1}{n+1} + \cdots + \dfrac{1}{3n-1} \right)$

(d) $\lim_{n\to+\infty} \sum_{k=1}^{n} \dfrac{1}{n} \cos^2 \dfrac{(2k-1)\pi}{2n}$

Solution. (a) Consider the regular partition $\{0, \dfrac{1}{n}, \dfrac{2}{n}, \ldots, \dfrac{n}{n}\}$ of $[0,1]$ with right endpoints as sample points: $\omega_k = \dfrac{k}{n} \in [\dfrac{k-1}{n}, \dfrac{k}{n}]$. Each subinterval has width $\Delta x = \dfrac{1}{n}$, so

$$\sum_{k=1}^{n} \frac{1}{n} \ln\left(1 + \frac{k}{n}\right) = \sum_{k=1}^{n} f(\omega_k)\, \Delta x$$

is precisely the right Riemann sum of $f(x) = \ln(1+x)$ on $[0,1]$. Since f is continuous on $[0,1]$, it is integrable, so

$$\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{1}{n} \ln\left(1 + \frac{k}{n}\right) = \int_0^1 \ln(1+x)\, dx.$$

(b) The given sum has n terms and rewrites as

$$\sum_{k=1}^{n} \frac{n^2+k^2}{(n+k)^3} = \sum_{k=1}^{n} \left[\frac{1 + (\frac{k}{n})^2}{(1+\frac{k}{n})^3} \cdot \frac{1}{n} \right].$$

(As k runs from 1 to n, $\dfrac{k}{n}$ runs from $\dfrac{1}{n} \to 0$ up to $\dfrac{n}{n} = 1$.) As in (a), this is the right Riemann sum of $f(x) = \dfrac{1+x^2}{(1+x)^3}$ with respect to the regular partition of $[0,1]$ into n subintervals. Since f is continuous on $[0,1]$,

$$\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{n^2+k^2}{(n+k)^3} = \lim_{n\to+\infty} \sum_{k=1}^{n} \left[\frac{1 + (\frac{k}{n})^2}{(1+\frac{k}{n})^3} \cdot \frac{1}{n} \right] = \int_0^1 \frac{1+x^2}{(1+x)^3}\, dx.$$

(c) With a similar analysis, write

$$\frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{3n-1} = \left(\frac{1}{\frac{n}{n}} + \frac{1}{\frac{n+1}{n}} + \cdots + \frac{1}{\frac{3n-1}{n}} \right) \cdot \frac{1}{n}.$$

It is the Riemann sum of $f(x) = \dfrac{1}{x}$ with respect to the partition $\{\dfrac{n}{n}, \dfrac{n+1}{n}, \ldots, \dfrac{3n}{n}\}$ of $[1,3]$ into $2n$ subintervals of width $\dfrac{1}{n}$, with left endpoints as sample points:

$$\omega_1 = \frac{n}{n} \in \left[\tfrac{n}{n}, \tfrac{n+1}{n}\right],\ \omega_2 = \tfrac{n+1}{n} \in \left[\tfrac{n+1}{n}, \tfrac{n+2}{n}\right],\ \ldots,\ \omega_{2n} = \tfrac{3n-1}{n} \in \left[\tfrac{3n-1}{n}, \tfrac{3n}{n}\right].$$

(As k varies, the sample points run from $\dfrac{n}{n} = 1$ to $\dfrac{3n-1}{n} \to 3$.) Since f is continuous on $[1,3]$,

$$\lim_{n\to+\infty} \left(\frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{3n-1} \right) = \int_1^3 \frac{1}{x}\, dx.$$

(d) Write the given sum as

$$\sum_{k=1}^{n} \frac{1}{n} \cos^2 \frac{(2k-1)\pi}{2n} = \sum_{k=1}^{n} \left[\left(\frac{1}{\pi} \cos^2 \frac{(2k-1)\pi}{2n} \right) \cdot \frac{\pi}{n} \right].$$

It is the Riemann sum of $f(x) = \dfrac{1}{\pi} \cos^2 x$ with respect to the partition $\{0, \dfrac{\pi}{n}, \dfrac{2\pi}{n}, \ldots, \dfrac{n\pi}{n}\}$ of $[0, \pi]$ into n subintervals of width $\dfrac{\pi}{n}$, with mid-points as sample points:

$$\omega_1 = \frac{\pi}{2n} \in \left[0, \tfrac{\pi}{n}\right], \ \omega_2 = \frac{3\pi}{2n} \in \left[\tfrac{\pi}{n}, \tfrac{2\pi}{n}\right], \ \ldots, \ \omega_n = \frac{(2n-1)\pi}{2n} \in \left[\tfrac{(n-1)\pi}{n}, \tfrac{n\pi}{n}\right].$$

Since f is continuous on $[0, \pi]$,

$$\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{1}{n} \cos^2 \frac{(2k-1)\pi}{2n} = \frac{1}{\pi} \int_0^{\pi} \cos^2 x \, dx.$$

Alternative solution: the given sum

$$\sum_{k=1}^{n} \frac{1}{n} \cos^2 \frac{(2k-1)\pi}{2n} = \sum_{k=1}^{n} \left[\cos^2 \left(\pi \cdot \frac{2k-1}{2n} \right) \cdot \frac{1}{n} \right]$$

can also be regarded as the Riemann sum of $f(x) = \cos^2(\pi x)$ with respect to the regular partition $\{0, \dfrac{1}{n}, \dfrac{2}{n}, \ldots, \dfrac{n}{n}\}$ of $[0, 1]$ with mid-points as sample points:

$$\omega_1 = \frac{1}{2n} \in \left[0, \tfrac{1}{n}\right], \ \omega_2 = \frac{3}{2n} \in \left[\tfrac{1}{n}, \tfrac{2}{n}\right], \ \ldots, \ \omega_n = \frac{2n-1}{2n} \in \left[\tfrac{n-1}{n}, \tfrac{n}{n}\right].$$

Since f is continuous on $[0, 1]$,

$$\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{1}{n} \cos^2 \frac{(2k-1)\pi}{2n} = \int_0^{1} \cos^2(\pi x) \, dx.$$

Our next goal is to compute integrals of some elementary functions from the definition.

Example — $\displaystyle\int_0^1 x^2 \, dx$ from the definition

Evaluate $\displaystyle\int_0^1 x^2 \, dx$ from the definition of the Riemann integral.

Solution. Let $f(x) = x^2$. Since f is a polynomial, it is continuous and thus integrable on $[0, 1]$. For each $n \in \mathbb{N}$, consider the right Riemann sum with respect to the regular partition $P = \{0, \dfrac{1}{n}, \dfrac{2}{n}, \ldots, 1\}$ of $[0, 1]$. Then $\|P\| = \dfrac{1}{n} \to 0$ as $n \to +\infty$, so

$$\begin{aligned} \int_0^1 x^2 \, dx &= \lim_{n\to+\infty} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \left(\frac{1}{n}\right) = \lim_{n\to+\infty} \left(\frac{1}{n^3} \sum_{k=1}^{n} k^2 \right) \\ &= \lim_{n\to+\infty} \left[\frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \right] = \lim_{n\to+\infty} \frac{1}{6} \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right) = \frac{1}{3}. \end{aligned}$$

Example — $\int_0^a e^x\,dx$ from the definition

Let $a > 0$. Evaluate $\displaystyle\int_0^a e^x\,dx$ from the definition of the Riemann integral.

Solution. Let $f(x) = e^x$; then f is continuous and thus integrable on $[0, a]$. For each positive integer n, consider the left Riemann sum with respect to the regular partition $P = \{0, \frac{a}{n}, \frac{2a}{n}, \ldots, \frac{na}{n}\}$ of $[0, a]$. Then $\|P\| = \frac{a}{n} \to 0$ as $n \to +\infty$, so

$$\int_0^a e^x\,dx = \lim_{n\to+\infty} \sum_{k=0}^{n-1} \left(e^{\frac{ka}{n}}\right)\left(\frac{a}{n}\right) = \lim_{n\to+\infty} \frac{a}{n} \sum_{k=0}^{n-1} \left(e^{\frac{a}{n}}\right)^k.$$

Using the geometric sum $\displaystyle\sum_{k=0}^{n-1} r^k = \frac{r^n - 1}{r - 1}$ $(r \neq 1)$ with $r = e^{\frac{a}{n}}$,

$$\int_0^a e^x\,dx = \lim_{n\to+\infty} \frac{a}{n} \cdot \frac{e^a - 1}{e^{\frac{a}{n}} - 1} = \frac{e^a - 1}{\displaystyle\lim_{t\to 0^+} \frac{e^t - 1}{t}} = e^a - 1,$$

where we substituted $t = \frac{a}{n}$ when computing the limit.

Example — $\int_1^a \frac{1}{x}\,dx$ from the definition

Let $a > 1$. Evaluate from the definition of the Riemann integral $\displaystyle\int_1^a \frac{1}{x}\,dx$.

Solution. Let $f(x) = \frac{1}{x}$, which is continuous and thus integrable on $[1, a]$. For each $n \in \mathbb{N}$, consider the left Riemann sum with respect to the (non-regular!) partition $P = \{1, a^{\frac{1}{n}}, a^{\frac{2}{n}}, a^{\frac{3}{n}}, \ldots, a^{\frac{n}{n}}\}$ of $[1, a]$ into n subintervals. Then $\|P\| = a^{\frac{n}{n}} - a^{\frac{n-1}{n}} \to 0$ as $n \to +\infty$, so

$$\begin{aligned}\int_1^a \frac{1}{x}\,dx &= \lim_{n\to+\infty} \sum_{k=1}^{n} \left(\frac{1}{a^{\frac{k-1}{n}}}\right)\left(a^{\frac{k}{n}} - a^{\frac{k-1}{n}}\right) \\ &= \lim_{n\to+\infty} \sum_{k=1}^{n} \left(a^{\frac{1}{n}} - 1\right) = \lim_{n\to+\infty} n\left(a^{\frac{1}{n}} - 1\right) = \lim_{t\to 0^+} \frac{a^t - 1}{t} = \ln a.\end{aligned}$$

Example — The Dirichlet function is not integrable

Let $f : [0, 1] \to \mathbb{R}$ be the Dirichlet function

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}.$$

Show that f is not integrable on $[0, 1]$ (although it is obviously bounded on $[0, 1]$).

Solution. For each $n \in \mathbb{N}$, consider the regular partition $P_n = \{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\}$ of $[0, 1]$. Since each subinterval $[\frac{k-1}{n}, \frac{k}{n}]$ contains a rational number, we can choose all sample points ω_k rational and get a Riemann sum (which is just the upper Darboux

sum)

$$U(P_n; f) = \sum_{k=1}^{n} f(\omega_k) \left(\frac{1}{n}\right) = \sum_{k=1}^{n} (1) \left(\frac{1}{n}\right) = 1.$$

Since each subinterval also contains an irrational number, we can also choose all sample points irrational and obtain another Riemann sum (just the lower Darboux sum)

$$L(P_n; f) = \sum_{k=1}^{n} f(\omega_k) \left(\frac{1}{n}\right) = \sum_{k=1}^{n} (0) \left(\frac{1}{n}\right) = 0.$$

Now $\|P_n\| = \dfrac{1}{n} \to 0$ as $n \to +\infty$, but

$$\lim_{n\to+\infty} U(P_n; f) = 1 \qquad \text{and} \qquad \lim_{n\to+\infty} L(P_n; f) = 0.$$

Thus there are two sequences of Riemann sums of f with different limits as $\|P_n\| \to 0$, so f is not integrable on $[0, 1]$. ■

5.3 Properties of the Riemann Integral

Remark — Conventions on limits of integration

Similar to the area property, for Riemann integrals we define

$$\int_a^a f(x)\,dx = 0 \qquad \text{and} \qquad \int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

Theorem (Area property)

Let a, b, c be real numbers and let f be integrable on the closed intervals relevant to the following integrals. Then

(i) $\displaystyle\int_a^b c\,dx = c(b-a),$

(ii) $\displaystyle\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$

Proof. For (i), we may assume $a < b$ without loss of generality, since the case $a \geq b$ then follows from the convention above. For every partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ and every choice of sample points $\omega_k \in [x_{k-1}, x_k]$, the Riemann sum of the constant function c is always

$$S(P; c) = \sum_{k=1}^{n} c\,(x_k - x_{k-1}) = c(x_n - x_0) = c(b-a);$$

so

$$\int_a^b c\,dx = \lim_{\|P\|\to 0} S(P; c) = \lim_{\|P\|\to 0} c(b-a) = c(b-a).$$

The proof of (ii) is too difficult and is omitted here. ■

Theorem (Linearity)

Let a, b, c be real numbers and let f, g be integrable on $[a, b]$. Then $f + g$, $f - g$ and cf are also integrable on $[a, b]$, with

(i) $\displaystyle\int_a^b (f \pm g)(x)\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx,$

(ii) $\displaystyle\int_a^b (cf)(x)\,dx = c\int_a^b f(x)\,dx.$

Proof. Using the definition of the Riemann integral, these equalities follow immediately from the corresponding linearity properties of summation. ■

Example — Combining known integrals

Let f and g be functions such that

$$\int_{-1}^{3} f(x)\,dx = -2, \qquad \int_{-1}^{6} f(x)\,dx = 4, \qquad \int_{3}^{6} g(x)\,dx = -5.$$

Compute (a) $\displaystyle\int_4^4 f(x)g(x)\,dx$; (b) $\displaystyle\int_6^3 [f(x)+g(x)]\,dx$; (c) $\displaystyle\int_3^6 -4f(y)\,dy$.

Solution. (a) $\displaystyle\int_4^4 f(x)g(x)\,dx = 0.$

(b)

$$\begin{aligned}\int_6^3 [f(x)+g(x)]\,dx &= \int_6^3 f(x)\,dx + \int_6^3 g(x)\,dx \\ &= \left[\int_6^{-1} f(x)\,dx + \int_{-1}^{3} f(x)\,dx\right] + \int_6^3 g(x)\,dx \\ &= -\int_{-1}^{6} f(x)\,dx + \int_{-1}^{3} f(x)\,dx - \int_3^6 g(x)\,dx = -4 + (-2) - (-5) = -1.\end{aligned}$$

(c)

$$\begin{aligned}\int_3^6 -4f(y)\,dy = \int_3^6 -4f(x)\,dx = -4\int_3^6 f(x)\,dx &= -4\left[\int_3^{-1} f(x)\,dx + \int_{-1}^{6} f(x)\,dx\right] \\ &= -4\left[-\int_{-1}^{3} f(x)\,dx + \int_{-1}^{6} f(x)\,dx\right] = -4[-(-2)+4] = -24.\end{aligned}$$

Theorem (Integral inequality)

Let $a < b$ and let f, g be integrable on $[a,b]$. If $f(x) \le g(x)$ for every $x \in [a,b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Proof. First note that if h is a non-negative integrable function on $[a,b]$, then every term of every Riemann sum of h is non-negative, so $\displaystyle\int_a^b h(x)\,dx \ge 0$. Applying this to the non-negative integrable function $h = g - f$, followed by linearity, the desired result follows. ■

Remark — Strict inequality

Unlike the situation for limits, the inequality can surprisingly be strengthened: if $a < b$ and $f(x) > 0$ for every $x \in [a, b]$, then actually $\displaystyle\int_a^b f(x)\,dx > 0$. The proof is not easy when f is merely integrable, but the easier case of continuous f is a nice exercise.

Theorem (Cauchy–Schwarz inequality)

Let $a < b$ and let f, g be integrable on $[a, b]$. Then f^2, g^2 and fg are all integrable on $[a, b]$ and

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \leq \left(\int_a^b f(x)^2\,dx\right)\left(\int_a^b g(x)^2\,dx\right).$$

Proof. We skip the difficult proof of integrability of the three functions (done in MATH2033). To prove the inequality, note that for every real number t,

$$(tf(x) - g(x))^2 \geq 0 \qquad \text{for every } x \in [a, b].$$

Therefore by the integral inequality,

$$\int_a^b (tf(x) - g(x))^2\,dx \geq 0,$$

i.e.

$$\underbrace{\left(\int_a^b f(x)^2\,dx\right)}_{A} t^2 + \underbrace{\left(-2\int_a^b f(x)g(x)\,dx\right)}_{B} t + \underbrace{\left(\int_a^b g(x)^2\,dx\right)}_{C} \geq 0.$$

This holds for every $t \in \mathbb{R}$. Since the left-hand side is a quadratic polynomial in t, its discriminant must be non-positive, i.e. $B^2 - 4AC \leq 0$. This is exactly the inequality to be proved. ■

Theorem (Triangle inequality)

Let $a < b$ and let f be integrable on $[a, b]$. Then $|f|$ is also integrable on $[a, b]$ and

$$\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx.$$

(Compare with the triangle inequality $|x + y| \leq |x| + |y|$ from chapter 1.)

Proof. We skip the difficult proof of integrability of $|f|$ (done in MATH2033). For every $x \in [a, b]$,

$$-|f(x)| \leq f(x) \leq |f(x)|.$$

So by the integral inequality,

$$\int_a^b -|f(x)|\,dx \leq \int_a^b f(x)\,dx \leq \int_a^b |f(x)|\,dx,$$

$$\text{i.e.} \qquad -\int_a^b |f(x)|\,dx \leq \int_a^b f(x)\,dx \leq \int_a^b |f(x)|\,dx.$$

This is just the same as $\displaystyle\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx.$ ■

Example — A limit of integrals

Show that, as the limit of a sequence,

$$\lim_{n\to+\infty}\int_0^1 \frac{\cos nx}{n(1+x^2)}\,dx = 0.$$

Solution. For each $n \in \mathbb{N}$, denote $a_n = \displaystyle\int_0^1 \frac{\cos nx}{n(1+x^2)}\,dx$. By the triangle inequality,

$$|a_n| = \left|\int_0^1 \frac{\cos nx}{n(1+x^2)}\,dx\right| = \frac{1}{n}\left|\int_0^1 \frac{\cos nx}{1+x^2}\,dx\right| \le \frac{1}{n}\int_0^1 \left|\frac{\cos nx}{1+x^2}\right|dx = \frac{1}{n}\int_0^1 \frac{|\cos nx|}{1+x^2}\,dx.$$

Since $|\cos nx| \le 1$ for every $x \in [0,1]$, by the integral inequality,

$$\int_0^1 \frac{|\cos nx|}{1+x^2}\,dx \le \int_0^1 \frac{1}{1+x^2}\,dx.$$

Since $\dfrac{1}{1+x^2}$ is continuous on $[0,1]$, it is integrable there, so $\displaystyle\int_0^1 \frac{1}{1+x^2}\,dx$ is a finite real number; denote it by I. The above shows $|a_n| \le \dfrac{I}{n}$ for every $n \in \mathbb{N}$. Finally, since $\displaystyle\lim_{n\to+\infty}\frac{I}{n} = 0$, we have $\displaystyle\lim_{n\to+\infty} a_n = 0$ by the Squeeze Theorem. ■

Theorem (Mean Value Theorem for integrals)

Let $a < b$ and let f be continuous on $[a,b]$. Then there exists $c \in (a,b)$ such that

$$\int_a^b f(x)\,dx = f(c)(b-a).$$

The number $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$ is often called the average value of f over $[a,b]$. (Figure: the horizontal line $y = f(c)$ cuts off a rectangle over $[a,b]$ whose area equals the area under the curve $y = f(x)$.)

Proof. Since f is continuous on $[a,b]$, it attains a global maximum and a global minimum by the Extreme Value Theorem, i.e. there exist $c_1, c_2 \in [a,b]$ such that

$$f(c_1) \le f(x) \le f(c_2) \qquad \text{for every } x \in [a,b].$$

By the integral inequality, $f(c_1)(b-a) \le \displaystyle\int_a^b f(x)\,dx \le f(c_2)(b-a)$, i.e.

$$f(c_1) \le \frac{1}{b-a}\int_a^b f(x)\,dx \le f(c_2).$$

Since f is continuous on $[a,b]$, by the Intermediate Value Theorem there exists c between c_1 and c_2 such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Such c belongs to (a,b) also. ■

Example — Generalized MVT for integrals

(a) Let $a < b$ and let $f, g : [a, b] \to \mathbb{R}$ be continuous. If $g(x) \geq 0$ for every $x \in [a, b]$, show that there exists $c \in (a, b)$ such that

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

(b) Using (a), show that there exists $c \in (1, e)$ such that $\displaystyle\int_1^e \frac{\sin x}{x}\,dx = \sin c$.

Solution. (a) We mimic the proof of the Mean Value Theorem for integrals. Since f is continuous on $[a, b]$, by the Extreme Value Theorem there exist $c_1, c_2 \in [a, b]$ such that

$$f(c_1) \leq f(x) \leq f(c_2) \qquad \text{for every } x \in [a, b].$$

Since $g(x) \geq 0$,

$$f(c_1)g(x) \leq f(x)g(x) \leq f(c_2)g(x) \qquad \text{for every } x \in [a, b].$$

By the integral inequality,

$$f(c_1)\int_a^b g(x)\,dx \leq \int_a^b f(x)g(x)\,dx \leq f(c_2)\int_a^b g(x)\,dx,$$

so there exists $m \in [f(c_1), f(c_2)]$ such that

$$\int_a^b f(x)g(x)\,dx = m\int_a^b g(x)\,dx.$$

Now f is continuous and $f(c_1) \leq m \leq f(c_2)$, so by the Intermediate Value Theorem there exists c between c_1 and c_2 with $f(c) = m$, i.e.

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

Such c belongs to (a, b) also. ■

(b) Let $f(x) = \sin x$ and $g(x) = \dfrac{1}{x}$. Then f, g are both continuous on $[1, e]$ and $g(x) > 0$ for every $x \in [1, e]$. By (a), there exists $c \in (1, e)$ such that

$$\int_1^e \frac{\sin x}{x}\,dx = (\sin c) \underbrace{\int_1^e \frac{1}{x}\,dx}_{=1 \text{ by an earlier example}} = \sin c.$$

Remark

The result of part (a) above is called the generalized Mean Value Theorem for integrals. It reduces to the original Mean Value Theorem for integrals when g is the constant function $g(x) = 1$.

5.4 Fundamental Theorem of Calculus

Calculating integrals from the Riemann sum definition is in general difficult and tedious. The Fundamental Theorem of Calculus simplifies the task enormously, providing an important link between the seemingly unrelated notions of antiderivatives (chapter 4) and integrals (chapter 5).

Definition. Let $a < b$ be fixed real numbers and let $f : [a, b] \to \mathbb{R}$ be an integrable function. The area function of f (starting from a) is the function $F : [a, b] \to \mathbb{R}$ defined

by

$$F(x) = \int_a^x f(t)\,dt.$$

The function F gives the "net area" under the graph of f over the interval $[a, x]$ with a variable right endpoint. F is a function of the upper limit x, while t is just a dummy variable which disappears after evaluation. (Figure: the net area under $y = f(t)$ from a to x is shaded and labeled $F(x)$.)

Example — An area function of a step function

Let $f : [0, 2] \to \mathbb{R}$ be the function

$$f(x) = \begin{cases} 1 & \text{if } 0 \le x < 1 \\ 2 & \text{if } 1 \le x \le 2 \end{cases}.$$

Then the area function of f starting from 0 is

$$\begin{aligned} F(x) &= \int_0^x f(t)\,dt \\ &= \begin{cases} \displaystyle\int_0^x 1\,dt & \text{if } 0 \le x < 1 \\ \displaystyle\int_0^1 1\,dt + \int_1^x 2\,dt & \text{if } 1 \le x \le 2 \end{cases} \\ &= \begin{cases} x & \text{if } 0 \le x < 1 \\ 1 + 2(x-1) & \text{if } 1 \le x \le 2 \end{cases}, \end{aligned}$$

which is a continuous function on $[0, 2]$. (Figures: the graph of f is a step function jumping from 1 to 2 at $x = 1$; the graph of F is continuous, piecewise linear, with slope 1 then 2.)

In the previous example, although f is not continuous, its area function F is still continuous. An area function of an integrable function is continuous in general.

Lemma (Area functions are continuous)

Let $f : [a, b] \to \mathbb{R}$ be integrable. Then the area function $F : [a, b] \to \mathbb{R}$ defined by

$$F(x) = \int_a^x f(t)\,dt$$

is continuous on $[a, b]$.

Proof. Since f is integrable on $[a, b]$, it is bounded, so there exists $M > 0$ such that $|f(x)| \le M$ for every $x \in [a, b]$. Thus for every $x \in [a, b)$ and every small $h > 0$,

$$\begin{aligned} |F(x+h) - F(x)| &= \left| \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt \right| = \left| \int_x^{x+h} f(t)\,dt \right| \\ &\le \int_x^{x+h} |f(t)|\,dt \qquad \text{(triangle inequality)} \\ &\le Mh \qquad \text{(integral inequality)}; \end{aligned}$$

so $\lim_{h \to 0^+} |F(x+h) - F(x)| = 0$ by the Squeeze Theorem, i.e. $\lim_{h \to 0^+} F(x+h) = F(x)$. Similarly, $\lim_{h \to 0^-} F(x+h) = F(x)$ for every $x \in (a, b]$. ∎

(Figure: the increment $F(x+h) - F(x)$ is the thin strip of net area under $y = f(t)$ between x and $x + h$.)

If f is continuous, then its area function F is not only continuous but also differentiable — and it turns out to be an antiderivative of f! This justifies the earlier claim that every continuous function has an antiderivative.

Theorem (Fundamental Theorem of Calculus, first version)

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then the area function $F : [a, b] \to \mathbb{R}$ defined by

$$F(x) = \int_a^x f(t)\,dt$$

is an antiderivative of f on (a, b). In other words, F is differentiable on (a, b) and

$$F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \text{for every } x \in (a, b).$$

Proof. Let $x \in (a, b)$ be arbitrary. For each small $h > 0$, f is continuous on $[x, x + h]$, so by the Mean Value Theorem for integrals there exists $c \in (x, x + h)$ such that

$$\int_x^{x+h} f(t)\,dt = f(c)(x + h - x) = f(c)h.$$

Thus

$$\frac{F(x + h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt = f(c).$$

Since f is continuous and $c \in (x, x + h)$,

$$F'_+(x) = \lim_{h\to 0^+} \frac{F(x + h) - F(x)}{h} = \lim_{h\to 0^+} f(c) = f\left(\lim_{h\to 0^+} c\right) = f(x)$$

by the Squeeze Theorem. Considering every small $h < 0$, one can also prove $F'_-(x) = f(x)$, and so $F'(x) = f(x)$. ■

Remark — The lower limit does not matter

If we change the lower limit a to any other number $c \in [a, b]$, then

$$\int_c^x f(t)\,dt = \int_c^a f(t)\,dt + \int_a^x f(t)\,dt = \text{constant} + F(x)$$

is also an antiderivative of f.

The first version of the FTC is often useful for computing derivatives of area functions.

Example — Derivatives of area functions

Evaluate:

(a) $\dfrac{d}{dx}\displaystyle\int_0^x \sqrt{1 - t^2}\,dt$

(b) $\dfrac{d}{dx}\displaystyle\int_0^1 e^{x^2}\,dx$

(c) $\dfrac{d}{dx}\displaystyle\int_0^{\sqrt{x}} \sin(1 + t^2)\,dt$

(d) $\dfrac{d}{dx}\displaystyle\int_1^x \frac{t}{x}\,dt$

Solution. (a) By the Fundamental Theorem of Calculus,

$$\frac{d}{dx}\int_0^x \sqrt{1-t^2}\,dt = \sqrt{1-x^2}.$$

(b) The symbol x inside the integral is just a dummy variable; $\displaystyle\int_0^1 e^{x^2}\,dx$ is in fact a constant, so

$$\frac{d}{dx}\int_0^1 e^{x^2}\,dx = 0.$$

(c) By the chain rule,

$$\begin{aligned}
&\frac{d}{dx}\int_0^{\sqrt{x}} \sin(1+t^2)\,dt \\
&= \left[\frac{d}{d\sqrt{x}}\int_0^{\sqrt{x}} \sin(1+t^2)\,dt\right]\left(\frac{d}{dx}\sqrt{x}\right) \\
&= \left[\sin\left(1+(\sqrt{x})^2\right)\right]\left(\frac{1}{2\sqrt{x}}\right) = \frac{\sin(1+x)}{2\sqrt{x}}.
\end{aligned}$$

(d) Note that x is independent of the dummy variable t, so we can take $\dfrac{1}{x}$ out of the integral:

$$\begin{aligned}
\frac{d}{dx}\int_1^x \frac{t}{x}\,dt &= \frac{d}{dx}\left(\frac{1}{x}\int_1^x t\,dt\right) = \left(\frac{d}{dx}\frac{1}{x}\right)\left(\int_1^x t\,dt\right) + \left(\frac{1}{x}\right)\left(\frac{d}{dx}\int_1^x t\,dt\right) \\
&= \left(-\frac{1}{x^2}\right)\left[\frac{1}{2}(1+x)(x-1)\right] + \left(\frac{1}{x}\right)(x) = \frac{x^2+1}{2x^2}.
\end{aligned}$$

Example — A functional equation

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$2\int_0^x tf(t)\,dt = x\int_0^x f(t)\,dt \qquad \text{for every } x \in \mathbb{R}.$$

Show that f is a constant function.

Solution. By the first version of the FTC, differentiating both sides with respect to x gives

$$2xf(x) = (1)\left(\int_0^x f(t)\,dt\right) + (x)(f(x)) = \int_0^x f(t)\,dt + xf(x),$$

so

$$xf(x) = \int_0^x f(t)\,dt \qquad \text{for every } x \in \mathbb{R}.$$

(We cannot directly differentiate this equality further: the product rule does not apply on the left-hand side since f may not be differentiable.) Let $F(x) = \displaystyle\int_0^x f(t)\,dt$ be the area function of f starting from 0. Then the above becomes

$$xF'(x) = F(x) \qquad \text{for every } x \in \mathbb{R}.$$

Now for every $x \in \mathbb{R} \setminus \{0\}$,

$$\frac{d}{dx}\frac{F(x)}{x} = \frac{xF'(x) - F(x)}{x^2} = \frac{0}{x^2} = 0,$$

so there exist constants $c_1, c_2 \in \mathbb{R}$ such that

$$\frac{F(x)}{x} = \begin{cases} c_1 & \text{if } x > 0 \\ c_2 & \text{if } x < 0 \end{cases}$$

(we cannot directly write one constant on $\mathbb{R} \setminus \{0\}$ since the domain is disconnected).

This implies

$$F(x) = \begin{cases} c_1 x & \text{if } x > 0 \\ c_2 x & \text{if } x < 0 \end{cases}, \qquad \text{and so} \qquad f(x) = F'(x) = \begin{cases} c_1 & \text{if } x > 0 \\ c_2 & \text{if } x < 0 \end{cases}.$$

But since f is continuous at 0, we must have $c_1 = c_2$, which shows that f is in fact a constant function on $\mathbb{R}$. ■

Theorem (Fundamental Theorem of Calculus, second version)

Let $f : [a, b] \to \mathbb{R}$ be an integrable function. If $F : [a, b] \to \mathbb{R}$ is an antiderivative of f on (a, b) and is continuous on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof. For each positive integer n, consider a regular partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ into n subintervals, i.e. let $x_k := a + \dfrac{k(b-a)}{n}$ for each $k \in \{0, 1, \ldots, n\}$. Then

$$\begin{aligned} F(b) - F(a) &= F(b) - F(x_{n-1}) + F(x_{n-1}) - F(x_{n-2}) + F(x_{n-2}) + \cdots \\ &\quad - F(x_1) + F(x_1) - F(a) \\ &= [F(x_n) - F(x_{n-1})] + [F(x_{n-1}) - F(x_{n-2})] + \cdots + [F(x_1) - F(x_0)] \\ &= \sum_{k=1}^{n} [F(x_k) - F(x_{k-1})]. \end{aligned}$$

For each k, since F is continuous on $[x_{k-1}, x_k]$ and differentiable on (x_{k-1}, x_k), there exists $c_k \in (x_{k-1}, x_k)$ such that $F(x_k) - F(x_{k-1}) = F'(c_k)(x_k - x_{k-1})$ by the Mean Value Theorem for derivatives. Since F is an antiderivative of f on (a, b),

$$F(b) - F(a) = \sum_{k=1}^{n} F'(c_k)(x_k - x_{k-1}) = \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1}).$$

Finally, since f is integrable on $[a, b]$, taking limits as $n \to +\infty$ gives

$$F(b) - F(a) = \lim_{n \to +\infty} \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1}) = \int_a^b f(x)\,dx.$$

■

Remark — Practical use

The second version says we can evaluate integrals using any antiderivative, so in practice we use the simplest one by dropping the arbitrary constant C.

Remark — Differentiation and integration are inverse operations

- (First version) Starting from a continuous function f, integrate to get its area func-

tion $F(x) = \int_a^x f(t)\,dt$, then differentiate to get $F'(x) = f(x)$, recovering f.

- (Second version) Starting from a (nice) function F, differentiate to get its derivative f, then integrate to get $\int_a^x f(t)\,dt = F(x) - F(a)$, recovering F up to a constant.

Remark — Evaluation notation

The commonly used notations

$$[F(x)]_a^b, \qquad F(x)\Big|_a^b, \qquad F(x)\Big]_a^b$$

all mean $F(b) - F(a)$. So the formula of the second version can be written as

$$\int_a^b f(x)\,dx = [F(x)]_a^b.$$

Remark — Indefinite integrals

At the beginning, antiderivatives and integrals seemed totally unrelated. The FTC reveals the intimate relation between them, and explains why we adopted the notation $\int f(x)\,dx$ for antiderivatives, which looks very similar to that of integrals. Some people therefore call antiderivatives indefinite integrals.

With the second version of the FTC, we can now compute integrals using antiderivatives, without referring to the tedious limit of Riemann sums.

Example — Revisiting earlier integrals via FTC

- Since $\frac{d}{dx}\left(\frac{1}{3}x^3\right) = x^2$, we have
$$\int_0^1 x^2\,dx = \left[\frac{1}{3}x^3\right]_0^1 = \frac{1}{3}(1)^3 - \frac{1}{3}(0)^3 = \frac{1}{3}.$$
- Since $\frac{d}{dx}e^x = e^x$, we have, for each $a > 0$,
$$\int_0^a e^x\,dx = [e^x]_0^a = e^a - e^0 = e^a - 1.$$
- Since $\frac{d}{dx}\ln x = \frac{1}{x}$, we have, for each $a > 1$,
$$\int_1^a \frac{1}{x}\,dx = [\ln x]_1^a = \ln a - \ln 1 = \ln a.$$
- Since $\frac{d}{dx}(-\cos x) = \sin x$, we have, for each $a > 0$,
$$\int_0^a \sin x\,dx = [-\cos x]_0^a = (-\cos a) - (-\cos 0) = 1 - \cos a.$$

- Since $\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$, we have

$$\int_0^1 \frac{1}{1+x^2}\,dx = [\arctan x]_0^1 = \arctan 1 - \arctan 0 = \frac{\pi}{4}.$$

Trap — Misusing FTC part 2

Be careful when computing integrals using the second version of the FTC.

- Although $\dfrac{d}{dx}\left(-\dfrac{1}{x}\right) = \dfrac{1}{x^2}$, it is INCORRECT to say that

$$\int_{-1}^1 \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_{-1}^1 = -2. \qquad \times$$

  Reasons: (i) $-\dfrac{1}{x}$ is not an antiderivative of $\dfrac{1}{x^2}$ on $(-1,1)$, since $-\dfrac{1}{x}$ is not differentiable at $0 \in (-1,1)$; (ii) the integrand $\dfrac{1}{x^2}$ is unbounded near 0, so it is not integrable on $[-1,1]$ — the integral does not exist.

- Similarly, although $f(x) = \dfrac{1}{x}$ is odd, it is INCORRECT to say that

$$\int_{-1}^1 \frac{1}{x}\,dx = 0. \qquad \times$$

  The integrand is unbounded near 0, so f is not integrable on $[-1,1]$; the integral does not exist.

Trap — The integrability assumption is not redundant

In the second version of the FTC, the assumption that f is integrable on $[a,b]$ is not redundant. Even if f has an antiderivative, f may still fail to be integrable.

- Consider $F : \mathbb{R} \to \mathbb{R}$ defined by

$$F(x) = \begin{cases} x^2 \sin\dfrac{1}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

  Then F is differentiable everywhere, but its derivative

$$f(x) = \begin{cases} 2x\sin\dfrac{1}{x^2} - \dfrac{2}{x}\cos\dfrac{1}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

  is unbounded near 0, so f is not integrable on any closed interval containing 0.

- Even worse, there exist (MATH2033) functions $F : \mathbb{R} \to \mathbb{R}$ whose derivative f is bounded on $\mathbb{R}$ but still not integrable.

Example — Direct evaluation

Evaluate (a) $\displaystyle\int_1^4 (5x+3)\sqrt{x}\,dx$; (b) $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}\right) d\theta$.

Solution. (a)

$$\int_1^4 (5x+3)\sqrt{x}\,dx = \int_1^4 \left(5x^{\frac{3}{2}} + 3x^{\frac{1}{2}}\right) dx = \left[2x^{\frac{5}{2}} + 2x^{\frac{3}{2}}\right]_1^4$$
$$= \left[2(4)^{\frac{5}{2}} + 2(4)^{\frac{3}{2}}\right] - \left[2(1)^{\frac{5}{2}} + 2(1)^{\frac{3}{2}}\right] = 80 - 4 = 76.$$

(b)

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}\right) d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\sec^2\theta + \csc^2\theta\right) d\theta = \left[\tan\theta - \cot\theta\right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$
$$= \left(\tan\frac{\pi}{3} - \cot\frac{\pi}{3}\right) - \left(\tan\frac{\pi}{4} - \cot\frac{\pi}{4}\right)$$
$$= \left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) - (1-1) = \frac{2}{\sqrt{3}}.$$

When handling an integrand that involves absolute values or other piecewise-defined functions, we do not want to find its antiderivative directly! Instead, we break the interval into pieces so that the integrand is elementary on each piece, before applying the FTC.

Example — Integrands with absolute values

Evaluate (a) $\int_{-1}^{3} |3x^2 - 6x|\,dx$; (b) $\int_0^2 e^{|1-x|}\,dx$; (c) $\int_0^{\frac{3\pi}{2}} |\sin x|\,dx$.

Solution. (a) Since

$$3x^2 - 6x = 3x(x-2) \quad \begin{cases} \geq 0 & \text{if } x \geq 2 \text{ or } x \leq 0 \\ < 0 & \text{if } 0 < x < 2 \end{cases}$$

(Figure: the parabola $y = 3x^2 - 6x$ dipping below the x-axis on $(0, 2)$), we have

$$\int_{-1}^{3} |3x^2 - 6x|\,dx = \int_{-1}^{0} (3x^2 - 6x)\,dx + \int_0^2 -(3x^2 - 6x)\,dx + \int_2^3 (3x^2 - 6x)\,dx$$
$$= \left[x^3 - 3x^2\right]_{-1}^{0} + \left[-x^3 + 3x^2\right]_0^2 + \left[x^3 - 3x^2\right]_2^3$$
$$= 0 - (-4) + 4 - 0 + 0 - (-4) = 12.$$

(b) Since

$$1 - x \quad \begin{cases} \geq 0 & \text{if } x \leq 1 \\ < 0 & \text{if } x > 1 \end{cases}$$

(Figure: the graph consists of $y = e^{1-x}$ on $[0, 1]$ and $y = e^{x-1}$ on $[1, 2]$), we have

$$\int_0^2 e^{|1-x|}\,dx = \int_0^1 e^{1-x}\,dx + \int_1^2 e^{x-1}\,dx = \left[-e^{1-x}\right]_0^1 + \left[e^{x-1}\right]_1^2$$
$$= -1 - (-e) + e - 1 = 2e - 2.$$

(c) Since

$$\sin x \quad \begin{cases} \geq 0 & \text{if } 0 \leq x \leq \pi \\ < 0 & \text{if } \pi < x \leq \dfrac{3\pi}{2} \end{cases}$$

(Figure: the sine arch above the axis on $[0, \pi]$ and below it on $[\pi, \frac{3\pi}{2}]$), we have

$$\int_0^{\frac{3\pi}{2}} |\sin x|\,dx = \int_0^{\pi} \sin x\,dx + \int_{\pi}^{\frac{3\pi}{2}} -\sin x\,dx = \left[-\cos x\right]_0^{\pi} + \left[\cos x\right]_{\pi}^{\frac{3\pi}{2}}$$
$$= 1 - (-1) + 0 - (-1) = 3.$$

6 Techniques of Integration

In chapter 5 we learnt the Fundamental Theorem of Calculus, which evaluates an integral using an antiderivative. But in chapter 4 we only learnt the simplest rules of antidifferentiation (the sum/difference rule and the constant multiple rule), which just mirror the corresponding rules of differentiation. To evaluate integrals efficiently we now need a more complete set of rules and techniques of antidifferentiation and integration.

6.1 Substitution Rule

The following rule of antidifferentiation corresponds to the chain rule of differentiation.

Theorem — Substitution rule for antiderivatives

Let g be a differentiable function and let f be a function which is well-defined on the range of g. If F is an antiderivative of f, then

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C,$$

where C is an arbitrary constant.

Proof. If F is an antiderivative of f, then $F' = f$, so by the chain rule of differentiation,

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x)$$

for every x in the domain of g. ■

Remark — Remembering the substitution rule

Let $u = g(x)$. Then the differential of u is $du = g'(x)dx$, so

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du = F(u) + C = F(g(x)) + C.$$

Computing differentials is therefore very useful.

Corollary — Linear substitutions

If $\int f(x)\,dx = F(x) + C$, then for every $a \neq 0$ and every $b \in \mathbb{R}$,

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C,$$

where C is an arbitrary constant.

Proof. Apply the substitution rule with $u = ax + b$. ■

Example — Basic substitutions

Evaluate the following antiderivatives:

(a) $\displaystyle\int \frac{1}{\sqrt{2x-5}}\,dx$;

(b) $\displaystyle\int e^x \cos(e^x)\,dx$;

(c) $\displaystyle\int \frac{(\ln x)^2}{x}\,dx.$

Solution. (a) Let $u = 2x - 5$. Then $du = 2\,dx$, so

$$\int \frac{1}{\sqrt{2x-5}}\,dx = \int \frac{1}{\sqrt{u}} \cdot \frac{du}{2} = \frac{1}{2} \cdot 2\sqrt{u} + C = \sqrt{2x-5} + C.$$

(b) Let $u = e^x$. Then $du = e^x\,dx$, so

$$\int e^x \cos(e^x)\,dx = \int \cos u\,du = \sin u + C = \sin(e^x) + C.$$

(c) Let $u = \ln x$. Then $du = \dfrac{1}{x}\,dx$, so

$$\int \frac{(\ln x)^2}{x}\,dx = \int u^2\,du = \frac{1}{3}u^3 + C = \frac{1}{3}(\ln x)^3 + C.$$

In each case C is an arbitrary constant.

Example — Antiderivative of $\tan x$

Evaluate $\displaystyle\int \tan x\,dx$.

Solution. Let $u = \cos x$. Then $du = -\sin x\,dx$, so by the substitution rule

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C,$$

where C is an arbitrary constant. After a substitution one must return to the original variable x.

Remark — Substitution without a new symbol

The solution may also be presented without introducing a new symbol u. For instance,

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{1}{\cos x}\,d\cos x = -\ln|\cos x| + C = \ln|\sec x| + C.$$

If the substitution is simple, this direct presentation is more convenient; if it is complicated, it is better to denote the substitution by a new symbol u.

Example — Antiderivative of $\sec x$

Evaluate $\displaystyle\int \sec x\,dx$.

Solution. Let $u = \sec x + \tan x$. Then $du = (\sec x \tan x + \sec^2 x)dx$, so by the substitution rule

$$\begin{aligned}\int \sec x\,dx &= \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x}\,dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx \\ &= \int \frac{du}{u} = \ln|u| + C,\end{aligned}$$

i.e.

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C,$$

where C is an arbitrary constant.

Remark — Common substitutions to remember

$$\frac{1}{x^2}\,dx = -d\left(\frac{1}{x}\right), \qquad \frac{1}{x}\,dx = d\ln|x|,$$
$$\frac{1}{\sqrt{x}}\,dx = 2\,d\sqrt{x}, \qquad x^r\,dx = \frac{1}{r+1}\,dx^{r+1} \quad (r \neq -1),$$
$$\cos x\,dx = d\sin x, \qquad \sin x\,dx = -d\cos x,$$
$$\sec^2 x\,dx = d\tan x, \qquad \sec x\tan x\,dx = d\sec x.$$

These are just the corresponding formulas for antiderivatives.

Example — Substitutions involving roots

Evaluate the following antiderivatives:

(a) $\displaystyle\int \frac{x^3}{1+x^4}\,dx$;

(b) $\displaystyle\int \frac{1}{\sqrt{x-x^2}}\,dx$.

Solution. (a) Let $u = 1 + x^4$. Then $du = 4x^3\,dx$, so

$$\int \frac{x^3}{1+x^4}\,dx = \int \frac{1}{u}\cdot\frac{du}{4} = \frac{1}{4}\int\frac{du}{u} = \frac{1}{4}\ln|u| + C = \frac{1}{4}\ln(1+x^4) + C.$$

(b) First solution. Let $u = \sqrt{x}$. Then $x = u^2$ and $du = \dfrac{1}{2\sqrt{x}}\,dx$, so

$$\int \frac{1}{\sqrt{x-x^2}}\,dx = \int \frac{1}{\sqrt{1-x}}\cdot\frac{1}{\sqrt{x}}\,dx = \int \frac{1}{\sqrt{1-u^2}}\,(2\,du)$$
$$= 2\int\frac{du}{\sqrt{1-u^2}} = 2\arcsin u + C = 2\arcsin\sqrt{x} + C.$$

Second solution. Let $u = \sqrt{1-x}$. Then $x = 1 - u^2$ and $du = \dfrac{-1}{2\sqrt{1-x}}\,dx$, so

$$\int \frac{1}{\sqrt{x-x^2}}\,dx = \int \frac{1}{\sqrt{x}}\cdot\frac{1}{\sqrt{1-x}}\,dx = \int \frac{1}{\sqrt{1-u^2}}\,(-2\,du)$$
$$= 2\int\frac{-du}{\sqrt{1-u^2}} = 2\arccos u + C = 2\arccos\sqrt{1-x} + C.$$

Third solution. Complete the squares under the square root sign:

$$\int \frac{1}{\sqrt{x-x^2}}\,dx = \int \frac{1}{\sqrt{\frac{1}{4}-(x-\frac{1}{2})^2}}\,dx = \int \frac{2}{\sqrt{1-(2x-1)^2}}\,dx.$$

Now let $u = 2x - 1$, so that $du = 2\,dx$ and

$$\int \frac{1}{\sqrt{x-x^2}}\,dx = \int \frac{du}{\sqrt{1-u^2}} = \arcsin u + C = \arcsin(2x-1) + C.$$

In each case C is an arbitrary constant.

Think — Three antiderivatives of the same function

In the second part above we obtained three different sets of antiderivatives of $f(x) = \dfrac{1}{\sqrt{x - x^2}}$. There is nothing wrong: they differ only by constants, because

$$2\arccos\sqrt{1-x} = 2\arcsin\sqrt{x} = \arcsin(2x-1) + \frac{\pi}{2} \qquad \text{for every } x \in [0,1],$$

where the first equality follows from $\sin^2\theta + \cos^2\theta = 1$ and the second from $\sin\left(2\theta - \dfrac{\pi}{2}\right) = -\cos 2\theta = 2\sin^2\theta - 1$.

Example — A rationalizing substitution

Using the substitution $u = \sqrt{x+1} + \sqrt{x}$, evaluate

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x} + 1}\, dx.$$

Solution. Unlike the previous examples, du is not directly available in the integrand, so we compute $\dfrac{dx}{du}$. By rationalization,

$$\frac{1}{u} = \frac{1}{\sqrt{x+1} + \sqrt{x}} \cdot \frac{\sqrt{x+1} - \sqrt{x}}{\sqrt{x+1} - \sqrt{x}} = \sqrt{x+1} - \sqrt{x}$$

for every $x > 0$. Hence $u - \dfrac{1}{u} = 2\sqrt{x}$, so $x = \dfrac{1}{4}\left(u - \dfrac{1}{u}\right)^2$ and

$$\frac{dx}{du} = \frac{1}{4} \cdot 2\left(u - \frac{1}{u}\right) \cdot \left(1 + \frac{1}{u^2}\right) = \frac{(u^2-1)(u^2+1)}{2u^3}.$$

Since $\sqrt{x+1} + \sqrt{x} + 1 = u + 1$, we get

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x} + 1}\, dx = \int \frac{1}{u+1} \cdot \frac{(u^2-1)(u^2+1)}{2u^3}\, du = \int \frac{u^3 - u^2 + u - 1}{2u^3}\, du$$

$$= \frac{1}{2}\int \left(1 - \frac{1}{u} + \frac{1}{u^2} - \frac{1}{u^3}\right) du = \frac{1}{2}\left(u - \ln|u| - \frac{1}{u} + \frac{1}{2u^2}\right) + C.$$

Finally, since $\dfrac{1}{u} = \sqrt{x+1} - \sqrt{x}$, we have

$$\frac{1}{2}u - \frac{1}{2u} = \sqrt{x}, \qquad -\frac{1}{2}\ln u = \frac{1}{2}\ln(\sqrt{x+1} - \sqrt{x}), \qquad \frac{1}{4u^2} = \frac{1}{4}\left(\sqrt{x+1} - \sqrt{x}\right)^2,$$

so

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x} + 1}\, dx = \sqrt{x} + \frac{1}{2}\ln\left(\sqrt{x+1} - \sqrt{x}\right) + \frac{1}{4}\left(\sqrt{x+1} - \sqrt{x}\right)^2 + C,$$

where C is an arbitrary constant.

Theorem — Substitution rule for definite integrals

Let g be a differentiable function such that g' is integrable on $[a, b]$, and let f be a function which is continuous on the range of g. Then

$$\int_a^b f(g(x))g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du.$$

Proof. Both integrals exist by the assumptions, so it remains to show that they are equal. Since f is continuous on the range of g, it has an antiderivative F on the range of g (the range of g is an interval, by the Intermediate Value Theorem). By the chain rule, $F(g(x))$ is an antiderivative of $f(g(x))g'(x)$, so by the Fundamental Theorem of Calculus

$$\int_a^b f(g(x))g'(x)\,dx = [F(g(x))]_a^b = F(g(b)) - F(g(a)).$$

On the other hand, the Fundamental Theorem of Calculus again gives

$$\int_{g(a)}^{g(b)} f(u)\,du = [F(u)]_{g(a)}^{g(b)} = F(g(b)) - F(g(a)).$$

The two integrals are therefore equal. ■

Example — Substitution in a definite integral

Evaluate

$$\int_0^2 \frac{8x^3}{\sqrt{2x^2+1}}\,dx.$$

Solution. Let $u = 2x^2 + 1$. Then $du = 4x\,dx$, $u = 1$ when $x = 0$, and $u = 9$ when $x = 2$. Hence

$$\int_0^2 \frac{8x^3}{\sqrt{2x^2+1}}\,dx = \int_0^2 \frac{2x^2}{\sqrt{2x^2+1}}\,(4x\,dx) = \int_1^9 \frac{u-1}{\sqrt{u}}\,du = \int_1^9 \left(\sqrt{u} - \frac{1}{\sqrt{u}}\right) du$$

$$= \left[\frac{2}{3}u^{\frac{3}{2}} - 2u^{\frac{1}{2}}\right]_1^9 = (18 - 6) - \left(\frac{2}{3} - 2\right) = \frac{40}{3}.$$

In some integrals it is very difficult to find an antiderivative of the integrand directly, but the integral can still be computed by exploiting some symmetry of the integrand using the substitution rule.

Example — Exploiting symmetry by substitution

(a) Let $a > 0$ and let $f : [0, a] \to \mathbb{R}$ be a continuous function. Show that

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx.$$

(b) Using the result from (a), compute

$$\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx.$$

Solution. (a) Let $u = a - x$. Then $du = -dx$, $u = a$ when $x = 0$, and $u = 0$ when $x = a$. Hence

$$\int_0^a f(a-x)\,dx = \int_a^0 f(u)\,(-du) = \int_0^a f(u)\,du = \int_0^a f(x)\,dx.$$

(b) Let $f : [0, \pi] \to \mathbb{R}$ be the function $f(x) = \dfrac{x\sin x}{1+\cos^2 x}$, which is continuous on $[0, \pi]$. Applying (a) to f,

$$\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx = \int_0^\pi \frac{(\pi - x)\sin(\pi - x)}{1+\cos^2(\pi - x)}\,dx = \int_0^\pi \frac{(\pi - x)\sin x}{1+\cos^2 x}\,dx$$

$$= \pi \int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx - \int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx.$$

Rearranging,

$$\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx = \frac{\pi}{2}\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx.$$

Now the substitution $u = \cos x$ gives

$$\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx = \int_1^{-1} \frac{1}{1+u^2}\,(-du) = \int_{-1}^1 \frac{du}{1+u^2} = [\arctan u]_{-1}^1 = \frac{\pi}{2}.$$

Therefore

$$\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx = \frac{\pi}{2}\cdot\frac{\pi}{2} = \frac{\pi^2}{4}.$$

Remark — Trigonometric identities: products to sums

Antiderivatives of products are often hard to evaluate, and trigonometric identities are often useful in computing antiderivatives involving products of trigonometric functions.

- The product-to-sum formulae

  $$\sin x\cos y = \frac{1}{2}\left[\sin(x+y)+\sin(x-y)\right], \qquad \cos x\sin y = \frac{1}{2}\left[\sin(x+y)-\sin(x-y)\right],$$
  $$\cos x\cos y = \frac{1}{2}\left[\cos(x+y)+\cos(x-y)\right], \qquad \sin x\sin y = \frac{1}{2}\left[\cos(x-y)-\cos(x+y)\right]$$

  convert products of sines and cosines into sums, which are much easier to handle.
- In particular, the double angle formulae

  $$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}, \qquad \sin x\cos x = \frac{\sin 2x}{2}$$

  reduce products of sines and cosines of the same angle.

Example — Products of sines and cosines

Evaluate the following integrals:

(a) $\displaystyle\int_0^{\frac{\pi}{2}} \sin 4x\cos 3x\,dx$;

(b) $\displaystyle\int_0^{\frac{\pi}{2}} \cos x\cos 2x\cos 3x\,dx$.

Solution. (a) Using the product-to-sum formula,

$$\int_0^{\frac{\pi}{2}} \sin 4x\cos 3x\,dx = \int_0^{\frac{\pi}{2}} \frac{1}{2}\left(\sin 7x+\sin x\right)dx = \frac{1}{2}\left[-\frac{\cos 7x}{7}-\cos x\right]_0^{\frac{\pi}{2}}$$
$$= \frac{1}{2}\left(0-\left(-\frac{1}{7}-1\right)\right) = \frac{1}{2}\cdot\frac{8}{7} = \frac{4}{7}.$$

(b) Using the product-to-sum formula twice,

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos x \cos 2x \cos 3x \, dx &= \int_0^{\frac{\pi}{2}} \frac{1}{2}(\cos 3x + \cos x)\cos 3x \, dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{1}{2}\cos^2 3x + \frac{1}{2}\cos x \cos 3x\right) dx \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{1}{2}\left(\frac{1+\cos 6x}{2}\right) + \frac{1}{4}(\cos 4x + \cos 2x)\right] dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{1}{4} + \frac{1}{4}\cos 2x + \frac{1}{4}\cos 4x + \frac{1}{4}\cos 6x\right) dx \\ &= \left[\frac{x}{4} + \frac{1}{8}\sin 2x + \frac{1}{16}\sin 4x + \frac{1}{24}\sin 6x\right]_0^{\frac{\pi}{2}} = \frac{\pi}{8}.\end{aligned}$$

Example — Integral of $\sqrt{1+\cos 2x}$

Let n be a positive integer. Evaluate

$$\int_0^{n\pi} \sqrt{1+\cos 2x}\, dx \qquad \text{and} \qquad \int_{-n\pi}^{n\pi} \sqrt{1+\cos 2x}\, dx.$$

Solution. Note that $|\cos x|$ is periodic with period π. Therefore

$$\int_0^{n\pi} \sqrt{1+\cos 2x}\, dx = \int_0^{n\pi} \sqrt{2\cos^2 x}\, dx = \sqrt{2}\int_0^{n\pi} |\cos x|\, dx$$

$$= \sqrt{2}\left(\int_0^{\pi} |\cos x|\, dx + \int_{\pi}^{2\pi} |\cos x|\, dx + \cdots + \int_{(n-1)\pi}^{n\pi} |\cos x|\, dx\right)$$

$$= \sqrt{2}\left(\int_0^{\pi} |\cos x|\, dx + \cdots + \int_0^{\pi} |\cos x|\, dx\right) = \sqrt{2}\, n \int_0^{\pi} |\cos x|\, dx,$$

using the periodicity of $|\cos x|$. Next, by the symmetry of $|\cos x|$ about $\dfrac{\pi}{2}$,

$$\int_0^{\pi} |\cos x|\, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\cos x|\, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\, dx = [\sin x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2,$$

so

$$\int_0^{n\pi} \sqrt{1+\cos 2x}\, dx = 2\sqrt{2}\, n.$$

Finally, since $\sqrt{1+\cos 2x}$ is an even function,

$$\int_{-n\pi}^{n\pi} \sqrt{1+\cos 2x}\, dx = 2\int_0^{n\pi} \sqrt{1+\cos 2x}\, dx = 4\sqrt{2}\, n.$$

Trigonometric Integrals

When we encounter antiderivatives or integrals of trigonometric functions, we often need to decide between using the substitution rule and using trigonometric identities.

Example — Powers of $\cos x$

Evaluate the following antiderivatives:

(a) $\displaystyle\int \cos^2 x\,dx$;

(b) $\displaystyle\int \cos^3 x\,dx$;

(c) $\displaystyle\int \cos^4 x\,dx$;

(d) $\displaystyle\int \cos^5 x\,dx$.

Solution. (a) Using the double angle formula,

$$\int \cos^2 x\,dx = \int \frac{1+\cos 2x}{2}\,dx = \frac{1}{2}\int dx + \frac{1}{2}\int \cos 2x\,dx = \frac{x}{2} + \frac{1}{4}\sin 2x + C.$$

(b) Using the substitution $u = \sin x$, we have $du = \cos x\,dx$ and

$$\int \cos^3 x\,dx = \int \cos^2 x\,(\cos x\,dx) = \int (1-\sin^2 x)\,d\sin x = \sin x - \frac{1}{3}\sin^3 x + C.$$

(c) Using the double angle formula twice,

$$\int \cos^4 x\,dx = \int (\cos^2 x)^2\,dx = \int \left(\frac{1+\cos 2x}{2}\right)^2 dx = \frac{1}{4}\int \left(1 + 2\cos 2x + \cos^2 2x\right) dx$$

$$= \frac{1}{4}\int \left(1 + 2\cos 2x + \frac{1+\cos 4x}{2}\right) dx = \frac{3}{8}\int dx + \frac{1}{2}\int \cos 2x\,dx + \frac{1}{8}\int \cos 4x\,dx$$

$$= \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C.$$

(d) Using the substitution $u = \sin x$, we have $du = \cos x\,dx$ and

$$\int \cos^5 x\,dx = \int \cos^4 x\,(\cos x\,dx) = \int (1-\sin^2 x)^2\,d\sin x$$

$$= \int (1 - 2\sin^2 x + \sin^4 x)\,d\sin x = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C.$$

In each case C is an arbitrary constant.

Think — Why the two strategies do not mix

Why should we not use the substitution $\cos x\,dx = d\sin x$ in (a) or (c) above? And what happens if we use the double angle formula in (b) or (d)?

Example — Products of powers of $\sin x$ and $\cos x$

Evaluate the following antiderivatives:

(a) $\displaystyle\int \sin^2 x \cos^3 x\,dx$;

(b) $\displaystyle\int \sin^3 x \cos^4 x\,dx$;

(c) $\displaystyle\int \sin^2 x \cos^4 x\,dx$.

Solution. (a) Using the substitution $u = \sin x$, we have $du = \cos x\,dx$ and

$$\int \sin^2 x \cos^3 x\,dx = \int \sin^2 x \cos^2 x\,(\cos x\,dx) = \int \sin^2 x(1-\sin^2 x)\,d\sin x$$

$$= \int (\sin^2 x - \sin^4 x)\, d\sin x = \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C.$$

(b) Using the substitution $u = \cos x$, we have $du = -\sin x\, dx$ and

$$\int \sin^3 x \cos^4 x\, dx = \int -\sin^2 x \cos^4 x\, (-\sin x\, dx) = \int (\cos^2 x - 1)\cos^4 x\, d\cos x$$

$$= \int (\cos^6 x - \cos^4 x)\, d\cos x = \frac{1}{7}\cos^7 x - \frac{1}{5}\cos^5 x + C.$$

(c) Using the double angle formula and the product-to-sum formula,

$$\int \sin^2 x \cos^4 x\, dx = \int \frac{1-\cos 2x}{2}\left(\frac{1+\cos 2x}{2}\right)^2 dx = \frac{1}{8}\int (1-\cos^2 2x)(1+\cos 2x)\, dx$$

$$= \frac{1}{8}\int \left(1 - \frac{1+\cos 4x}{2}\right)(1+\cos 2x)\, dx = \frac{1}{16}\int (1+\cos 2x - \cos 4x - \cos 4x \cos 2x)\, dx$$

$$= \frac{1}{16}\int \left[1 + \cos 2x - \cos 4x - \frac{1}{2}(\cos 6x + \cos 2x)\right] dx$$

$$= \frac{1}{16}\int dx + \frac{1}{32}\int \cos 2x\, dx - \frac{1}{16}\int \cos 4x\, dx - \frac{1}{32}\int \cos 6x\, dx$$

$$= \frac{1}{16}x + \frac{1}{64}\sin 2x - \frac{1}{64}\sin 4x - \frac{1}{192}\sin 6x + C,$$

where C is an arbitrary constant.

The two examples above show that $\int \sin^m x \cos^n x\, dx$ can be evaluated easily by substitution if m is odd or n is odd, whereas if both m and n are even it is better to use trigonometric identities or another method (studied in the next section).

Example — Powers of $\sec x$ and $\tan x$

Evaluate the following.

(a) $\int \sec^4 x\, dx$;

(b) $\int \sec^4 x \tan^2 x\, dx$;

(c) $\int \sec^4 x \tan^3 x\, dx$.

Solution. (a) Let $u = \tan x$. Then $du = \sec^2 x\, dx$, so

$$\int \sec^4 x\, dx = \int \sec^2 x\, (\sec^2 x\, dx) = \int (1 + \tan^2 x)\, d\tan x = \tan x + \frac{1}{3}\tan^3 x + C.$$

(b) Let $u = \tan x$. Then $du = \sec^2 x\, dx$, so

$$\int \sec^4 x \tan^2 x\, dx = \int \sec^2 x \tan^2 x\, (\sec^2 x\, dx) = \int (1 + \tan^2 x)\tan^2 x\, d\tan x$$

$$= \int (\tan^2 x + \tan^4 x)\, d\tan x = \frac{1}{3}\tan^3 x + \frac{1}{5}\tan^5 x + C.$$

(c) Let $u = \sec x$. Then $du = \sec x \tan x \, dx$, so

$$\int \sec^4 x \tan^3 x \, dx = \int \sec^3 x \tan^2 x \, (\sec x \tan x \, dx) = \int \sec^3 x (\sec^2 x - 1) \, d \sec x$$

$$= \int (\sec^5 x - \sec^3 x) \, d \sec x = \frac{1}{6} \sec^6 x - \frac{1}{4} \sec^4 x + C.$$

Alternatively, let $u = \tan x$. Then $du = \sec^2 x \, dx$, so

$$\int \sec^4 x \tan^3 x \, dx = \int \sec^2 x \tan^3 x \, (\sec^2 x \, dx) = \int (1 + \tan^2 x) \tan^3 x \, d \tan x$$

$$= \int (\tan^3 x + \tan^5 x) \, d \tan x = \frac{1}{4} \tan^4 x + \frac{1}{6} \tan^6 x + C.$$

In each case C is an arbitrary constant.

In general, $\int \sec^m x \tan^n x \, dx$ can be evaluated easily by substitution if m is even or n is odd. To handle antiderivatives like $\int \sec^3 x \, dx$, $\int \sec^5 x \, dx$ or $\int \sec^7 x \, dx$, we need the last rule of antidifferentiation, to be studied in the next section.

6.2 Integration by Parts

The following rule of antidifferentiation corresponds to the product rule of differentiation.

Theorem — Antidifferentiation by parts

Let f and g be differentiable functions. Then

$$\int f(x) g'(x) \, dx = f(x) g(x) - \int f'(x) g(x) \, dx.$$

If we let $u = f(x)$ and $v = g(x)$, then $du = f'(x) dx$ and $dv = g'(x) dx$, so the above can be written as

$$\int u \, dv = uv - \int v \, du.$$

Proof. The derivative of the product fg is $(fg)' = f'g + fg'$, so fg is an antiderivative of $f'g + fg'$. In other words,

$$\int [f'(x) g(x) + f(x) g'(x)] \, dx = f(x) g(x) + C,$$

where C is an arbitrary constant, and hence $\int f(x) g'(x) \, dx = f(x) g(x) - \int f'(x) g(x) \, dx$. ■

Corollary — Integration by parts

Let f and g be differentiable functions such that f' and g' are integrable on $[a, b]$. Then

$$\int_a^b f(x) g'(x) \, dx = [f(x) g(x)]_a^b - \int_a^b f'(x) g(x) \, dx.$$

Example — $\int xe^x\,dx$

Evaluate $\displaystyle\int xe^x\,dx$.

Solution. Let $f(x) = x$ and $g'(x) = e^x$ (so that we may take $g(x) = e^x$). Then

$$\int \underbrace{x}_{f(x)}\underbrace{e^x}_{g'(x)}\,dx = \underbrace{x}_{f(x)}\underbrace{e^x}_{g(x)} - \int \underbrace{1}_{f'(x)}\underbrace{e^x}_{g(x)}\,dx = xe^x - e^x + C,$$

where C is an arbitrary constant.

Example — $\int \ln x\,dx$

Evaluate $\displaystyle\int \ln x\,dx$.

Solution. Let $u = \ln x$ and $dv = dx$ (so that we may take $v = x$). Then

$$\int \underbrace{\ln x}_{u}\underbrace{dx}_{dv} = \underbrace{x}_{v}\underbrace{\ln x}_{u} - \int \underbrace{x}_{v}\underbrace{\frac{1}{x}\,dx}_{du} = x\ln x - \int dx = x\ln x - x + C,$$

where C is an arbitrary constant.

Example — A definite integral by parts

Evaluate

$$\int_0^1 \frac{1}{(1+x^2)^{\frac{3}{2}}}\,dx.$$

Solution. First observe that

$$\int_0^1 \frac{1}{(1+x^2)^{\frac{3}{2}}}\,dx = \int_0^1 \frac{1+x^2-x^2}{(1+x^2)^{\frac{3}{2}}}\,dx = \int_0^1 \frac{1}{(1+x^2)^{\frac{1}{2}}}\,dx + \int_0^1 x\cdot\frac{-x}{(1+x^2)^{\frac{3}{2}}}\,dx.$$

We keep the first integral unchanged and handle the second using integration by parts. Let $f(x) = x$ and $g'(x) = \dfrac{-x}{(1+x^2)^{\frac{3}{2}}}$ (so that we may take $g(x) = \dfrac{1}{(1+x^2)^{\frac{1}{2}}}$, since $\dfrac{d}{dx}(1+x^2)^{-\frac{1}{2}} = -x(1+x^2)^{-\frac{3}{2}}$). Then

$$\int_0^1 \frac{1}{(1+x^2)^{\frac{3}{2}}}\,dx = \int_0^1 \frac{1}{(1+x^2)^{\frac{1}{2}}}\,dx + \left[x\cdot\frac{1}{(1+x^2)^{\frac{1}{2}}}\right]_0^1$$

$$- \int_0^1 \frac{1}{(1+x^2)^{\frac{1}{2}}}\,dx = \left[\frac{x}{(1+x^2)^{\frac{1}{2}}}\right]_0^1 = \frac{1}{\sqrt{2}}.$$

Example — Repeated integration by parts

Evaluate

$$\int_0^\pi x^2\cos x\,dx.$$

Solution. Let $f(x) = x^2$ and $g'(x) = \cos x$ (so that we may take $g(x) = \sin x$). Integrating by parts,

$$\begin{aligned}\int_0^\pi x^2 \cos x \, dx &= \left[x^2 \sin x\right]_0^\pi - \int_0^\pi 2x \sin x \, dx \\ &= \left(\pi^2 \sin \pi - 0^2 \sin 0\right) - \int_0^\pi 2x \sin x \, dx \\ &= \int_0^\pi -2x \sin x \, dx.\end{aligned}$$

To deal with the last integral we integrate by parts again, this time with $f(x) = 2x$ and $g'(x) = -\sin x$ (so that we may take $g(x) = \cos x$):

$$\begin{aligned}\int_0^\pi x^2 \cos x \, dx &= \int_0^\pi -2x \sin x \, dx \\ &= [2x \cos x]_0^\pi - \int_0^\pi 2 \cos x \, dx \\ &= [2x \cos x - 2 \sin x]_0^\pi = -2\pi.\end{aligned}$$

Remark — The tabular method

When we need to integrate by parts several times, there is an efficient tabular method. To compute $\int fg \, dx$ by parts four times, construct the table

sign	D	I
$+$	f	g
$-$	f'	$\int g$
$+$	f''	$\iint g$
$-$	f'''	$\iiint g$
$+$	$f^{(4)}$	$\iiiint g$

Then multiply along the diagonals (with the signs) and sum the terms, and finally add the integral of the product of the two entries of the last row. This gives the correct formula for integration by parts.

Example — $\int x^4 \sin x \, dx$ by the tabular method

Evaluate $\int x^4 \sin x \, dx$.

Solution. Let $f(x) = x^4$ and $g(x) = \sin x$, and construct the table

sign	D	I
$+$	x^4	$\sin x$
$-$	$4x^3$	$-\cos x$
$+$	$12x^2$	$-\sin x$
$-$	$24x$	$\cos x$
$+$	24	$\sin x$

from which

$$\int x^4 \sin x \, dx = -x^4 \cos x + 4x^3 \sin x + 12x^2 \cos x - 24x \sin x + \int 24 \sin x \, dx$$

$$= -x^4 \cos x + 4x^3 \sin x + 12x^2 \cos x - 24x \sin x - 24 \cos x + C,$$

where C is an arbitrary constant.

Example — $\int e^x \cos x \, dx$

Evaluate

$$\int_0^\pi e^x \cos x \, dx.$$

Solution. Let $f(x) = e^x$ and $g(x) = \cos x$, and construct the table

sign	D	I
$+$	e^x	$\cos x$
$-$	e^x	$\sin x$
$+$	e^x	$-\cos x$

from which

$$\int_0^\pi e^x \cos x \, dx = [e^x \sin x + e^x \cos x]_0^\pi - \int_0^\pi e^x \cos x \, dx = -e^\pi - 1 - \int_0^\pi e^x \cos x \, dx.$$

Now the required integral appears on both sides, so rearranging we get $2\int_0^\pi e^x \cos x \, dx = -e^\pi - 1$, i.e.

$$\int_0^\pi e^x \cos x \, dx = -\frac{e^\pi + 1}{2}.$$

Reduction Formulas

Example — A reduction formula for $\int \sin^n x \, dx$

For each non-negative integer n, denote

$$I_n := \int_0^\pi \sin^n x \, dx.$$

Show that $I_n = \dfrac{n-1}{n} I_{n-2}$ for every integer $n \geq 2$, and hence evaluate

$$\int_0^\pi \sin^8 x \, dx.$$

Solution. For each integer $n \geq 2$,

$$I_n = \int_0^\pi \sin^n x \, dx = \int_0^\pi \sin^{n-1} x \sin x \, dx.$$

Let $f(x) = \sin^{n-1} x$ and $g'(x) = \sin x$ (so that we may take $g(x) = -\cos x$). Integrating by parts,

$$\begin{aligned} I_n &= \left[-\sin^{n-1} x \cos x\right]_0^\pi - \int_0^\pi (-\cos x)(n-1)\sin^{n-2} x \cos x \, dx \\ &= 0 + (n-1)\int_0^\pi \sin^{n-2} x \cos^2 x \, dx, \end{aligned}$$

where the boundary term $\left[-\sin^{n-1} x \cos x\right]_0^\pi = 0$ because $n \geq 2$. Hence

$$\begin{aligned} I_n &= (n-1)\int_0^\pi \sin^{n-2} x(1-\sin^2 x)\, dx \\ &= (n-1)\int_0^\pi \sin^{n-2} x \, dx - (n-1)\int_0^\pi \sin^n x \, dx \\ &= (n-1)I_{n-2} - (n-1)I_n. \end{aligned}$$

Since I_n appears on both sides, rearranging gives

$$I_n = \frac{n-1}{n} I_{n-2}.$$

Finally, by this formula,

$$\begin{aligned} \int_0^\pi \sin^8 x \, dx = I_8 &= \frac{7}{8} I_6 = \frac{7}{8}\cdot\frac{5}{6} I_4 = \frac{7}{8}\cdot\frac{5}{6}\cdot\frac{3}{4} I_2 = \frac{7}{8}\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2} I_0 \\ &= \frac{7}{8}\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\int_0^\pi dx = \frac{7}{8}\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\pi = \frac{35\pi}{128}. \end{aligned}$$

Remark — Reduction formulas

Above, the recurrence relation $I_n = \dfrac{n-1}{n} I_{n-2}$ (for every integer $n \geq 2$) played a key role in reducing the complexity of the integrand. In general, a recurrence relation that expresses I_n in terms of other I_k's with smaller indices $k < n$ is called a reduction formula.

Example — Reducing $\int (1-x)^m (1+x)^n \, dx$

For each pair of non-negative integers m and n, denote

$$I_{m,n} := \int_{-1}^1 (1-x)^m (1+x)^n \, dx.$$

(a) Show that $I_{m,n} = \dfrac{m}{n+1} I_{m-1,n+1}$ for every pair of integers $m \geq 1$ and $n \geq 0$.

(b) Hence evaluate $\displaystyle\int_{-1}^1 (1-x^2)^n \, dx$ for each non-negative integer n.

Solution. (a) Let $m \geq 1$ and $n \geq 0$ be integers, and let $f(x) = (1-x)^m$ and $g'(x) = (1+x)^n$ (so that we may take $g(x) = \dfrac{1}{n+1}(1+x)^{n+1}$). Integrating by parts,

$$I_{m,n} = \left[\frac{1}{n+1}(1-x)^m (1+x)^{n+1}\right]_{-1}^1 - \int_{-1}^1 \left[-m(1-x)^{m-1}\right]\left[\frac{1}{n+1}(1+x)^{n+1}\right] dx$$

$$= 0 + \frac{m}{n+1}\int_{-1}^{1}(1-x)^{m-1}(1+x)^{n+1}dx = \frac{m}{n+1}I_{m-1,n+1},$$

where the boundary term $\left[(1-x)^m(1+x)^{n+1}\right]_{-1}^{1} = 0$ because $m \geq 1$ and $n \geq 0$.
(b) By the reduction formula above,

$$\int_{-1}^{1}(1-x^2)^n\,dx = \int_{-1}^{1}(1-x)^n(1+x)^n\,dx = I_{n,n} = \frac{n}{n+1}I_{n-1,n+1} = \frac{n}{n+1}\cdot\frac{n-1}{n+2}I_{n-2,n+2}$$

$$= \frac{n}{n+1}\cdot\frac{n-1}{n+2}\cdot\frac{n-2}{n+3}I_{n-3,n+3} = \cdots = \frac{n(n-1)(n-2)\cdots 1}{(n+1)(n+2)(n+3)\cdots(2n)}I_{0,2n}$$

$$= \frac{n!}{(n+1)(n+2)\cdots(2n)}\int_{-1}^{1}(1+x)^{2n}dx = \frac{n!}{(n+1)(n+2)\cdots(2n)}\cdot\frac{2^{2n+1}}{2n+1} = \frac{2^{2n+1}(n!)^2}{(2n+1)!}$$

for every non-negative integer n.

Example — A reduction formula for $\int \sec^n x\,dx$

Evaluate $\int \sec^5 x\,dx$.

Solution. We first obtain a reduction formula. For each positive integer n denote

$$I_n(x) := \int \sec^n x\,dx,$$

so that I_n is defined up to addition by a constant function. For each integer $n > 2$ let $f(x) = \sec^{n-2} x$ and $g'(x) = \sec^2 x$ (so that we may take $g(x) = \tan x$). Then

$$f'(x) = (n-2)\sec^{n-3} x(\sec x \tan x) = (n-2)\sec^{n-2} x \tan x,$$

so integrating by parts,

$$I_n(x) = \int(\sec^{n-2} x)(\sec^2 x)\,dx = \sec^{n-2} x \tan x - \int\left[(n-2)\sec^{n-2} x \tan x\right](\tan x)\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x(\sec^2 x - 1)\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)\int \sec^n x\,dx + (n-2)\int \sec^{n-2} x\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)I_n(x) + (n-2)I_{n-2}(x).$$

Since $I_n(x)$ appears on both sides, rearranging gives, for $n > 2$,

$$I_n(x) = \frac{1}{n-1}\sec^{n-2} x \tan x + \frac{n-2}{n-1}I_{n-2}(x).$$

Finally, by this reduction formula,

$$\begin{aligned}\int \sec^5 x\,dx = I_5(x) &= \frac{1}{4}\sec^3 x\tan x + \frac{3}{4}I_3(x)\\ &= \frac{1}{4}\sec^3 x\tan x + \frac{3}{4}\left[\frac{1}{2}\sec x\tan x + \frac{1}{2}I_1(x)\right]\\ &= \frac{1}{4}\sec^3 x\tan x + \frac{3}{8}\sec x\tan x + \frac{3}{8}\int \sec x\,dx\\ &= \frac{1}{4}\sec^3 x\tan x + \frac{3}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x+\tan x| + C,\end{aligned}$$

where C is an arbitrary constant.

Example — A reduction formula for $\int (x^2+a^2)^{-n}dx$

Let a be a non-zero real number. For each positive integer n, denote

$$I_n(x) := \int \frac{1}{(x^2+a^2)^n}\,dx,$$

so that I_n is defined up to addition by a constant function. Show that

$$I_n(x) = \frac{1}{(2n-2)a^2}\frac{x}{(x^2+a^2)^{n-1}} + \frac{2n-3}{(2n-2)a^2}I_{n-1}(x)$$

for every integer $n \geq 2$. Hence evaluate the antiderivative $\displaystyle\int \frac{1}{(x^2+a^2)^3}\,dx$.

Solution. For each integer $n \geq 1$ let $f(x) = \dfrac{1}{(x^2+a^2)^n}$ and $g'(x) = 1$ (so that we may take $g(x) = x$). Then $f'(x) = -\dfrac{2nx}{(x^2+a^2)^{n+1}}$, so integrating by parts,

$$\begin{aligned}I_n(x) &= \frac{x}{(x^2+a^2)^n} - \int x\left[-\frac{2nx}{(x^2+a^2)^{n+1}}\right]dx\\ &= \frac{x}{(x^2+a^2)^n} + 2n\int \frac{(x^2+a^2)-a^2}{(x^2+a^2)^{n+1}}\,dx\\ &= \frac{x}{(x^2+a^2)^n} + 2n\left[\int \frac{1}{(x^2+a^2)^n}\,dx - a^2\int \frac{1}{(x^2+a^2)^{n+1}}\,dx\right]\\ &= \frac{x}{(x^2+a^2)^n} + 2nI_n(x) - 2na^2 I_{n+1}(x).\end{aligned}$$

Since $I_n(x)$ appears on both sides, rearranging gives, for $n \geq 1$,

$$I_{n+1}(x) = \frac{1}{2na^2}\frac{x}{(x^2+a^2)^n} + \frac{2n-1}{2na^2}I_n(x).$$

Replacing n by $n-1$ we obtain the stated formula for $n \geq 2$. Finally, by this reduction formula,

$$\begin{aligned}\int \frac{1}{(x^2+a^2)^3}\,dx = I_3(x) &= \frac{1}{4a^2}\frac{x}{(x^2+a^2)^2} + \frac{3}{4a^2}I_2(x)\\ &= \frac{1}{4a^2}\frac{x}{(x^2+a^2)^2} + \frac{3}{4a^2}\left[\frac{1}{2a^2}\frac{x}{x^2+a^2} + \frac{1}{2a^2}I_1(x)\right]\end{aligned}$$

$$= \frac{1}{4a^2}\frac{x}{(x^2+a^2)^2} + \frac{3}{8a^4}\frac{x}{x^2+a^2} + \frac{3}{8a^4}\int \frac{1}{x^2+a^2}\,dx$$
$$= \frac{1}{4a^2}\frac{x}{(x^2+a^2)^2} + \frac{3}{8a^4}\frac{x}{x^2+a^2} + \frac{3}{8a^5}\arctan\frac{x}{a} + C,$$

where C is an arbitrary constant.

6.3 Trigonometric Substitution

So far we have applied the substitution rule by making explicit substitutions, where an expression $g(x)$ in the original integral is replaced by a new variable u: "Let $u = g(x)$. Then $du = g'(x)dx$." In fact, the substitution rule also allows implicit substitutions, where a variable x in the original integral is replaced by any one-to-one function $g(t)$ of a new variable t: "Let $x = g(t)$. Then $dx = g'(t)dt$."

Corollary — Implicit substitution

Let g be a one-to-one differentiable function and let f be a function which is well-defined on the range of g. If $\int f(g(t))g'(t)\,dt = H(t) + C$, then

$$\int f(x)\,dx = H(g^{-1}(x)) + C,$$

where C is an arbitrary constant.

Proof. This follows immediately from the substitution rule. Note that g being one-to-one guarantees that one can directly replace $t = g^{-1}(x)$ in terms of x. (Sometimes the implicit substitution $x = g(t)$ can still be applied even if g is not one-to-one, as long as there is a valid replacement of the expressions involved in $H(t)$ in terms of x.) ■

Example — Substitution $x = t^6$

Using the substitution $x = t^6$, evaluate

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\,dx.$$

Solution. The function $g(t) = t^6$ is one-to-one and differentiable on $(0, +\infty)$, and on its range the function $\dfrac{1}{\sqrt{x} + \sqrt[3]{x}}$ is well-defined; so we may make the implicit substitution $x = t^6$. Then $\sqrt{x} = t^3$, $\sqrt[3]{x} = t^2$ and $dx = 6t^5\,dt$, so

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\,dx = \int \frac{6t^5}{t^3 + t^2}\,dt = 6\int \frac{t^3}{t+1}\,dt = 6\int \left(t^2 - t + 1 - \frac{1}{t+1}\right)dt$$
$$= 2t^3 - 3t^2 + 6t - 6\ln|t+1| + C = 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln\left(1 + \sqrt[6]{x}\right) + C,$$

where C is an arbitrary constant.

In practice, when the integrand involves a quadratic polynomial under a square root, it is often useful to make implicit substitutions of the trigonometric type.

Remark — Which trigonometric substitution?

Suppose that $a > 0$.

- If the integrand involves $\sqrt{x^2+a^2}$, we apply the implicit substitution $x = a\tan t$.
- If the integrand involves $\sqrt{x^2-a^2}$, we apply the implicit substitution $x = a\sec t$.
- If the integrand involves $\sqrt{a^2-x^2}$, we apply the implicit substitution $x = a\sin t$.

When applying such substitutions, we focus only on an interval where the trigonometric function is one-to-one.

Example — Substitution $x = a\sec t$

Evaluate

$$\int \frac{\sqrt{x^2-25}}{x}\,dx$$

for $x > 5$.

Solution. Let $x = 5\sec t$ where $t \in \left(0, \frac{\pi}{2}\right)$. Then

$$\sqrt{x^2-25} = \sqrt{(5\sec t)^2 - 25} = |5\tan t| = 5\tan t$$

and $dx = 5\sec t\tan t\,dt$, so

$$\int \frac{\sqrt{x^2-25}}{x}\,dx = \int \frac{5\tan t}{5\sec t} 5\sec t\tan t\,dt = 5\int \tan^2 t\,dt = 5\int (\sec^2 t - 1)\,dt$$

$$= 5\tan t - 5t + C = \sqrt{x^2-25} - 5\operatorname{arcsec}\frac{x}{5} + C = \sqrt{x^2-25} - 5\arccos\frac{5}{x} + C,$$

where C is an arbitrary constant.

Example — Substitution $x = a\tan t$

Evaluate

$$\int \frac{1}{(x^2+9)^{\frac{3}{2}}}\,dx.$$

Solution. Let $x = 3\tan t$ where $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $dx = 3\sec^2 t\,dt$ and

$$\sqrt{x^2+9} = \sqrt{(3\tan t)^2 + 9} = |3\sec t| = 3\sec t,$$

so

$$\int \frac{1}{(x^2+9)^{\frac{3}{2}}}\,dx = \int \frac{1}{(3\sec t)^3} 3\sec^2 t\,dt = \frac{1}{9}\int \cos t\,dt = \frac{1}{9}\sin t + C = \frac{x}{9\sqrt{x^2+9}} + C,$$

where C is an arbitrary constant. (Figure: a right triangle with angle t, opposite side x, adjacent side 3 and hypotenuse $\sqrt{x^2+9}$, so that $\sin t = \frac{x}{\sqrt{x^2+9}}$.)

Example — Substitution $x = a\sin t$

Evaluate

$$\int \frac{1}{x\sqrt{4-x^2}}\,dx.$$

Solution. Let $x = 2\sin t$ where $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \setminus \{0\}$. Then

$$\sqrt{4-x^2} = \sqrt{4 - (2\sin t)^2} = |2\cos t| = 2\cos t$$

and $dx = 2\cos t\,dt$, so

$$\int \frac{1}{x\sqrt{4-x^2}}\,dx = \int \frac{1}{(2\sin t)(2\cos t)} 2\cos t\,dt = \frac{1}{2}\int \csc t\,dt = \frac{1}{2}\ln|\csc t - \cot t| + C$$
$$= \frac{1}{2}\ln\left|\frac{2-\sqrt{4-x^2}}{x}\right| + C,$$

where C is an arbitrary constant. (Restricting the domain to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ ensures that sin is one-to-one. Without specifying this domain, $\sqrt{4-x^2}$ may equal $2\cos t$ or $-2\cos t$, and $\cot t$ may equal $\frac{\sqrt{4-x^2}}{x}$ or $-\frac{\sqrt{4-x^2}}{x}$. Since $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ we must have $\cot t = \frac{\sqrt{4-x^2}}{x}$.)

Example — Completing the square

Evaluate

$$\int \frac{1}{x^2+2x+2}\,dx.$$

Solution. Let $x = \tan t - 1$ where $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $dx = \sec^2 t\,dt$, so

$$\int \frac{1}{x^2+2x+2}\,dx = \int \frac{1}{(x+1)^2+1}\,dx$$
$$= \int \frac{1}{(\tan t)^2+1}\sec^2 t\,dt = \int dt$$
$$= t + C = \arctan(x+1) + C,$$

where C is an arbitrary constant.

Example — Completing the square and $x = a\sin t$

Evaluate

$$\int \frac{1}{\sqrt{9+16x-4x^2}}\,dx.$$

Solution. Let $x = \frac{5}{2}\sin t + \frac{5}{2}$ where $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then

$$\sqrt{9+16x-4x^2} = \sqrt{25 - 4\left(x-\frac{5}{2}\right)^2} = \sqrt{25 - 4\left(\frac{5}{2}\sin t\right)^2} = |5\cos t| = 5\cos t$$

and $dx = \frac{5}{2}\cos t\,dt$, so

$$\int \frac{1}{\sqrt{9+16x-4x^2}}\,dx = \int \frac{1}{5\cos t}\cdot\frac{5}{2}\cos t\,dt$$
$$= \frac{1}{2}\int dt = \frac{1}{2}t + C$$
$$= \frac{1}{2}\arcsin\frac{2\left(x-\frac{5}{2}\right)}{5} + C,$$

where C is an arbitrary constant.

Corollary — Definite integrals by implicit substitution

Let I be an open interval, let $g : I \to \mathbb{R}$ be a one-to-one differentiable function such that g' is integrable on each closed subinterval of I, and let f be a function which is continuous on the range of g. Then for every pair of real numbers c and d in the range of g,

$$\int_c^d f(u)\,du = \int_{g^{-1}(c)}^{g^{-1}(d)} f(g(x))g'(x)\,dx.$$

Proof. This follows immediately from the substitution rule for definite integrals. Note that the numbers $g^{-1}(c)$ and $g^{-1}(d)$ on the right-hand side are well-defined since g is one-to-one. ■

Example — Definite integral by a trigonometric substitution

Evaluate

$$\int_0^{\sqrt{3}} \frac{1}{(x^2+9)^{\frac{3}{2}}}\,dx.$$

Solution. Let $x = 3\tan t$ where $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then

$$\sqrt{x^2+9} = \sqrt{(3\tan t)^2+9} = |3\sec t| = 3\sec t,$$

$dx = 3\sec^2 t\,dt$, $t = 0$ when $x = 0$, and $t = \frac{\pi}{6}$ when $x = \sqrt{3}$. So

$$\int_0^{\sqrt{3}} \frac{1}{(x^2+9)^{\frac{3}{2}}}\,dx = \int_0^{\frac{\pi}{6}} \frac{1}{(3\sec t)^3} 3\sec^2 t\,dt = \frac{1}{9}\int_0^{\frac{\pi}{6}} \cos t\,dt = \frac{1}{9}\left[\sin t\right]_0^{\frac{\pi}{6}} = \frac{1}{18}.$$

We emphasize the importance of working on an interval where the implicit substitution is one-to-one, as shown in the following two examples.

Example — A trigonometric substitution on the wrong interval

To evaluate the integral

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx,$$

we may use the substitution $x = \sin t$, so $dx = \cos t\,dt$. If we incorrectly take $t = \frac{\pi}{6}$ when $x = \frac{1}{2}$, and $t = \pi$ when $x = 0$, then we obtain

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx \overset{\times}{=} \int_\pi^{\frac{\pi}{6}} \cos t\,dt = \int_\pi^{\frac{\pi}{6}} dt = -\frac{5\pi}{6}.$$

In fact, on $\left[\frac{\pi}{6}, \pi\right]$ we have

$$\sqrt{1-x^2} = \sqrt{1-\sin^2 t} = |\cos t| = \begin{cases} \cos t & \text{if } t \in \left[\frac{\pi}{6}, \frac{\pi}{2}\right], \\ -\cos t & \text{if } t \in \left[\frac{\pi}{2}, \pi\right], \end{cases}$$

which is not always $\cos t$: the function sin is not one-to-one on $\left[\frac{\pi}{6}, \pi\right]$.

A correct way of using the implicit substitution $x = \sin t$ is to work on an interval

such as $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, on which sin is one-to-one. There we may take $t = \frac{\pi}{6}$ when $x = \frac{1}{2}$ and $t = 0$ when $x = 0$, and for $t \in \left[0, \frac{\pi}{6}\right]$ we always have $\sqrt{1-x^2} = \sqrt{1-\sin^2 t} = \cos t$, so

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx = \int_0^{\frac{\pi}{6}} \frac{1}{\cos t}\cos t\,dt = \int_0^{\frac{\pi}{6}} dt = \frac{\pi}{6}.$$

Example — A substitution not defined on the whole interval

With the implicit substitution $x = \frac{1}{t}$, one may incorrectly obtain

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx \overset{\times}{=} \int_{-1}^{1} \frac{1}{1+\frac{1}{t^2}}\left(-\frac{1}{t^2}\right)dt = -\int_{-1}^{1} \frac{1}{t^2+1}\,dt = -\int_{-1}^{1} \frac{1}{1+x^2}\,dx,$$

which would lead to the conclusion that $\int_{-1}^{1} \frac{1}{1+x^2}\,dx = 0$. This is incorrect, since

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx = [\arctan x]_{-1}^{1} = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}.$$

The mistake is that the function $g(t) = \frac{1}{t}$ is undefined at 0, so g is not well-defined on the whole interval $[-1, 1]$, although it is one-to-one on its domain.

If we insist that the lower limit of an integral has to be smaller than the upper limit, then the rule for implicit substitution takes the following form, which involves an absolute value in the factor $g'(t)$.

Theorem — Implicit substitution with absolute value

Let $g : [a, b] \to [c, d]$ be a bijective function which is continuously differentiable (i.e. of class C^1) on $[a, b]$, and let $f : [c, d] \to \mathbb{R}$ be a continuous function. Then

$$\int_c^d f(x)\,dx = \int_a^b f(g(t))\,|g'(t)|\,dt.$$

Proof. This follows from the substitution rule for definite integrals. Since g is one-to-one and continuous, it is either strictly increasing or strictly decreasing on $[a, b]$ ("bijective" means that every number in $[a, b]$ is sent to some number in $[c, d]$, every number in $[c, d]$ comes from some number in $[a, b]$, and different numbers in $[a, b]$ are sent to different numbers in $[c, d]$).
(i) If g is strictly increasing on $[a, b]$, then $g(a) = c$, $g(b) = d$ and $g' > 0$ on (a, b); hence the substitution $x = g(t)$ gives

$$\int_a^b f(g(t))g'(t)\,dt = \int_{g(a)}^{g(b)} f(x)\,dx = \int_c^d f(x)\,dx.$$

(ii) If g is strictly decreasing on $[a, b]$, then $g(a) = d$, $g(b) = c$ and $g' < 0$ on (a, b); hence the substitution $x = g(t)$ gives

$$\int_a^b f(g(t))g'(t)\,dt = \int_{g(a)}^{g(b)} f(x)\,dx = -\int_{g(b)}^{g(a)} f(x)\,dx = -\int_c^d f(x)\,dx$$

instead, which is compensated by the absolute value $|g'(t)| = -g'(t)$. ■

6.4 Integration of Rational Functions

Recall that a rational function is a quotient of polynomials. We have already seen examples of finding antiderivatives of a rational function. In this section we develop a systematic way of finding antiderivatives of all rational functions. The following important example covers the four types of building blocks of all rational functions.

Example — Building blocks

Evaluate the following:

(a) $\displaystyle\int \frac{5}{x-2}\,dx$;

(b) $\displaystyle\int \frac{5}{(x-2)^4}\,dx$;

(c) $\displaystyle\int \frac{4x-3}{x^2-2x+5}\,dx$;

(d) $\displaystyle\int \frac{4x-3}{(x^2-2x+5)^2}\,dx$.

Solution. (a)

$$\int \frac{5}{x-2}\,dx = 5\ln|x-2| + C,$$

where C is an arbitrary constant.

(b)

$$\int \frac{5}{(x-2)^4}\,dx = 5\frac{(x-2)^{-3}}{-3} + C = -\frac{5}{3(x-2)^3} + C,$$

where C is an arbitrary constant.

(c) Let $u = x^2-2x+5$. Then $du = (2x-2)dx$; we match the x term in the numerator with a multiple of du.

$$\begin{aligned}\int \frac{4x-3}{x^2-2x+5}\,dx &= \int \frac{4x-4}{x^2-2x+5}\,dx + \int \frac{1}{x^2-2x+5}\,dx \\ &= \int \frac{2}{u}\,du + \int \frac{1}{(x-1)^2+4}\,dx \\ = 2\ln|u| + \frac{1}{2}\arctan\frac{x-1}{2} + C &= 2\ln(x^2-2x+5) + \frac{1}{2}\arctan\frac{x-1}{2} + C,\end{aligned}$$

where C is an arbitrary constant.

(d) Again let $u = x^2-2x+5$, so $du = (2x-2)dx$, and match the x term in the numerator with a multiple of du:

$$\begin{aligned}\int \frac{4x-3}{(x^2-2x+5)^2}\,dx &= \int \frac{4x-4}{(x^2-2x+5)^2}\,dx + \int \frac{1}{(x^2-2x+5)^2}\,dx \\ &= \int \frac{2}{u^2}\,du + \int \frac{1}{[(x-1)^2+4]^2}\,dx \\ &= \frac{2u^{-1}}{-1} + \left[\frac{1}{2\cdot 4}\frac{x-1}{(x-1)^2+4} + \frac{1}{2\cdot 4}\int \frac{1}{(x-1)^2+4}\,dx\right] \\ &= \frac{x-17}{8(x^2-2x+5)} + \frac{1}{16}\arctan\frac{x-1}{2} + C,\end{aligned}$$

where C is an arbitrary constant. (We have used the reduction formula from the example on $\displaystyle\int (x^2+a^2)^{-n}dx$ above.)

Example — A first partial fraction decomposition

Evaluate

$$\int \frac{1}{1-x^2}\,dx.$$

Solution. Since $\dfrac{1}{1-x^2} = \dfrac{1}{(1-x)(1+x)}$, we hope to find two numbers a and b such that

$$\frac{1}{1-x^2} = \frac{a}{1-x} + \frac{b}{1+x}$$

if possible. If such numbers exist, they have to satisfy

$$\frac{1}{(1-x)(1+x)} = \frac{a(1+x)+b(1-x)}{(1-x)(1+x)} = \frac{(a-b)x+(a+b)}{(1-x)(1+x)}$$

for every x. In other words, we need $a-b=0$ and $a+b=1$. On solving this pair of equations we get $a=b=\dfrac{1}{2}$, so

$$\frac{1}{1-x^2} = \frac{1}{2}\left(\frac{1}{1-x}+\frac{1}{1+x}\right).$$

Therefore

$$\begin{aligned}\int \frac{1}{1-x^2}\,dx &= \frac{1}{2}\int\left(\frac{1}{1-x}+\frac{1}{1+x}\right)dx \\ &= -\frac{1}{2}\ln|1-x| + \frac{1}{2}\ln|1+x| + C \\ &= \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right| + C,\end{aligned}$$

where C is an arbitrary constant.

Remark — Partial fractions

Splitting a rational function such as $\dfrac{1}{1-x^2}$ into a sum of simpler "building blocks" $\dfrac{1}{2}\left(\dfrac{1}{1-x}+\dfrac{1}{1+x}\right)$ is called the method of partial fractions.

Example — A natural antiderivative of $\sec x$

Evaluate $\displaystyle\int \sec x\,dx$.

Solution. (Previously we began by writing $\sec x = \dfrac{\sec x(\sec x+\tan x)}{\sec x+\tan x}$ and then substituting $u=\sec x+\tan x$. The following gives a more natural solution.) Let $u=\sin x$. Then $du=\cos x\,dx$, so by the substitution rule and the previous example,

$$\begin{aligned}\int \sec x\,dx &= \int \frac{1}{\cos x}\,dx = \int \frac{\cos x}{\cos^2 x}\,dx = \int \frac{1}{1-\sin^2 x}\cos x\,dx \\ &= \int \frac{1}{1-u^2}\,du = \frac{1}{2}\ln\left|\frac{1+u}{1-u}\right| + C = \frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right| + C,\end{aligned}$$

where C is an arbitrary constant.

Think — Two formulas for $\int \sec x\,dx$

Why does the formula $\frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right|$ agree with $\ln|\sec x+\tan x|$ obtained earlier?

Example — Long division before partial fractions

Evaluate

$$\int \frac{x^3}{x^2-1}\,dx.$$

Solution. By "long division", $\dfrac{x^3}{x^2-1} = x + \dfrac{x}{x^2-1}$. Partial fraction decomposition of the latter term gives

$$\frac{x^3}{x^2-1} = x + \frac{1}{2}\cdot\frac{1}{x-1} + \frac{1}{2}\cdot\frac{1}{x+1}.$$

Therefore

$$\begin{aligned}\int \frac{x^3}{x^2-1}\,dx &= \int\left(x + \frac{1}{2}\cdot\frac{1}{x-1} + \frac{1}{2}\cdot\frac{1}{x+1}\right)dx \\ &= \frac{1}{2}x^2 + \frac{1}{2}\ln|x-1| + \frac{1}{2}\ln|x+1| + C = \frac{1}{2}x^2 + \frac{1}{2}\ln|x^2-1| + C,\end{aligned}$$

where C is an arbitrary constant.

Remark — Steps for a partial fraction decomposition

Consider a rational function $f(x) = \dfrac{p(x)}{q(x)}$, where p and q are polynomials. To obtain a partial fraction decomposition of f, adopt the following steps.
Step 0A: Cancel all common factors of p and q. From now on we may assume without loss of generality that p and q have no common factors.
Step 0B: Make sure that $\deg p < \deg q$. If $\deg p \geq \deg q$, perform "long division" and rewrite

$$f(x) = Q(x) + \frac{r(x)}{q(x)},$$

where the "quotient" Q is a polynomial and the "remainder" r is a polynomial with $\deg r < \deg q$.
Step 1: Factorize the denominator q into a product of linear factors and/or irreducible quadratic factors (degree 2 with negative discriminant). A typical example is

$$q(x) = a(x-b)(x-c)^3(x^2+dx+e)(x^2+mx+n)^2.$$

Step 2: Determine the form of the partial fraction decomposition according to the factorization of q.

- For each linear factor $(x-b)^k$ of q there corresponds a sum of the form

$$\frac{A_1}{x-b} + \frac{A_2}{(x-b)^2} + \cdots + \frac{A_k}{(x-b)^k}.$$

- For each quadratic factor $(x^2-bx+c)^k$ of q there corresponds a sum of the form

$$\frac{A_1x+B_1}{x^2-bx+c} + \frac{A_2x+B_2}{(x^2-bx+c)^2} + \cdots + \frac{A_kx+B_k}{(x^2-bx+c)^k}.$$

For example, the factorization $q(x) = a(x-b)(x-c)^3(x^2+dx+e)(x^2+mx+n)^2$ results in

$$f(x) = \frac{A}{x-b} + \frac{B}{x-c} + \frac{C}{(x-c)^2} + \frac{D}{(x-c)^3} + \frac{Ex+F}{x^2+dx+e} + \frac{Gx+H}{x^2+mx+n} + \frac{Ix+J}{(x^2+mx+n)^2},$$

where $A, B, C, \ldots, I, J$ are unknown numbers to be determined in the next step.
Step 3: Clear the denominator of the partial fraction decomposition to obtain a polynomial identity from which the unknowns can be determined. There are various ways to find these unknowns, such as comparing coefficients, substituting convenient particular values of x, or differentiating both sides of the identity.

After obtaining the partial fraction decomposition of the rational function f, we can evaluate $\int f(x)\,dx$ using the "building blocks" seen above.

Example — A repeated linear factor

Evaluate the integral

$$\int \frac{x+1}{x(x-1)^3}\,dx.$$

Solution. To get the partial fraction decomposition of $\dfrac{x+1}{x(x-1)^3}$, write

$$\frac{x+1}{x(x-1)^3} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{D}{(x-1)^3}.$$

This gives

$$\frac{x+1}{x(x-1)^3} = \frac{A(x-1)^3 + Bx(x-1)^2 + Cx(x-1) + Dx}{x(x-1)^3},$$

so we require that

$$\begin{aligned} x+1 &= A(x-1)^3 + Bx(x-1)^2 + Cx(x-1) + Dx \\ &= (A+B)x^3 + (-3A-2B+C)x^2 + (3A+B-C+D)x - A. \end{aligned}$$

Comparing coefficients we obtain the system

$$\begin{cases} A+B=0, \\ -3A-2B+C=0, \\ 3A+B-C+D=1, \\ -A=1, \end{cases}$$

and solving it gives $A=-1$, $B=1$, $C=-1$ and $D=2$. Therefore

$$\frac{x+1}{x(x-1)^3} = -\frac{1}{x} + \frac{1}{x-1} - \frac{1}{(x-1)^2} + \frac{2}{(x-1)^3},$$

and so

$$\begin{aligned} \int \frac{x+1}{x(x-1)^3}\,dx &= \int \left[-\frac{1}{x} + \frac{1}{x-1} - \frac{1}{(x-1)^2} + \frac{2}{(x-1)^3}\right] dx \\ &= -\ln|x| + \ln|x-1| + \frac{1}{x-1} - \frac{1}{(x-1)^2} + C \\ &= \ln\left|\frac{x-1}{x}\right| + \frac{x-2}{(x-1)^2} + C, \end{aligned}$$

where C is an arbitrary constant.

Example — Two irreducible quadratic factors

Evaluate the integral

$$\int \frac{x^5+4x^4+6x^3+8x^2+9x}{(x^2+1)(x^2+4x+5)}\,dx.$$

Solution. Long division gives

$$\frac{x^5+4x^4+6x^3+8x^2+9x}{(x^2+1)(x^2+4x+5)} = \frac{x^5+4x^4+6x^3+8x^2+9x}{x^4+4x^3+6x^2+4x+5} = x+\frac{4x^2+4x}{(x^2+1)(x^2+4x+5)},$$

so we first aim to get the partial fraction decomposition of the latter term. We write

$$\frac{4x^2+4x}{(x^2+1)(x^2+4x+5)} = \frac{Ax+B}{x^2+1}+\frac{Cx+D}{x^2+4x+5}.$$

This gives

$$\begin{aligned} 4x^2+4x &= (Ax+B)(x^2+4x+5)+(Cx+D)(x^2+1)\\ &= (A+C)x^3+(4A+B+D)x^2+(5A+4B+C)x+(5B+D), \end{aligned}$$

so we require that

$$\begin{cases} A+C=0,\\ 4A+B+D=4,\\ 5A+4B+C=4,\\ 5B+D=0, \end{cases}$$

and on solving this system we get $A=1$, $B=0$, $C=-1$ and $D=0$. Therefore

$$\begin{aligned} \int \frac{x^5+4x^4+6x^3+8x^2+9x}{(x^2+1)(x^2+4x+5)}\,dx &= \int\left(x+\frac{x}{x^2+1}-\frac{x}{x^2+4x+5}\right)dx\\ &= \int\left[x+\frac{1}{2}\frac{2x}{x^2+1}-\frac{1}{2}\frac{(2x+4)-2}{x^2+4x+5}\right]dx \end{aligned}$$

$$= \int x\,dx+\frac{1}{2}\int\frac{1}{x^2+1}\,d(x^2+1)-\frac{1}{2}\int\frac{1}{x^2+4x+5}\,d(x^2+4x+5)+2\int\frac{1}{(x+2)^2+1}\,dx$$

$$\begin{aligned} &= \frac{1}{2}x^2+\frac{1}{2}\ln(x^2+1)-\frac{1}{2}\ln(x^2+4x+5)+2\arctan(x+2)+C\\ &= \frac{1}{2}x^2+\frac{1}{2}\ln\frac{x^2+1}{x^2+4x+5}+2\arctan(x+2)+C, \end{aligned}$$

where C is an arbitrary constant.

6.5 Trigonometric Rational Functions

In the previous section we saw how to find antiderivatives of rational functions. Here we study a powerful method to reduce integrals of trigonometric rational functions into integrals of rational functions. A trigonometric rational function is a combination of the six trigonometric functions (sin, cos, tan, cot, sec, csc) using the four arithmetic operations $(+,-,\times,\div)$. Such a function can always be reduced to a combination of just $\cos x$ and $\sin x$; for example,

$$\frac{\sin x\sec^2 x+2\csc x}{3\cot x-1} = \frac{\sin^2 x+2\cos^2 x}{3\cos^3 x-\sin x\cos^2 x}.$$

Remark — Which substitution for $R(\cos x, \sin x)$?

Let $f(x) = R(\cos x, \sin x)$ be a trigonometric rational function.

- If $R(-\cos x, -\sin x) = R(\cos x, \sin x)$, i.e. the expression is unchanged when we replace $\cos x$ and $\sin x$ by $-\cos x$ and $-\sin x$ simultaneously, then to evaluate $\displaystyle\int f(x)\,dx$ we let
$$t = \tan x.$$
With this (explicit) substitution we have $dt = \sec^2 x\,dx$ and $dx = \dfrac{1}{1+t^2}\,dt$, which reduces the problem to finding the antiderivative of a rational function in t.
- If $R(-\cos x, \sin x) = -R(\cos x, \sin x)$, i.e. the expression is odd in the $\cos x$ variable, then we may let $u = \sin x$.
- If $R(\cos x, -\sin x) = -R(\cos x, \sin x)$, i.e. the expression is odd in the $\sin x$ variable, then we may let $u = \cos x$.

Example — Substitution $t = \tan x$

Evaluate
$$\int \frac{1}{\cos^2 x + 3\sin x \cos x + 2\sin^2 x}\,dx.$$

Solution. Let $t = \tan x$. Then $dt = \sec^2 x\,dx$, so
$$\begin{aligned}\int \frac{1}{\cos^2 x + 3\sin x \cos x + 2\sin^2 x}\,dx &= \int \frac{\sec^2 x}{1 + 3\tan x + 2\tan^2 x}\,dx \\ &= \int \frac{1}{1+3t+2t^2}\,dt = \int \left(\frac{2}{1+2t} - \frac{1}{1+t}\right) dt \\ = \ln|1+2t| - \ln|1+t| + C &= \ln\left|\frac{1+2\tan x}{1+\tan x}\right| + C = \ln\left|\frac{\cos x + 2\sin x}{\cos x + \sin x}\right| + C,\end{aligned}$$
where C is an arbitrary constant.

The following Weierstrass t-substitution is useful for any trigonometric rational function in general, regardless of whether it is odd in either the $\cos x$ or $\sin x$ variable.

Remark — Weierstrass' t-substitution

Let $f(x) = R(\cos x, \sin x)$ be any trigonometric rational function. To find $\displaystyle\int f(x)\,dx$ we may let
$$t = \tan\frac{x}{2}$$
where $x \in (-\pi, \pi)$. With this (explicit) substitution,
$$\cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}, \qquad \text{and} \qquad dx = \frac{2}{1+t^2}\,dt,$$
which reduces the problem to finding the antiderivative of a rational function in t.

Example — A Weierstrass substitution

Evaluate
$$\int \frac{1}{2+\cos x}\,dx$$
for $x \in (-\pi, \pi)$.

Solution. Let $t = \tan\dfrac{x}{2}$ where $x \in (-\pi, \pi)$. Then $\cos x = \dfrac{1-t^2}{1+t^2}$ and $dx = \dfrac{2}{1+t^2}\,dt$, so

$$\begin{aligned}\int \frac{1}{2+\cos x}\,dx &= \int \frac{1}{2+\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2}\,dt = \int \frac{2}{t^2+3}\,dt \\ &= \frac{2}{\sqrt{3}} \arctan\frac{t}{\sqrt{3}} + C = \frac{2}{\sqrt{3}} \arctan\left(\frac{1}{\sqrt{3}}\tan\frac{x}{2}\right) + C,\end{aligned}$$

where C is an arbitrary constant.

Remark — A discontinuous antiderivative

Although the function $(2+\cos x)^{-1}$ is continuous on $\mathbb{R}$, the antiderivative found above is discontinuous at $(2n+1)\pi$ for each $n \in \mathbb{Z}$. This subtle problem arises because the substitution $t = \tan(\dfrac{x}{2})$ is differentiable only on intervals such as $(-\pi, \pi)$, but not on $\mathbb{R}$. To alleviate this problem, we may observe (using trigonometric identities from chapter 1) that

$$\arctan\left(\frac{1}{\sqrt{3}}\tan\frac{x}{2}\right) = \frac{x}{2} - \arctan\frac{\sin x}{2+\sqrt{3}+\cos x} \qquad \text{for } x \in (-\pi, \pi);$$

this provides an antiderivative of $(2+\cos x)^{-1}$ that is valid on $\mathbb{R}$:

$$\int \frac{1}{2+\cos x}\,dx = \frac{x}{\sqrt{3}} - \frac{2}{\sqrt{3}} \arctan\frac{\sin x}{2+\sqrt{3}+\cos x} + C,$$

where C is an arbitrary constant.

Example — Simplifying before the Weierstrass substitution

Evaluate

$$\int \frac{\cos x + 2}{\cos x + \sin x}\,dx.$$

Solution. If we apply the Weierstrass substitution $t = \tan(\dfrac{x}{2})$ right away, we may end up with a very complicated rational function in t, so we simplify first. Let $u = \cos x + \sin x$. Then $du = (\cos x - \sin x)dx$, so matching the non-constant terms in the numerator with multiples of u or du,

$$\begin{aligned}\int \frac{\cos x + 2}{\cos x + \sin x}\,dx &= \int \frac{\frac{1}{2}(\cos x + \sin x) + \frac{1}{2}(\cos x - \sin x) + 2}{\cos x + \sin x}\,dx \\ &= \frac{1}{2}\int dx + \frac{1}{2}\int \frac{du}{u} + 2\int \frac{1}{\cos x + \sin x}\,dx.\end{aligned}$$

The first two terms are easy; the last term can be handled using the Weierstrass substitution $t = \tan(\dfrac{x}{2})$:

$$\int \frac{2}{\cos x + \sin x}\,dx = \int \frac{2}{\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2}\,dt = -4\int \frac{1}{(t-1)^2 - 2}\,dt$$

$$= -\frac{4}{2\sqrt{2}} \int \left(\frac{1}{t-1-\sqrt{2}} - \frac{1}{t-1+\sqrt{2}}\right) dt = \sqrt{2}\ln\left|\frac{t-1+\sqrt{2}}{t-1-\sqrt{2}}\right| + C,$$

where C is an arbitrary constant. Consequently,

$$\int \frac{\cos x + 2}{\cos x + \sin x}\,dx = \frac{1}{2}x + \frac{1}{2}\ln|u| + \sqrt{2}\ln\left|\frac{t-1+\sqrt{2}}{t-1-\sqrt{2}}\right| + C$$

$$= \frac{1}{2}x + \frac{1}{2}\ln|\cos x + \sin x| + \sqrt{2}\ln\left|\frac{\tan(\frac{x}{2})-1+\sqrt{2}}{\tan(\frac{x}{2})-1-\sqrt{2}}\right| + C,$$

where C is an arbitrary constant.

Remark — Antiderivatives that are not elementary

Up to this point we have learnt various techniques of finding antiderivatives: trigonometric identities, explicit and implicit substitutions, integration by parts, partial fractions, and so on. But there are functions which are constructed from elementary functions (via $+, -, \times, \div, \circ$) whose antiderivatives can never be expressed using elementary functions. Some notorious examples are

$$\int e^{x^2}\,dx, \qquad \int x^x\,dx, \qquad \int \frac{\sin x}{x}\,dx, \qquad \int \sqrt{1+x^4}\,dx,$$

$$\int \frac{1}{\ln x}\,dx, \qquad \int e^{e^x}\,dx, \qquad \int \ln\ln x\,dx, \qquad \int \sqrt{4-\sin^2 x}\,dx.$$

This makes antidifferentiation very different from differentiation, as the rules of differentiation are always sufficient for computing the derivative of a function constructed from elementary functions. Note however that if we only need the integral of such a function over a certain interval, then numerical approximation becomes useful (see the supplement on numerical integration below).

6.6 Improper Integrals

So far we have studied the integrals of an integrable (bounded) function on a bounded closed interval. The notion of integrals can be slightly extended to the cases of an unbounded interval or an unbounded function, provided that the "area" of such an "unbounded region" is still finite.

Definition (Unbounded interval). Let a be a real number and let f be a function.

- If f is integrable on $[a,t]$ for every $t \geq a$, then the improper integral of f over $[a,+\infty)$ is defined by the limit

$$\int_a^{+\infty} f(x)\,dx := \lim_{t\to+\infty}\int_a^t f(x)\,dx.$$

- Similarly, if f is integrable on $[t,a]$ for every $t \leq a$, then we also define

$$\int_{-\infty}^a f(x)\,dx := \lim_{t\to-\infty}\int_t^a f(x)\,dx.$$

- Each of the above improper integrals is said to converge if the limit exists as a real number; otherwise we say that the improper integral diverges.
- If the improper integrals $\displaystyle\int_a^{+\infty} f(x)\,dx$ and $\displaystyle\int_{-\infty}^a f(x)\,dx$ both converge, then we define

$$\int_{-\infty}^{+\infty} f(x)\,dx := \int_{-\infty}^a f(x)\,dx + \int_a^{+\infty} f(x)\,dx = \lim_{t\to-\infty}\int_t^a f(x)\,dx + \lim_{s\to+\infty}\int_a^s f(x)\,dx.$$

Note that the sum on the right is independent of the choice of the "intermediate point"

a.

(Figure: the region under $y = f(x)$ over $[a, +\infty)$, whose area is the improper integral $\int_a^{+\infty} f(x)\,dx$.)

Example — Two basic improper integrals

Evaluate each of the following if it converges:

(a) $\int_0^{+\infty} 1\,dx$;

(b) $\int_0^{+\infty} e^{-x}dx$.

Solution. (a) It is obvious that

$$\lim_{t\to+\infty}\int_0^t 1\,dx = \lim_{t\to+\infty} t = +\infty,$$

so the improper integral $\int_0^{+\infty} 1\,dx$ diverges.

(b) Since

$$\lim_{t\to+\infty}\int_0^t e^{-x}dx = \lim_{t\to+\infty}\left[-e^{-x}\right]_0^t = \lim_{t\to+\infty}\left(1-e^{-t}\right) = 1,$$

the improper integral $\int_0^{+\infty} e^{-x}dx$ converges and equals 1.

Example — p-test

Let p be a real number. Evaluate

$$\int_1^{+\infty} \frac{1}{x^p}\,dx$$

if it converges.

Solution. For each positive number t, we have

$$\int_1^t \frac{1}{x^p}\,dx = \begin{cases} \left[\dfrac{x^{1-p}}{1-p}\right]_1^t = \dfrac{1}{p-1}\left(1-\dfrac{1}{t^{p-1}}\right) & \text{if } p\neq 1,\\ [\ln x]_1^t = \ln t & \text{if } p = 1,\end{cases}$$

so

$$\lim_{t\to+\infty}\int_1^t \frac{1}{x^p}\,dx = \begin{cases}\dfrac{1}{p-1} & \text{if } p>1,\\ +\infty & \text{if } p\le 1.\end{cases}$$

So the improper integral $\int_1^{+\infty} \frac{1}{x^p}\,dx$ converges only when $p > 1$, and in this case it equals $\dfrac{1}{p-1}$.

Example — An integral improper at both ends

Evaluate

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2}\,dx$$

if it converges.

Solution. The integral is improper both at $-\infty$ and at $+\infty$; consider the "intermediate point" 0. We have

$$\lim_{t\to+\infty}\int_0^t \frac{1}{1+x^2}\,dx = \lim_{t\to+\infty}\left[\arctan x\right]_0^t = \lim_{t\to+\infty}\arctan t = \frac{\pi}{2}$$

and

$$\lim_{s\to-\infty}\int_s^0 \frac{1}{1+x^2}\,dx = \lim_{s\to-\infty}\left[\arctan x\right]_s^0 = \lim_{s\to-\infty}(-\arctan s) = -\left(-\frac{\pi}{2}\right) = \frac{\pi}{2},$$

so $\displaystyle\int_0^{+\infty}\frac{1}{1+x^2}\,dx = \frac{\pi}{2}$ and $\displaystyle\int_{-\infty}^0\frac{1}{1+x^2}\,dx = \frac{\pi}{2}$. Therefore

$$\int_{-\infty}^{+\infty}\frac{1}{1+x^2}\,dx = \frac{\pi}{2}+\frac{\pi}{2} = \pi.$$

Remark — Cauchy principal value

In the example above we do NOT write

$$\int_{-\infty}^{+\infty}\frac{1}{1+x^2}\,dx \overset{\times}{=} \lim_{t\to+\infty}\int_{-t}^{t}\frac{1}{1+x^2}\,dx$$

although this happens to still give the correct answer. The limit $\displaystyle\lim_{t\to+\infty}\int_{-t}^{t} f(x)\,dx$ is called the Cauchy principal value of the improper integral $\displaystyle\int_{-\infty}^{+\infty} f(x)\,dx$. In general, even when $\displaystyle\int_{-\infty}^{+\infty} f(x)\,dx$ diverges, its Cauchy principal value may still exist.

Remark — Integration by parts for improper integrals

Integration by parts is still valid for improper integrals. For example,

$$\begin{aligned}\int_{-\infty}^0 xe^x\,dx &= \lim_{t\to-\infty}\int_t^0 xe^x\,dx\\ &= \lim_{t\to-\infty}\left(\left[xe^x\right]_t^0 - \int_t^0 e^x\,dx\right)\\ &= \lim_{t\to-\infty}\left(-te^t - 1 + e^t\right) = -1,\end{aligned}$$

in which the first two steps can also be formally written as

$$\int_{-\infty}^0 xe^x\,dx = \left[xe^x\right]_{-\infty}^0 - \int_{-\infty}^0 e^x\,dx$$

without causing any confusion. Note however that in a notation like $[f(x)]_{-\infty}^{+\infty}$, the limits at the two infinities have to be taken separately and independently:

$$[f(x)]_{-\infty}^{+\infty} = \lim_{s\to+\infty} f(s) - \lim_{t\to-\infty} f(t).$$

Example — Substitution rule for improper integrals

Let $a > 1$ and let $p \in \mathbb{R}$. Then for every $t > a$, the substitution $u = \ln x$ gives

$$\int_a^t \frac{1}{x(\ln x)^p}\,dx = \int_{\ln a}^{\ln t} \frac{1}{u^p}\,du,$$

so

$$\int_a^{+\infty} \frac{1}{x(\ln x)^p}\,dx = \lim_{t\to+\infty}\int_a^t \frac{1}{x(\ln x)^p}\,dx = \lim_{t\to+\infty}\int_{\ln a}^{\ln t} \frac{1}{u^p}\,du = \int_{\ln a}^{+\infty} \frac{1}{u^p}\,du.$$

By the p-test above, the improper integral $\displaystyle\int_{\ln a}^{+\infty} \frac{1}{u^p}\,du$ converges if and only if $p > 1$; so $\displaystyle\int_a^{+\infty} \frac{1}{x(\ln x)^p}\,dx$ also converges if and only if $p > 1$. Moreover, when $p > 1$,

$$\int_a^{+\infty} \frac{1}{x(\ln x)^p}\,dx = \int_{\ln a}^{+\infty} \frac{1}{u^p}\,du = \left[\frac{1}{1-p}u^{1-p}\right]_{\ln a}^{+\infty} = \frac{1}{(p-1)(\ln a)^{p-1}}.$$

This example shows that the substitution rule for integration is still valid for improper integrals.

Example — A periodic integrand on an unbounded domain

Evaluate

$$\int_0^{2\pi} \frac{1}{2+\cos x}\,dx.$$

Solution. Let $t = \tan\dfrac{x}{2}$. We have seen in the Weierstrass example above that

$$\int \frac{1}{2+\cos x}\,dx = \int \frac{2}{t^2+3}\,dt.$$

However, the interval of integration $[0, 2\pi]$ does not lie completely within any possible domain of the substitution. To alleviate this problem we evaluate the given integral separately over $[0, \pi)$ and $(\pi, 2\pi]$, so that on each of these intervals the substitution is valid. Thus

$$\begin{aligned}
\int_0^{2\pi} \frac{1}{2+\cos x}\,dx &= \int_0^{\pi} \frac{1}{2+\cos x}\,dx + \int_{\pi}^{2\pi} \frac{1}{2+\cos x}\,dx \\
&= \int_0^{+\infty} \frac{2}{t^2+3}\,dt + \int_{-\infty}^{0} \frac{2}{t^2+3}\,dt \\
&= \int_{-\infty}^{+\infty} \frac{2}{t^2+3}\,dt = \frac{2}{\sqrt{3}}\int_{-\infty}^{+\infty} \frac{1}{u^2+1}\,du \\
&= \frac{2\pi}{\sqrt{3}},
\end{aligned}$$

where the last step uses the substitution $u = \dfrac{t}{\sqrt{3}}$ and the result of the example on $\displaystyle\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$ above.

Definition (Unbounded integrand). Let $a < b$ be real numbers and let f be a function.

- If f is integrable on $[t, b]$ for every $a < t \leq b$ and if f is unbounded at a, then the

improper integral of f over $(a, b]$ is defined by

$$\int_a^b f(x)\,dx := \lim_{t\to a^+} \int_t^b f(x)\,dx.$$

- Similarly, if f is integrable on $[a, t]$ for every $a \le t < b$ and if f is unbounded at b, then we also define

$$\int_a^b f(x)\,dx := \lim_{t\to b^-} \int_a^t f(x)\,dx.$$

- Each of the above improper integrals is said to converge if the limit exists as a real number; otherwise we say that it diverges.
- If f is unbounded at $c \in (a, b)$ and if the improper integrals $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ both converge, then we define

$$\int_a^b f(x)\,dx := \int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \lim_{t\to c^-} \int_a^t f(x)\,dx + \lim_{s\to c^+} \int_s^b f(x)\,dx.$$

- If f is unbounded at a and at b, and if the improper integrals $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ both converge for some $c \in (a, b)$, then we also define

$$\int_a^b f(x)\,dx := \int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \lim_{t\to a^+} \int_t^c f(x)\,dx + \lim_{s\to b^-} \int_c^s f(x)\,dx.$$

(Figure: the region under $y = f(x)$ over $[a, b]$, where f blows up at a; its area is the improper integral $\int_a^b f(x)\,dx$.)

Example — Unbounded integrands

Evaluate each of the following improper integrals if it converges:

(a) $\displaystyle\int_0^1 \frac{1}{x}\,dx$;

(b) $\displaystyle\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$.

Solution. (a) With the substitution $u = \dfrac{1}{x}$,

$$\int_0^1 \frac{1}{x}\,dx = \int_{+\infty}^1 \frac{1}{\frac{1}{u}}\left(-\frac{1}{u^2}\right)du = \int_1^{+\infty} \frac{1}{u}\,du$$

diverges by the p-test above.

(b) Since the integrand $\dfrac{1}{(x-1)^{\frac{2}{3}}}$ is undefined at 1, the improper integral is defined as

$$\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = \int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx + \int_1^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx.$$

Now on the right-hand side,

$$\begin{aligned}\int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx &= \lim_{t\to 1^-}\int_0^t \frac{1}{(x-1)^{\frac{2}{3}}}\,dx \\ &= \lim_{t\to 1^-}\left[3(x-1)^{\frac{1}{3}}\right]_0^t \\ &= \lim_{t\to 1^-}\left[3(t-1)^{\frac{1}{3}}+3\right] = 3\end{aligned}$$

and

$$\begin{aligned}\int_1^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx &= \lim_{s\to 1^+}\int_s^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx \\ &= \lim_{s\to 1^+}\left[3(x-1)^{\frac{1}{3}}\right]_s^3 \\ &= \lim_{s\to 1^+}\left[3\cdot 2^{\frac{1}{3}} - 3(s-1)^{\frac{1}{3}}\right] = 3\cdot 2^{\frac{1}{3}}.\end{aligned}$$

Therefore the improper integral converges and

$$\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = 3 + 3\cdot 2^{\frac{1}{3}}.$$

Remark — The integral $\displaystyle\int_{-1}^1 \frac{dx}{x}$

Consider the integral $\displaystyle\int_{-1}^1 \frac{1}{x}\,dx$.

- It is certainly incorrect to write

$$\int_{-1}^1 \frac{1}{x}\,dx \overset{\times}{=} [\ln|x|]_{-1}^1 = \ln|1| - \ln|-1| = 0,$$

  since the function is unbounded near 0 and not integrable on $[-1,1]$.
- More importantly, it is still incorrect to write

$$\begin{aligned}\int_{-1}^1 \frac{1}{x}\,dx &\overset{\times}{=} \lim_{t\to 0}\left(\int_{-1}^t \frac{1}{x}\,dx + \int_t^1 \frac{1}{x}\,dx\right) = \lim_{t\to 0}\left([\ln|x|]_{-1}^t + [\ln|x|]_t^1\right) \\ &= \lim_{t\to 0}(\ln|t| - \ln 1 + \ln 1 - \ln|t|) = \lim_{t\to 0} 0 = 0,\end{aligned}$$

  as the improper integrals $\displaystyle\int_{-1}^0 \frac{1}{x}\,dx$ and $\displaystyle\int_0^1 \frac{1}{x}\,dx$ should be treated as two independent limits. Using the same variable t in both limits is incorrect.

Sometimes it is difficult or impossible to find the exact value of a given improper integral, but we still want tools to determine whether it converges or diverges. These tools are called convergence tests for improper integrals.

Theorem — Comparison test

Let (a,b) be a (bounded or unbounded) interval, and let $f, g : (a,b) \to [0,+\infty)$ be continuous functions such that $f(x) \geq g(x) \geq 0$ for all $x \in (a,b)$.

(i) If $\displaystyle\int_a^b f(x)\,dx$ converges, then $\displaystyle\int_a^b g(x)\,dx$ also converges.

(ii) If $\int_a^b g(x)\,dx$ diverges, then $\int_a^b f(x)\,dx$ also diverges.

Proof. Graphically obvious (the region under g is contained in the region under f). A rigorous proof will be given in MATH2033. ■

Example — Convergence of the Gaussian integral

Determine whether the improper integral

$$\int_0^{+\infty} e^{-x^2}dx$$

converges or not.

Solution. Since the exponential function is increasing on $\mathbb{R}$, it follows that $e^{-x^2} \geq e^{-x} > 0$ for every $x \geq 1$. We know from the basic improper integral example above that $\int_1^{+\infty} e^{-x}dx$ converges, so $\int_1^{+\infty} e^{-x^2}dx$ converges by the comparison test. Finally,

$$\int_0^{+\infty} e^{-x^2}dx = \int_0^1 e^{-x^2}dx + \int_1^{+\infty} e^{-x^2}dx$$

also converges.

Remark — The exact value of the Gaussian integral

In fact we have the exact value

$$\int_0^{+\infty} e^{-x^2}dx = \frac{\sqrt{\pi}}{2}.$$

This can be proved using multivariable calculus (alternatively, see the supplement on Feynman's trick below).

Example — A divergent integral by comparison

Evaluate

$$\int_0^{\frac{\pi}{2}} (\csc x)^{\frac{3}{2}}dx$$

if it converges.

Solution. The given integral is improper at 0 only. Since $\sin x < x$ for every $x \in \left(0, \frac{\pi}{2}\right]$ (cf. chapter 2),

$$(\csc x)^{\frac{3}{2}} = \frac{1}{(\sin x)^{\frac{3}{2}}} > \frac{1}{x^{\frac{3}{2}}} > 0 \qquad \text{for every } x \in \left(0, \frac{\pi}{2}\right].$$

But the improper integral

$$\int_0^{\frac{\pi}{2}} \frac{1}{x^{\frac{3}{2}}}\,dx = \int_{+\infty}^{\frac{2}{\pi}} u^{\frac{3}{2}} \left(-\frac{1}{u^2}\right) du = \int_{\frac{2}{\pi}}^{+\infty} \frac{1}{u^{\frac{1}{2}}}\,du$$

diverges by the p-test (with $p = \frac{1}{2} \leq 1$; here we used the substitution $u = \frac{1}{x}$), so $\int_0^{\frac{\pi}{2}} (\csc x)^{\frac{3}{2}}dx$ also diverges by the comparison test.

Example — Another divergent integral

Let n be a positive integer. Evaluate

$$\int_0^1 \frac{1}{1-x^n}\,dx$$

if it converges.

Solution. The given integral is improper at 1 only. For every $x \in [0,1)$ we have

$$\frac{1}{1-x^n} = \frac{1}{1-x}\cdot\frac{1}{1+x+x^2+\cdots+x^{n-1}} > \frac{1}{1-x}\cdot\frac{1}{1+1+1+\cdots+1} = \frac{1}{n}\cdot\frac{1}{1-x} > 0.$$

Since the improper integral

$$\int_0^1 \frac{1}{n}\cdot\frac{1}{1-x}\,dx = \frac{1}{n}\int_1^{+\infty}\frac{1}{u}\,du$$

(the substitution $u = 1-x$) diverges by the example on $\displaystyle\int_0^1 \frac{1}{x}\,dx$ above, it follows that $\displaystyle\int_0^1 \frac{1}{1-x^n}\,dx$ also diverges by the comparison test.

Corollary — Tail test for divergence

Let $a \in \mathbb{R}$ and let $f : [a,+\infty) \to \mathbb{R}$ be a continuous function. If $\lim\limits_{x\to+\infty} f(x)$ exists and is non-zero, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ diverges.

Proof. Denote $\lim\limits_{x\to+\infty} f(x) = L \neq 0$ and suppose that $L > 0$. Let m be a number in the interval $(0, L)$ (if $L = +\infty$ we may choose m to be any positive real number; if $L \in (0,+\infty)$ we may simply choose $m = \dfrac{L}{2}$). Then by the definition of limit there exists $b \geq a$ such that

$$f(x) \geq m > 0 \qquad \text{for every } x \geq b.$$

Since the improper integral $\displaystyle\int_b^{+\infty} m\,dx$ diverges, $\displaystyle\int_b^{+\infty} f(x)\,dx$ also diverges by the comparison test. Thus $\displaystyle\int_a^{+\infty} f(x)\,dx = \int_a^b f(x)\,dx + \int_b^{+\infty} f(x)\,dx$ also diverges. The case $L < 0$ is similar. ■

Theorem — Absolute convergence test

Let (a,b) be a (bounded or unbounded) interval and let $f : (a,b) \to \mathbb{R}$ be a continuous function. If the improper integral $\displaystyle\int_a^b |f(x)|\,dx$ converges, then the improper integral $\displaystyle\int_a^b f(x)\,dx$ also converges.

Proof. Let $f^+ := \max\{f, 0\}$ and $f^- := \max\{-f, 0\}$. Then f^+ and f^- are both non-negative continuous functions. Since $|f| \geq f^+ \geq 0$ and $\displaystyle\int_a^b |f(x)|\,dx$ converges,

$\int_a^b f^+(x)\,dx$ converges by the comparison test. In a similar way, $\int_a^b f^-(x)\,dx$ also converges. Finally, since $f = f^+ - f^-$,

$$\int_a^b f(x)\,dx = \int_a^b f^+(x)\,dx - \int_a^b f^-(x)\,dx$$

also converges. ■

Remark — Absolute convergence

We say that the improper integral $\int_a^b f(x)\,dx$ converges absolutely if $\int_a^b |f(x)|\,dx$ converges. Thus the absolute convergence test simply says that "if an improper integral converges absolutely, then it converges".

Example — Tail test in action

Determine whether the improper integral

$$\int_1^{+\infty} x \sin\frac{1}{x}\,dx$$

converges or not.

Solution. The integral is improper at $+\infty$ only. With a change of variable $t = \dfrac{1}{x}$,

$$\lim_{x\to+\infty} x\sin\frac{1}{x} = \lim_{t\to 0^+}\frac{\sin t}{t} = 1.$$

This limit exists and is non-zero, so $\int_1^{+\infty} x\sin\frac{1}{x}\,dx$ diverges by the tail test.

Example — Absolute convergence of an oscillatory integral

Determine whether the improper integral

$$\int_1^{+\infty} \frac{\sin x}{x^2}\,dx$$

converges or not.

Solution. The integral is improper at $+\infty$ only. Since $|\sin x| \le 1$ for every x,

$$0 \le \left|\frac{\sin x}{x^2}\right| \le \frac{1}{x^2} \qquad \text{for every } x \ge 1.$$

We know that $\int_1^{+\infty} \frac{1}{x^2}\,dx$ converges by the p-test, so $\int_1^{+\infty} \left|\frac{\sin x}{x^2}\right| dx$ converges by the comparison test. Consequently $\int_1^{+\infty} \frac{\sin x}{x^2}\,dx$ also converges by the absolute convergence test.

Example — $\displaystyle\int_1^{+\infty} \sin(x^2)\,dx$

Determine whether the improper integral

$$\int_1^{+\infty} \sin(x^2)\,dx$$

converges or not.

Solution. For every $t > 1$, after using the substitution $u = x^2$, integration by parts gives

$$\int_1^t \sin(x^2)\,dx = \int_1^{t^2} \frac{\sin u}{2\sqrt{u}}\,du = \left[-\frac{\cos u}{2\sqrt{u}}\right]_1^{t^2} - \int_1^{t^2} (-\cos u)\left(-\frac{1}{4u^{\frac{3}{2}}}\right) du$$

$$= \frac{\cos 1}{2} - \frac{\cos(t^2)}{2t} - \frac{1}{4}\int_1^{t^2} \frac{\cos u}{u^{\frac{3}{2}}}\,du.$$

Now since $|\cos u| \leq 1$ for every u,

$$0 \leq \left|\frac{\cos u}{u^{\frac{3}{2}}}\right| \leq \frac{1}{u^{\frac{3}{2}}} \qquad \text{for every } u \geq 1.$$

We know that $\displaystyle\int_1^{+\infty} \frac{1}{u^{\frac{3}{2}}}\,du$ converges by the p-test, so $\displaystyle\int_1^{+\infty} \left|\frac{\cos u}{u^{\frac{3}{2}}}\right| du$ converges by the comparison test. Thus $\displaystyle\int_1^{+\infty} \frac{\cos u}{u^{\frac{3}{2}}}\,du$ also converges by the absolute convergence test. Finally, the limit

$$\lim_{t\to+\infty} \int_1^t \sin(x^2)\,dx = \frac{\cos 1}{2} - \lim_{t\to+\infty} \frac{\cos(t^2)}{2t} - \frac{1}{4}\lim_{t\to+\infty} \int_1^{t^2} \frac{\cos u}{u^{\frac{3}{2}}}\,du$$

$$= \frac{\cos 1}{2} - \frac{1}{4}\int_1^{+\infty} \frac{\cos u}{u^{\frac{3}{2}}}\,du$$

exists as a real number, so the improper integral $\displaystyle\int_1^{+\infty} \sin(x^2)\,dx$ converges.

Remark — When the tail test fails

The example above shows that the tail test does not work if $\lim\limits_{x\to+\infty} f(x)$ does not exist. Even if $\lim\limits_{x\to+\infty} f(x)$ does not exist, $\displaystyle\int_a^{+\infty} f(x)\,dx$ can still possibly converge.

Theorem — Limit comparison test

Let $f, g : [a, +\infty) \to (0, +\infty)$ be positive continuous functions and suppose that the limit

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = L$$

exists.

(i) If $L \in [0, +\infty)$ and if $\displaystyle\int_a^{+\infty} g(x)\,dx$ converges, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ also converges.

(ii) If $L \in (0, +\infty]$ and if $\displaystyle\int_a^{+\infty} f(x)\,dx$ converges, then $\displaystyle\int_a^{+\infty} g(x)\,dx$ also converges.

(Idea: $L = 0$ means that g "grows much faster" than f; $L = +\infty$ means that f "grows much faster" than g; $L \in (0, +\infty)$ means that f and g "have similar growth".)

Proof. We only need to prove (i), because (ii) follows from (i) simply by swapping f and g. Suppose that $L \in [0, +\infty)$. Then by the definition of limit, there exists $r \geq a$ such that

$$f(x) < (L+1)g(x) \qquad \text{for every } x \geq r.$$

If $\displaystyle\int_a^{+\infty} g(x)\,dx$ converges, then $\displaystyle\int_r^{+\infty} g(x)\,dx$ also converges; so $\displaystyle\int_r^{+\infty} f(x)\,dx$ converges by the comparison test, and hence $\displaystyle\int_a^{+\infty} f(x)\,dx = \int_a^r f(x)\,dx + \int_r^{+\infty} f(x)\,dx$ also converges. ■

Remark — Limit comparison for the other types

By appropriately modifying $x \to +\infty$ to other kinds of limits, the limit comparison test also works for all the other types of improper integrals.

Example — $\displaystyle\int_0^1 \frac{dx}{1-x^n}$ by limit comparison

Let n be a positive integer. Evaluate

$$\int_0^1 \frac{1}{1-x^n}\,dx$$

if it converges.

Solution. The integral is improper at 1 only, and the solution is simpler if we use the limit comparison test instead of the comparison test. Now the limit

$$\lim_{x\to 1^-} \frac{\frac{1}{1-x^n}}{\frac{1}{1-x}} = \lim_{x\to 1^-} \frac{1-x}{1-x^n} = \lim_{x\to 1^-} \frac{1}{1+x+x^2+\cdots+x^{n-1}} = \frac{1}{n}$$

exists and is finite. We know from the example on $\displaystyle\int_0^1 \frac{1}{x}\,dx$ above that the improper integral $\displaystyle\int_0^1 \frac{1}{1-x}\,dx$ diverges, so $\displaystyle\int_0^1 \frac{1}{1-x^n}\,dx$ also diverges by the limit comparison test.

Example — $\displaystyle\int_1^{+\infty} \frac{\ln x}{x^p}\,dx$

Let p be a real number. Show that the improper integral

$$\int_1^{+\infty} \frac{\ln x}{x^p}\,dx$$

diverges if $p \leq 1$ and converges if $p > 1$.

Proof. The integral is improper at $+\infty$. For every sufficiently large x (namely for every $x \geq e$),

$$\frac{\ln x}{x^p} \geq \frac{1}{x^p} > 0.$$

- If $p \leq 1$, then $\displaystyle\int_e^{+\infty} \frac{1}{x^p}\,dx$ diverges by the p-test, so $\displaystyle\int_e^{+\infty} \frac{\ln x}{x^p}\,dx$, and hence $\displaystyle\int_1^{+\infty} \frac{\ln x}{x^p}\,dx$, also diverges by the comparison test.
- If $p > 1$, then the "lower bound" $\displaystyle\int_e^{+\infty} \frac{1}{x^p}\,dx$ converges by the p-test, so the comparison test gives no useful conclusion; therefore let us use the limit comparison test instead. Let q be a number between 1 and p, i.e. $p > q > 1$ (recall that $\ln x \ll x^\varepsilon$ as $x \to +\infty$ for every $\varepsilon > 0$). Then

$$\lim_{x\to+\infty} \frac{\frac{\ln x}{x^p}}{\frac{1}{x^q}} = \lim_{x\to+\infty} \frac{\ln x}{x^{p-q}} = \lim_{x\to+\infty} \frac{\frac{1}{x}}{(p-q)x^{p-q-1}} = \lim_{x\to+\infty} \frac{1}{(p-q)x^{p-q}} = 0$$

by l'Hôpital's rule, since $p > q$. This limit exists and is finite. Since $q > 1$, the improper integral $\displaystyle\int_1^{+\infty} \frac{1}{x^q}\,dx$ converges by the p-test; so $\displaystyle\int_1^{+\infty} \frac{\ln x}{x^p}\,dx$ also converges by the limit comparison test.

■

6.7 Summary of Chapter 6

The following is what you need to know in this chapter in order to get a pass (a distinction) in this course.

✓ Rules of antidifferentiation and integration
- Sum rule, difference rule, constant multiple rule (from chapter 4);
- Substitution rule;
- Integration by parts.

✓ Techniques of antidifferentiation and integration
- Using trigonometric identities, e.g. the double angle formula or the product-to-sum formula;
- Implicit substitutions in general;
- Implicit trigonometric substitutions, e.g. $x = a + b\sin t$, $a + b\tan t$ or $a + b\sec t$;
- Reduction formulas;
- Integration of rational functions: partial fraction decomposition;
- Integration of trigonometric rational functions $R(\cos x, \sin x)$: $t = \tan x$, $\cos x$, $\sin x$ when appropriate, or Weierstrass' $t = \tan\dfrac{x}{2}$ in any situation.

✓ Improper integrals
- "Unbounded interval" type and "unbounded value" type;
- Cauchy principal value.

✓ Convergence tests for improper integrals
- p-test;
- Comparison test;
- Tail test for divergence;
- Absolute convergence test;
- Limit comparison test.

6.8 Supplement: Numerical Integration

Sometimes an integral is too difficult (if not impossible) to be computed exactly. Numerical approximations become very useful in such a situation.

Mid-point rule

Let $f : [a, b] \to \mathbb{R}$ be an integrable function, let $n \in \mathbb{N}$ and let $\{x_0, x_1, \ldots, x_n\}$ be a regular partition of $[a, b]$, i.e. $x_k = a + \dfrac{k}{n}(b - a)$. Then

$$\int_a^b f(x)\,dx \approx \frac{b-a}{n} \sum_{k=1}^{n} f\left(\frac{x_{k-1} + x_k}{2}\right).$$

Reasoning. The right-hand side is the mid-point Riemann sum of f with respect to the partition $\{x_0, x_1, \ldots, x_n\}$. Since f is integrable on $[a, b]$, this sum is approximately equal to the Riemann integral of f over $[a, b]$.

Example — Mid-point rule for e^{-x^2}

Find an approximation to the integral

$$\int_0^1 e^{-x^2} dx$$

using the mid-point rule with 5 subintervals of $[0, 1]$ of equal width.
Solution. The regular partition of $[0, 1]$ into 5 subintervals is $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$. By the mid-point rule,

$$\int_0^1 e^{-x^2} dx \approx \frac{1-0}{5}\left(e^{-0.1^2} + e^{-0.3^2} + e^{-0.5^2} + e^{-0.7^2} + e^{-0.9^2}\right) \approx 0.7481.$$

Trapezoidal rule

Let $f : [a, b] \to \mathbb{R}$ be an integrable function. For $n = 1$,

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\left[f(a) + f(b)\right].$$

(Figure: the region under $y = f(x)$ over $[a, b]$ approximated by the single trapezoid with vertices $(a, 0)$, $(a, f(a))$, $(b, f(b))$, $(b, 0)$.)
In general, let $n \in \mathbb{N}$ and let $\{x_0, x_1, \ldots, x_n\}$ be a regular partition of $[a, b]$, i.e. $x_k = a + \dfrac{k}{n}(b - a)$. Then

$$\int_a^b f(x)\,dx \approx \frac{1}{2} \cdot \frac{b-a}{n}\left[f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b)\right].$$

(Figure: the region under $y = f(x)$ over $[a, b]$ approximated by the n trapezoids over the subintervals $[x_{k-1}, x_k]$.) Reasoning. Since f is integrable on $[a, b]$, it is integrable on each closed subinterval $[x_{k-1}, x_k]$. So by the $(n = 1)$ case,

$$\int_{x_{k-1}}^{x_k} f(x)\,dx \approx \frac{x_k - x_{k-1}}{2}\left[f(x_{k-1}) + f(x_k)\right].$$

Therefore

$$\int_a^b f(x)\,dx = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f(x)\,dx \approx \sum_{k=1}^{n} \frac{x_k - x_{k-1}}{2}\left[f(x_{k-1}) + f(x_k)\right]$$

$$= \sum_{k=1}^{n} \frac{b-a}{2n} \left[f(x_{k-1}) + f(x_k) \right] = \frac{b-a}{2n} \sum_{k=1}^{n} \left[f(x_{k-1}) + f(x_k) \right]$$
$$= \frac{1}{2} \cdot \frac{b-a}{n} \left[f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b) \right].$$

Example — Trapezoidal rule for e^{-x^2}

Find an approximation to the integral

$$\int_0^1 e^{-x^2} dx$$

using the trapezoidal rule with 5 subintervals of $[0, 1]$ of equal width.
Solution. The regular partition of $[0, 1]$ into 5 subintervals is $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$. By the trapezoidal rule,

$$\int_0^1 e^{-x^2} dx \approx \frac{1}{2} \cdot \frac{1}{5} \left(e^{-0^2} + 2e^{-0.2^2} + 2e^{-0.4^2} + 2e^{-0.6^2} + 2e^{-0.8^2} + e^{-1^2} \right) \approx 0.7444.$$

An immediate question when applying an approximation rule is how good the approximation is. The following theorem estimates the size of the error of approximation by the trapezoidal rule. (Can you mimic the proof and find a similar result for the mid-point rule?)

Theorem — Error bound for the trapezoidal rule

Let f be a function such that f'' is continuous on $[a, b]$, let n be a positive integer and let $\{x_0, x_1, \ldots, x_n\}$ be a regular partition of $[a, b]$, i.e. $x_k = a + \frac{k}{n}(b - a)$. Then there exists $c \in (a, b)$ such that

$$\int_a^b f(x)\, dx - \frac{1}{2} \cdot \frac{b-a}{n} \left[f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b) \right]$$
$$= -\frac{(b-a)^3}{12n^2} f''(c).$$

Proof. We first prove the theorem for $n = 1$. Integrating twice by parts gives

$$\int_a^b f(x)\, dx = \frac{b-a}{2} \left[f(a) + f(b) \right] - \int_a^b \left(x - \frac{a+b}{2} \right) f'(x)\, dx$$
$$= \frac{b-a}{2} \left[f(a) + f(b) \right] - 0 + \int_a^b \underbrace{\left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right]}_{\text{always } <0} f''(x)\, dx,$$

where the boundary term vanishes because $\left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] = 0$ at $x = a$ and at $x = b$. Now since f'' is continuous on $[a, b]$, it attains its absolute minimum m and absolute maximum M in $[a, b]$ by the Extreme Value Theorem, i.e. $m \le f''(x) \le M$ for every $x \in [a, b]$. So

$$M \int_a^b \left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] dx \le \int_a^b \left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] f''(x)\, dx$$

$$\leq m \int_a^b \left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] dx,$$

i.e. (the bracket integrates to $-\dfrac{(b-a)^3}{12}$)

$$m \leq -\frac{12}{(b-a)^3} \int_a^b \left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] f''(x)\, dx \leq M.$$

Applying the Intermediate Value Theorem to the continuous function f'', there exists $c \in [a, b]$ such that

$$f''(c) = -\frac{12}{(b-a)^3} \int_a^b \left[\frac{1}{2} \left(x - \frac{a+b}{2} \right)^2 - \frac{(b-a)^2}{8} \right] f''(x)\, dx.$$

Consequently,

$$\int_a^b f(x)\, dx - \frac{b-a}{2} \left[f(a) + f(b) \right] = -\frac{(b-a)^3}{12} f''(c).$$

For $n > 1$, the above proof can be carried over to each subinterval of the partition $\{x_0, x_1, \ldots, x_n\}$. We obtain

$$m \leq -\frac{12}{(x_k - x_{k-1})^3} \int_{x_{k-1}}^{x_k} \left[\frac{1}{2} \left(x - \frac{x_{k-1} + x_k}{2} \right)^2 - \frac{(x_k - x_{k-1})^2}{8} \right] f''(x)\, dx \leq M$$

for each k, so summing from $k = 1$ to n we get

$$m \leq -\frac{12n^2}{(b-a)^3} \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left[\frac{1}{2} \left(x - \frac{x_{k-1} + x_k}{2} \right)^2 - \frac{(x_k - x_{k-1})^2}{8} \right] f''(x)\, dx \leq M.$$

Applying the Intermediate Value Theorem to the continuous function f'', there exists $c \in [a, b]$ such that

$$f''(c) = -\frac{12n^2}{(b-a)^3} \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left[\frac{1}{2} \left(x - \frac{x_{k-1} + x_k}{2} \right)^2 - \frac{(x_k - x_{k-1})^2}{8} \right] f''(x)\, dx.$$

Consequently,

$$\begin{aligned}
&\int_a^b f(x)\, dx - \frac{1}{2} \cdot \frac{b-a}{n} \left[f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b) \right] \\
&\quad = \sum_{k=1}^{n} \left\{ \int_{x_{k-1}}^{x_k} f(x)\, dx - \frac{x_k - x_{k-1}}{2} \left[f(x_{k-1}) + f(x_k) \right] \right\} \\
&\quad = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left[\frac{1}{2} \left(x - \frac{x_{k-1} + x_k}{2} \right)^2 - \frac{(x_k - x_{k-1})^2}{8} \right] f''(x)\, dx = -\frac{(b-a)^3}{12n^2} f''(c).
\end{aligned}$$

■

Corollary — Concavity and the trapezoidal rule

Let f be a function such that f'' is continuous. If the graph of f is concave upward, then the trapezoidal rule overestimates the integrals of f; if the graph of f is concave downward, then the trapezoidal rule underestimates the integrals of f.

Example — How many subintervals for the trapezoidal rule?

If we want to estimate the integral

$$\int_0^1 e^{-x^2}\,dx$$

using the trapezoidal rule so that the error of approximation is less than 0.00005, how many subintervals of equal width do we need to partition $[0,1]$ into?

Solution. Let $f(x) = e^{-x^2}$. Then $f''(x) = 2(2x^2-1)e^{-x^2}$ for every x, so that

$$|f''(x)| \leq 2$$

for all $x \in [0,1]$. To ensure that the error is less than 0.00005 when we partition $[0,1]$ into n subintervals of equal width, we require that

$$\frac{(1-0)^3}{12n^2} \cdot 2 < 0.00005,$$

i.e. $n > \dfrac{100}{\sqrt{3}}$. Therefore 58 subintervals of $[0,1]$ will be enough.

The following method improves the approximation further. The idea is to approximate the given function f by a quadratic polynomial g, whose integral can be evaluated easily. A parabola can be determined by three points it passes through.

(Figure: the graph of f over $[a,b]$ and the parabola $y = g(x)$ passing through the three points $(a, f(a))$, $\left(\dfrac{a+b}{2}, f\left(\dfrac{a+b}{2}\right)\right)$ and $(b, f(b))$.)

Simpson's $\frac{1}{3}$-rule (2 subintervals)

Let $f : [a,b] \to \mathbb{R}$ be an integrable function. Then

$$\int_a^b f(x)\,dx \approx \frac{1}{3} \cdot \frac{b-a}{2}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right].$$

Reasoning. Let g be the quadratic polynomial

$$g(x) = \frac{f(a) - 2f\left(\frac{a+b}{2}\right) + f(b)}{\frac{(b-a)^2}{2}}\left(x - \frac{a+b}{2}\right)^2 + \frac{f(b)-f(a)}{b-a}\left(x - \frac{a+b}{2}\right) + f\left(\frac{a+b}{2}\right).$$

Then $g(a) = f(a)$, $g\left(\dfrac{a+b}{2}\right) = f\left(\dfrac{a+b}{2}\right)$ and $g(b) = f(b)$, so g gives a quadratic polynomial approximation of f over the interval $[a,b]$. Therefore we expect that

$$\int_a^b f(x)\,dx \approx \int_a^b g(x)\,dx = \frac{1}{3} \cdot \frac{b-a}{2}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right].$$

Simpson's $\frac{1}{3}$-rule (general)

Let $f : [a,b] \to \mathbb{R}$ be an integrable function, let $n \in \mathbb{N}$ and let $\{x_0, x_1, \ldots, x_{2n}\}$ be a

regular partition of $[a,b]$ into $2n$ subintervals, i.e. $x_k = a + \dfrac{k}{2n}(b-a)$. Then

$$\int_a^b f(x)\,dx \approx \frac{1}{3}\cdot\frac{b-a}{2n}\left[f(a)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+\cdots\right.$$
$$\left.+4f(x_{2n-1})+f(b)\right].$$

Reasoning. Since f is integrable on $[a,b]$, it is integrable on each closed subinterval $[x_{2k-2},x_{2k}]$. So by the (2 subintervals) case,

$$\int_{x_{2k-2}}^{x_{2k}} f(x)\,dx \approx \frac{x_{2k}-x_{2k-2}}{6}\left[f(x_{2k-2})+4f(x_{2k-1})+f(x_{2k})\right].$$

Therefore

$$\int_a^b f(x)\,dx = \sum_{k=1}^{n}\int_{x_{2k-2}}^{x_{2k}} f(x)\,dx \approx \sum_{k=1}^{n}\frac{x_{2k}-x_{2k-2}}{6}\left[f(x_{2k-2})+4f(x_{2k-1})+f(x_{2k})\right]$$

$$= \sum_{k=1}^{n}\frac{b-a}{6n}\left[f(x_{2k-2})+4f(x_{2k-1})+f(x_{2k})\right]$$
$$= \frac{1}{3}\cdot\frac{b-a}{2n}\left[f(a)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+\cdots\right.$$
$$\left.+4f(x_{2n-1})+f(b)\right].$$

Remark — An even number of subintervals is needed

We must divide $[a,b]$ into an even number of subintervals of equal width in order to approximate $\displaystyle\int_a^b f(x)\,dx$ using Simpson's $\dfrac{1}{3}$-rule.

Example — Simpson's rule for e^{-x^2}

Find an approximation to the integral

$$\int_0^1 e^{-x^2}dx$$

using Simpson's $\dfrac{1}{3}$-rule with 4 subintervals of $[0,1]$ of equal width.

Solution. The partition of $[0,1]$ into 4 subintervals of equal width is $\{0, 0.25, 0.5, 0.75, 1\}$. By Simpson's rule,

$$\int_0^1 e^{-x^2}dx \approx \frac{1}{3}\cdot\frac{1}{4}\left(e^{-0^2}+4e^{-0.25^2}+2e^{-0.5^2}+4e^{-0.75^2}+e^{-1^2}\right)\approx 0.7469.$$

The following theorem provides a way to estimate the size of the error of approximation by Simpson's rule. We omit its proof.

Theorem — Error bound for Simpson's rule

Let f be a function such that its fourth derivative $f^{(4)}$ is continuous on $[a,b]$, let n be a positive integer and let $\{x_0, x_1, \ldots, x_{2n}\}$ be a regular partition of $[a,b]$ into $2n$

subintervals, i.e. $x_k = a + \dfrac{k}{2n}(b-a)$. If $|f^{(4)}(x)| \le M$ for every $x \in [a,b]$, then

$$\left| \int_a^b f(x)\,dx - \frac{1}{3} \cdot \frac{b-a}{2n} \left[f(a) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{2n-1}) + f(b) \right] \right| \le \frac{M(b-a)}{180} \left(\frac{b-a}{2n} \right)^4.$$

Example — How many subintervals for Simpson's rule?

If we want to estimate the integral

$$\int_0^1 e^{-x^2} dx$$

using Simpson's $\dfrac{1}{3}$-rule so that the error of approximation is less than 0.00005, how many subintervals of equal width do we need to partition $[0,1]$ into?

Solution. Let $f(x) = e^{-x^2}$. Then $f^{(4)}(x) = (16x^4 - 48x^2 + 12)e^{-x^2}$ for every x, so that

$$\left| f^{(4)}(x) \right| \le (16 + 48 + 12) \cdot 1 = 76$$

for all $x \in [0,1]$. To ensure that the error is less than 0.00005 when we partition $[0,1]$ into $2n$ subintervals of equal width, we require that

$$\frac{(1-0)^5}{180(2n)^4} \cdot 76 < 0.00005,$$

i.e. $2n > \left(\dfrac{76000}{9} \right)^{\frac{1}{4}} \approx 9.59$. Therefore 10 $(= 2n)$ subintervals of $[0,1]$ will be enough.

Remark — A sharper bound for $|f^{(4)}|$

In the example above we have only used a rough upper bound of $|f^{(4)}|$ obtained using the triangle inequality. In fact, by studying the absolute maximum of $f^{(4)}$ on $[0,1]$ (using techniques from chapter 3, such as the first derivative test), one obtains the more precise upper bound

$$\left| f^{(4)}(x) \right| \le 12$$

for all $x \in [0,1]$. So it turns out that 7 $(= 2n$ with $2n \ge 6)$ subintervals of $[0,1]$ will already be enough.

6.9 Supplement: Feynman's Trick

The following supplementary material is for those who know about partial derivatives (knowing how to compute them is enough).

One is often curious about whether "differentiating under the integral sign" is valid. The following theorem gives a sufficient condition for this to be valid; you will learn its proof in MATH3033/3043.

Theorem — Leibniz's integral rule

Let $a < b$ and $c < d$ be real numbers, and let $f(x,t)$ be a function in two variables

such that both f and $\dfrac{\partial f}{\partial t}$ are continuous for $x \in [a,b]$ and $t \in [c,d]$. Then the function

$$F(t) = \int_a^b f(x,t)\,dx$$

is differentiable on (c,d) and

$$F'(t) = \frac{d}{dt}\int_a^b f(x,t)\,dx = \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx.$$

Remark — Extension to improper integrals

This result can also be extended to the case of improper integrals, provided that the improper integral of $f(x,t)$ converges for every t and that $\left|\dfrac{\partial f}{\partial t}(x,t)\right|$ is bounded from above by a function $g(x)$ whose improper integral converges.

Remark — The idea of Feynman's trick

To compute a certain difficult integral

$$I = \int_a^b g(x)\,dx,$$

one tries to introduce a new parameter t so that one of the integrals

$$\int_a^b f(x,t)\,dx \qquad \text{or} \qquad \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx$$

is related to I, and so that one of these integrals can be evaluated easily. This technique is often called "Feynman's trick" by physicists, because Feynman popularized the usage of this method in computing difficult integrals.

Example — The Dirichlet integral

In a problem-set exercise we have already seen that the improper integral

$$\int_0^{+\infty} \frac{\sin x}{x}\,dx$$

converges. Now show that

$$\int_0^{+\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

Proof. For each $t \in [0,+\infty)$, let

$$I(t) = \int_0^{+\infty} \frac{\sin x}{x} e^{-xt} dx.$$

Then by Leibniz's integral rule and integration by parts, we have

$$\begin{aligned} I'(t) &= \int_0^{+\infty} \frac{\partial}{\partial t}\left(\frac{\sin x}{x} e^{-xt}\right) dx \\ &= \int_0^{+\infty} -e^{-xt} \sin x \, dx = -\int_0^{+\infty} e^{-xt} \sin x \, dx \\ &= -\left(1 - t\int_0^{+\infty} e^{-xt} \cos x \, dx\right) \\ &= -1 + t\int_0^{+\infty} e^{-xt} \cos x \, dx \\ &= -1 + t\left(0 + t\int_0^{+\infty} e^{-xt} \sin x \, dx\right) \\ &= -1 + t^2 \int_0^{+\infty} e^{-xt} \sin x \, dx = -1 - t^2 I'(t), \end{aligned}$$

so

$$I'(t) = -\frac{1}{1+t^2} \qquad \text{for every } t > 0.$$

This implies that

$$I(t) = -\arctan t + C$$

for some constant C. Now since $\lim\limits_{t\to+\infty} I(t) = 0$ (why?), we have $C = \dfrac{\pi}{2}$ and so

$$I(t) = \frac{\pi}{2} - \arctan t \qquad \text{for every } t \geq 0.$$

Therefore

$$\int_0^{+\infty} \frac{\sin x}{x} \, dx = I(0) = \frac{\pi}{2}.$$

■

Example — The Gaussian integral

We have already seen that the improper integral

$$\int_0^{+\infty} e^{-x^2} dx$$

converges. Now show that

$$\int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Proof. For each $t \in [0, +\infty)$, let

$$I(t) = \int_0^{+\infty} \frac{e^{-t(1+x^2)}}{1+x^2} \, dx.$$

Then by Leibniz's integral rule and the substitution $u = \sqrt{t}\,x$, we have

$$\begin{aligned} I'(t) &= \int_0^{+\infty} \frac{\partial}{\partial t}\left(\frac{e^{-t(1+x^2)}}{1+x^2}\right) dx \\ &= \int_0^{+\infty} -e^{-t(1+x^2)} dx = -e^{-t}\int_0^{+\infty} e^{-tx^2} dx \\ &= -\frac{e^{-t}}{\sqrt{t}}\int_0^{+\infty} e^{-u^2} du \end{aligned}$$

for every $t > 0$. Now it is obvious that

$$I(0) = \int_0^{+\infty} \frac{1}{1+x^2}\, dx = \frac{\pi}{2} \qquad \text{and} \qquad \lim_{t\to+\infty} I(t) = 0$$

(why?), so

$$\int_0^{+\infty} I'(t)\, dt = \lim_{t\to+\infty} I(t) - I(0) = -\frac{\pi}{2}.$$

But on the other hand, we also have

$$\int_0^{+\infty} I'(t)\, dt = -\int_0^{+\infty} \frac{e^{-t}}{\sqrt{t}}\, dt \int_0^{+\infty} e^{-u^2} du = -2\left(\int_0^{+\infty} e^{-v^2} dv\right)^2,$$

where we have used the substitution $v = \sqrt{t}$. Therefore

$$\int_0^{+\infty} e^{-x^2} dx = \sqrt{-\frac{1}{2}\int_0^{+\infty} I'(t)\, dt} = \frac{\sqrt{\pi}}{2}.$$

■

Interested readers are suggested to read further about this computational technique from the following reference.

Reference. Z. Burghelea, "Feynman's Trick", https://zackyzz.github.io/feynman.

7 Vectors and Geometry

In this chapter, we use algebra and calculus as tools to study geometry in the "plane" and in "space". The study of plane geometry using algebra has been done with the rectangular coordinate system since high school: each point P in the plane is labelled by a unique ordered pair of real numbers (a, b), called the (rectangular) coordinates of P, and these numbers a and b are called the x-coordinate and the y-coordinate of P respectively.

Definition. The set of all points in the coordinate plane is denoted by

$$\mathbb{R}^2 := \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}.$$

7.1 Conic Sections

In high school we have already studied two special kinds of planar curves in $\mathbb{R}^2$, namely lines and circles. We now enrich our inventory of planar curves by studying conic sections.

Definition. A conic section is a set of points in $\mathbb{R}^2$ whose coordinates (x, y) satisfy a polynomial equation of degree two.

The three generic types of non-degenerate conic sections

In each defining equation below, a and b are non-zero real constants.

- **Ellipse:** $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$. The coefficients of x^2 and y^2 have the same sign. (Special case: circle.)
- **Parabola:** one of the coefficients of x^2 and y^2 is zero.
- **Hyperbola:** $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$. The coefficients of x^2 and y^2 have opposite signs. Its asymptotes are $y=\pm\dfrac{b}{a}x$.

Degenerate cases: an ellipse can degenerate to a point, a parabola to a line, and a hyperbola to a pair of intersecting lines.

Remark: why "conic section"

A conic section can also be thought of as a planar curve obtained by intersecting a plane with the surface of a (double) right circular cone. This explains the terminology.

Definition. Let F_1 and F_2 be two points in $\mathbb{R}^2$. An ellipse is a curve in $\mathbb{R}^2$ which consists of all points P such that the distances between P and F_1 and between P and F_2 have a fixed sum. Each of the points F_1 and F_2 is called a focus (pl. foci) of the ellipse.

Example 7.5 (equation of an ellipse)

Let $F_1(-3,0)$ and $F_2(3,0)$ be points in $\mathbb{R}^2$. Find the equation of the ellipse consisting of points whose sum of distances from F_1 and from F_2 is 10.

Solution. Every point P on the required ellipse has coordinates (x,y) which satisfy

$$\underbrace{\sqrt{(x+3)^2+(y-0)^2}}_{\text{distance between } P \text{ and } F_1}+\underbrace{\sqrt{(x-3)^2+(y-0)^2}}_{\text{distance between } P \text{ and } F_2}=10.$$

Squaring both sides we get

$$(x+3)^2+y^2+(x-3)^2+y^2+2\sqrt{(x+3)^2+y^2}\sqrt{(x-3)^2+y^2}=100,$$

i.e.

$$\sqrt{(x+3)^2+y^2}\sqrt{(x-3)^2+y^2}=41-x^2-y^2.$$

Squaring both sides again, we obtain

$$(x^2+y^2+9)^2-36x^2=(41-x^2-y^2)^2,$$

i.e.

$$\frac{x^2}{25}+\frac{y^2}{16}=1.$$

Definition. Let L be a line in $\mathbb{R}^2$ and F be a point in $\mathbb{R}^2$ which does not lie on L. A parabola is a curve in $\mathbb{R}^2$ which consists of all points P equidistant from L and F. F is called the focus of the parabola and L is called the directrix of the parabola.

Example 7.7 (equation of a parabola)

Let $F(4,0)$ be a point in $\mathbb{R}^2$. Find the equation of the parabola consisting of points equidistant from F and from the y-axis.

Solution. Every point P on the required parabola has coordinates (x,y) which

satisfy

$$\underbrace{|x|}_{\text{distance between } P \text{ and } y\text{-axis}} = \underbrace{\sqrt{(x-4)^2+(y-0)^2}}_{\text{distance between } P \text{ and } F}.$$

Squaring both sides, we obtain

$$x^2 = (x-4)^2 + y^2,$$

i.e.

$$y^2 = 8x - 16.$$

Definition. Let F_1 and F_2 be two points in $\mathbb{R}^2$. A hyperbola is a curve in $\mathbb{R}^2$ which consists of all points P such that the distances between P and F_1 and between P and F_2 have a fixed difference. Each of the points F_1 and F_2 is called a focus of the hyperbola.

Example 7.9 (equation of a hyperbola)

Let $F_1(0,3)$ and $F_2(0,-3)$ be points in $\mathbb{R}^2$. Find the equation of the hyperbola consisting of points whose difference of distances from F_1 and from F_2 is 4. What are the equations of the asymptotes of this hyperbola?

Solution. Every point P on the required hyperbola has coordinates (x,y) which satisfy

$$\left| \underbrace{\sqrt{(x-0)^2+(y-3)^2}}_{\text{distance between } P \text{ and } F_1} - \underbrace{\sqrt{(x-0)^2+(y+3)^2}}_{\text{distance between } P \text{ and } F_2} \right| = 4.$$

Squaring both sides we get

$$x^2+(y-3)^2+x^2+(y+3)^2-2\sqrt{x^2+(y-3)^2}\sqrt{x^2+(y+3)^2} = 16,$$

i.e.

$$\sqrt{x^2+(y-3)^2}\sqrt{x^2+(y+3)^2} = x^2+y^2+1.$$

Squaring both sides again, we obtain $(x^2+y^2+9)^2 - 36y^2 = (x^2+y^2+1)^2$, i.e.

$$\frac{y^2}{4} - \frac{x^2}{5} = 1.$$

The equations of its asymptotes are $\dfrac{y^2}{4} - \dfrac{x^2}{5} = 0$, i.e. $y = \pm\dfrac{2}{\sqrt{5}}x$.

Example 7.10 (classifying conics)

For each of the following quadratic equations, what kind of conic section in $\mathbb{R}^2$ does the equation represent?

(a) $x^2 - 4y^2 + 2x + 8y - 7 = 0$

(b) $4x^2 + y^2 + 8x - 2y + 1 = 0$

(c) $x^2 + 2x + 4y - 3 = 0$

Solution. Note that there are no "cross-terms" xy in each of the given equations.

(a) Since the coefficients of x^2 and y^2 have opposite signs, the equation represents a hyperbola. By completing the squares, it can be put in the standard form

$$\frac{(x+1)^2}{2^2} - (y-1)^2 = 1,$$

whose asymptotes are $y - 1 = \pm\dfrac{1}{2}(x+1)$.

(b) Since the coefficients of x^2 and y^2 have the same sign, the equation represents an

ellipse. By completing the squares, it can be put in the standard form

$$(x+1)^2 + \frac{(y-1)^2}{2^2} = 1.$$

(c) Since the coefficient of y^2 is zero, the equation represents a parabola. By completing the squares, it can be put in the standard form

$$\frac{(x+1)^2}{2^2} = -(y-1).$$

7.2 Polar Coordinates

Apart from the rectangular coordinate system, there is another coordinate system in $\mathbb{R}^2$ which is also useful for calculus.

Definition. Let P be a point in $\mathbb{R}^2$ with rectangular coordinates (x, y). The ordered pair of real numbers (r, θ) is called polar coordinates of P if

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$

Remark: polar coordinates are not unique

Each point in $\mathbb{R}^2$ has one and only one pair of rectangular coordinates (x, y). However, each point in $\mathbb{R}^2$ has infinitely many possible polar coordinates (r, θ).

For example, both $(r,\theta) = \left(\sqrt{2}, \frac{\pi}{4}\right)$ and $(r,\theta) = \left(-\sqrt{2}, \frac{5\pi}{4}\right)$ are polar coordinates of the point in $\mathbb{R}^2$ with rectangular coordinates $(x, y) = (1, 1)$; and (r, θ) represents the same point if we add to θ any integral multiple of 2π.

Lemma 7.13 (geometric meaning of polar coordinates)

Let P be a point in $\mathbb{R}^2$ with polar coordinates (r, θ), and let P be different from the origin O. Then

- the distance between P and O is $|r|$, and
- the angle from the positive x-axis to the line segment OP is

$$\begin{cases} \theta, & \text{if } r > 0, \\ \theta - \pi, & \text{if } r < 0, \end{cases}$$

measured counterclockwise.

Remark 7.14 (simple polar equations)

The following are some geometric objects in $\mathbb{R}^2$ whose defining equations are simple in polar coordinates.

- For each $k > 0$, the equation $r = k$ represents a circle with radius k centred at the origin.
- For each $k \in \mathbb{R}$, the equation $\theta = k$ (with $r \geq 0$) represents a half-line (or a ray) emanating from the origin making an angle of k radians with the positive x-axis, measured counterclockwise.

Many geometric objects in $\mathbb{R}^2$ can be conveniently expressed using a polar equation $r = f(\theta)$, where $f : \mathbb{R} \to \mathbb{R}$ is some function.

Example 7.15 (cardioid)

Consider the curve in $\mathbb{R}^2$ which consists of all points whose polar coordinates (r, θ) satisfy the equation $r = 1 - \sin\theta$. Such a curve is called a cardioid as it looks like a "heart". Find the equation of this curve in rectangular coordinates.

Solution. The rectangular coordinates (x, y) and the polar coordinates (r, θ) of a point in $\mathbb{R}^2$ satisfy the relation

$$x^2 + y^2 = (r\cos\theta)^2 + (r\sin\theta)^2 = r^2.$$

Now the given equation $r = 1 - \sin\theta$ can be rewritten as $r^2 = r - r\sin\theta$, so it becomes

$$x^2 + y^2 = \sqrt{x^2 + y^2} - y$$

in rectangular coordinates, or simply

$$x^2 + y^2 = (x^2 + y^2 + y)^2.$$

Example 7.16 (p-orbital)

Consider the curve in $\mathbb{R}^2$ which consists of all points whose polar coordinates (r, θ) satisfy the equation $r = 2|\cos\theta|$.

Solution. This equation can be rewritten as $r^2 = 2|r\cos\theta|$, so it becomes

$$x^2 + y^2 = 2|x|$$

in rectangular coordinates. It is the union of the two circles $x^2 + y^2 = 2x$ and $x^2 + y^2 = -2x$, which touch each other at the origin. This curve is the cross-sectional boundary of one of the "p-orbitals" studied in chemistry.

Example 7.17 (d-orbital)

Consider the curve in $\mathbb{R}^2$ which consists of all points whose rectangular coordinates (x, y) satisfy the equation

$$|2x^2 - y^2| = (x^2 + y^2)^{\frac{3}{2}}.$$

Solution. In polar coordinates, it becomes $|2(r\cos\theta)^2 - (r\sin\theta)^2| = r^3$, or simply

$$r = |3\cos^2\theta - 1|.$$

This curve is the cross-sectional boundary of one of the "d-orbitals" studied in chemistry.

Example 7.18 (a line in polar form)

Consider the curve in $\mathbb{R}^2$ consisting of all points whose polar coordinates (r, θ) satisfy the equation $r = 2\sec\left(\theta - \frac{\pi}{3}\right)$. Find the equation of this curve in rectangular coordinates.

Solution. The equation can be rewritten as $r\left(\cos\theta\cos\frac{\pi}{3} + \sin\theta\sin\frac{\pi}{3}\right) = 2$, i.e.

$$\frac{1}{2}r\cos\theta + \frac{\sqrt{3}}{2}r\sin\theta = 2.$$

Therefore the equation becomes

$$x + \sqrt{3}y = 4$$

in rectangular coordinates. It is the equation of a line in $\mathbb{R}^2$.

Example 7.19 (eccentricity and conics)

Let ε and l be positive real numbers. Then a conic section is a curve C in $\mathbb{R}^2$ consisting of all points whose polar coordinates (r, θ) satisfy the equation

$$r = \frac{l}{1 + \varepsilon \cos\theta}.$$

ε is called the eccentricity of C and l is called the semi-latus rectum of C. If L is the vertical line in the right-half plane with equation $x = \dfrac{l}{\varepsilon}$, then C collects all the points P such that

$$\frac{\text{distance from the origin to } P}{\text{perpendicular distance from } P \text{ to } L} = \varepsilon.$$

Moreover,

- C is an ellipse if $0 < \varepsilon < 1$,
- C is a parabola if $\varepsilon = 1$, and
- C is a hyperbola if $\varepsilon > 1$.

Example 7.20 (spiral)

The curve in $\mathbb{R}^2$ which consists of all the points with polar coordinates (r, θ) that satisfy the equation $r = \theta$ is called a spiral.

7.3 Euclidean Vectors

In order to use calculus in real-world situations, we often need a mathematical description of "three-dimensional" space. To describe it, we introduce another coordinate axis called the z-axis, which is perpendicular to both the x- and y-axes, so as to form the Cartesian coordinate system.

Remark 7.21 (Cartesian coordinate system in space)

The Cartesian coordinate system in (three-dimensional) space consists of three pairwise perpendicular axes. This coordinate system is right-handed by convention: if you point your right hand toward $+x$ and bend your fingers toward $+y$, then your thumb will point toward $+z$. Each point P in the space is labelled by a unique ordered triple of real numbers (a, b, c), called the coordinates of P; these numbers are called the x-, y- and z-coordinate of P respectively.

Definition. The set of all points in (three-dimensional) space is denoted by

$$\mathbb{R}^3 := \{(x, y, z) : x \in \mathbb{R} \text{ and } y \in \mathbb{R} \text{ and } z \in \mathbb{R}\}.$$

In general, for each positive integer n, the set of points

$$\mathbb{R}^n := \{(x_1, x_2, \ldots, x_n) : x_1, x_2, \ldots, x_n \in \mathbb{R}\}$$

is called the n-dimensional (real) Euclidean space.

To understand the geometry of space or to extend our study of calculus to higher dimensions, Euclidean vectors will be a very useful tool. The word "vector" in this course always refers to "Euclidean vectors".

Definition. An n-dimensional (Euclidean) vector is an ordered n-tuple of real numbers. A vector can be denoted by a bold-face upright letter, e.g. $\mathbf{v}$ (in print), a letter with an arrow above it, e.g. $\vec{v}$ (in writing), or an underlined letter, e.g. $\underline{v}$ (in writing). For example, if $v_1, v_2, \ldots, v_n$ are real numbers, then

$$\mathbf{v} = \langle v_1, v_2, \ldots, v_n \rangle$$

is an n-dimensional vector. The components of a vector are often written inside angled parentheses, so as to be distinguished from a point in $\mathbb{R}^n$.

Definition. Let P be a point in $\mathbb{R}^n$ and $(x_1, x_2, \ldots, x_n)$ be its coordinates. Then the n-dimensional vector $\langle x_1, x_2, \ldots, x_n \rangle$ is called the position vector of P. In particular, the point $(x, y) \in \mathbb{R}^2$ has position vector $\langle x, y \rangle$. Because of the one-to-one correspondence between each point and its position vector, $\mathbb{R}^n$ can also be regarded as the set of all n-dimensional vectors, for instance

$$\mathbb{R}^2 = \{\langle x, y \rangle : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}, \qquad \mathbb{R}^3 = \{\langle x, y, z \rangle : x \in \mathbb{R} \text{ and } y \in \mathbb{R} \text{ and } z \in \mathbb{R}\}.$$

Real numbers are also called scalars when we are dealing with vectors in $\mathbb{R}^n$.

Definition. Let $u_1, u_2, \ldots, u_n, v_1, v_2, \ldots, v_n$ and a be real numbers. Then

(i) (Addition) The sum of the two vectors $\mathbf{u} = \langle u_1, u_2, \ldots, u_n \rangle$ and $\mathbf{v} = \langle v_1, v_2, \ldots, v_n \rangle$ is defined to be the vector

$$\mathbf{u} + \mathbf{v} := \langle u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n \rangle;$$

(ii) (Scalar multiplication) The multiple of the vector $\mathbf{v} = \langle v_1, v_2, \ldots, v_n \rangle$ by the scalar a is defined to be the vector

$$a\mathbf{v} := \langle av_1, av_2, \ldots, av_n \rangle.$$

Definition. The (n-dimensional) zero vector is the vector $\mathbf{0} := \langle \underbrace{0, 0, \ldots, 0}_{n \text{ components}} \rangle$. Given a vector $\mathbf{v} = \langle v_1, v_2, \ldots, v_n \rangle$, the negative of $\mathbf{v}$ is the vector $-\mathbf{v} := \langle -v_1, -v_2, \ldots, -v_n \rangle$.

Theorem 7.27 (vector space properties)

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors of the same dimension, let $\mathbf{0}$ denote the zero vector, also of the same dimension, and let a and b be scalars. Then

(i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
(ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
(iii) $\mathbf{u} + \mathbf{0} = \mathbf{u}$
(iv) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
(v) $(ab)\mathbf{u} = a(b\mathbf{u})$
(vi) $1\mathbf{u} = \mathbf{u}$
(vii) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
(viii) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$

All these are equalities of vectors, i.e. each corresponding component on both sides are equal.

Remark 7.28 (geometric interpretation of vectors)

(i) A vector can be represented by a directed line segment (an arrow), characterized by an initial point and a terminal point. A vector with initial point $A = (a_1, a_2, \ldots, a_n)$ and terminal point $B = (b_1, b_2, \ldots, b_n)$ is denoted by

$$\overrightarrow{AB} := \langle b_1 - a_1, b_2 - a_2, \ldots, b_n - a_n \rangle.$$

Given $P = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, the position vector of P is $\overrightarrow{OP} = \langle x_1, x_2, \ldots, x_n \rangle$, where $O = (0, 0, \ldots, 0)$ is the origin.

(ii) Two vectors are equal if and only if they are represented by arrows having the same length and the same direction, regardless of their initial points. In other words, if A, B, C, D are distinct points, then $\overrightarrow{AB} = \overrightarrow{CD}$ if and only if $ABDC$ is a parallelogram.

(iii) The zero vector $\mathbf{0}$ is represented by any arrow whose initial point and terminal point are the same.

(iv) Given any vector $\mathbf{v} = \overrightarrow{AB}$, the negative of $\mathbf{v}$ is $-\mathbf{v} = \overrightarrow{BA}$.

(v) (Vector addition) Given two vectors $\mathbf{u}$ and $\mathbf{v}$ of the same dimension, if $\mathbf{u} = \overrightarrow{AB}$, then there is a unique point C such that $\mathbf{v} = \overrightarrow{BC}$; in this case $\mathbf{u} + \mathbf{v} = \overrightarrow{AC}$.

(vi) (Vector addition) Given two vectors $\mathbf{u}$ and $\mathbf{v}$ of the same dimension, if $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AC}$, then $\mathbf{u} + \mathbf{v} = \overrightarrow{AD}$, where D is the unique point such that $ABDC$ is a parallelogram.

(vii) (Scalar multiplication) Given a non-zero real number a, multiplication by a is represented by stretching or compressing the length of the arrow representing a vector by the magnification factor $|a|$, keeping its direction unchanged if $a > 0$, and reversing its direction if $a < 0$.

(viii) Two vectors are parallel if they are scalar multiples of each other.

Example 7.29 (section formula)

Let P and Q be two points in the same space, with position vectors $\mathbf{u}$ and $\mathbf{v}$ respectively. Then

- the mid-point of the line segment PQ has position vector $\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v}$;
- more generally, let $r \in (0, 1)$ and let R be the point on the line segment PQ such that
$$\frac{\text{length of } PR}{\text{length of } RQ} = \frac{r}{1-r}.$$
Then R has position vector $(1-r)\mathbf{u} + r\mathbf{v}$.

Example 7.30 (collinearity)

Are the three points $A(0, 1, 0)$, $B(2, -1, 3)$ and $C(-4, 5, -6)$ collinear in $\mathbb{R}^3$?
Solution. The three points A, B, C are collinear when they lie on the same line, i.e. when the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel. Now
$$\overrightarrow{AB} = \langle 2 - 0, -1 - 1, 3 - 0\rangle = \langle 2, -2, 3\rangle$$
and
$$\overrightarrow{AC} = \langle -4 - 0, 5 - 1, -6 - 0\rangle = \langle -4, 4, -6\rangle.$$
Since $\overrightarrow{AC} = -2\,\overrightarrow{AB}$, it follows that $\overrightarrow{AC}$ and $\overrightarrow{AB}$ are parallel, and so A, B and C are collinear.

7.4 Dot Product of Vectors

The following is another operation on two vectors which produces a scalar.

Definition. Let $u_1, u_2, \ldots, u_n, v_1, v_2, \ldots, v_n$ be real numbers. Then the dot product of the two vectors $\mathbf{u} = \langle u_1, u_2, \ldots, u_n\rangle$ and $\mathbf{v} = \langle v_1, v_2, \ldots, v_n\rangle$ is defined to be the real number
$$\mathbf{u} \cdot \mathbf{v} := u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

Theorem 7.32 (properties of the dot product)

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors of the same dimension, and let a be a scalar. Then

(i) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

(ii) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(iii) $(a\mathbf{u}) \cdot \mathbf{v} = a(\mathbf{u} \cdot \mathbf{v})$

(iv) $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ only when $\mathbf{v} = \mathbf{0}$.

All these are equalities of scalars.

Lemma 7.33 (Cauchy–Schwarz inequality)

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors of the same dimension. Then

$$(\mathbf{u} \cdot \mathbf{v})^2 \leq (\mathbf{u} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{v}).$$

Proof. If $\mathbf{v} = \mathbf{0}$, then both sides equal 0 and the inequality holds trivially. If $\mathbf{v} \neq \mathbf{0}$, then for every real number t, we consider the vector $\mathbf{u} - t\mathbf{v}$. By Theorem 7.32 (iv) we always have

$$\begin{aligned} 0 &\leq (\mathbf{u} - t\mathbf{v}) \cdot (\mathbf{u} - t\mathbf{v}) \\ &= (\mathbf{u} \cdot \mathbf{u}) - t(\mathbf{u} \cdot \mathbf{v}) - t(\mathbf{v} \cdot \mathbf{u}) + t^2(\mathbf{v} \cdot \mathbf{v}) \\ &= \underbrace{(\mathbf{v} \cdot \mathbf{v})}_{A} t^2 + \underbrace{(-2(\mathbf{u} \cdot \mathbf{v}))}_{B} t + \underbrace{(\mathbf{u} \cdot \mathbf{u})}_{C}. \end{aligned}$$

Now the above inequality holds for every $t \in \mathbb{R}$. Since the right-hand side is a quadratic polynomial in t, its discriminant must be non-positive, i.e. $B^2 - 4AC \leq 0$. This is exactly the inequality that needs to be proved. ■

Theorem 7.32 (iv) enables us to make the following definition.

Definition. The length (or magnitude, or norm) of a vector $\mathbf{v} = \langle v_1, v_2, \ldots, v_n \rangle$ is defined by the non-negative real number

$$\|\mathbf{v}\| := \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

A unit vector is a vector whose length equals 1.

Remark 7.35

Geometrically, the length of a vector is the same as the length of each directed line segment representing it. This follows from Pythagoras' theorem.

Remark 7.36 (standard basis vectors)

With $\mathbf{i} := \langle 1, 0 \rangle$ and $\mathbf{j} := \langle 0, 1 \rangle$, the vector $\langle x, y \rangle \in \mathbb{R}^2$ is sometimes written as $\langle x, y \rangle = x\mathbf{i} + y\mathbf{j}$. With $\mathbf{i} := \langle 1, 0, 0 \rangle$, $\mathbf{j} := \langle 0, 1, 0 \rangle$ and $\mathbf{k} := \langle 0, 0, 1 \rangle$, the vector $\langle x, y, z \rangle \in \mathbb{R}^3$ is also written as $\langle x, y, z \rangle = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. These vectors are called standard basis vectors: they are unit vectors pointing in the directions of the positive coordinate axes. In general, the standard basis vectors in $\mathbb{R}^n$ are denoted by $\mathbf{e}_i := \langle 0, \ldots, 0, \underbrace{1}_{\text{only the } i\text{th component is 1}}, 0, \ldots, 0 \rangle$ for each $i \in \{1, 2, \ldots, n\}$.

Theorem 7.37 (properties of the length)

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors of the same dimension, let $\mathbf{0}$ denote the zero vector, also of the same dimension, and let a be a scalar. Then

(i) $\|\mathbf{u}\| \geq 0$, and $\mathbf{0}$ is the only vector having zero length;
(ii) $\|a\mathbf{u}\| = |a| \, \|\mathbf{u}\|$;
(iii) (Triangle inequality) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. (i) and (ii) are direct consequences of the definition of the length and the definition of scalar multiplication. To prove (iii), we note that the Cauchy–Schwarz

inequality (Lemma 7.33) implies that $\mathbf{u}\cdot\mathbf{v} \le \|\mathbf{u}\|\|\mathbf{v}\|$. Therefore

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\|^2 &= (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v})\\ &= \mathbf{u}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v}+2(\mathbf{u}\cdot\mathbf{v})\\ &= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2+2(\mathbf{u}\cdot\mathbf{v})\\ &\le \|\mathbf{u}\|^2+\|\mathbf{v}\|^2+2\|\mathbf{u}\|\|\mathbf{v}\|\\ &= (\|\mathbf{u}\|+\|\mathbf{v}\|)^2,\end{aligned}$$

which proves the triangle inequality. ■

Definition. Let $\mathbf{v}$ be a non-zero vector. The normalization of $\mathbf{v}$ is the unit vector $\dfrac{1}{\|\mathbf{v}\|}\mathbf{v}$.

Example 7.39 (length and direction)

Every non-zero vector $\mathbf{v}$ can be expressed as a scalar multiple

$$\mathbf{v} = \|\mathbf{v}\|\left(\frac{1}{\|\mathbf{v}\|}\mathbf{v}\right),$$

in which $\|\mathbf{v}\|$ represents the length of $\mathbf{v}$ and the unit vector $\dfrac{1}{\|\mathbf{v}\|}\mathbf{v}$ represents the direction of $\mathbf{v}$.
For example, the vector $\mathbf{v} = 9\mathbf{i}-2\mathbf{j}+6\mathbf{k}$ has length

$$\|\mathbf{v}\| = \sqrt{9^2+(-2)^2+6^2} = 11,$$

so we may express it as a product of its length and its direction by

$$\mathbf{v} = 11\left(\frac{9}{11}\mathbf{i}-\frac{2}{11}\mathbf{j}+\frac{6}{11}\mathbf{k}\right).$$

Example 7.40 (unit vectors and parallel vectors)

Consider the points $P(6,4)$ and $Q(-6,9)$ in $\mathbb{R}^2$. Find the component forms of

(a) the vector $\overrightarrow{PQ}$,

(b) two unit vectors that are parallel to $\overrightarrow{PQ}$, and

(c) two vectors of length 3 that are parallel to $\overrightarrow{PQ}$.

Solution.

(a) $\overrightarrow{PQ} = \langle -6-6, 9-4\rangle = \langle -12, 5\rangle$.

(b) The length of $\overrightarrow{PQ}$ is $\|\overrightarrow{PQ}\| = \sqrt{(-12)^2+5^2} = 13$, so the unit vectors parallel to $\overrightarrow{PQ}$ are

$$\frac{1}{\|\overrightarrow{PQ}\|}\overrightarrow{PQ} = \frac{1}{13}\langle -12,5\rangle = \left\langle -\frac{12}{13}, \frac{5}{13}\right\rangle$$

and

$$-\frac{1}{\|\overrightarrow{PQ}\|}\overrightarrow{PQ} = -\frac{1}{13}\langle -12,5\rangle = \left\langle \frac{12}{13}, -\frac{5}{13}\right\rangle.$$

(c) The required vectors are 3 times the two unit vectors found in (b):

$$3\left\langle -\frac{12}{13}, \frac{5}{13}\right\rangle = \left\langle -\frac{36}{13}, \frac{15}{13}\right\rangle \qquad \text{and} \qquad 3\left\langle \frac{12}{13}, -\frac{5}{13}\right\rangle = \left\langle \frac{36}{13}, -\frac{15}{13}\right\rangle.$$

Let us consider two non-zero vectors $\mathbf{u} = \overrightarrow{OP}$ and $\mathbf{v} = \overrightarrow{OQ}$. Then the angle $\theta \in [0,\pi]$

between the directed line segments $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ is $\theta = \angle POQ$. According to the Cosine Law, this angle θ satisfies

$$\cos\theta = \frac{\|\overrightarrow{OP}\|^2 + \|\overrightarrow{OQ}\|^2 - \|\overrightarrow{PQ}\|^2}{2\|\overrightarrow{OP}\|\|\overrightarrow{OQ}\|} = \frac{\mathbf{u}\cdot\mathbf{u} + \mathbf{v}\cdot\mathbf{v} - (\mathbf{v}-\mathbf{u})\cdot(\mathbf{v}-\mathbf{u})}{2\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

Note that the right-hand side is indeed in $[-1, 1]$ according to the Cauchy–Schwarz inequality. This motivates the following definition.

Definition. Let $\mathbf{u}$ and $\mathbf{v}$ be non-zero vectors of the same dimension. Then the angle between $\mathbf{u}$ and $\mathbf{v}$ is the number $\theta \in [0, \pi]$ defined by

$$\theta = \arccos \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

Remark 7.42

Geometrically, the dot product $\mathbf{u}\cdot\mathbf{v}$ is the product of the lengths of the arrows representing $\mathbf{u}$ and $\mathbf{v}$ with the cosine of the angle between these two arrows, i.e.

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta.$$

Definition. Two vectors $\mathbf{u}$ and $\mathbf{v}$ of the same dimension are said to be orthogonal if $\mathbf{u}\cdot\mathbf{v} = 0$. (For Euclidean vectors this is the same as saying that the arrows representing $\mathbf{u}$ and $\mathbf{v}$ are perpendicular to each other, i.e. the angle between these arrows is $\frac{\pi}{2}$.)

Example 7.44 (orthogonality)

(a) The vectors $\mathbf{u} = \langle 3, 2, -1\rangle$ and $\mathbf{v} = \langle 0, -2, -4\rangle$ are orthogonal because $\mathbf{u}\cdot\mathbf{v} = (3)(0) + (2)(-2) + (-1)(-4) = 0$.

(b) The zero vector $\mathbf{0}$ is orthogonal to every vector of the same dimension.

(c) The vectors $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathbb{R}^3$ are pairwise orthogonal to each other. In general, every distinct pair of standard basis vectors $\mathbf{e}_i$ and $\mathbf{e}_j$ in the same dimension are orthogonal.

Example 7.45 (angle in a semicircle)

Let A and B be the end-points of a diameter of a circle, and let C be a point different from A and B, also lying on the circumference of the circle. Show that the line segments AC and BC are perpendicular to each other.

Proof. Let O denote the centre of the circle, let $\mathbf{u} = \overrightarrow{OA}$ and let $\mathbf{v} = \overrightarrow{OC}$. Then $\|\mathbf{u}\| = \|\mathbf{v}\|$, as they are radii of the same circle, and

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \mathbf{v} - \mathbf{u}, \qquad \overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \overrightarrow{OC} + \overrightarrow{OA} = \mathbf{v} + \mathbf{u}.$$

Thus

$$\begin{aligned}\overrightarrow{AC}\cdot\overrightarrow{BC} &= (\mathbf{v}-\mathbf{u})\cdot(\mathbf{v}+\mathbf{u})\\ &= \mathbf{v}\cdot\mathbf{v} - \mathbf{u}\cdot\mathbf{v} + \mathbf{v}\cdot\mathbf{u} - \mathbf{u}\cdot\mathbf{u}\\ &= \|\mathbf{v}\|^2 - \|\mathbf{u}\|^2\\ &= 0,\end{aligned}$$

and so $\overrightarrow{AC}$ and $\overrightarrow{BC}$ are orthogonal vectors. In other words, the line segments AC and BC are perpendicular to each other. ■

Lemma 7.46 (decomposition along a vector)

Let $\mathbf{v}$ be a non-zero vector. Then every vector $\mathbf{u}$ of the same dimension can be decomposed as a sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

Proof. Let $\mathbf{u}_1 := \dfrac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}$ and let $\mathbf{u}_2 := \mathbf{u} - \mathbf{u}_1$. Then $\mathbf{u}$ is decomposed as a sum $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$, in which $\mathbf{u}_1$ is parallel to $\mathbf{v}$ (because $\dfrac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}$ is just a scalar). Finally, since

$$\begin{aligned}\mathbf{u}_2\cdot\mathbf{v} &= (\mathbf{u}-\mathbf{u}_1)\cdot\mathbf{v} = \mathbf{u}\cdot\mathbf{v} - \mathbf{u}_1\cdot\mathbf{v}\\ &= \mathbf{u}\cdot\mathbf{v} - \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\right)\cdot\mathbf{v} = \mathbf{u}\cdot\mathbf{v} - \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}(\mathbf{v}\cdot\mathbf{v}) = 0,\end{aligned}$$

it follows that $\mathbf{u}_2$ is orthogonal to $\mathbf{v}$. ■

We give a special name to the portion of $\mathbf{u}$ that is parallel to $\mathbf{v}$.

Definition. Let $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{u}$ be two vectors of the same dimension. The orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ is the vector defined by

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} := \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}.$$

Note that the direction of $\operatorname{proj}_{\mathbf{v}}\mathbf{u}$ is parallel to $\mathbf{v}$.

Example 7.48 (dot products and projections)

Let $\mathbf{u} = 4\mathbf{i} - 4\mathbf{j}$ and $\mathbf{v} = 2\mathbf{j} - 2\mathbf{k}$.

(a) Find $\mathbf{u}\cdot\mathbf{u}$, $\mathbf{v}\cdot\mathbf{v}$ and $\mathbf{u}\cdot\mathbf{v}$.

(b) Find the angle between $\mathbf{u}$ and $\mathbf{v}$.

(c) Find the vectors $\operatorname{proj}_{\mathbf{v}}\mathbf{u}$ and $\operatorname{proj}_{\mathbf{u}}\mathbf{v}$.

Solution.

(a) We have $\mathbf{u}\cdot\mathbf{u} = 4^2 + (-4)^2 + 0^2 = 32$, $\mathbf{v}\cdot\mathbf{v} = 0^2 + 2^2 + (-2)^2 = 8$ and $\mathbf{u}\cdot\mathbf{v} = (4)(0) + (-4)(2) + (0)(-2) = -8$.

(b) Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Since $\|\mathbf{u}\| = \sqrt{\mathbf{u}\cdot\mathbf{u}} = 4\sqrt{2}$ and $\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}} = 2\sqrt{2}$, we have

$$\theta = \arccos\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \arccos\frac{-8}{(4\sqrt{2})(2\sqrt{2})} = \arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}.$$

(c) $\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \dfrac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v} = \dfrac{-8}{8}(2\mathbf{j} - 2\mathbf{k}) = -2\mathbf{j} + 2\mathbf{k}$ and $\operatorname{proj}_{\mathbf{u}}\mathbf{v} = \dfrac{\mathbf{v}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u} = \dfrac{-8}{32}(4\mathbf{i} - 4\mathbf{j}) = -\mathbf{i} + \mathbf{j}$.

7.5 Cross Product of Vectors in $\mathbb{R}^3$

The cross product is defined only for three-dimensional vectors.

Definition. Let $u_1, u_2, u_3, v_1, v_2, v_3$ be real numbers. Then the cross product of the three-dimensional vectors $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ is the three-dimensional vector given by

$$\mathbf{u}\times\mathbf{v} := \langle u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1\rangle.$$

Remark 7.50 (determinants and the cross product)

In linear algebra, the determinant of a 2×2 matrix of numbers is defined by the

number

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} := ad - bc,$$

while the determinant of a 3×3 matrix of numbers is defined by the number

$$\begin{aligned} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} &= a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1 \\ &= a_1(b_2c_3 - b_3c_2) + a_2(b_3c_1 - b_1c_3) + a_3(b_1c_2 - b_2c_1) \\ &= a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \end{aligned}$$

Because of this computation rule, some people prefer to remember the cross product using a formal determinant:

$$\mathbf{u} \times \mathbf{v} := \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Applying the same computation rule to this formal determinant recovers the original definition:

$$\begin{aligned} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} &= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \\ &= (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k} \\ &= \mathbf{u} \times \mathbf{v}. \end{aligned}$$

Note that this is not really a determinant (the first row does not consist of numbers!), but just a formal way to remember the definition. $\mathbf{u} \times \mathbf{v}$ is a vector, not a scalar.

Theorem 7.51

Let $\mathbf{u}$ and $\mathbf{v}$ be three-dimensional vectors. Then $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

Proof. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. Then

$$\begin{aligned} \mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) &= \langle u_1, u_2, u_3 \rangle \cdot \langle u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1 \rangle \\ &= u_1(u_2v_3 - u_3v_2) + u_2(u_3v_1 - u_1v_3) + u_3(u_1v_2 - u_2v_1) \\ &= 0, \end{aligned}$$

and the proof that $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0$ is similar. ■

Example 7.52 (a unit vector orthogonal to two vectors)

Find a unit vector in $\mathbb{R}^3$ which is orthogonal to both the vectors $\mathbf{u} = \mathbf{i} + 2\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - 2\mathbf{j}$.

Solution. We know that the cross product $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$. Now

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 2 & -2 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 2 & -2 \end{vmatrix} \mathbf{k} \\ &= [(0)(0) - (2)(-2)]\mathbf{i} - [(1)(0) - (2)(2)]\mathbf{j} + [(1)(-2) - (0)(2)]\mathbf{k} \\ &= 4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}. \end{aligned}$$

Any scalar multiple of this vector is also orthogonal to both $\mathbf{u}$ and $\mathbf{v}$. Among all

such vectors, a unit vector orthogonal to both is given by either

$$\frac{1}{\|\mathbf{u}\times\mathbf{v}\|}(\mathbf{u}\times\mathbf{v}) = \frac{1}{\sqrt{4^2+4^2+(-2)^2}}(4\mathbf{i}+4\mathbf{j}-2\mathbf{k}) = \frac{2}{3}\mathbf{i}+\frac{2}{3}\mathbf{j}-\frac{1}{3}\mathbf{k}$$

or

$$-\frac{1}{\|\mathbf{u}\times\mathbf{v}\|}(\mathbf{u}\times\mathbf{v}) = -\frac{2}{3}\mathbf{i}-\frac{2}{3}\mathbf{j}+\frac{1}{3}\mathbf{k}.$$

Remark 7.53 (right-hand rule)

If $\mathbf{u}$ and $\mathbf{v}$ are represented by directed line segments with the same initial point, then the vector $\mathbf{u}\times\mathbf{v}$ points in a direction perpendicular to the plane containing both $\mathbf{u}$ and $\mathbf{v}$; its direction is indicated by the right-hand rule. Hence $\mathbf{u}\times\mathbf{v}$ and $\mathbf{v}\times\mathbf{u}$ point in opposite directions, and in general

$$\mathbf{u}\times\mathbf{v} \neq \mathbf{v}\times\mathbf{u}, \qquad \mathbf{u}\times(\mathbf{v}\times\mathbf{w}) \neq (\mathbf{u}\times\mathbf{v})\times\mathbf{w}.$$

Theorem 7.54 (properties of the cross product)

Let $\mathbf{u},\mathbf{v},\mathbf{w}$ be three-dimensional vectors and c be a scalar. Then

(i) $\mathbf{u}\times\mathbf{v} = -(\mathbf{v}\times\mathbf{u})$
(ii) $(c\mathbf{u})\times\mathbf{v} = c(\mathbf{u}\times\mathbf{v}) = \mathbf{u}\times(c\mathbf{v})$
(iii) $\mathbf{u}\times(\mathbf{v}+\mathbf{w}) = \mathbf{u}\times\mathbf{v}+\mathbf{u}\times\mathbf{w}$
(iv) $(\mathbf{u}+\mathbf{v})\times\mathbf{w} = \mathbf{u}\times\mathbf{w}+\mathbf{v}\times\mathbf{w}$
(v) $\mathbf{u}\times\mathbf{u} = \mathbf{0}$

Theorem 7.55

$\mathbf{i}\times\mathbf{j} = \mathbf{k}$, $\mathbf{j}\times\mathbf{k} = \mathbf{i}$ and $\mathbf{k}\times\mathbf{i} = \mathbf{j}$.

Theorem 7.56 (length of the cross product)

Let $\mathbf{u}$ and $\mathbf{v}$ be three-dimensional non-zero vectors and let $\theta\in[0,\pi]$ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then

$$\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin\theta.$$

Proof. Let $\mathbf{u} = \langle u_1,u_2,u_3\rangle$ and $\mathbf{v} = \langle v_1,v_2,v_3\rangle$, and recall that $\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$. Now

$$\begin{aligned}
\|\mathbf{u}\times\mathbf{v}\|^2 &= (u_2v_3-u_3v_2)^2+(u_3v_1-u_1v_3)^2+(u_1v_2-u_2v_1)^2\\
&= (u_1^2+u_2^2+u_3^2)(v_1^2+v_2^2+v_3^2)-(u_1v_1+u_2v_2+u_3v_3)^2\\
&= \|\mathbf{u}\|^2\|\mathbf{v}\|^2-(\mathbf{u}\cdot\mathbf{v})^2\\
&= \|\mathbf{u}\|^2\|\mathbf{v}\|^2-(\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta)^2\\
&= \|\mathbf{u}\|^2\|\mathbf{v}\|^2(1-\cos^2\theta)\\
&= \|\mathbf{u}\|^2\|\mathbf{v}\|^2\sin^2\theta.
\end{aligned}$$

We always have $\sin\theta\geq 0$ as $\theta\in[0,\pi]$, so taking square roots on both sides gives $\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin\theta$. ■

Corollary 7.57

Let $\mathbf{u}$ and $\mathbf{v}$ be three-dimensional vectors. Then $\mathbf{u}\times\mathbf{v} = \mathbf{0}$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel.

Proof. The statement is trivial if $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$. If $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$, then from Theorem 7.56, $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ if and only if the angle $\theta \in [0, \pi]$ between $\mathbf{u}$ and $\mathbf{v}$ satisfies $\sin\theta = 0$, i.e. either $\theta = 0$ or $\theta = \pi$, which happens exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel. ■

Corollary 7.58 (area of a parallelogram)

Let $\mathbf{u}$ and $\mathbf{v}$ be three-dimensional vectors. Then the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$ has area equal to $\|\mathbf{u} \times \mathbf{v}\|$.

Proof. Let $\theta \in [0, \pi]$ be the angle between $\mathbf{u}$ and $\mathbf{v}$ (what if one or both of them are $\mathbf{0}$?). Then the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$ has a base of length $\|\mathbf{u}\|$ and the corresponding altitude of length $\|\mathbf{v}\| \sin\theta$. So its area is

$$A = (\|\mathbf{u}\|)(\|\mathbf{v}\| \sin\theta) = \|\mathbf{u} \times \mathbf{v}\|$$

by Theorem 7.56. ■

Example 7.59 (area of a parallelogram)

Find the area of the parallelogram with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$, $R(0, 3, 1)$ and $S(-1, 1, 2)$.
Solution. In fact, we only need the coordinates of three of the vertices, as the fourth vertex is determined by the other three. Now

$$\overrightarrow{PQ} = \langle 2-1, 1-(-1), -1-0\rangle = \langle 1, 2, -1\rangle$$
$$\text{and} \qquad \overrightarrow{PS} = \langle -1-1, 1-(-1), 2-0\rangle = \langle -2, 2, 2\rangle,$$

so

$$\overrightarrow{PQ} \times \overrightarrow{PS} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \langle 6, 0, 6\rangle.$$

The area of the parallelogram is given by the length of $\overrightarrow{PQ} \times \overrightarrow{PS}$, which is

$$\|\overrightarrow{PQ} \times \overrightarrow{PS}\| = \sqrt{6^2 + 0^2 + 6^2} = 6\sqrt{2}.$$

Theorem 7.60 (scalar triple product)

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three-dimensional vectors. Then

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}).$$

Proof. Let $\mathbf{u} = \langle u_1, u_2, u_3\rangle$, $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ and $\mathbf{w} = \langle w_1, w_2, w_3\rangle$. Then

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = u_1(v_2w_3 - v_3w_2) + u_2(v_3w_1 - v_1w_3) + u_3(v_1w_2 - v_2w_1)$$
$$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = v_1(w_2u_3 - w_3u_2) + v_2(w_3u_1 - w_1u_3) + v_3(w_1u_2 - w_2u_1)$$
$$\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}) = w_1(u_2v_3 - u_3v_2) + w_2(u_3v_1 - u_1v_3) + w_3(u_1v_2 - u_2v_1).$$

The right-hand sides are all equal to the same number

$$u_1v_2w_3 + u_2v_3w_1 + u_3v_1w_2 - u_1v_3w_2 - u_2v_1w_3 - u_3v_2w_1,$$

and so they are all equal. ■

Remark 7.61 (scalar triple product as a determinant)

Each of the above expressions $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$, $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{u})$ and $\mathbf{w}\cdot(\mathbf{u}\times\mathbf{v})$ is called the scalar

triple product of the vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$, and they are all equal to the determinant

$$\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

Proposition 7.62 (volume of a parallelepiped)

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three-dimensional vectors. Then the parallelepiped determined by $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ has volume equal to the absolute value of their scalar triple product, i.e. $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$.

Proof. Regard the base of the parallelepiped as the parallelogram determined by $\mathbf{v}$ and $\mathbf{w}$. We knew from Corollary 7.58 that this base has area

$$A = \|\mathbf{v} \times \mathbf{w}\|.$$

Now let $\theta \in [0, \pi]$ be the angle between the vectors $\mathbf{u}$ and $\mathbf{v} \times \mathbf{w}$. Then the corresponding height of the parallelepiped is $h = \|\mathbf{u}\||\cos\theta|$ (we need the absolute value because $\cos\theta < 0$ if $\theta \in (\frac{\pi}{2}, \pi]$). Therefore its volume is the product of its base area and its height,

$$V = Ah = \|\mathbf{v} \times \mathbf{w}\|(\|\mathbf{u}\||\cos\theta|) = |\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|,$$

as asserted. ■

Corollary 7.63 (coplanarity)

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three-dimensional vectors. Then $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ if and only if $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ are coplanar (i.e. they lie in the same plane).

7.6 Parametrized Curves

In high school, a line L in $\mathbb{R}^2$ is the set of points whose coordinates satisfy a linear equation, namely $L = \{(x, y) \in \mathbb{R}^2 : Ax + By + C = 0\}$. To generalise this to $\mathbb{R}^n$, note that a line is determined by a point lying on it together with a non-zero vector indicating its direction.

Definition. Let P_0 be a point with position vector $\mathbf{r}_0 := \overrightarrow{OP_0}$, and let $\mathbf{v}$ be a non-zero vector. The line passing through P_0 and parallel to $\mathbf{v}$ is the set of all points P whose position vector $\mathbf{r} := \overrightarrow{OP}$ satisfies the vector equation

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

for some real number $t \in (-\infty, +\infty)$.

Figure. A point P_0 with an arrow $\mathbf{v}$ issuing from it, and the line L drawn through P_0 in the direction of $\mathbf{v}$; a general point P on L satisfies $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$.

Remark 7.65 (facts about Definition 7.64)

(i) We may regard the definition as requiring that $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$; in other words, $\mathbf{r} - \mathbf{r}_0$ is a scalar multiple of $\mathbf{v}$ for every point P on the line.
(ii) It does not matter if we replace P_0 by any other point lying on the line.
(iii) It does not matter if we replace $\mathbf{v}$ by $k\mathbf{v}$ for a non-zero real k. The vector $\mathbf{v}$ only indicates the direction of the line; its length does not matter.

Lemma 7.66 (parametric equations of a line)

The line in $\mathbb{R}^3$ passing through (x_0, y_0, z_0) and parallel to $\langle a, b, c\rangle$ is the set of all points $(x, y, z) \in \mathbb{R}^3$ satisfying

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$

for some real number $t \in (-\infty, +\infty)$. A similar statement is true with $\mathbb{R}^3$ replaced by $\mathbb{R}^n$.

Proof. Focus only on the $\mathbb{R}^3$ case. Writing the vectors in Definition 7.64 in component form, $\mathbf{r}_0 = \langle x_0, y_0, z_0\rangle$, $\mathbf{v} = \langle a, b, c\rangle$ and $\mathbf{r} = \langle x, y, z\rangle$. So $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ becomes

$$\langle x, y, z\rangle = \langle x_0, y_0, z_0\rangle + t\langle a, b, c\rangle = \langle x_0 + at,\ y_0 + bt,\ z_0 + ct\rangle.$$

Equating corresponding components gives the required system. ■

Remark 7.67 (recovering the equation of a line in $\mathbb{R}^2$)

A line in $\mathbb{R}^2$ consists of all points (x, y) satisfying

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \end{cases}.$$

Eliminating t gives

$$bx - ay + ay_0 - bx_0 = 0,$$

which is the "high school" form $L = \{(x, y) \in \mathbb{R}^2 \colon Ax + By + C = 0\}$ with $A = b$, $B = -a$ and $C = ay_0 - bx_0$.

Remark 7.68 (terminology)

The equation $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$, with variable vector $\mathbf{r}$, is called the vector equation of a line, while the system $x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$ is called the parametric equations of a line. In both forms, $t \in (-\infty, +\infty)$ is called a parameter.

Example 7.69 (line through two points in $\mathbb{R}^3$)

Let L be the line in $\mathbb{R}^3$ that passes through the points $P(2, 0, 1)$ and $Q(1, 1, 3)$. (a) Find the parametric equations of L. (b) At which point does L intersect the yz-plane? (c) At which point does L intersect the plane with equation $4x - 3y - z = 16$?

Solution. (a) The required line L passes through $P(2, 0, 1)$ and is parallel to the vector

$$\mathbf{v} = \overrightarrow{PQ} = \langle 1 - 2,\ 1 - 0,\ 3 - 1\rangle = \langle -1, 1, 2\rangle,$$

so its parametric equations are given by

$$\begin{cases} x = 2 + (-1)t \\ y = 0 + 1t \\ z = 1 + 2t \end{cases}, \quad \text{i.e.} \quad x = 2 - t,\ y = t,\ z = 1 + 2t,$$

where $t \in (-\infty, +\infty)$. (To find the equation of a line, we just need a point and a direction vector.)

(b) At the point (x, y, z) where L intersects the yz-plane, we must have $x = 0$. This means $2 - t = 0$, so $t = 2$. Therefore the point of intersection has coordinates $(0, 2, 5)$.

(c) At the point where L intersects $4x - 3y - z = 16$, we must have

$$4(2 - t) - 3(t) - (1 + 2t) = 16,$$

and so $t = -1$. Therefore the point of intersection has coordinates $(3, -1, -1)$.

The vector equation $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ of a line, or equivalently $\mathbf{r}(t) = \langle x_0 + at,\ y_0 + bt,\ z_0 + ct\rangle$, can be regarded as a special case of a vector-valued function. The three components of this vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ are the linear polynomials

$$f(t) = x_0 + at, \qquad g(t) = y_0 + bt \qquad \text{and} \qquad h(t) = z_0 + ct.$$

Definition. A vector-valued function is a function $\mathbf{r}$ whose codomain is $\mathbb{R}^n$. In other words, to each object t in the domain it assigns a vector $\mathbf{r}(t)$ which can be written as

$$\mathbf{r}(t) = \langle r_1(t), r_2(t), \ldots, r_n(t)\rangle,$$

where $r_1, r_2, \ldots, r_n$ are scalar-valued (real-valued) functions. These functions are called the component functions of $\mathbf{r}$.

Definition. Let I be an interval in $\mathbb{R}$ and let $n \geq 2$. Then in geometry, a (parametrized) curve in $\mathbb{R}^n$ is a vector-valued function $\mathbf{r} : I \to \mathbb{R}^n$ whose component functions are all continuous.

Figure. The image of a curve $\mathbf{r} : [a, b] \to \mathbb{R}^3$. Here $\mathbf{r}(a)$ and $\mathbf{r}(b)$ are the position vectors of the end-points, and as t increases from a to b, the tip of the moving vector $\mathbf{r}(t)$ traces the image of $\mathbf{r}$.

Remark 7.72 (describing the image of a curve)

Similar to lines, we can express the image of a curve in $\mathbb{R}^n$ in various ways:
- Use the vector equation $\mathbf{r}(t) = \langle r_1(t), r_2(t), \ldots, r_n(t)\rangle$.
- Use parametric equations $x_1 = r_1(t),\ x_2 = r_2(t),\ \ldots,\ x_n = r_n(t)$.

Example (a line segment as a curve)

The function $\mathbf{r} : [0, 1] \to \mathbb{R}^3$ defined by $\mathbf{r}(t) = \langle 2 - t,\ t,\ 1 + 2t\rangle$ is a curve in $\mathbb{R}^3$. Its image is the line segment in $\mathbb{R}^3$ joining $(2, 0, 1)$ and $(1, 1, 3)$. (If the domain $[0, 1]$ were replaced by the whole $\mathbb{R}$, the image would be the whole line joining the two points.)

Example 7.73 (a parabola)

The function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = t\,\mathbf{i} + t^2\,\mathbf{j}$ is a curve in $\mathbb{R}^2$, with parametric equations

$$\begin{cases} x = t \\ y = t^2 \end{cases},$$

where $t \in (-\infty, +\infty)$. Eliminating the parameter t, the image of this curve is the branch of the parabola in $\mathbb{R}^2$ defined by $y = x^2$.

Example 7.74 (circle, ellipse and hyperbola)

Let a, b be positive real numbers.
- The function $\mathbf{r} : [0, 2\pi] \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle a\cos t,\ a\sin t\rangle$ is a curve in $\mathbb{R}^2$ with parametric equations $x = a\cos t,\ y = a\sin t$ $(t \in [0, 2\pi])$. Eliminating t gives $x^2 + y^2 = a^2$, so its image is the circle centred at the origin with radius a.
- The function $\mathbf{r} : [0, 2\pi] \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle a\cos t,\ b\sin t\rangle$ parametrizes the ellipse in $\mathbb{R}^2$ with equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$.
- The function $\mathbf{r} : \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle a\sec t,\ b\tan t\rangle$ parametrizes

(the right branch of) the hyperbola in $\mathbb{R}^2$ with equation $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.

Example 7.75 (the graph of a function)

Let I be an interval and let $f : I \to \mathbb{R}$ be a continuous function. Then the graph of f, defined by the equation $y = f(x)$, is the curve in $\mathbb{R}^2$ described by the vector-valued function $\mathbf{r} : I \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle t,\ f(t)\rangle$.

Example 7.76 (a polar curve)

Let I be an interval and let $f : I \to \mathbb{R}$ be a continuous function. The polar curve in $\mathbb{R}^2$ defined by the polar equation $r = f(\theta)$ is described by the vector-valued function $\mathbf{r} : I \to \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = \langle f(t)\cos t,\ f(t)\sin t\rangle.$$

Example 7.77 (a circular helix)

The function $\mathbf{r} : [0, 4\pi] \to \mathbb{R}^3$ defined by $\mathbf{r}(t) = (\cos t)\,\mathbf{i} + (\sin t)\,\mathbf{j} + t\,\mathbf{k}$ is a curve in $\mathbb{R}^3$, with parametric equations

$$\begin{cases} x = \cos t \\ y = \sin t \\ z = t \end{cases},$$

where $t \in [0, 4\pi]$. The image of $\mathbf{r}$ is called a circular helix.

Example 7.78 (intersection of a cylinder and a plane)

The function $\mathbf{r} : [0, 2\pi] \to \mathbb{R}^3$ defined by $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ 2 - \cos t\rangle$ is a curve in $\mathbb{R}^3$, with parametric equations

$$\begin{cases} x = \cos t \\ y = \sin t \\ z = 2 - \cos t \end{cases},$$

where $t \in [0, 2\pi]$. Eliminating t gives $x^2 + y^2 = 1$ and $x + z = 2$, so its image is a loop in $\mathbb{R}^3$: the curve of intersection of the circular cylinder $x^2 + y^2 = 1$ and the plane $x + z = 2$.

Definition. Let $\mathbf{r}$ be a vector-valued function.

(i) Let a be a real number. Then the derivative of $\mathbf{r}$ at a is the limit

$$\mathbf{r}'(a) := \lim_{h\to 0} \frac{1}{h}\,[\mathbf{r}(a + h) - \mathbf{r}(a)].$$

(ii) The derivative of $\mathbf{r}$ is the vector-valued function $\mathbf{r}'$ defined by

$$\mathbf{r}'(t) := \lim_{h\to 0} \frac{1}{h}\,[\mathbf{r}(t + h) - \mathbf{r}(t)].$$

Theorem 7.80 (componentwise differentiation)

Let $\mathbf{r}$ be a vector-valued function given by $\mathbf{r}(t) = \langle r_1(t), r_2(t), \ldots, r_n(t)\rangle$. If the component functions $r_1, r_2, \ldots, r_n$ are all differentiable at t, then

$$\mathbf{r}'(t) = \langle r_1'(t), r_2'(t), \ldots, r_n'(t)\rangle.$$

Proof. By Definition 7.79, form the difference quotient componentwise:

$$\frac{1}{h}[\mathbf{r}(t+h)-\mathbf{r}(t)] = \left\langle \frac{r_1(t+h)-r_1(t)}{h},\ \ldots,\ \frac{r_n(t+h)-r_n(t)}{h} \right\rangle.$$

As $h \to 0$, each component tends to $r_j'(t)$; a limit of vectors is taken componentwise, so

$$\mathbf{r}'(t) = \langle r_1'(t), r_2'(t), \ldots, r_n'(t)\rangle.$$

■

Remark 7.81 (smooth curves, tangent vectors and tangent lines)

In geometry, a curve $\mathbf{r} : I \to \mathbb{R}^n$ is said to be smooth (or $\mathbf{C}^1$) if its derivative $\mathbf{r}'$ has continuous component functions and $\mathbf{r}'$ is never $\mathbf{0}$. In this case, for each $t \in I$,

- the derivative $\mathbf{r}'(t)$ is a tangent vector to the curve at t;
- the tangent line to the curve at t is the line which passes through the point with position vector $\mathbf{r}(t)$ and is parallel to $\mathbf{r}'(t)$.

Example (tangent vector to a graph)

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function and let the graph of f be the curve $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle t, f(t)\rangle$. At each $t \in \mathbb{R}$ the tangent vector is $\mathbf{r}'(t) = \langle 1, f'(t)\rangle$; the direction of this vector agrees with the "slope" $f'(t)$ of the tangent line.

Example (slope of the tangent to a polar curve)

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function and consider the polar curve $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle f(t)\cos t,\ f(t)\sin t\rangle$. At each $t \in \mathbb{R}$ the tangent vector is

$$\mathbf{r}'(t) = \langle f'(t)\cos t - f(t)\sin t,\ f'(t)\sin t + f(t)\cos t\rangle;$$

so the slope of the tangent line at the point $\mathbf{r}(t)$ is

$$\frac{f'(t)\sin t + f(t)\cos t}{f'(t)\cos t - f(t)\sin t}.$$

Example 7.77 (tangent line to a circular helix)

Let $\mathbf{r} : [0, 4\pi] \to \mathbb{R}^3$ be the circular helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t\rangle$. Find the parametric equations of the tangent line to the curve $\mathbf{r}$ at the point P with coordinates $(0, 1, \frac{\pi}{2})$.

Solution. The derivative $\mathbf{r}' : (0, 4\pi) \to \mathbb{R}^3$ is given by

$$\mathbf{r}'(t) = \langle -\sin t,\ \cos t,\ 1\rangle.$$

The point $P(0, 1, \frac{\pi}{2})$ corresponds to $t = \frac{\pi}{2}$, so a tangent vector at P is

$$\mathbf{r}'\left(\frac{\pi}{2}\right) = \langle -\sin\tfrac{\pi}{2},\ \cos\tfrac{\pi}{2},\ 1\rangle = \langle -1, 0, 1\rangle.$$

The parametric equations of the tangent line are therefore

$$\begin{cases} x = 0 + (-1)u \\ y = 1 + (0)u \\ z = \dfrac{\pi}{2} + (1)u \end{cases}, \quad \text{i.e.} \quad \begin{cases} x = -u \\ y = 1 \\ z = \dfrac{\pi}{2} + u \end{cases},$$

where $u \in (-\infty, +\infty)$. (We use different symbols t and u for parameters correspond-

ing to different curves.)

7.7 Geometric Applications of Integrals

Similar to estimating an area by a Riemann sum, we can estimate the length of a curve by dividing it into small portions.

Figure. A curve $\mathbf{r} : [a, b] \to \mathbb{R}^n$ subdivided at $a = t_0 < t_1 < t_2 < t_3 < t_4 = b$ into small pieces; the points $\mathbf{r}(t_0), \ldots, \mathbf{r}(t_4)$ are joined by chords whose lengths approximate the corresponding pieces.

Let $\mathbf{r} : [a, b] \to \mathbb{R}^n$ be a curve so that as t increases from a to b, the image is traced exactly once. Consider a regular partition $P = \{t_0, t_1, \ldots, t_m\}$ of $[a, b]$ into m subintervals, so that the curve is divided into m small pieces. As $m \to +\infty$, each small piece is approximately a line segment, so the length of the curve is the sum of the lengths of the segments.

Definition. A curve $\mathbf{r} : [a, b] \to \mathbb{R}^n$ is said to be rectifiable if the limit

$$l = \lim_{m \to +\infty} \sum_{k=1}^{m} \|\mathbf{r}(t_k) - \mathbf{r}(t_{k-1})\|$$

exists as a real number. This limit l is called the arc-length of the curve $\mathbf{r}$.

Theorem 7.83 (arc-length of a smooth curve)

Let $\mathbf{r} : [a, b] \to \mathbb{R}^n$ be a smooth (i.e. C^1) curve. Then $\mathbf{r}$ is rectifiable and its arc-length is given by

$$l = \int_a^b \|\mathbf{r}'(t)\| \, dt.$$

Proof. For each $k \in \{1, 2, \ldots, m\}$,

$$\|\mathbf{r}(t_k) - \mathbf{r}(t_{k-1})\| = \frac{\|\mathbf{r}(t_k) - \mathbf{r}(t_{k-1})\|}{t_k - t_{k-1}} (t_k - t_{k-1}).$$

As $m \to +\infty$ we have $t_k - t_{k-1} \to 0$ and

$$\frac{\|\mathbf{r}(t_k) - \mathbf{r}(t_{k-1})\|}{t_k - t_{k-1}} \longrightarrow \|\mathbf{r}'(t_{k-1})\|.$$

Since $\mathbf{r}'$ is continuous, it follows that

$$\lim_{m \to +\infty} \sum_{k=1}^{m} \|\mathbf{r}(t_k) - \mathbf{r}(t_{k-1})\| = \lim_{m \to +\infty} \sum_{k=1}^{m} \|\mathbf{r}'(t_{k-1})\| \, (t_k - t_{k-1}) = \int_a^b \|\mathbf{r}'(t)\| \, dt.$$

■

Example 7.77 (arc-length of a circular helix)

Let $\mathbf{r}$ be the curve in $\mathbb{R}^3$ defined by $\mathbf{r}(t) = \langle \cos t, \ \sin t, \ t \rangle$ for $0 \leq t \leq 4\pi$, a circular helix. Find its arc-length.

Solution. We have $\mathbf{r}'(t) = \langle -\sin t, \ \cos t, \ 1 \rangle$ and so

$$\|\mathbf{r}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

for every $t \in (0, 4\pi)$. Therefore

$$l = \int_0^{4\pi} \|\mathbf{r}'(t)\|\, dt = \int_0^{4\pi} \sqrt{2}\, dt = 4\sqrt{2}\,\pi.$$

Example 7.84 (arc-length of a portion of a curve)

Let $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ be defined by $\mathbf{r}(t) = \langle t^3,\ 0,\ t^2 \rangle$ for $-\infty < t < +\infty$. Find the arc-length of the portion of the curve whose coordinates (x, y, z) satisfy $z \geq \sqrt{x^2 + y^2}$.
Solution. The parametric equations are $x = t^3$, $y = 0$, $z = t^2$. The required portion satisfies $z \geq \sqrt{x^2 + y^2}$, i.e.

$$t^2 \geq \sqrt{(t^3)^2 + 0^2} = |t|^3,$$

which rearranges to $t^2(|t| - 1) \leq 0$. Since $t^2 \geq 0$ always, this is simply $|t| - 1 \leq 0$, so

$$-1 \leq t \leq 1.$$

Therefore the required portion has arc-length

$$\begin{aligned} l = \int_{-1}^{1} \|\mathbf{r}'(t)\|\, dt &= \int_{-1}^{1} \sqrt{(3t^2)^2 + 0^2 + (2t)^2}\, dt \\ &= \int_{-1}^{1} \sqrt{9t^4 + 4t^2}\, dt = \int_{-1}^{1} |t|\sqrt{9t^2 + 4}\, dt. \end{aligned}$$

Since the integrand $|t|\sqrt{9t^2 + 4}$ is even, the last integral equals

$$2\int_0^1 |t|\sqrt{9t^2 + 4}\, dt = \int_0^1 2t\sqrt{9t^2 + 4}\, dt.$$

With the substitution $u = 9t^2 + 4$, $du = 18t\, dt$,

$$l = \int_0^1 2t\sqrt{9t^2 + 4}\, dt = \frac{1}{9}\int_4^{13} \sqrt{u}\, du = \frac{1}{9}\left[\frac{2}{3}u^{\frac{3}{2}}\right]_4^{13} = \frac{2}{27}\left(13\sqrt{13} - 8\right).$$

As in Example 7.75, the graph of a function $f : [a, b] \to \mathbb{R}$ can be regarded as the image of a vector-valued function $\mathbf{r} : [a, b] \to \mathbb{R}^2$.

Corollary 7.85 (arc-length of a graph)

Let $f : [a, b] \to \mathbb{R}$ be a continuously differentiable function. Then the arc-length of the graph of f is

$$l = \int_a^b \sqrt{1 + [f'(t)]^2}\, dt.$$

Proof. The graph of f is the curve $\mathbf{r} : [a, b] \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle t, f(t) \rangle$, whose derivative is $\mathbf{r}'(t) = \langle 1, f'(t) \rangle$ for every $t \in (a, b)$. So

$$l = \int_a^b \|\mathbf{r}'(t)\|\, dt = \int_a^b \sqrt{1 + [f'(t)]^2}\, dt.$$

■

Example 7.86 (length of a hanging cable)

Let $a > 0$ and $b > 0$. A hanging cable has the shape of the graph of

$$f(x) = \frac{1}{2a}\left(e^{ax} + e^{-ax}\right)$$

for $x \in [-b, b]$. Compute the length of the cable, i.e. the arc-length of the graph of f.

Solution. For each $x \in (-b, b)$, $f'(x) = \frac{1}{2}\left(e^{ax} - e^{-ax}\right)$. So

$$\begin{aligned} l = \int_{-b}^{b} \sqrt{1 + [f'(x)]^2}\, dx &= \int_{-b}^{b} \sqrt{1 + \left[\tfrac{1}{2}\left(e^{ax} - e^{-ax}\right)\right]^2}\, dx \\ &= \int_{-b}^{b} \sqrt{1 + \tfrac{1}{4}\left(e^{2ax} - 2 + e^{-2ax}\right)}\, dx \\ &= \int_{-b}^{b} \sqrt{\tfrac{1}{4}\left(e^{2ax} + 2 + e^{-2ax}\right)}\, dx \\ &= \int_{-b}^{b} \sqrt{\tfrac{1}{4}\left(e^{ax} + e^{-ax}\right)^2}\, dx \\ &= \int_{-b}^{b} \tfrac{1}{2}\left(e^{ax} + e^{-ax}\right) dx = \frac{1}{a}\left(e^{ab} - e^{-ab}\right). \end{aligned}$$

As in Example 7.76, a polar curve can also be regarded as the image of a vector-valued function $\mathbf{r} : [a, b] \to \mathbb{R}^2$.

Corollary 7.87 (arc-length of a polar curve)

Let $f : [a, b] \to \mathbb{R}$ be a continuously differentiable function. Then the arc-length of the polar curve defined by $r = f(\theta)$ is

$$l = \int_a^b \sqrt{[f(\theta)]^2 + [f'(\theta)]^2}\, d\theta.$$

Proof. As in Example 7.76, the polar curve $r = f(\theta)$ is the curve $\mathbf{r} : [a, b] \to \mathbb{R}^2$ defined by $\mathbf{r}(t) = \langle f(t)\cos t,\ f(t)\sin t\rangle$, whose derivative is

$$\mathbf{r}'(t) = \langle f'(t)\cos t - f(t)\sin t,\ f'(t)\sin t + f(t)\cos t\rangle$$

for every $a < t < b$. Its length is

$$\begin{aligned} \|\mathbf{r}'(t)\| &= \sqrt{[f'(t)\cos t - f(t)\sin t]^2 + [f'(t)\sin t + f(t)\cos t]^2} \\ &= \sqrt{[f(t)]^2 + [f'(t)]^2}, \end{aligned}$$

so the arc-length is

$$l = \int_a^b \|\mathbf{r}'(t)\|\, dt = \int_a^b \sqrt{[f(t)]^2 + [f'(t)]^2}\, dt.$$

■

Example 7.88 (arc-length of a cardioid)

Find the arc-length of the cardioid in $\mathbb{R}^2$ defined by the polar equation $r = 1 - \cos\theta$.

Solution. Let $f(\theta) = 1 - \cos\theta$. The whole curve is traced once as θ increases from

0 to 2π. So

$$l = \int_0^{2\pi} \sqrt{[f(\theta)]^2 + [f'(\theta)]^2}\, d\theta = \int_0^{2\pi} \sqrt{(1-\cos\theta)^2 + (\sin\theta)^2}\, d\theta$$
$$= \int_0^{2\pi} \sqrt{2 - 2\cos\theta}\, d\theta = \int_0^{2\pi} \sqrt{4\sin^2\tfrac{\theta}{2}}\, d\theta$$
$$= \int_0^{2\pi} \left|2\sin\tfrac{\theta}{2}\right| d\theta$$
$$= \int_0^{2\pi} 2\sin\tfrac{\theta}{2}\, d\theta = \left[-4\cos\tfrac{\theta}{2}\right]_0^{2\pi} = 8.$$

Example 7.89 (arc-length of another curve)

Find the arc-length of the curve defined by the polar equation $r = 1 + \cos\dfrac{\theta}{2}$.

Solution. Let $f(\theta) = 1 + \cos\dfrac{\theta}{2}$. The whole curve is traced once as θ increases from 0 to 4π. So

$$l = \int_0^{4\pi} \sqrt{[f(\theta)]^2 + [f'(\theta)]^2}\, d\theta = \int_0^{4\pi} \sqrt{\left(1+\cos\tfrac{\theta}{2}\right)^2 + \left(-\tfrac{1}{2}\sin\tfrac{\theta}{2}\right)^2}\, d\theta$$
$$= \frac{1}{2}\int_0^{4\pi} \sqrt{5 + 8\cos\tfrac{\theta}{2} + 3\cos^2\tfrac{\theta}{2}}\, d\theta$$
$$= \frac{1}{2}\int_0^{4\pi} \sqrt{\left(1+\cos\tfrac{\theta}{2}\right)\left(5+3\cos\tfrac{\theta}{2}\right)}\, d\theta.$$

The integrand $g(\theta)$ satisfies $g(4\pi - \theta) = g(\theta)$, so

$$l = \int_0^{2\pi} \sqrt{\left(1+\cos\tfrac{\theta}{2}\right)\left(5+3\cos\tfrac{\theta}{2}\right)}\, d\theta = \int_0^{2\pi} \sqrt{\underbrace{2\cos^2\tfrac{\theta}{4}}_{\geq 0}\left(8 - 6\sin^2\tfrac{\theta}{4}\right)}\, d\theta,$$

using $1 + \cos\frac{\theta}{2} = 2\cos^2\frac{\theta}{4}$ and $5 + 3\cos\frac{\theta}{2} = 8 - 6\sin^2\frac{\theta}{4}$. Hence

$$l = 2\int_0^{2\pi} \cos\tfrac{\theta}{4}\sqrt{4 - 3\sin^2\tfrac{\theta}{4}}\, d\theta.$$

Let $u = \sin\dfrac{\theta}{4}$; then $du = \dfrac{1}{4}\cos\dfrac{\theta}{4}\, d\theta$, and as θ runs from 0 to 2π, u runs from 0 to 1. Thus

$$l = 8\int_0^1 \sqrt{4 - 3u^2}\, du.$$

Finally let $u = \dfrac{2}{\sqrt{3}}\sin t$ with $t \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$. Then $du = \dfrac{2}{\sqrt{3}}\cos t\, dt$ and $\sqrt{4 - 3u^2} = \sqrt{4 - 4\sin^2 t} = |2\cos t| = 2\cos t$, so

$$l = 8\int_0^{\frac{\pi}{3}} (2\cos t)\left(\frac{2}{\sqrt{3}}\cos t\, dt\right) = \frac{8}{\sqrt{3}}\int_0^{\frac{\pi}{3}} (2 + 2\cos 2t)\, dt$$
$$= \frac{8}{\sqrt{3}}\Big[2t + \sin 2t\Big]_0^{\frac{\pi}{3}} = 4 + \frac{16\pi}{3\sqrt{3}}.$$

In Chapter 5 we learnt how to use integrals to find the area of the region in $\mathbb{R}^2$ under the graph of a function. This section elaborates the method so that we can also find areas

of general regions in $\mathbb{R}^2$, volumes of solids in $\mathbb{R}^3$, and areas of surfaces of revolution in $\mathbb{R}^3$.

Theorem 7.90 (Cavalieri's principle in $\mathbb{R}^2$)

Let X and Y be two bounded regions in $\mathbb{R}^2$. If for each vertical line L in $\mathbb{R}^2$, the line segments $L \cap X$ and $L \cap Y$ have equal length, then X and Y have the same area.

Corollary 7.91 (area between two graphs)

Let $f, g : [a, b] \to \mathbb{R}$ be continuous functions such that $f(x) \geq g(x)$ for every $x \in [a, b]$. Then the region $\{(x, y) \in \mathbb{R}^2 : a \leq x \leq b \text{ and } g(x) \leq y \leq f(x)\}$ has area

$$A = \int_a^b (f(x) - g(x))\, dx.$$

Proof. A vertical line at x meets the region in a segment of length $f(x) - g(x)$. By Cavalieri's principle and the definition of the integral as a limit of Riemann sums of these segment lengths, the area is $\displaystyle\int_a^b (f(x) - g(x))\, dx$. ■

Remark 7.92 (the signed-area version)

Let $f, g : [a, b] \to \mathbb{R}$ be continuous. No matter which of them has larger values on $[a, b]$, the region between their graphs over $[a, b]$ has area

$$A = \int_a^b |f(x) - g(x)|\, dx.$$

The integrand is obtained by always subtracting the function with the lower graph from the function with the higher graph.

Remark 7.93 (horizontal slices)

Cavalieri's principle is still true if "vertical lines" are replaced by "horizontal lines". As a result, if $f(y) \geq g(y)$ for every $y \in [a, b]$, then the region $\{(x, y) \in \mathbb{R}^2 : a \leq y \leq b \text{ and } g(y) \leq x \leq f(y)\}$ has area

$$A = \int_a^b (f(y) - g(y))\, dy,$$

the region between the curves $x = f(y)$ and $x = g(y)$ over $[a, b]$ in the y-axis. The integrand is obtained by always subtracting the function corresponding to the left curve from the function corresponding to the right curve.

Example 7.94 (area between two trigonometric graphs)

Find the area of the region in $\mathbb{R}^2$ bounded between the curves $y = -\sin x$ and $y = \cos x$ on the interval $[0, \pi]$.

Solution. The two curves intersect when $\cos x = -\sin x$, i.e. $\tan x = -1$, i.e. $x = \frac{3\pi}{4}$. On $[0, \frac{3\pi}{4}]$, $\cos x$ lies above $-\sin x$; on $[\frac{3\pi}{4}, \pi]$ the reverse holds. Hence the

required area is

$$A = \int_0^{\frac{3\pi}{4}} [\cos x - (-\sin x)]\,dx + \int_{\frac{3\pi}{4}}^{\pi} [(-\sin x) - \cos x]\,dx$$
$$= \Big[\sin x - \cos x\Big]_0^{\frac{3\pi}{4}} + \Big[\cos x - \sin x\Big]_{\pi}^{\frac{3\pi}{4}} = 2\sqrt{2}.$$

Example 7.95 (area bounded by two graphs and the y-axis)

Find the area of the region in $\mathbb{R}^2$ bounded by the graphs of $f(x) = \dfrac{\pi}{2}$ and $g(x) = \arcsin x$, and the y-axis.

Solution. The graphs of f and g intersect only at $(1, \dfrac{\pi}{2})$. Since $f(x) \geq g(x)$ for $x \in [0, 1]$, the required area is

$$A = \int_0^1 \Big(\frac{\pi}{2} - \arcsin x\Big)\,dx$$
$$= \int_0^1 \frac{\pi}{2}\,dx - \Big[x \arcsin x\Big]_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx = \Big[\frac{\pi}{2}x - x\arcsin x - \sqrt{1-x^2}\Big]_0^1 = 1.$$

Alternative solution. The region $\{(x, y) \in \mathbb{R}^2 \colon 0 \leq x \leq 1 \text{ and } \arcsin x \leq y \leq \frac{\pi}{2}\}$ can be rewritten as $\{(x, y) \in \mathbb{R}^2 \colon 0 \leq y \leq \frac{\pi}{2} \text{ and } 0 \leq x \leq \sin y\}$. Hence

$$A = \int_0^{\frac{\pi}{2}} \sin y\,dy = 1.$$

Instead of the area between graphs, we can study more generally the area of a region in $\mathbb{R}^2$ bounded by a parametrized curve.

Theorem 7.96 (area bounded by a parametrized curve)

Let $\mathbf{r} : [a, b] \to \mathbb{R}^2$ be a smooth curve defined by $\mathbf{r}(t) = \langle f(t), g(t)\rangle$. Then

- the (net) area under the curve and above the x-axis is

$$\int_a^b g(t)\,f'(t)\,dt;$$

- the (net) area to the left of the curve and to the right of the y-axis is

$$\int_a^b f(t)\,g'(t)\,dt.$$

If $\mathbf{r} : [a, b] \to \mathbb{R}^2$ is a simple closed curve (a loop without self-intersection) oriented counterclockwise, then the area of the region enclosed is

$$A = \int_a^b f(t)\,g'(t)\,dt = -\int_a^b g(t)\,f'(t)\,dt.$$

Proof (derivation of the net-area formula). Under the curve, $y = g(t)$ and $x = f(t)$. Substituting $x = f(t)$, $dx = f'(t)\,dt$ into $A = \displaystyle\int y\,dx$ gives

$$A = \int_a^b g(t)\,f'(t)\,dt,$$

with the orientation as in the figure; swapping the initial and terminal points $\mathbf{r}(a)$ and $\mathbf{r}(b)$ introduces a minus sign. Integrating instead with respect to y gives the "left-of-the-curve" formula $\displaystyle\int_a^b f(t)\,g'(t)\,dt$. For a counterclockwise simple closed curve the two expressions agree in absolute value and together give the enclosed area. ■

Example 7.97 (area of an ellipse)

Let $a, b > 0$ and consider the ellipse defined by $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, parametrized by $\mathbf{r} : [0, 2\pi] \to \mathbb{R}^2$ with $\mathbf{r}(t) = \langle f(t), g(t)\rangle = \langle a\cos t,\ b\sin t\rangle$. This curve is oriented counterclockwise, so the area inside the ellipse is

$$A = \int_0^{2\pi} f(t)g'(t)\,dt = \int_0^{2\pi} (a\cos t)(b\cos t)\,dt = ab\int_0^{2\pi} \cos^2 t\,dt = \pi ab.$$

Theorem 7.98 (area bounded by a polar curve)

Let $a < b \le a + 2\pi$ and let $f : [a, b] \to [0, +\infty)$ be a non-negative continuous function. Then the region $\{(r, \theta)\colon a \le \theta \le b \text{ and } 0 \le r \le f(\theta)\}$, in polar coordinates, has area

$$A = \int_a^b \frac{1}{2}[f(\theta)]^2\,d\theta.$$

Proof. Let $P = \{\theta_0, \theta_1, \ldots, \theta_n\}$ be a partition of $[a, b]$. For each $1 \le k \le n$, the part of the region at angle θ_k is approximately a thin circular sector of area

$$\frac{1}{2}[f(\theta_k)]^2(\theta_k - \theta_{k-1}).$$

Summing these sector areas and letting $\|P\| \to 0$ in the resulting right Riemann sum,

$$A = \lim_{\|P\|\to 0} \sum_{k=1}^{n} \frac{1}{2}[f(\theta_k)]^2(\theta_k - \theta_{k-1}) = \int_a^b \frac{1}{2}[f(\theta)]^2\,d\theta.$$

■

Example 7.99 (area bounded by the outer loop of $r = 1 + 2\cos\theta$)

Compute the area of the region in $\mathbb{R}^2$ bounded by (the outer loop of) the curve with polar equation $r = 1 + 2\cos\theta$.

Solution. The polar coordinates (r, θ) of each point in the region satisfy $0 \le r \le 1 + 2\cos\theta$. Both inequalities can hold only when $1 + 2\cos\theta \ge 0$, i.e. when $\theta \in \left[-\dfrac{2\pi}{3}, \dfrac{2\pi}{3}\right]$. So the region is $\left\{(r, \theta)\colon\ -\dfrac{2\pi}{3} \le \theta \le \dfrac{2\pi}{3} \text{ and } 0 \le r \le 1 + 2\cos\theta\right\}$, and its area is

$$\begin{aligned} A &= \int_{-\frac{2\pi}{3}}^{\frac{2\pi}{3}} \frac{1}{2}(1 + 2\cos\theta)^2\,d\theta = \int_{-\frac{2\pi}{3}}^{\frac{2\pi}{3}} \frac{1}{2}\left(1 + 4\cos\theta + 4\cos^2\theta\right)d\theta \\ &= \int_{-\frac{2\pi}{3}}^{\frac{2\pi}{3}} \frac{1}{2}\Big[1 + 4\cos\theta + 2(1 + \cos 2\theta)\Big]d\theta = 2\pi + \frac{3\sqrt{3}}{2}. \end{aligned}$$

Think

If we had forgotten to restrict the interval for θ and had instead taken $[0, 2\pi]$, what area would $\displaystyle\int_0^{2\pi} \frac{1}{2}(1+2\cos\theta)^2\,d\theta$ represent? Why must f be non-negative in Theorem 7.98? What if we want only the area inside the outer loop but outside the inner loop?

Example 7.100 (area enclosed by one loop of a four-leaved rose)

Find the area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$.

Solution. Consider the region enclosed by the "right loop", $\left\{(r,\theta)\colon -\frac{\pi}{4} \le \theta \le \frac{\pi}{4} \text{ and } 0 \le r \le \cos 2\theta\right\}$. Its area is

$$A = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2}(\cos 2\theta)^2\,d\theta = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{4}(1+\cos 4\theta)\,d\theta = \frac{\pi}{8}.$$

Example 7.101 (area bounded by a polar parabola and a line)

Let $l > 0$ and $a \in (0, \pi)$. Let C be the parabola with polar equation $r = \dfrac{l}{1+\cos\theta}$, and let L be the line through the origin with slope $\tan a$ (so L is the y-axis if $a = \frac{\pi}{2}$). Compute the area of the region in $\mathbb{R}^2$ bounded by C and L.

Solution. The region is $\left\{(r,\theta)\colon a-\pi \le \theta \le a \text{ and } 0 \le r \le \dfrac{l}{1+\cos\theta}\right\}$, so its area is

$$A = \int_{a-\pi}^{a} \frac{1}{2}\left(\frac{l}{1+\cos\theta}\right)^2 d\theta.$$

Since the integrand is a trigonometric rational function, we use Weierstrass' substitution $t = \tan\dfrac{\theta}{2}$ (where $\theta \in (-\pi, \pi)$): $\cos\theta = \dfrac{1-t^2}{1+t^2}$ and $d\theta = \dfrac{2}{1+t^2}\,dt$. As θ runs from $a-\pi$ to a, t runs from $-\cot\dfrac{a}{2}$ to $\tan\dfrac{a}{2}$. Then

$$\begin{aligned} A &= \int_{-\cot(\frac{a}{2})}^{\tan(\frac{a}{2})} \frac{1}{2}\left(\frac{l(1+t^2)}{2}\right)^2 \frac{2}{1+t^2}\,dt = \frac{l^2}{4}\int_{-\cot(\frac{a}{2})}^{\tan(\frac{a}{2})} (1+t^2)\,dt \\ &= \frac{l^2}{4}\left[t + \frac{t^3}{3}\right]_{-\cot(\frac{a}{2})}^{\tan(\frac{a}{2})} = \frac{2l^2}{3\sin^3 a}. \end{aligned}$$

Corollary 7.102 (area between two polar curves)

Let $a < b \le a + 2\pi$ and let $f, g : [a, b] \to [0, +\infty)$ be non-negative continuous functions. Then the region $\{(r,\theta)\colon a \le \theta \le b \text{ and } g(\theta) \le r \le f(\theta)\}$ has area

$$A = \int_a^b \frac{1}{2}\left(f(\theta)^2 - g(\theta)^2\right) d\theta.$$

Proof. Write the region as the difference of the two sectors $0 \le r \le f(\theta)$ and $0 \le r \le g(\theta)$ and apply Theorem 7.98 to each:

$$A = \int_a^b \frac{1}{2}[f(\theta)]^2 d\theta - \int_a^b \frac{1}{2}[g(\theta)]^2 d\theta = \int_a^b \frac{1}{2}\left(f(\theta)^2 - g(\theta)^2\right) d\theta.$$

■

Example 7.103 (area inside one polar curve and outside another)

Compute the area of the region in $\mathbb{R}^2$ which lies in the interior of (the outer loop of) the curve $r = 1 + 2\cos\theta$, but in the exterior of the curve $r = 2$.
Solution. The polar coordinates (r, θ) of each point in the region satisfy $2 \le r \le 1 + 2\cos\theta$. Both inequalities can hold only when $1 + 2\cos\theta \ge 2$, i.e. when $\theta \in \left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$. So the region is $\left\{(r,\theta)\colon\ -\frac{\pi}{3} \le \theta \le \frac{\pi}{3} \text{ and } 2 \le r \le 1 + 2\cos\theta\right\}$, and its area is

$$A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{1}{2}\left[(1+2\cos\theta)^2 - 2^2\right] d\theta = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \left(2\cos^2\theta + 2\cos\theta - \frac{3}{2}\right) d\theta$$
$$= \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \left[(1+\cos 2\theta) + 2\cos\theta - \frac{3}{2}\right] d\theta = \frac{5\sqrt{3}}{2} - \frac{\pi}{3}.$$

Theorem 7.104 (Cavalieri's principle in $\mathbb{R}^3$)

Let X and Y be two bounded solids in $\mathbb{R}^3$. If for each horizontal plane P in $\mathbb{R}^3$, the cross-sectional regions $P \cap X$ and $P \cap Y$ have equal area, then X and Y have the same volume.
Again, "horizontal planes" can be replaced by "all planes perpendicular to a given direction".

Corollary 7.105 (volume by cross-sections)

Let $a < b$ and let D be a solid in $\mathbb{R}^3$ lying between planes perpendicular to a given axis at a and at b. If at each $x \in [a, b]$ the cross section of D perpendicular to the axis has area $A(x)$, then D has volume

$$V = \int_a^b A(x)\, dx.$$

Proof. Partition $[a, b]$ by $a = x_0 < x_1 < \cdots < x_n = b$. For each k, the slab of D over $[x_{k-1}, x_k]$ is approximately a cylinder of cross-sectional area $A(x_k)$ and thickness $x_k - x_{k-1}$, hence of volume $\approx A(x_k)(x_k - x_{k-1})$. Summing and letting $\|P\| \to 0$,

$$V = \lim_{\|P\|\to 0} \sum_{k=1}^{n} A(x_k)(x_k - x_{k-1}) = \int_a^b A(x)\, dx.$$

■

Example 7.106 (volume of a pyramid)

Let $h > 0$ and $A > 0$. Compute the volume of a pyramid whose height is h and whose base area is A.
Solution. For each $x \in [0, h]$, the cross section of the pyramid at a distance x from the vertex has area $(\frac{x}{h})^2 A$, by the area ratio of similar figures. Therefore

$$V = \int_0^h \left(\frac{x}{h}\right)^2 A\, dx = \frac{A}{h^2}\left[\frac{x^3}{3}\right]_0^h = \frac{1}{3}Ah.$$

Corollary 7.107 (disk method: revolution about the x-axis)

Let $f : [a, b] \to [0, +\infty)$ be a continuous function, and consider the region in the plane bounded by the graph of f, the x-axis and the lines $x = a$ and $x = b$. Then the solid obtained by revolving this region about the x-axis has volume

$$V = \int_a^b \pi f(x)^2 \, dx.$$

Proof. By Corollary 7.105 the cross section perpendicular to the x-axis at x is a disk of radius $f(x)$, hence area $A(x) = \pi f(x)^2$, so $V = \displaystyle\int_a^b \pi f(x)^2 \, dx$. ■

Example 7.108 (volume of a ball)

Let $r > 0$. Find the volume of a ball in $\mathbb{R}^3$ with radius r.
Solution. Let $f : [-r, r] \to [0, +\infty)$ be $f(x) = \sqrt{r^2 - x^2}$. A ball of radius r is the solid bounded by the surface obtained by revolving the graph of f about the x-axis. Therefore

$$V = \int_{-r}^{r} \pi [f(x)]^2 \, dx = \int_{-r}^{r} \pi (r^2 - x^2) \, dx = \left[\pi r^2 x - \frac{\pi}{3} x^3 \right]_{-r}^{r} = \frac{4\pi}{3} r^3.$$

Example 7.109 (revolution about a horizontal line)

Consider the region in the plane bounded by the graph of $f(x) = \sin x$ and the lines $x = 0$, $x = \pi$ and $y = 2$. Find the volume of the solid obtained by revolving this region about the line $y = 2$.
Solution. At each $x \in [0, \pi]$, the cross section of the solid is a disk of radius $2 - \sin x$, whose area is $A(x) = \pi(2 - \sin x)^2$. Therefore

$$\begin{aligned} V &= \int_0^{\pi} \pi (2 - \sin x)^2 \, dx = \pi \int_0^{\pi} \left(4 - 4 \sin x + \sin^2 x\right) dx \\ &= \pi \int_0^{\pi} \left(\frac{9}{2} - 4 \sin x - \frac{1}{2} \cos 2x \right) dx = \pi \left[\frac{9}{2} x + 4 \cos x - \frac{1}{4} \sin 2x \right]_0^{\pi} = \frac{9\pi^2}{2} - 8\pi. \end{aligned}$$

Example 7.110 (revolution about the y-axis)

Let $f, g : [0, 1] \to [0, +\infty)$ be defined by $f(x) = 2x^2$ and $g(x) = x^2$, and consider the region in the plane bounded by the graphs of f and g and the line $y = 1$. Find the volume of the solid obtained by revolving this region about the y-axis.
Solution. The graphs have equations $y = 2x^2$ and $y = x^2$. The given region can also be regarded as bounded by the curves $x = \sqrt{\dfrac{y}{2}}$ and $x = \sqrt{y}$ and the line $y = 1$. Slicing perpendicular to the y-axis (disks minus disks),

$$V = \int_0^1 \pi (\sqrt{y})^2 \, dy - \int_0^1 \pi \left(\sqrt{\frac{y}{2}} \right)^2 dy = \frac{\pi}{4}.$$

Corollary 7.111 (washer method: revolution about the x-axis)

Let $f, g : [a, b] \to [0, +\infty)$ be continuous with $f(x) \geq g(x) \geq 0$ for every $x \in [a, b]$, and consider the region bounded by the graphs of f and g and the lines $x = a$ and $x = b$.

Then the solid obtained by revolving this region about the x-axis has volume

$$V = \int_a^b \pi \left(f(x)^2 - g(x)^2 \right) dx.$$

Proof. By Corollary 7.105 the cross section perpendicular to the x-axis at x is a washer (an annulus) of inner radius $g(x)$ and outer radius $f(x)$, hence of area $A(x) = \pi(f(x)^2 - g(x)^2)$. So $V = \int_a^b \pi(f(x)^2 - g(x)^2)\, dx$. ■

Example 7.112 (volume of a torus)

Let $a > b > 0$ and consider the region in the plane bounded by the circle $x^2 + (y - a)^2 = b^2$. Find the volume of the solid torus obtained by revolving this region about the x-axis.

Solution. The region is bounded between the graphs of the two functions $f, g : [-b, b] \to [0, +\infty)$ defined by

$$f(x) = a + \sqrt{b^2 - x^2} \qquad \text{and} \qquad g(x) = a - \sqrt{b^2 - x^2}.$$

So

$$V = \int_{-b}^{b} \pi \left(f(x)^2 - g(x)^2 \right) dx = \int_{-b}^{b} \pi \left[\left(a + \sqrt{b^2 - x^2} \right)^2 - \left(a - \sqrt{b^2 - x^2} \right)^2 \right] dx$$
$$= \int_{-b}^{b} 4\pi a \sqrt{b^2 - x^2}\, dx = 4\pi a \cdot (\text{area of a half-disk of radius } b) = (2\pi a)(\pi b^2).$$

The solids above are solids of revolution. There is an alternative method: think of cutting the solid into thin shells, each shaped like a circular cylinder of very small thickness.

Theorem 7.113 (shell method: revolution about the y-axis)

Let $0 \le a < b$, let $f : [a, b] \to [0, +\infty)$ be continuous, and consider the region in the plane bounded by the graph of f, the x-axis and the lines $x = a$ and $x = b$. Then the solid obtained by revolving this region about the y-axis has volume

$$V = \int_a^b 2\pi x f(x)\, dx.$$

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. For each $1 \le k \le n$, a thin shell with outer radius x_k, parallel to the y-axis, is approximately a hollow circular cylinder with circumference $2\pi x_k$, height $f(x_k)$ and thickness $x_k - x_{k-1}$, hence of volume approximately $2\pi x_k f(x_k)(x_k - x_{k-1})$. Summing and letting $\|P\| \to 0$,

$$V = \lim_{\|P\| \to 0} \sum_{k=1}^{n} 2\pi x_k f(x_k)(x_k - x_{k-1}) = \int_a^b 2\pi x f(x)\, dx.$$

■

Example 7.114 (shell method)

Let $f : [0, 1] \to \mathbb{R}$ be $f(x) = e^{-\frac{x^2}{2}}$ and consider the region bounded by the graph of f, the x-axis, the y-axis and the line $x = 1$. Revolve this region about the y-axis;

the resulting solid has volume

$$V = \int_0^1 2\pi x f(x)\,dx = \int_0^1 2\pi x e^{-\frac{x^2}{2}}\,dx = 2\pi \int_0^{\frac{1}{2}} e^{-u}\,du = 2\pi\left(1 - \frac{1}{\sqrt{e}}\right),$$

where $u = \dfrac{x^2}{2}$.

Example 7.115 (volume of a cone)

Let $h > 0$ and $r > 0$, and consider the triangular region in the coordinate plane with vertices $(0,0)$, $(0,h)$ and (r,h). Find the volume of the cone obtained by revolving this region about the y-axis.

Solution. First consider the rectangular region bounded by the x-axis, the y-axis, the line $y = h$ and the line $x = r$. The solid obtained by revolving it about the y-axis has volume

$$V_0 = \int_0^r 2\pi x h\,dx = \pi r^2 h.$$

Next consider the triangular region with vertices $(0,0)$, $(r,0)$ and (r,h). The solid obtained by revolving it about the y-axis has volume

$$V_1 = \int_0^r 2\pi x\left(\frac{h}{r}x\right)dx = \frac{2}{3}\pi r^2 h.$$

Therefore the required cone has volume

$$V = V_0 - V_1 = \pi r^2 h - \frac{2}{3}\pi r^2 h = \frac{1}{3}\pi r^2 h.$$

Remark 7.116 (methods for volumes of solids of revolution)

The following table summarises the available methods. Understand the methods instead of memorising the table.

Curve(s)	Revolved about the x-axis	Revolved about the y-axis
$y = f(x)$	Slice: $V = \int_a^b \pi f(x)^2\,dx$	Shell: $V = \int_a^b 2\pi x f(x)\,dx$
$y = f_1(x),\ y = f_2(x)$	Slice: $V = \int_a^b \pi(f_1(x)^2 - f_2(x)^2)\,dx$	Shell: $V = \int_a^b 2\pi x(f_1(x) - f_2(x))\,dx$
$x = g(y)$	Shell: $V = \int_a^b 2\pi y g(y)\,dy$	Slice: $V = \int_a^b \pi g(y)^2\,dy$
$x = g_1(y),\ x = g_2(y)$	Shell: $V = \int_a^b 2\pi y(g_1(y) - g_2(y))\,dy$	Slice: $V = \int_a^b \pi(g_1(y)^2 - g_2(y)^2)\,dy$

Theorem 7.117 (surface area of a surface of revolution)

Let $f : [a,b] \to [0,+\infty)$ be a continuously differentiable function. The surface obtained

by revolving the graph of f about the x-axis has surface area

$$S = \int_a^b 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx.$$

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$. For each $1 \le k \le n$, the portion of the graph of f on $[x_{k-1}, x_k]$ is approximated by the tangent line segment joining $\big(x_{k-1},\ f(m_k) - f'(m_k)(m_k - x_{k-1})\big)$ and $\big(x_k,\ f(m_k) + f'(m_k)(x_k - m_k)\big)$, where $m_k = \dfrac{x_{k-1}+x_k}{2}$. This segment has length

$$\sqrt{(x_k - x_{k-1})^2 + [f'(m_k)(x_k - x_{k-1})]^2} = \sqrt{1+[f'(m_k)]^2}\,(x_k - x_{k-1}),$$

and it sweeps out a surface of area approximately $2\pi f(m_k)\sqrt{1+[f'(m_k)]^2}\,(x_k - x_{k-1})$ when revolved about the x-axis. Summing and letting $\|P\| \to 0$ in the resulting mid-point Riemann sum,

$$S = \lim_{\|P\|\to 0} \sum_{k=1}^{n} 2\pi f(m_k)\sqrt{1+[f'(m_k)]^2}\,(x_k - x_{k-1}) = \int_a^b 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx.$$

■

Example 7.118 (surface area of a sphere)

Let $r > 0$. Find the surface area of a sphere with radius r.

Solution. Let $f : [-r, r] \to [0, +\infty)$ be $f(x) = \sqrt{r^2 - x^2}$. A sphere of radius r is the surface obtained by revolving the graph of f about the x-axis. Since

$$f'(x) = \frac{-x}{\sqrt{r^2 - x^2}}$$

for every $x \in (-r, r)$, its surface area is

$$\begin{aligned} S &= \int_{-r}^{r} 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx \\ &= \int_{-r}^{r} 2\pi\sqrt{r^2-x^2}\sqrt{1+\frac{x^2}{r^2-x^2}}\,dx \\ &= 2\pi r\int_{-r}^{r} dx = 4\pi r^2. \end{aligned}$$

Example 7.119 (surface area of a torus)

Let $a > b > 0$ and consider the circle $x^2 + (y-a)^2 = b^2$ in the plane. Find the surface area of the torus obtained by revolving this circle about the x-axis.

Solution. The circle consists of the graphs of the two functions $f, g : [-b, b] \to [0, +\infty)$ defined by

$$f(x) = a + \sqrt{b^2 - x^2} \qquad \text{and} \qquad g(x) = a - \sqrt{b^2 - x^2}.$$

Since

$$f'(x) = \frac{-x}{\sqrt{b^2 - x^2}} \qquad \text{and} \qquad g'(x) = \frac{x}{\sqrt{b^2 - x^2}}$$

for every $x \in (-b, b)$, we have

$$\sqrt{1+[f'(x)]^2} = \frac{b}{\sqrt{b^2 - x^2}} = \sqrt{1+[g'(x)]^2}$$

for every $x \in (-b, b)$. Therefore

$$\begin{aligned}
S &= \int_{-b}^{b} 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx + \int_{-b}^{b} 2\pi g(x)\sqrt{1+[g'(x)]^2}\,dx \\
&= 2\pi \int_{-b}^{b} \left[(a+\sqrt{b^2-x^2})\frac{b}{\sqrt{b^2-x^2}} + (a-\sqrt{b^2-x^2})\frac{b}{\sqrt{b^2-x^2}}\right] dx \\
&= 2\pi \int_{-b}^{b} \frac{2ab}{\sqrt{b^2-x^2}}\,dx \\
&= 2\pi \left(\lim_{t\to b^-} \int_0^t \frac{2ab}{\sqrt{b^2-x^2}}\,dx + \lim_{s\to -b^+} \int_s^0 \frac{2ab}{\sqrt{b^2-x^2}}\,dx\right) \\
&= 4\pi ab \left(\lim_{t\to b^-} \arcsin\frac{t}{b} - \lim_{s\to -b^+} \arcsin\frac{s}{b}\right) = 4\pi^2 ab = (2\pi a)(2\pi b).
\end{aligned}$$

8 Series

In chapter 2 of MATH1013 we have studied sequences and their limits; now in this chapter we would like to investigate what happens if we try to add up all the terms in a sequence. In another point of view, we would like to study the sum of infinitely many objects.

8.1 Recap: Key Facts about Sequences

Definition 2.7, 2.87 (Limit of a sequence). Let (a_n) be a sequence of real numbers.

(i) If the number a_n gets closer and closer to a unique real number L when n gets larger and larger, then L is called the limit of the sequence (a_n).
Rigorously, a real number L is called a limit of the sequence (a_n) if for each $\varepsilon > 0$, there exists $N > 0$ such that

$$n \text{ is an integer and } n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

It turns out that L must be unique if it exists. In symbols we write

$$\lim_{n\to+\infty} a_n = L.$$

(ii) If (a_n) has a limit L which is a finite real number, then we say that (a_n) converges (to L). Otherwise we say that (a_n) diverges.

(iii) In case (a_n) diverges, we further say that (a_n) diverges to $+\infty$ (resp. diverges to $-\infty$) if the number a_n (resp. $-a_n$) gets arbitrarily large, and keeps getting larger and larger as n increases.
Rigorously speaking, (a_n) diverges to $+\infty$ (resp. diverges to $-\infty$) if for each $r \in \mathbb{R}$, there exists $N > 0$ such that

$$n \text{ is an integer and } n \geq N \quad \Longrightarrow \quad a_n > r \text{ (resp. } a_n < r).$$

In symbols we write

$$\lim_{n\to+\infty} a_n = +\infty \qquad \left(\text{resp. } \lim_{n\to+\infty} a_n = -\infty\right).$$

(Figure: a plot of the points (n, a_n); $\lim_{n\to+\infty} a_n = L$ if no matter how narrow the shaded horizontal strip around $y = L$ is, the points eventually fall within it.)

Theorem 2.19 (Squeeze Theorem)

Let (a_n), (b_n) and (c_n) be sequences of real numbers such that

$$a_n \leq b_n \leq c_n$$

for every sufficiently large n. If (a_n) and (c_n) both converge and if

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} c_n,$$

then (b_n) also converges to the same number.

Theorem 2.29 (Monotone Sequence Theorem)

Let (a_n) be a sequence of real numbers.

(i) If (a_n) is increasing, then (a_n) converges if and only if it is bounded from above.

(ii) If (a_n) is decreasing, then (a_n) converges if and only if it is bounded from below.

Theorem 2.103

Let b be a real number, let (a_n) be a sequence of real numbers that converges to b and let f be a function which is well-defined near b and is continuous at b. Then the sequence $(f(a_n))$ converges and

$$\lim_{n\to+\infty} f(a_n) = f\left(\lim_{n\to+\infty} a_n\right) = f(b).$$

Example 4.28

Evaluate each of the following limits of sequences, if it exists.

$$\text{(a)} \lim_{n\to+\infty} \frac{\ln n}{n} \qquad \text{(b)} \lim_{n\to+\infty} \sqrt[n]{n}$$

Solution. (a) Note that for every $x > 0$, we have $\ln x < x$. So for every positive integer n, we have

$$0 \leq \frac{\ln n}{n} = \frac{2\ln\sqrt{n}}{n} < \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}}.$$

Since $\lim\limits_{n\to+\infty} \dfrac{2}{\sqrt{n}} = 0$, we have

$$\lim_{n\to+\infty} \frac{\ln n}{n} = 0$$

by Squeeze Theorem.

(b) Note that $\sqrt[n]{n} = e^{\frac{\ln n}{n}}$ for every positive integer n. Since $\lim\limits_{n\to+\infty} \dfrac{\ln n}{n} = 0$ according to (a) and since the exponential function is continuous at 0, we have

$$\lim_{n\to+\infty} \sqrt[n]{n} = \lim_{n\to+\infty} e^{\frac{\ln n}{n}} = e^{\lim_{n\to+\infty} \frac{\ln n}{n}} = e^0 = 1$$

by Theorem 2.103.

8.2 Series

Definition 8.1. Let (a_n) be a sequence. A series (pl. series) is the sequence $\left(\sum_{k=1}^{n} a_k\right)_{n\in\mathbb{N}}$ whose nth term is the "nth partial sum", i.e. the sequence of the form

$$(a_1,\ a_1+a_2,\ a_1+a_2+a_3,\ \dots).$$

Example 8.2

Let $a_n = \dfrac{1}{n!}$ for every $n \in \mathbb{N}$. The series of numbers $\left(\sum_{k=1}^{n} a_k\right)$ is the sequence

$$\left(\frac{1}{1!},\ \frac{1}{1!}+\frac{1}{2!},\ \frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!},\ \frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!},\ \dots\right),$$

i.e. $\left(1, \dfrac{3}{2}, \dfrac{5}{3}, \dfrac{41}{24}, \dots\right)$.

Example 8.3

Let $a_n = 2^n$ for every $n \geq 0$. The series of numbers $\left(\sum_{k=0}^{n} a_k\right)$ is the sequence

$$(1,\ 1+2,\ 1+2+4,\ 1+2+4+8,\ \dots),$$

i.e. $(1, 3, 7, 15, \dots)$.

Remark 8.4

Let (a_n) be a sequence of numbers $(a_1, a_2, a_3, \dots)$. Also let $b_1 = a_1$ and let $b_n = a_n - a_{n-1}$ for every integer $n \geq 2$. Then we have

$$a_n = a_1 + (a_2 - a_1) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) = \sum_{k=1}^{n} b_k$$

for every positive integer n. This way, every sequence (a_n) can be regarded conversely as a series $\left(\sum_{k=1}^{n} b_k\right)$. Therefore, sequences and series are equivalent concepts.

Definition 8.5. Let $\left(\sum_{k=1}^{n} a_k\right)$ be a series of real numbers.

(i) We say that $\left(\sum_{k=1}^{n} a_k\right)$ converges if it has some limit $L \in \mathbb{R}$ in the sense of a sequence of real numbers (as in Definition 2.7 or 2.82), i.e. for each $\varepsilon > 0$, there exists $N > 0$ such that

$$n \text{ is an integer and } n \geq N \quad \Longrightarrow \quad |(a_1 + a_2 + \cdots + a_n) - L| < \varepsilon.$$

Otherwise, we say that the series $\left(\sum_{k=1}^{n} a_k\right)$ diverges.

(ii) If a series $\left(\sum_{k=1}^{n} a_k\right)$ converges, then its limit (or sum) is denoted by

$$\sum_{k=1}^{+\infty} a_k := \lim_{n\to+\infty} \sum_{k=1}^{n} a_k.$$

(iii) Note that we usually abuse the notations for a series and its sum, so we may simply say that "the series $\sum_{k=1}^{+\infty} a_k$ converges" or "the series $\sum_{k=1}^{+\infty} a_k$ diverges".

Example 8.6 — Geometric series

Let r be a real number and consider the geometric series

$$\sum_{k=0}^{+\infty} r^k.$$

Solution.
- If $r = 1$, then the series is actually the sequence $(1, 2, 3, \dots)$, which diverges to $+\infty$.
- If $r \neq 1$, then for each $n \in \mathbb{N}$, the nth partial sum is (cf. Theorem 0.79)

$$s_n := \sum_{k=0}^{n-1} r^k = 1 + r + r^2 + \cdots + r^{n-1} = \frac{1 - r^n}{1 - r}.$$

Recall (cf. Example 2.11) that the geometric sequence (r^n) converges to 0 if $-1 < r < 1$, and diverges if $r > 1$ or $r \leq -1$. Therefore:
- If $-1 < r < 1$, then

$$\sum_{k=0}^{+\infty} r^k = \lim_{n\to+\infty} s_n = \frac{1}{1 - r}.$$

- If $r \geq 1$ or $r \leq -1$, then $\sum_{k=0}^{+\infty} r^k$ diverges.

Example 8.7

Let f be the function defined by

$$f(x) = \sum_{k=0}^{+\infty} e^{-kx}.$$

What is the natural domain of f?
Solution. The natural domain of f is the set that collects all those $x \in \mathbb{R}$ such that the series $f(x)$ converges. Note that

$$f(x) = \sum_{k=0}^{+\infty} e^{-kx} = \sum_{k=0}^{+\infty} (e^{-x})^k$$

is a geometric series, which converges if and only if $e^{-x} < 1$, i.e. $x > 0$. Therefore the natural domain of f is $(0, +\infty)$. Moreover, we can conclude that

$$f(x) = \frac{1}{1 - e^{-x}}$$

for every $x \in (0, +\infty)$.

Example 8.8

Determine whether each of the following series converges or not.

$$\text{(a) } \sum_{k=1}^{+\infty}(-1)^k \qquad \text{(b) } \sum_{k=2}^{+\infty}\frac{1}{k(k-1)}$$

Solution. (a) The given series is actually the sequence $(-1, 0, -1, 0, -1, 0, \dots)$, which diverges.
(b) For each integer $n \geq 2$, we have

$$\sum_{k=2}^{n}\frac{1}{k(k-1)} = \sum_{k=2}^{n}\left(\frac{1}{k-1} - \frac{1}{k}\right) = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = 1 - \frac{1}{n}.$$

So the given series $\displaystyle\sum_{k=2}^{+\infty}\frac{1}{k(k-1)}$ is the same as the sequence $\left(1 - \frac{1}{n}\right)$, which converges to 1.

Remark 8.9 — Telescoping series

The series in Example 8.8(b) is a telescoping series, i.e. a series of the form

$$\sum_{k=1}^{+\infty}(a_k - a_{k-1}).$$

For each positive integer n, the nth partial sum of this telescoping series is given by

$$\sum_{k=1}^{n}(a_k - a_{k-1}) = (a_1 - a_0) + (a_2 - a_1) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) = a_n - a_0;$$

so the telescoping series converges if and only if the sequence (a_n) converges.

Remark 8.10 — Riemann sums are NOT series

In chapter 5, we have learnt that the integral of a function can be defined as the limit of (a sequence of) Riemann sums of the form

$$\lim_{n\to+\infty}\sum_{k=1}^{n} f(\omega_k)(x_k - x_{k-1}).$$

- In spite of the similar notations, the above sequence of Riemann sums is NOT a series. In a series, each term a_k should not change with the total number of terms n. However in a Riemann sum, the partition points x_k as well as the sample points ω_k all depend on n (although their notations do not highlight this dependence). When n changes, every term in a Riemann sum also changes!
- For the same reason, DO NOT write the above Riemann sum as

$$\sum_{k=1}^{+\infty} f(\omega_k)(x_k - x_{k-1}).$$

Recall that the main machineries for computing the limit of a sequence are

- Arithmetic operations,
- Squeeze Theorem, and
- Monotone Sequence Theorem.

Example 8.11

Show that the series

$$\sum_{k=0}^{+\infty} \frac{1}{k!}$$

converges and its sum is e.

Solution. For each $n \in \mathbb{N} \cup \{0\}$, let $s_n = \sum_{k=0}^{n} \frac{1}{k!}$ be the nth partial sum. As in the proof of Theorem 2.31,

$$\begin{aligned}\left(1+\frac{1}{n}\right)^n &= \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = 1 + 1 + \sum_{k=2}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!} \frac{1}{n^k} \\ &= 1 + 1 + \sum_{k=2}^{n} \frac{1}{k!} \left(1-\frac{1}{n}\right) \cdots \left(1-\frac{k-1}{n}\right).\end{aligned}$$

Now for every integer $n \geq 2$, we observe the following.

- $\left(1+\frac{1}{n}\right)^n \leq 1 + 1 + \sum_{k=2}^{n} \frac{1}{k!} = s_n$ since $1-\frac{1}{n}, \ldots, 1-\frac{k-1}{n} \in (0,1)$.
- For every integer $m > n$, we have

$$\begin{aligned}\left(1+\frac{1}{m}\right)^m &= 1 + 1 + \sum_{k=2}^{m} \frac{1}{k!} \left(1-\frac{1}{m}\right) \cdots \left(1-\frac{k-1}{m}\right) \\ &> 1 + 1 + \sum_{k=2}^{n} \frac{1}{k!} \left(1-\frac{1}{m}\right) \cdots \left(1-\frac{k-1}{m}\right).\end{aligned}$$

  Now n is fixed. Letting $m \to +\infty$ on both sides, we obtain $e \geq 1 + 1 + \sum_{k=2}^{n} \frac{1}{k!} = s_n$.

Therefore $\left(1+\frac{1}{n}\right)^n \leq s_n \leq e$ for every integer $n \geq 2$. Since $\lim_{n\to+\infty} \left(1+\frac{1}{n}\right)^n = e$, we have

$$\sum_{k=0}^{+\infty} \frac{1}{k!} = \lim_{n\to+\infty} s_n = e$$

by Squeeze Theorem.

The sum of a series can often be very difficult to compute, but it is much easier just to determine whether it converges or diverges (cf. Q7, Problem Set 3 of MATH1013).

Remark 8.12

If $a_n \geq 0$ for every n, then the sequence $\left(\sum_{k=1}^{n} a_k\right)$ is increasing. Therefore a series $\sum_{k=1}^{+\infty} a_k$ consisting of non-negative terms either converges or diverges to $+\infty$.

Example 8.13

Show that the series

$$\sum_{k=1}^{+\infty} \frac{1}{k^2}$$

converges.

Solution. For each $n \in \mathbb{N}$, let $s_n = \sum_{k=1}^{n} \frac{1}{k^2}$ be the nth partial sum. Then

- The sequence (s_n) is increasing because each term $\frac{1}{k^2}$ is positive.
- For every integer $n \geq 2$, we have $\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$, so

$$\begin{aligned} s_n &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \\ &\leq 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) = 2 - \frac{1}{n} \leq 2 \end{aligned}$$

  which shows that the sequence (s_n) is bounded from above by 2.

Therefore (s_n) converges by the Monotone Sequence Theorem (Theorem 2.29). ■

Remark 8.14 — Basel problem

It turns out that we have the sum

$$\sum_{k=1}^{+\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

The computation of this sum is a famous problem called the Basel problem, which was solved by Euler in 1734. We will compute this sum in the next chapter.

Example 8.15 — Harmonic series

Show that the harmonic series

$$\sum_{k=1}^{+\infty} \frac{1}{k}$$

diverges.

Solution. For each $n \in \mathbb{N}$, let $s_n = \sum_{k=1}^{n} \frac{1}{k}$ be the nth partial sum. Then

- The sequence (s_n) is increasing because each term $\frac{1}{k}$ is positive.

- Moreover, for each $m \in \mathbb{N}$ we have

$$\begin{aligned} s_{2^m} &= 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{2 \text{ terms}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{4 \text{ terms}} + \cdots + \underbrace{\left(\frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m}\right)}_{2^{m-1} \text{ terms}} \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \underbrace{\left(\frac{1}{2^m} + \cdots + \frac{1}{2^m}\right)}_{2^{m-1} \text{ terms}} \\ &= 1 + \underbrace{\frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}}_{m \text{ terms}} = 1 + \frac{m}{2}, \end{aligned}$$

  which shows that (s_n) is unbounded.

Therefore (s_n) diverges (to $+\infty$) by the Monotone Sequence Theorem. ■

Remark 8.16

Inserting, removing or modifying finitely many terms in a series does not affect whether the series converges or not (cf. Remark 2.21). (But of course it changes the sum.)

Lemma 8.17

Let c be a real number and suppose that the series $\sum_{k=1}^{+\infty} a_k$ and $\sum_{k=1}^{+\infty} b_k$ both converge.

Then

(i) $\sum_{k=1}^{+\infty}(a_k + b_k)$ also converges and

$$\sum_{k=1}^{+\infty}(a_k + b_k) = \sum_{k=1}^{+\infty} a_k + \sum_{k=1}^{+\infty} b_k;$$

(ii) $\sum_{k=1}^{+\infty} ca_k$ also converges and

$$\sum_{k=1}^{+\infty} ca_k = c \sum_{k=1}^{+\infty} a_k.$$

Proof. Lemma 0.75 and Theorem 2.14 (arithmetic operations on limits of sequences). ■

Remark 8.18

It is important to note that the following equalities are in general NOT true:

$$\sum_{k=1}^{+\infty} a_k b_k \neq \left(\sum_{k=1}^{+\infty} a_k\right)\left(\sum_{k=1}^{+\infty} b_k\right) \qquad \text{and} \qquad \sum_{k=1}^{+\infty} \frac{a_k}{b_k} \neq \frac{\sum_{k=1}^{+\infty} a_k}{\sum_{k=1}^{+\infty} b_k}.$$

8.3 Convergence Tests for Series

Similar to the treatment for improper integrals, we next study some convergence tests for series which allows us to just determine whether it converges or diverges, without computing the sum. These tests are classified into the following categories:

- The term test (similar to the tail test of improper integrals),
- Related to improper integrals: Integral test, p-test, comparison test, limit comparison test
- Related to the geometric series: Root test, ratio test
- Related to absolute convergence: Absolute convergence test, alternating series test

Lemma 8.19 (Term test for divergence)

If a series $\sum_{k=1}^{+\infty} a_k$ converges, then $\lim_{n\to+\infty} a_n = 0$.

Consequently, if the sequence (a_n) does not converge to 0, then the series $\sum_{k=1}^{+\infty} a_k$ diverges.

Proof. Let $s_n := \sum_{k=1}^{n} a_k$ for each $n \in \mathbb{N}$. If $\sum_{k=1}^{+\infty} a_k$ converges, then (s_n) converges to a certain real number L. In other words $\lim_{n\to+\infty} s_n = L$ and so $\lim_{n\to+\infty} s_{n-1} = L$. Therefore

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} (s_n - s_{n-1}) = \lim_{n\to+\infty} s_n - \lim_{n\to+\infty} s_{n-1} = L - L = 0.$$

■

Remark 8.20

For a series with possibly both positive and negative terms, it is enough if we check that $\lim_{n\to+\infty} |a_n| \neq 0$. This is because $\lim_{n\to+\infty} |a_n| = 0$ if and only if $\lim_{n\to+\infty} a_n = 0$.

Example 8.21

Determine whether the series $\sum_{k=1}^{+\infty} \frac{k^2-1}{k^2+1}$ converges or not.

Solution. Since

$$\lim_{n\to+\infty} \frac{n^2-1}{n^2+1} = 1 \neq 0,$$

the series $\sum_{k=1}^{+\infty} \frac{k^2-1}{k^2+1}$ diverges by the term test.

Remark 8.22 — Converse of the term test is false

The converse of the term test is false. That is, even if $\lim_{n\to+\infty} a_n = 0$, the series $\sum_{k=1}^{+\infty} a_k$ may still converge (e.g. $\sum_{k=1}^{+\infty} \frac{1}{k^2}$ in Example 8.13) or diverge ($\sum_{k=1}^{+\infty} \frac{1}{k}$ in Example 8.15).

Remark 8.23

Note that the term test for series is "stronger" than the tail test for improper integrals (Corollary 6.78).

- If $\lim_{x\to+\infty} f(x)$ does not exist (e.g. $f(x)$ oscillates), tail test cannot tell whether $\int_0^{+\infty} f(x)\,dx$ converges or not (cf. Example 6.83).
- However, even in the case when (a_n) diverges but not to infinity (e.g. (a_n) oscillates), we can already apply the term test to conclude that the series $\sum_{k=1}^{+\infty} a_k$ diverges.

The convergence of some series are closely related to that of improper integrals.

Theorem 8.24 (Integral test)

Let N be an integer and let $f : [N, +\infty) \to [0, +\infty)$ be a decreasing function. Let (a_n) be a sequence of real numbers such that $a_n = f(n)$ for every $n \geq N$. Then

$$\sum_{k=N}^{+\infty} a_k \text{ converges} \qquad \text{if and only if} \qquad \int_N^{+\infty} f(x)\,dx \text{ converges.}$$

Proof. Let $n \geq N$ be an integer. For every integer $k \in [N, n]$ and every number $x \in [k, k+1]$, we have

$$f(k+1) \leq f(x) \leq f(k)$$

since f is decreasing. (Figure: the graph $y = f(x)$ over $[k, k+1]$, sandwiched between two rectangles of heights $f(k+1)$ and $f(k)$.) Then the integral inequality gives

$$\int_k^{k+1} f(k+1)\,dx \leq \int_k^{k+1} f(x)\,dx \leq \int_k^{k+1} f(k)\,dx,$$

i.e. $f(k+1) \leq \int_k^{k+1} f(x)\,dx \leq f(k)$. This implies that

$$\begin{aligned}\int_N^{n+1} f(x)\,dx &= \sum_{k=N}^{n} \int_k^{k+1} f(x)\,dx \leq \sum_{k=N}^{n} f(k) = \sum_{k=N}^{n} a_k = a_N + \sum_{k=N}^{n-1} a_{k+1} \\ &= a_N + \sum_{k=N}^{n-1} f(k+1) \leq a_N + \sum_{k=N}^{n-1} \int_k^{k+1} f(x)\,dx = a_N + \int_N^n f(x)\,dx;\end{aligned}$$

or in short,

$$\int_N^{n+1} f(x)\,dx \leq \sum_{k=N}^{n} a_k \leq a_N + \int_N^n f(x)\,dx$$

for each integer $n \geq N$. Now f is non-negative, by the comparison test for improper integrals:

- if $\sum_{k=N}^{+\infty} a_k$ converges, then the left inequality above shows that $\int_N^{+\infty} f(x)\,dx$ also converges;
- if $\int_N^{+\infty} f(x)\,dx$ converges, then the right inequality implies that $\sum_{k=N}^{+\infty} a_k$ also converges.

■

Theorem 8.25 (p-test)

Let p be a real number. The series

$$\sum_{k=1}^{+\infty} \frac{1}{k^p}$$

converges if and only if $p > 1$.

We have already seen the cases $p = 2$ and $p = 1$ in Examples 8.13 and 8.15.

Proof.

- If $p \leq 0$, then $\left(\frac{1}{n^p}\right)$ does not converge to 0, so the series $\sum_{k=1}^{+\infty} \frac{1}{k^p}$ diverges by term test.
- If $p > 0$, then we let $f : [1, +\infty) \to [0, +\infty)$ be the function

  $$f(x) = \frac{1}{x^p},$$

  which is continuous, non-negative and decreasing on $[1, +\infty)$. Now according to the p-test for improper integrals, $\int_1^{+\infty} f(x)\,dx$ converges if and only if $p > 1$. So by integral test, the series $\sum_{k=1}^{+\infty} \frac{1}{k^p} = \sum_{k=1}^{+\infty} f(k)$ also converges if and only if $p > 1$.

From the above, we conclude that the series $\sum_{k=1}^{+\infty} \frac{1}{k^p}$ converges if and only if $p > 1$. ■

Example 8.26

Let p be a real number. Determine whether the series

$$\sum_{k=2}^{+\infty} \frac{1}{k(\ln k)^p}$$

converges or not.

Solution. Let $f : [2, +\infty) \to [0, +\infty)$ be the function defined by

$$f(x) = \frac{1}{x(\ln x)^p},$$

which is continuous, non-negative and decreasing on $[2, +\infty)$. In Example 6.68, we have already seen that the improper integral $\int_2^{+\infty} f(x)\,dx = \int_{\ln 2}^{+\infty} \frac{1}{u^p}\,du$ converges if and only if $p > 1$, by p-test for improper integrals. So by integral test, the series $\sum_{k=1}^{+\infty} \frac{1}{k(\ln k)^p}$ also converges if and only if $p > 1$.

Remark 8.27 — The decreasing assumption in the integral test

Note that in the integral test, it is important to assume that the non-negative function f is decreasing. Without this "decreasing" assumption, even if we further assume that f is integrable on every bounded closed interval and that $\lim_{x \to +\infty} f(x) = 0$, the

conclusion of the integral test may still fail in both directions.

- Let $f : [1, +\infty) \to [0, +\infty)$ be the function

$$f(x) = \begin{cases} \dfrac{1}{x} & \text{if } x \notin \mathbb{N} \\ 0 & \text{if } x \in \mathbb{N} \end{cases}.$$

  Then $\lim\limits_{x \to +\infty} f(x) = 0$. However, we have $\displaystyle\sum_{k=1}^{+\infty} f(k) = \sum_{k=1}^{+\infty} 0$ converges but $\displaystyle\int_1^{+\infty} f(x)\,dx = \int_1^{+\infty} \frac{1}{x}\,dx$ diverges.
- Let $f : [0, +\infty) \to [0, +\infty)$ be the continuous function whose graph at each $x = n \in \mathbb{N}$ coincides with an isosceles triangle with height $\dfrac{1}{n}$ and base width $\dfrac{1}{n}$. (Figure: a sequence of triangular spikes, one above each positive integer n, of height $\dfrac{1}{n}$ and base width $\dfrac{1}{n}$.) Then $\lim\limits_{x \to +\infty} f(x) = 0$. However, we have $\displaystyle\sum_{k=0}^{+\infty} f(k) = \sum_{k=1}^{+\infty} \frac{1}{n}$ diverges but $\displaystyle\int_0^{+\infty} f(x)\,dx = \sum_{k=1}^{+\infty} \frac{1}{2n^2}$ converges.

Theorem 8.28

Let N be an integer, let $f : [N, +\infty) \to [0, +\infty)$ be a decreasing function, and let (b_n) be the sequence defined by $b_n = \displaystyle\sum_{k=N}^{n} f(k) - \int_N^n f(x)\,dx$ for every integer $n \geq N$. Then (b_n) converges, i.e.

$$\lim_{n \to +\infty} \left(\sum_{k=N}^{n} f(k) - \int_N^n f(x)\,dx \right)$$

exists as a real number.

Proof. For each integer $n \geq N$, we observe the following:

- $b_{n+1} - b_n = f(n+1) - \displaystyle\int_n^{n+1} \underbrace{f(x)}_{\geq f(n+1)}\,dx \leq 0$, which shows that (b_n) is decreasing.
- As seen in the proof of integral test, we have $\displaystyle\int_N^{n+1} f(x)\,dx \leq \sum_{k=N}^{n} f(k)$. Thus

$$b_n = \sum_{k=N}^{n} f(k) - \int_N^n f(x)\,dx \geq \int_n^{n+1} \underbrace{f(x)}_{\geq 0}\,dx \geq 0,$$

  which shows that (b_n) is bounded from below by 0.

Therefore (b_n) converges by the Monotone Sequence Theorem (Theorem 2.29). ■

Remark 8.29

Note that Theorem 8.28 does NOT imply that

$$\lim_{n\to+\infty}\sum_{k=N}^{n} f(k) \qquad \text{or} \qquad \lim_{n\to+\infty}\int_{N}^{n} f(x)\,dx$$

is finite; it may happen that these two latter limits are both infinite.

Example 8.30 — Euler–Mascheroni constant

The series $\displaystyle\sum_{k=1}^{+\infty}\frac{1}{k}$ and the improper integral $\displaystyle\int_{1}^{+\infty}\frac{1}{x}\,dx$ both diverge, but the limit

$$\lim_{n\to+\infty}\left(\sum_{k=1}^{n}\frac{1}{k}-\int_{1}^{n}\frac{1}{x}\,dx\right)=\lim_{n\to+\infty}\left(\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}\right)-\ln n\right)$$

exists by Theorem 8.28. In fact, this limit equals

$$\gamma\approx 0.57721566490\ldots,$$

which is known as the Euler–Mascheroni constant.

Similar to improper integrals, we also have the useful comparison test and limit comparison test.

Theorem 8.31 (Comparison test)

Suppose that $a_n \geq b_n \geq 0$ for all sufficiently large n.

(i) If $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges, then $\displaystyle\sum_{k=1}^{+\infty} b_k$ also converges.

(ii) If $\displaystyle\sum_{k=1}^{+\infty} b_k$ diverges to $+\infty$, then $\displaystyle\sum_{k=1}^{+\infty} a_k$ also diverges to $+\infty$.

Proof. Suppose that N is a positive integer so that $a_n \geq b_n \geq 0$ for all $n \geq N$. If $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges, then $\displaystyle\sum_{k=N}^{+\infty} a_k$ also converges to some number L. Now the sequence $\displaystyle\left(\sum_{k=N}^{n} b_k\right)$ is increasing and bounded from above by L, so $\displaystyle\sum_{k=N}^{+\infty} b_k$ converges by Monotone Sequence Theorem. Therefore $\displaystyle\sum_{k=1}^{+\infty} b_k$ also converges. ■

Example 8.32

Determine whether each of the following series converges or not.

(a) $\displaystyle\sum_{k=1}^{+\infty}\frac{1}{2k-1}$ (b) $\displaystyle\sum_{k=0}^{+\infty}\frac{1}{2^k+\sqrt{k}}$ (c) $\displaystyle\sum_{k=1}^{+\infty}\frac{1+k\ln k}{k^2+2}$

Solution. (a) Note that the series $\displaystyle\sum_{k=1}^{+\infty}\frac{1}{2k}=\frac{1}{2}\sum_{k=1}^{+\infty}\frac{1}{k}$ diverges according to p-test. Since

$$\frac{1}{2n-1}>\frac{1}{2n}\qquad\text{for every positive integer } n,$$

the series $\displaystyle\sum_{k=1}^{+\infty}\frac{1}{2k-1}$ also diverges by comparison test.

(b) Note that the geometric series $\displaystyle\sum_{k=0}^{+\infty}\frac{1}{2^k}$ converges as its common ratio is $\dfrac{1}{2}\in(-1,1)$. Since

$$\frac{1}{2^n+\sqrt{n}}\le\frac{1}{2^n}\qquad\text{for every non-negative integer } n,$$

the series $\displaystyle\sum_{k=0}^{+\infty}\frac{1}{2^k+\sqrt{k}}$ also converges by comparison test.

(c) Since the series $\displaystyle\sum_{k=1}^{+\infty}\frac{1}{2k}=\frac{1}{2}\sum_{k=1}^{+\infty}\frac{1}{k}$ diverges by p-test and since

$$\frac{1+n\ln n}{n^2+2}>\frac{n\ln n}{n^2+2}>\frac{n\ln n}{n^2+n^2}=\frac{\ln n}{2n}>\frac{1}{2n}\qquad\text{for every integer } n\ge 3,$$

the series $\displaystyle\sum_{k=1}^{+\infty}\frac{1+k\ln k}{k^2+2}$ also diverges by comparison test.

Corollary 8.33 (Limit comparison test)

Suppose that a_n and b_n are non-negative for all sufficiently large n, and suppose that the limit

$$\lim_{n\to+\infty}\frac{a_n}{b_n}=L$$

exists.

(i) If $L\in[0,+\infty)$ and if the series $\displaystyle\sum_{k=1}^{+\infty}b_k$ converges, then the series $\displaystyle\sum_{k=1}^{+\infty}a_k$ also converges.

(ii) If $L\in(0,+\infty]$ and if the series $\displaystyle\sum_{k=1}^{+\infty}a_k$ converges, then the series $\displaystyle\sum_{k=1}^{+\infty}b_k$ also converges.

Proof. We only need to prove (i), because (ii) follows from (i) simply by swapping (a_n) and (b_n). Suppose that $L\in[0,+\infty)$. Then by the definition of limit, for every sufficiently large n we must have

$$\frac{a_n}{b_n}<L+1,$$

i.e. $0 \leq a_n < (L+1)b_n$. Now if $\sum_{k=1}^{+\infty} b_k$ converges, then $\sum_{k=1}^{+\infty}(L+1)b_k = (L+1)\sum_{k=1}^{+\infty} b_k$ also converges; so $\sum_{k=1}^{+\infty} a_k$ converges by comparison test. ■

Example 8.34

Determine whether the series

$$\sum_{k=1}^{+\infty} \frac{k^3}{k^4+k^2+1}$$

converges or not.

Solution. Since the harmonic series $\sum_{k=1}^{+\infty} \frac{1}{k}$ diverges and since

$$\lim_{n\to+\infty} \frac{\frac{n^3}{n^4+n^2+1}}{\frac{1}{n}} = \lim_{n\to+\infty} \frac{n^4}{n^4+n^2+1} = 1$$

which exists and is non-zero, the series $\sum_{k=1}^{+\infty} \frac{k^3}{k^4+k^2+1}$ also diverges by limit comparison test.

Example 8.35

Let p be a real number. Show that the series

$$\sum_{k=1}^{+\infty} \frac{\ln k}{k^p}$$

diverges if $p \leq 1$ and converges if $p > 1$. (Compare with Example 6.87.)
Solution. For every sufficiently large n (namely for every $n \geq 3$), we have

$$\frac{\ln n}{n^p} \geq \frac{1}{n^p} > 0.$$

- If $p \leq 1$, then $\sum_{k=1}^{+\infty} \frac{1}{k^p}$ diverges by p-test, so $\sum_{k=1}^{+\infty} \frac{\ln k}{k^p}$ also diverges by comparison test.
- If $p > 1$, then the "lower bound" $\sum_{k=1}^{+\infty} \frac{1}{k^p}$ converges by p-test, so the comparison test does not give any useful conclusion; therefore let's try using limit comparison test instead. Let q be some number between p and 1, i.e. $p > q > 1$. Then the limit

  $$\lim_{n\to+\infty} \frac{\frac{\ln n}{n^p}}{\frac{1}{n^q}} = \lim_{n\to+\infty} \frac{\ln n}{n^{p-q}} = 0$$

  exists and is finite. (Idea (Remark 4.40): for any $\varepsilon > 0$ we have $\ln n \ll n^\varepsilon$ as $n \to +\infty$.) Since $q > 1$, the series $\sum_{k=1}^{+\infty} \frac{1}{k^q}$ converges by p-test; so $\sum_{k=1}^{+\infty} \frac{\ln k}{k^p}$ also converges by the limit comparison test.

Alternative proof (for the case $p > 1$):

- If $p > 1$, then the function $f(x) = \dfrac{\ln x}{x^p}$ is non-negative and decreasing on $[1, +\infty)$. Since the improper integral $\displaystyle\int_1^{+\infty} \frac{\ln x}{x^p}\,dx$ converges by Example 6.87, the series $\displaystyle\sum_{k=1}^{+\infty} \frac{\ln k}{k^p}$ also converges by the integral test.

Definition 8.36. We say that a series $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges absolutely if the series

$$\sum_{k=1}^{+\infty} |a_k|$$

converges.

Remark

Do not call a series "diverge absolutely" if it does not converge absolutely!

Theorem 8.37 (Absolute convergence test)

If a series $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges absolutely, then it converges.

Proof. [We can of course mimic the proof of the absolute convergence test for improper integrals (Theorem 6.79), and consider the sequences

$$a_n^+ := \max\{a_n, 0\} \qquad \text{and} \qquad a_n^- := \max\{-a_n, 0\}$$

for every $n \in \mathbb{N}$. Now let's look at an alternative (perhaps easier) proof.]

Note that for every $n \in \mathbb{N}$ we have $-|a_n| \le a_n \le |a_n|$, so

$$0 \le a_n + |a_n| \le 2|a_n|.$$

Since $\displaystyle\sum_{k=1}^{+\infty} 2|a_k| = 2\sum_{k=1}^{+\infty} |a_k|$ converges, it follows that $\displaystyle\sum_{k=1}^{+\infty}(a_k + |a_k|)$ also converges by comparison test. Therefore

$$\sum_{k=1}^{+\infty} a_k = \sum_{k=1}^{+\infty}(a_k + |a_k|) - \sum_{k=1}^{+\infty} |a_k|$$

converges. ■

Example 8.38

Determine whether the series

$$\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^{\frac{3}{2}}}$$

converges or not.

Solution. The series of absolute values

$$\sum_{k=1}^{+\infty} \left| \frac{(-1)^k}{k^{\frac{3}{2}}} \right| = \sum_{k=1}^{+\infty} \frac{1}{k^{\frac{3}{2}}}$$

converges by p-test. Therefore the series $\displaystyle\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^{\frac{3}{2}}}$ converges absolutely, and so it converges.

The following two tests make use of the convergence of the geometric series.

Theorem 8.39 (Root test)

Let (a_n) be a sequence of real numbers.

(i) If $\lim\limits_{n\to+\infty} \sqrt[n]{|a_n|} < 1$, then $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges absolutely.

(ii) If $\lim\limits_{n\to+\infty} \sqrt[n]{|a_n|} > 1$, then $\displaystyle\sum_{k=1}^{+\infty} a_k$ diverges.

Proof. To prove (i), we suppose that $\lim\limits_{n\to+\infty} \sqrt[n]{|a_n|} = L < 1$. Then we pick a number $r \in (L, 1)$, so $\sqrt[n]{|a_n|} < r$ for every sufficiently large n, i.e.

$$|a_n| < r^n \qquad \text{for every sufficiently large } n.$$

Now the geometric series $\displaystyle\sum_{k=1}^{+\infty} r^k$ converges since $0 < r < 1$, and so $\displaystyle\sum_{k=1}^{+\infty} |a_k|$ converges by comparison test.

To prove (ii), we suppose that $\lim\limits_{n\to+\infty} \sqrt[n]{|a_n|} = L > 1$. Then we pick a number $r \in (1, L)$, so $\sqrt[n]{|a_n|} > r$ for every sufficiently large n, i.e.

$$|a_n| > r^n \qquad \text{for every sufficiently large } n.$$

This shows that $\lim\limits_{n\to+\infty} |a_n| = +\infty$, so $\displaystyle\sum_{k=1}^{+\infty} a_k$ diverges by term test. ■

Theorem 8.40 (Ratio test)

Suppose that $a_n \neq 0$ for every sufficiently large n.

(i) If $\lim\limits_{n\to+\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, then $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges absolutely.

(ii) If $\lim\limits_{n\to+\infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$, then $\displaystyle\sum_{k=1}^{+\infty} a_k$ diverges.

Proof. To prove (i), we suppose that $\lim\limits_{n\to+\infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$. Then we pick a number $r \in (L, 1)$, so $\left| \frac{a_{n+1}}{a_n} \right| < r$ for every sufficiently large n. More precisely, there is an integer

N such that $\left|\dfrac{a_{n+1}}{a_n}\right| < r$ for every $n \geq N$. This implies that

$$|a_{n+1}| < r|a_n| \qquad \text{for every } n \geq N;$$

so by induction, $|a_{N+n}| < |a_N| r^n$ for every $n \in \mathbb{N}$. Now the geometric series $|a_N| \displaystyle\sum_{k=1}^{+\infty} r^k$ converges since $0 < r < 1$, and so $\displaystyle\sum_{k=1}^{+\infty} |a_k|$ converges by comparison test.

To prove (ii), we now suppose that $\displaystyle\lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$. Then we pick a number $r \in (1, L)$, so there exists an integer N such that $\left|\dfrac{a_{n+1}}{a_n}\right| > r$ for every $n \geq N$. This implies that

$$|a_{n+1}| > r|a_n| \qquad \text{for every } n \geq N;$$

so by induction, $|a_{N+n}| > |a_N| r^n$ for every $n \in \mathbb{N}$. Since $r > 1$, we have $\displaystyle\lim_{n\to+\infty} |a_{N+n}| = +\infty$, and so $\displaystyle\sum_{k=1}^{+\infty} a_k$ diverges by term test. ■

Remark 8.41 — The case $L = 1$

No conclusion about convergence can be made using root test or ratio test if the relevant limit equals to 1. To see this, we consider the series $\displaystyle\sum_{k=1}^{+\infty} \frac{1}{k}$ and $\displaystyle\sum_{k=1}^{+\infty} \frac{1}{k^2}$. We knew that the first series diverges and the second series converges according to p-test. But for both of these series we have $\displaystyle\lim_{n\to+\infty} \sqrt[n]{a_n} = 1$ (cf. Example 4.28(b) on page 2) and $\displaystyle\lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = 1$.

Example 8.42

Determine whether each of the following series converges or not.

$$\text{(a)} \sum_{k=1}^{+\infty} \frac{k}{2^k} \qquad \text{(b)} \sum_{k=1}^{+\infty} \frac{(2k)!}{(k!)^2} \qquad \text{(c)} \sum_{k=1}^{+\infty} \frac{k!}{k^k} \qquad \text{(d)} \sum_{k=1}^{+\infty} \left(1 - \frac{1}{k}\right)^{k^2}$$

Solution. (a) Let $a_n = \dfrac{n}{2^n}$ for every n. Then

$$\lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to+\infty} \left(\frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n}\right) = \frac{1}{2} \lim_{n\to+\infty} \frac{n+1}{n} = \frac{1}{2} < 1,$$

so $\displaystyle\sum_{k=1}^{+\infty} \frac{k}{2^k} = \sum_{k=1}^{+\infty} a_k$ converges by ratio test.

(b) Let $a_n = \dfrac{(2n)!}{(n!)^2}$ for every n. Then

$$\lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to+\infty} \left[\frac{(2(n+1))!}{((n+1)!)^2} \cdot \frac{(n!)^2}{(2n)!}\right] = \lim_{n\to+\infty} \frac{(2n+2)(2n+1)}{(n+1)^2} = 4 > 1,$$

so $\displaystyle\sum_{k=1}^{+\infty} \frac{(2k)!}{(k!)^2} = \sum_{k=1}^{+\infty} a_k$ diverges by ratio test.

(c) Let $a_n = \dfrac{n!}{n^n}$ for every n. Then $a_n > 0$ and

$$\lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to+\infty} \left[\frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!}\right] = \lim_{n\to+\infty} \frac{n^n}{(n+1)^n} = \lim_{n\to+\infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1,$$

so $\displaystyle\sum_{k=1}^{+\infty} \frac{k!}{k^k} = \sum_{k=1}^{+\infty} a_k$ converges by ratio test.

(d) Let $a_n = \left(1 - \dfrac{1}{n}\right)^{n^2}$ for every n. Then $a_n > 0$ and

$$\lim_{n\to+\infty} \sqrt[n]{|a_n|} = \lim_{n\to+\infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e} < 1,$$

so $\displaystyle\sum_{k=1}^{+\infty} \left(1 - \frac{1}{k}\right)^{k^2} = \sum_{k=1}^{+\infty} a_k$ converges by root test.

Remark 8.43 — Root test is stronger than ratio test

In fact, root test is stronger than ratio test in the sense that if ratio test can be applied to a series, then root test can be applied to the same series as well. This is because for any sequence of non-zero numbers (a_n), if $\displaystyle\lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$, then $\displaystyle\lim_{n\to+\infty} \sqrt[n]{|a_n|} = L$. (Although it is a bit difficult, it is a good exercise to try proving this fact on your own using the ε-N definition.)

Example 8.44

Let

$$a_n = \begin{cases} \dfrac{n}{2^n} & \text{if } n \text{ is odd} \\ \dfrac{1}{2^n} & \text{if } n \text{ is even} \end{cases}$$

for each $n \in \mathbb{N}$. Determine whether the series $\displaystyle\sum_{k=1}^{+\infty} a_k$ converges or not.

Solution. Since

$$\sqrt[n]{|a_n|} = \begin{cases} \dfrac{\sqrt[n]{n}}{2} & \text{if } n \text{ is odd} \\ \dfrac{1}{2} & \text{if } n \text{ is even} \end{cases}$$

for every $n \in \mathbb{N}$, it follows that

$$\frac{1}{2} \le \sqrt[n]{|a_n|} \le \frac{\sqrt[n]{n}}{2} \qquad \text{for every } n \in \mathbb{N}.$$

Now $\displaystyle\lim_{n\to+\infty} \frac{\sqrt[n]{n}}{2} = \frac{1}{2}$ (cf. Example 4.28(b) on page 2 of this chapter), so by Squeeze Theorem we have

$$\lim_{n\to+\infty} \sqrt[n]{|a_n|} = \frac{1}{2} < 1.$$

Consequently, the series $\sum_{k=1}^{+\infty} a_k$ converges by root test. [Note however that the ratio test does not directly apply to this series since the sequence $\left(\frac{2a_{n+1}}{a_n}\right) = \left(1, 3, \frac{1}{3}, 5, \frac{1}{5}, 7, \frac{1}{7}, 9, \frac{1}{9}, \dots\right)$ diverges.]

Definition 8.45. We say that a series $\sum_{k=1}^{+\infty} a_k$ converges conditionally if it converges but does not converge absolutely.

Remark 8.46

There are altogether three possibilities for a series $\sum_{k=1}^{+\infty} a_k$:

- It may converge absolutely (i.e. $\sum_{k=1}^{+\infty} |a_k|$ converges and thus $\sum_{k=1}^{+\infty} a_k$ also converges),
- It may converge conditionally (i.e. $\sum_{k=1}^{+\infty} |a_k|$ diverges but $\sum_{k=1}^{+\infty} a_k$ still converges), or
- It may diverge.

Just in case $\sum_{k=1}^{+\infty} |a_k|$ diverges, DO NOT say that $\sum_{k=1}^{+\infty} a_k$ "diverges absolutely" and thus diverges!

Example 8.47

Determine whether the series

$$\sum_{k=1}^{+\infty} \frac{(-1)^k}{k}$$

converges absolutely, converges conditionally, or diverges.

Solution.

- Since the series of absolute values

$$\sum_{k=1}^{+\infty} \left|\frac{(-1)^k}{k}\right| = \sum_{k=1}^{+\infty} \frac{1}{k}$$

diverges, it follows that $\sum_{k=1}^{+\infty} \frac{(-1)^k}{k}$ does not converge absolutely.

- Let's denote $s_n := \sum_{k=1}^{n} \frac{(-1)^k}{k}$ for each $n \in \mathbb{N}$. Then the sequence of "even partial

sums"

$$\begin{aligned} s_{2n} &= \sum_{k=1}^{2n} \frac{(-1)^k}{k} \\ &= \left(-1+\frac{1}{2}\right) + \left(-\frac{1}{3}+\frac{1}{4}\right) + \cdots + \left(-\frac{1}{2n-1}+\frac{1}{2n}\right) \\ &= -\sum_{k=1}^{n} \frac{1}{2k(2k-1)} \end{aligned}$$

converges to some $L \in \mathbb{R}$ by conducting limit comparison test with $\displaystyle\sum_{k=1}^{+\infty} \frac{1}{k^2}$. On the other hand, the sequence of "odd partial sums"

$$s_{2n+1} = \sum_{k=1}^{2n+1} \frac{(-1)^k}{k} = \underbrace{s_{2n}}_{\to L} - \underbrace{\frac{1}{2n+1}}_{\to 0}$$

also converges to the same number L since $\displaystyle\lim_{n\to+\infty} \frac{1}{2n+1} = 0$. Thus $\displaystyle\sum_{k=1}^{+\infty} \frac{(-1)^k}{k}$ converges.

Therefore $\displaystyle\sum_{k=1}^{+\infty} \frac{(-1)^k}{k}$ converges conditionally.

In fact there is a useful convergence test for the type of series in Example 8.47, whose terms have alternating signs.

Theorem 8.48 (Alternating series test)

Let (a_n) be a decreasing sequence (of non-negative numbers) which converges to 0. Then the series

$$\sum_{k=0}^{+\infty} (-1)^k a_k$$

converges.

Proof. Let $s_n := \displaystyle\sum_{k=0}^{n} (-1)^k a_k$ for each $n \in \mathbb{N}$. Then the sequence (s_{2n+1}) is increasing and bounded from above by a_0, so it converges to some $L \in \mathbb{R}$ by the Monotone Sequence Theorem. Similarly, the sequence (s_{2n}) is decreasing and bounded from below by $a_0 - a_1$, so it converges to some $L' \in \mathbb{R}$ by the Monotone Sequence Theorem. Now since

$$s_{2n+1} = s_{2n} - a_{2n+1} \qquad \text{for every } n \in \mathbb{N}$$

and since $\displaystyle\lim_{n\to+\infty} a_{2n+1} = 0$, taking limits on both sides we get $L = L'$, i.e. the two sequences (s_{2n+1}) and (s_{2n}) converge to the same number. This implies that (s_n) converges, i.e. the series $\displaystyle\sum_{k=0}^{+\infty} (-1)^k a_k$ converges. ■

Example 8.49

Let p be a real number. Determine whether the series

$$\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^p}$$

converges absolutely, converges conditionally, or diverges.

Solution.

- If $p \leq 0$, then $\left(\frac{(-1)^n}{n^p}\right)$ does not converge to 0, so $\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^p}$ diverges by term test.
- If $p > 0$, then $\left(\frac{1}{n^p}\right)$ is a decreasing non-negative sequence that converges to 0; so by alternating series test, the series $\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^p}$ converges.
- Moreover, the series $\sum_{k=1}^{+\infty} \left|\frac{(-1)^k}{k^p}\right| = \sum_{k=1}^{+\infty} \frac{1}{k^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

From the above, we conclude that the series $\sum_{k=1}^{+\infty} \frac{(-1)^k}{k^p}$ converges absolutely if $p > 1$, converges conditionally if $0 < p \leq 1$, and diverges if $p \leq 0$.

Remark 8.50 — Rearrangement of a series

One cares about absolute convergence of a series not only because it is a convenient test of convergence, but more importantly because if a series converges absolutely, then the terms can be "rearranged" without changing the sum of the series. More precisely:

- If $\sum_{k=1}^{+\infty} a_k$ converges absolutely and $\sum_{k=1}^{+\infty} a_k = L$, then given any bijection $\sigma : \mathbb{N} \to \mathbb{N}$, the "rearrangement" $\sum_{k=1}^{+\infty} a_{\sigma(k)}$ also converges absolutely and $\sum_{k=1}^{+\infty} a_{\sigma(k)} = L$.
- On the other hand, if $\sum_{k=1}^{+\infty} a_k$ converges conditionally, then for each $L \in [-\infty, +\infty]$, there exists a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that the "rearrangement" $\sum_{k=1}^{+\infty} a_{\sigma(k)} = L$. (That is, by rearranging the terms, the sum can equal to any real number or $\pm\infty$ or have no limit.)
- As an example, consider the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \ln 2$ (cf. Q6 of Problem Set 7) which converges conditionally. The following rearrangement converges to a

different sum:

$$\begin{aligned}
&\underbrace{1-\frac{1}{2}}_{1+\frac{1}{2}-\frac{1}{2}\cdot 2}-\frac{1}{4}+\underbrace{\frac{1}{3}-\frac{1}{6}}_{1+\frac{1}{3}-\frac{1}{2}\cdot 3}-\frac{1}{8}+\underbrace{\frac{1}{5}-\frac{1}{10}}_{1+\frac{1}{5}-\frac{1}{2}\cdot 5}-\frac{1}{12}+\cdots\\
&=\left(1-\frac{1}{2}\right)-\frac{1}{4}+\left(\frac{1}{3}-\frac{1}{6}\right)-\frac{1}{8}+\left(\frac{1}{5}-\frac{1}{10}\right)-\frac{1}{12}+\cdots\\
&=\frac{1}{2}\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\cdots\right)=\frac{1}{2}\ln 2.
\end{aligned}$$

Remark 8.51 — Summary of convergence tests

The following is a quick summary on the useful tests for convergence and divergence of a series of real numbers that we have studied so far.

(i) For a series whose sum is easy to compute:

- Geometric series: $\sum_{k=1}^{+\infty} r^k$ converges if $|r|<1$ and diverges if $|r|\geq 1$.
- Telescoping series: $\sum_{k=1}^{+\infty}(a_{k+1}-a_k)$ converges if and only if (a_n) converges.

(ii) For a series whose terms a_n does not tend to 0:

- Term test: If (a_n) does not converge to 0, then $\sum_{k=1}^{+\infty} a_k$ diverges.

(iii) For a series with non-negative terms a_n:

- Integral test: If f is a non-negative decreasing function, then $\sum_{k=1}^{+\infty} f(k)$ converges if and only if $\int_1^{+\infty} f(x)\,dx$ converges.
- p-test: $\sum_{k=1}^{+\infty}\frac{1}{k^p}$ converges if $p>1$ and diverges if $p\leq 1$.
- Comparison test: Let $a_n\geq b_n\geq 0$. If $\sum_{k=1}^{+\infty} a_n$ converges, then $\sum_{k=1}^{+\infty} b_n$ converges.
- Limit comparison test: Let $a_n, b_n\geq 0$. If $\lim_{n\to+\infty}\frac{a_n}{b_n}\in(0,+\infty)$, then $\sum_{k=1}^{+\infty} a_n$ converges if and only if $\sum_{k=1}^{+\infty} b_n$ converges.

(iv) For a series with possibly both positive and negative terms a_n:

- Absolute convergence test: If $\sum_{k=1}^{+\infty}|a_k|$ converges, then $\sum_{k=1}^{+\infty} a_k$ also converges.
- Alternating series test: If (a_n) is a decreasing non-negative sequence that converges to 0, then $\sum_{k=1}^{+\infty}(-1)^k a_k$ converges.

- Root test: Let $L = \lim_{n\to+\infty} \sqrt[n]{|a_n|}$. Then $\sum_{k=1}^{+\infty} a_k$ converges absolutely if $L < 1$ and diverges if $L > 1$.
- Ratio test: Let $L = \lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right|$. Then $\sum_{k=1}^{+\infty} a_k$ converges absolutely if $L < 1$ and diverges if $L > 1$.

Root test and ratio test are related to the convergence of a geometric series.

A series "adds up a list of infinitely many objects". In the same way, an infinite product "multiplies a list of infinitely many objects". Interested readers are suggested to refer to Supplementary Note C for some information about infinite products.

8.4 Power Series

In the previous two sections, we have studied series of real numbers. In the remainder of this chapter, we will study functions defined using series. One can also regard them as "series of functions", i.e. each term of the series is a function.

Definition 8.52. Let a be a real number and let (c_n) be a sequence of real numbers. A power series is a function f defined by

$$f(x) = \sum_{k=0}^{+\infty} c_k(x-a)^k,$$

i.e. a series of monomials in $(x-a)$. The number a is called the center of such a power series, and the c_n's are called its coefficients.

A power series is like a polynomial with possibly "infinite degree".

Remark 8.53

The domain of a power series f can only consist of real numbers x at which the series converges. Note that the domain of f is not specified in Definition 8.52, although it is obvious that this domain must at least contain the center a. A main goal of this section is to study the natural domain of a power series.

Example 8.54

The function f defined by $f(x) = \sum_{k=0}^{+\infty} x^k$ is a power series centered at 0.

- For each fixed $x \in \mathbb{R}$, the value $f(x)$ is a geometric series with common ratio x. Therefore such a series converges if $-1 < x < 1$, and in this case we have
$$f(x) = \frac{1}{1-x}.$$
- On the other hand, we know that this geometric series $f(x)$ diverges if $x \geq 1$ or $x \leq -1$.

Therefore the natural domain of f is the open interval $(-1, 1)$.

Note that the equation $f(x) = \frac{1}{1-x}$ does not hold for any $x \in \mathbb{R} \setminus (-1, 1)$, e.g.

$$-1 = \frac{1}{1-2} \neq 1 + 2 + 4 + 8 + 16 + \cdots.$$

In fact, the natural domain of a power series must be an interval. Let's see why this is true.

Lemma 8.55

Let $a \in \mathbb{R}$, let $r \neq 0$, and let f be a power series centered at a, i.e.

$$f(x) = \sum_{k=0}^{+\infty} c_k (x-a)^k.$$

(i) If $f(a+r)$ converges, then $f(x)$ converges absolutely for every x with $|x-a| < |r|$.
(ii) If $f(a+r)$ diverges, then $f(x)$ diverges for every x with $|x-a| > |r|$.

Proof. We only need to prove (i), because (ii) is an immediate consequence of (i). To prove (i), we suppose that $f(a+r) = \sum_{k=0}^{+\infty} c_k r^k$ converges. Then the sequence $(c_n r^n)$ converges to 0; so for every sufficiently large n we have $|c_n r^n| < 1$, i.e. $|c_n| < \dfrac{1}{|r|^n}$. Thus we always have

$$0 \leq |c_n (x-a)^n| \leq \left|\frac{x-a}{r}\right|^n \qquad \text{for every sufficiently large } n.$$

Now if $|x-a| < |r|$, then the geometric series $\sum_{k=0}^{+\infty} \left|\dfrac{x-a}{r}\right|^k$ converges; so $\sum_{k=0}^{+\infty} |c_k (x-a)^k|$ converges by comparison test. Therefore $f(x) = \sum_{k=0}^{+\infty} c_k (x-a)^k$ converges absolutely. ■

Theorem 8.56

Let a be a real number and let f be a power series centered at a, i.e.

$$f(x) = \sum_{k=0}^{+\infty} c_k (x-a)^k.$$

Then there exists a unique $R \in [0, +\infty]$ such that the series $f(x)$
- converges absolutely if $|x-a| < R$, and
- diverges if $|x-a| > R$.

This number R is called the radius of convergence of the power series f.

Remark 8.57

Theorem 8.56 implies that the natural domain of the power series f must be an interval with end-points $a-R$ and $a+R$. However, it does not tell whether any of these end-points belongs to the domain, so the convergence at the end-points needs to be checked separately. The natural domain can possibly be any one of the following intervals:

$$(a-R,\ a+R), \quad (a-R,\ a+R], \quad [a-R,\ a+R) \text{ or } [a-R,\ a+R].$$

- Note that ratio test and root test will not be useful when checking the convergence at the end-points, because the relevant limits must equal to 1 at these points.

The natural domain of a power series is also called its interval of convergence.

Example 8.54 (revisited)

The radius of convergence of the power series

$$\sum_{k=0}^{+\infty} x^k$$

is 1. Its interval of convergence is $(-1, 1)$, which includes neither of the two end-points.

Example 8.58

Find the interval of convergence of each of the following power series.

(a) $f(x) = \displaystyle\sum_{k=1}^{+\infty} \frac{(x-1)^k}{4^k k}$ (b) $f(x) = \displaystyle\sum_{k=1}^{+\infty} \frac{x^k}{(2k)!}$ (c) $f(x) = \displaystyle\sum_{k=1}^{+\infty} (2k)!\,(x-5)^k$

Solution. (a) Since

$$\lim_{n\to+\infty} \left| \frac{(x-1)^{n+1}}{4^{n+1}(n+1)} \cdot \frac{4^n n}{(x-1)^n} \right| = \frac{|x-1|}{4} \lim_{n\to+\infty} \frac{n}{n+1} = \frac{|x-1|}{4},$$

the power series $f(x)$ converges absolutely if $\dfrac{|x-1|}{4} < 1$, i.e. if $|x-1| < 4$, and diverges if $|x-1| > 4$, according to ratio test. Therefore the radius of convergence of f is 4, and the natural domain of f is an interval with end-points $1-4=-3$ and $1+4=5$. Now

- at -3, the power series becomes $f(-3) = \displaystyle\sum_{k=1}^{+\infty} \frac{(-3-1)^k}{4^k k} = \sum_{k=1}^{+\infty} \frac{(-1)^k}{k}$ which converges by alternating series test;
- at 5, it becomes $f(5) = \displaystyle\sum_{k=1}^{+\infty} \frac{(5-1)^k}{4^k k} = \sum_{k=1}^{+\infty} \frac{1}{k}$ which diverges by p-test.

Therefore the natural domain of f is the interval $[-3, 5)$.
(b) Since

$$\lim_{n\to+\infty} \left| \frac{x^{n+1}}{(2(n+1))!} \cdot \frac{(2n)!}{x^n} \right| = |x| \lim_{n\to+\infty} \frac{1}{(2n+2)(2n+1)} = 0 \qquad \text{for every } x \in \mathbb{R},$$

the power series $f(x)$ converges absolutely for every $x \in \mathbb{R}$, according to ratio test. Therefore the natural domain of f is $\mathbb{R}$.
(c) Since

$$\lim_{n\to+\infty} \left| \frac{(2(n+1))!}{(2n)!} \frac{(x-5)^{n+1}}{(x-5)^n} \right| = \lim_{n\to+\infty} [(2n+2)(2n+1)|x-5|] = \begin{cases} +\infty & \text{if } x \neq 5 \\ 0 & \text{if } x = 5 \end{cases},$$

the power series $f(x)$ converges absolutely if $x = 5$, and diverges if $x \neq 5$, according to ratio test. Therefore the natural domain of f is $\{5\}$.

Remark 8.59 — Computing the radius of convergence

The radius of convergence of the power series $\sum_{k=0}^{+\infty} c_k(x-a)^k$ is given by

$$R = \frac{1}{\lim_{n\to+\infty} \sqrt[n]{|c_n|}} = \lim_{n\to+\infty} \left|\frac{c_n}{c_{n+1}}\right|,$$

if either one of these limits exists in $[0,+\infty]$. If $\lim_{n\to+\infty} \sqrt[n]{|c_n|} = \begin{cases} 0 \\ +\infty \end{cases}$, then $R = \begin{cases} +\infty \\ 0 \end{cases}$. Note that the radius of convergence still exists even if these limits do not exist. We will learn in more advanced courses about how to compute the radius of convergence when these limits do not exist.

The main motivation to studying power series is the important property that its derivatives and integrals can be taken "term-by-term", so they can be computed as easily as those for monomials.

Theorem 8.60 (Termwise differentiation and integration of power series)

Let a be a real number and let $f(x) = \sum_{k=0}^{+\infty} c_k(x-a)^k$ be a power series centered at a with radius of convergence $R \in (0,+\infty]$. Then f is differentiable on the interval $(a-R, a+R)$, its derivative is the power series

$$f'(x) = \sum_{k=1}^{+\infty} kc_k(x-a)^{k-1} = \sum_{k=0}^{+\infty} (k+1)c_{k+1}(x-a)^k$$

for every $x \in (a-R, a+R)$, and one of its antiderivatives is the power series

$$\int_a^x f(t)\,dt = \sum_{k=0}^{+\infty} \frac{c_k}{k+1}(x-a)^{k+1} = \sum_{k=1}^{+\infty} \frac{c_{k-1}}{k}(x-a)^k$$

for every $x \in (a-R, a+R)$.

Add an arbitrary constant $+C$ to get the collection of all antiderivatives.

Remark 8.61 — Verifying termwise differentiation for the geometric series

The actual proof of Theorem 8.60 about termwise differentiation of power series is out of the scope of this course and will be done in MATH2043/3033. But at this stage, we are still able to verify that the theorem is indeed true at least specifically for the geometric series. Let $f : (-1,1) \to \mathbb{R}$ be the geometric series defined by $f(x) = \sum_{k=0}^{+\infty} x^k$. We already knew that $f(x) = \dfrac{1}{1-x}$ for $x \in (-1,1)$, so $f'(x) = \dfrac{1}{(1-x)^2}$ for $x \in (-1,1)$. Now consider the power series

$$g(x) = \sum_{k=1}^{+\infty} kx^{k-1} = \sum_{k=0}^{+\infty} (k+1)x^k$$

which is constructed by differentiating f termwise. We can then verify that (i) the radius of convergence of g is also 1 and (ii) $g(a) = f'(a)$ for each $a \in (-1,1)$:

(i) For the power series g, we have

$$\lim_{n\to+\infty}\left|\frac{c_n}{c_{n+1}}\right| = \lim_{n\to+\infty}\left|\frac{n+1}{n+2}\right| = 1$$

which exists, so the radius of convergence of g is indeed 1.

(ii) At each $a \in (-1,1)$, the partial sum of g up to n terms is

$$\begin{aligned}\sum_{k=1}^{n} ka^{k-1} &= \sum_{k=1}^{n} \frac{d}{dx}x^k\bigg|_{x=a} = \frac{d}{dx}\left(\sum_{k=1}^{n} x^k\right)\bigg|_{x=a} = \frac{d}{dx}\frac{x-x^{n+1}}{1-x}\bigg|_{x=a} \\ &= \frac{na^{n+1}-(n+1)a^n+1}{(1-a)^2}.\end{aligned}$$

Now the power series g converges at a, so $\lim\limits_{n\to+\infty} na^{n-1} = 0$ by term test. Therefore

$$g(a) = \sum_{k=1}^{+\infty} ka^{k-1} = \lim_{n\to+\infty}\frac{na^{n+1}-(n+1)a^n+1}{(1-a)^2} = \frac{1}{(1-a)^2} = f'(a).$$

This shows that termwise differentiation of power series is valid at least for a geometric series.

Example 8.62

Let $x \in (-1,1)$. Evaluate the sum of the series

$$\sum_{k=1}^{+\infty} kx^{3k} \qquad \text{and} \qquad \sum_{k=1}^{+\infty} k^2x^{3k}.$$

Solution. Consider the power series $f(x) = \sum\limits_{k=0}^{+\infty} x^{3k}$, which has radius of convergence 1. By termwise differentiation of power series, we have

$$f'(x) = \sum_{k=1}^{+\infty} 3kx^{3k-1} \qquad \text{for } x \in (-1,1).$$

Now we have $f(x) = \dfrac{1}{1-x^3}$ for $x \in (-1,1)$ by the sum of a geometric series, so

$$\sum_{k=1}^{+\infty} kx^{3k} = \sum_{k=1}^{+\infty} 3kx^{3k-1}\cdot\frac{x}{3} = \frac{x}{3}f'(x) = \frac{x}{3}\frac{d}{dx}\left(\frac{1}{1-x^3}\right) = \frac{x^3}{(1-x^3)^2}.$$

Differentiating termwise once again, we obtain $\sum\limits_{k=1}^{+\infty} 3k^2x^{3k-1} = \dfrac{d}{dx}\dfrac{x^3}{(1-x^3)^2}$ for $x \in (-1,1)$, so

$$\sum_{k=1}^{+\infty} k^2x^{3k} = \sum_{k=1}^{+\infty} 3k^2x^{3k-1}\cdot\frac{x}{3} = \frac{x}{3}\frac{d}{dx}\frac{x^3}{(1-x^3)^2} = \frac{x^3+x^6}{(1-x^3)^3}.$$

Although a power series can be differentiated termwise and the derivative shares the same radius of convergence, it is in general unknown whether the derivative will converge at an end-point of the interval of convergence.

Example 8.63

The geometric series

$$1 + x + x^2 + x^3 + x^4 + \cdots = \frac{1}{1-x}$$

converges on $(-1, 1)$ only and diverges at both the end-points -1 and 1.

- The termwise derivative

$$1 + 2x + 3x^2 + 4x^3 + \cdots = \frac{1}{(1-x)^2} \qquad \text{for every } x \in (-1, 1),$$

  also diverges at both the end-points -1 and 1.
- On the other hand, the termwise integral from 0 to x is

$$[t]_0^x + \left[\frac{t^2}{2}\right]_0^x + \left[\frac{t^3}{3}\right]_0^x + \left[\frac{t^4}{4}\right]_0^x + \cdots = \int_0^x \frac{1}{1-t}\, dt,$$

  i.e.

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots = -\ln(1-x) \qquad \text{for every } x \in (-1, 1).$$

  Replacing x by $-x$, we obtain

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \ln(1+x) \qquad \text{for every } x \in (-1, 1).$$

  Note that this power series diverges at -1 but converges at 1 by alternating series test. This leads to a natural question: Does the above equation still hold if $x = 1$? Do we have

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln(1+1) = \ln 2?$$

Although a power series may converge or diverge at an end-point of the interval of convergence, the following theorem asserts that if a power series converges at an end-point, then it must also be continuous there. In other words, a power series is continuous throughout its interval of convergence, even at the end-points. Its proof is again out of the scope of this course.

Theorem 8.64 (Abel's limit theorem)

Let a be a real number and let $f(x) = \sum_{k=0}^{+\infty} c_k (x-a)^k$ be a power series centered at a with radius of convergence $R > 0$.

(i) If $f(x)$ converges at $a + R$, then

$$f(a+R) = \lim_{x \to (a+R)^-} f(x).$$

(ii) If $f(x)$ converges at $a - R$, then

$$f(a-R) = \lim_{x \to (a-R)^+} f(x).$$

Example 8.65

- In Example 8.63 we have obtained that

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \ln(1+x) \qquad \text{for every } x \in (-1, 1).$$

Since the power series on the left-hand side also converges at 1, we have

$$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots=\lim_{x\to1^-}\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots\right)=\lim_{x\to1^-}\ln(1+x)=\ln 2$$

according to Abel's limit theorem.

- In a similar way, the geometric series

$$1-x^2+x^4-x^6+\cdots=\frac{1}{1+x^2}\qquad\text{for every } x\in(-1,1)$$

can be integrated termwise from 0 to x so as to give

$$x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\cdots=\arctan x\qquad\text{for every } x\in(-1,1).$$

Since the power series on the left-hand side converges at 1 by alternating series test, we have

$$1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots=\lim_{x\to1^-}\left(x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\cdots\right)=\lim_{x\to1^-}\arctan x=\frac{\pi}{4}$$

according to Abel's limit theorem. This leads to a "series representation" of π:

$$\pi=4\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots\right).$$

8.5 Supplement: Infinite Products

A series "adds up a list of infinitely many objects". In the same way, an infinite product "multiplies a list of infinitely many objects".

Definition. Let (a_n) be a sequence. An infinite product is the sequence $\left(\prod_{k=1}^{n}a_k\right)_{n\in\mathbb{N}}$ whose nth term is the "nth partial product", i.e. a sequence of the form

$$(a_1,\ a_1a_2,\ a_1a_2a_3,\ \ldots).$$

Example

Let $a_n=\frac{1}{2}$. The infinite product of numbers $\left(\prod_{k=1}^{n}a_k\right)$ is the sequence of numbers

$$\left(\frac{1}{2},\ \frac{1}{2^2},\ \frac{1}{2^3},\ \cdots\right),$$

i.e. $\left(\frac{1}{2},\frac{1}{4},\frac{1}{8},\ldots\right)$.

Example

Let $a_n=\frac{1}{2^n}$. The infinite product of numbers $\left(\prod_{k=1}^{n}a_k\right)$ is the sequence of numbers

$$\left(\frac{1}{2},\ \frac{1}{2\cdot2^2},\ \frac{1}{2\cdot2^2\cdot2^3},\ \frac{1}{2\cdot2^2\cdot2^3\cdot2^4},\ \cdots\right),$$

i.e. $\left(\frac{1}{2^1}, \frac{1}{2^{1+2}}, \frac{1}{2^{1+2+3}}, \frac{1}{2^{1+2+3+4}}, \dots\right)$.

Definition. Let $\left(\prod_{k=1}^{n} a_k\right)$ be an infinite product of real numbers.

(i) We say that $\left(\prod_{k=1}^{n} a_k\right)$ converges if it has some non-zero limit $L \in \mathbb{R} \setminus \{0\}$ in the sense of a sequence of real numbers (as in Definition 2.7 or 2.87), i.e. for each $\varepsilon > 0$, there exists $N > 0$ such that

$$n \text{ is an integer and } n \geq N \implies |(a_1 a_2 \cdots a_n) - L| < \varepsilon.$$

Otherwise, we say that the infinite product $\left(\prod_{k=1}^{n} a_k\right)$ diverges.

(ii) If an infinite product $\left(\prod_{k=1}^{n} a_k\right)$ converges, then its limit is denoted by

$$\prod_{k=1}^{+\infty} a_k := \lim_{n \to +\infty} \prod_{k=1}^{n} a_k.$$

(iii) Note that we usually abuse the notations for an infinite product and its limit, so we may simply say that the infinite product $\prod_{k=1}^{+\infty} a_k$ converges or the infinite product $\prod_{k=1}^{+\infty} a_k$ diverges.

Remark — Divergence to 0

Note that instead of "converging to 0", we say that $\prod_{k=1}^{+\infty} a_k$ diverges to 0 if

$$\lim_{n \to +\infty} \prod_{k=1}^{n} a_k = 0.$$

The number zero interacts with multiplication of real numbers in a similar way as infinity interacts with addition. If $\sum_{k=1}^{+\infty} a_k$ diverges to $+\infty$, further adding finitely many real numbers to the series does not save the partial sums from diverging to $+\infty$; in a similar way, if an infinite product tends to 0, further multiplying some real numbers to the partial product does not save it from tending to 0 either.

Example

The infinite products $\prod_{k=1}^{+\infty} \frac{1}{2}$ and $\prod_{k=1}^{+\infty} \frac{1}{2^k}$ both diverge to 0.

Example

Consider the infinite product $\prod_{k=1}^{+\infty}\left(1+\frac{1}{k}\right)$. Since

$$\prod_{k=1}^{n}\left(1+\frac{1}{k}\right)=\prod_{k=1}^{n}\frac{k+1}{k}=\frac{2}{1}\cdot\frac{3}{2}\cdots\frac{n+1}{n}=n+1$$

for every $n\in\mathbb{N}$, it follows that $\prod_{k=1}^{+\infty}\left(1+\frac{1}{k}\right)$ diverges to $+\infty$.

Example

Consider the infinite product $\prod_{k=2}^{+\infty}\frac{k^2}{k^2-1}$. Since

$$\prod_{k=2}^{n}\frac{k^2}{k^2-1}=\prod_{k=2}^{n}\frac{k}{k-1}\cdot\frac{k}{k+1}=\frac{2}{1}\cdot\frac{3}{2}\cdots\frac{n}{n-1}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdots\frac{n}{n+1}=\frac{2n}{n+1}$$

for every $n\in\mathbb{N}$, it follows that $\prod_{k=2}^{+\infty}\frac{k^2}{k^2-1}$ converges to 2.

Example

In Q4(b) of Problem Set 3 in MATH1013, we have seen that

$$\prod_{k=2}^{+\infty}\left(1-\frac{1}{1+2+\cdots+k}\right)=\prod_{k=2}^{+\infty}\frac{(k-1)(k+2)}{k(k+1)}$$

converges to $\frac{1}{3}$.

Remark

Whether an infinite product converges or not is unaffected by inserting, removing or amending finitely many non-zero factors in it.

Lemma (Factor test for divergence)

If an infinite product of non-zero real numbers $\prod_{k=1}^{+\infty}a_k$ converges, then $\lim_{n\to+\infty}a_n=1$. Consequently, if the sequence (a_n) does not converge to 1, then the infinite product $\prod_{k=1}^{+\infty}a_k$ diverges.

The convergence of an infinite product is closely related to the convergence of a relevant series.

Theorem

Let (a_n) be a sequence of positive real numbers. Then

$$\prod_{k=1}^{+\infty} a_k \text{ converges} \qquad \text{if and only if} \qquad \sum_{k=1}^{+\infty} \ln a_k \text{ converges.}$$

Proof. For every $n \in \mathbb{N}$, we have $\prod_{k=1}^{n} a_k = e^{\sum_{k=1}^{n} \ln a_k}$. ■

Example

For every $x \in (-1, 1)$, we have

$$\prod_{k=1}^{+\infty} e^{\frac{x^k}{k}} = \frac{1}{1-x} \qquad \text{and} \qquad \prod_{k=0}^{+\infty} \left(1 + x^{2^k}\right) = \frac{1}{1-x}.$$

Example — Wallis' formula for π

For every $x \in \mathbb{R} \setminus \{0\}$, we have

$$\prod_{k=1}^{+\infty} \left(1 - \frac{x^2}{k^2}\right) = \frac{\sin \pi x}{\pi x}.$$

When $x = 0$ the infinite product is 1. In particular when $x = \frac{1}{2}$, we have $\frac{2}{\pi} = \prod_{k=1}^{+\infty} \left(1 - \frac{1}{4k^2}\right)$, which gives Wallis' formula for π:

$$\frac{\pi}{2} = \prod_{k=1}^{+\infty} \frac{4k^2}{4k^2 - 1} = \prod_{k=1}^{+\infty} \frac{(2k)(2k)}{(2k-1)(2k+1)}.$$

Example

For every non-negative integer n, we have

$$n! = \prod_{k=1}^{+\infty} \frac{1}{1 + \frac{n}{k}} \left(1 + \frac{1}{k}\right)^n.$$

8.6 Supplement: The Exponent a^b

Since high school, we have worked with the exponent notation a^b where $a > 0$ and $b \in \mathbb{R}$. However, in high school (and even in MATH1013-L01) it is usually not fully explained what this exponent notation really means.

When $a \in \mathbb{R}$ and $b \in \mathbb{N}$, we know perfectly what a^b is:

$$a^b = \underbrace{a \cdot a \cdot a \cdots a}_{b \text{ times}}.$$

With the law of exponents, we can extend the idea so as to understand a^b

- when $a \in \mathbb{R} \setminus \{0\}$ and $b \in \mathbb{Z}$, and even

- when $a > 0$ and $b \in \mathbb{Q}$,

without worrying about "complex numbers". But what actually does 2^π mean?

After studying calculus, we can now understand more about the actual meaning of the exponent. The following are some approaches to rigorously define what a^b is, when $a > 0$ and $b \in \mathbb{R}$. The first three approaches all sound straight-forward, but they all turn out to involve more mathematical analysis than we have learnt.

Approach I: Rational approximation (sounds the easiest, but a lot of gaps to be justified)

Let's take advantage of the fact that we already know what a^r is when $a > 0$ and $r \in \mathbb{Q}$. We use the fact (to be introduced in mathematical analysis) that every real number has a (possibly infinitely long) decimal expansion.

Lemma

For each $b \in \mathbb{R}$, there exists an increasing sequence (b_n) of rational numbers such that

$$\lim_{n\to+\infty} b_n = b.$$

Example

For $b = \pi$, we have the sequence

$$(b_n) = (3,\ 3.1,\ 3.14,\ 3.141,\ 3.1415,\ 3.14159,\ 3.141592,\ \dots).$$

Lemma

Let $a > 0$ and $b \in \mathbb{R}$, and let (b_n) be an increasing sequence of rational numbers such that $\lim_{n\to+\infty} b_n = b$. Then the sequence (a^{b_n}) converges (to a positive real number).

Proof. The sequence (a^{b_n}) is either increasing (in case $a \geq 1$) or decreasing (in case $a \leq 1$). In any case it is also bounded between the positive numbers $a^{\lfloor b \rfloor}$ and $a^{\lfloor b \rfloor + 1}$. So it converges by Monotone Sequence Theorem and has a positive limit between $a^{\lfloor b \rfloor}$ and $a^{\lfloor b \rfloor + 1}$. ■

Now we try to make the following definition.

Definition. Let $a > 0$ and $b \in \mathbb{R}$, and let (b_n) be a sequence of rational numbers such that $\lim_{n\to+\infty} b_n = b$. Then we define

$$a^b := \lim_{n\to+\infty} a^{b_n}.$$

Remark — Uniqueness issue

There is a uniqueness issue to sort out when making the above definition. If (b_n) and (c_n) are two sequences of rational numbers with $\lim_{n\to+\infty} b_n = \lim_{n\to+\infty} c_n = b$, we need to justify that

$$\lim_{n\to+\infty} a^{b_n} = \lim_{n\to+\infty} a^{c_n},$$

which is unfortunately not an easy task.

Under this definition, the law of exponents is then immediately valid for real exponents because we already knew that it holds for rational exponents.

Corollary

Let $a > 0$ and $x, y \in \mathbb{R}$. Then

$$a^{x+y} = a^x a^y.$$

Proof. Let (x_n) and (y_n) be increasing sequences of rational numbers with $\lim_{n\to+\infty} x_n = x$ and $\lim_{n\to+\infty} y_n = y$. Then $\lim_{n\to+\infty}(x_n + y_n) = x + y$ and so

$$a^x a^y = \lim_{n\to+\infty} a^{x_n} \lim_{n\to+\infty} a^{y_n} = \lim_{n\to+\infty} a^{x_n} a^{y_n} \overset{(*)}{=} \lim_{n\to+\infty} a^{x_n+y_n} = a^{x+y},$$

where the second last step $(*)$ follows from the law of exponents for rational exponents. ■

To complete the construction, we need to show that the function $f : \mathbb{R} \to \mathbb{R}$ with

$$f(x) = a^x$$

defined as above, is continuous and is one-to-one. Thanks to the law of exponents in the above corollary, this can be readily reduced to just proving the following, but the proof is still not that easy.

Lemma

Let $a > 0$. Then

(i) $\lim_{h\to 0} a^h = 1$, and

(ii) if $a^b = 1$, then $b = 0$.

Approach II: The limit $\left(1 + \frac{x}{n}\right)^n$

With a similar reasoning as in Theorem 2.31, we know that for each real number x, the sequence $\left(\left(1 + \frac{x}{n}\right)^n\right)$ converges to a positive number. Therefore, we can make the following definition.

Definition. Let $x \in \mathbb{R}$. Then we define

$$\exp x := \lim_{n\to+\infty} \left(1 + \frac{x}{n}\right)^n.$$

With this definition, one can then easily show that the law of exponents follows.

Lemma

For each $x, y \in \mathbb{R}$, we have

$$\exp(x + y) = \exp x \, \exp y.$$

Proof. For each $x, y \in \mathbb{R}$, we have

$$\begin{aligned}\exp x \, \exp y &= \lim_{n\to+\infty} \left(1 + \frac{x}{n}\right)^n \lim_{n\to+\infty} \left(1 + \frac{y}{n}\right)^n = \lim_{n\to+\infty} \left(1 + \frac{x}{n}\right)^n \left(1 + \frac{y}{n}\right)^n \\ &= \lim_{n\to+\infty} \left(1 + \frac{x+y}{n} + \frac{xy}{n^2}\right)^n = \exp(x + y),\end{aligned}$$

where the computation of the last limit follows from a similar trick as in Q10(c) of Problem Set 3 in MATH1013. ■

We need to show that the function $\exp : \mathbb{R} \to \mathbb{R}$ defined as above is continuous and is one-to-one. Similar to Approach I, the proof is not that easy.

Lemma

Let $\exp : \mathbb{R} \to \mathbb{R}$ be the function defined above. Then
(i) $\lim_{x\to 0} \exp x = 1$, and
(ii) if $\exp x = 1$, then $x = 0$.

Sloppy proof of (i).

$$\lim_{x\to 0} \exp x = \lim_{x\to 0} \lim_{n\to+\infty} \left(1 + \frac{x}{n}\right)^n \overset{(?)}{=} \lim_{n\to+\infty} \lim_{x\to 0} \left(1 + \frac{x}{n}\right)^n = \lim_{n\to+\infty} 1^n = 1.$$

■

Finally, since exp has an inverse $\ln : (0, +\infty) \to \mathbb{R}$, we can then define a^b using this function ln.

Definition. For each $a > 0$ and $b \in \mathbb{R}$, we define

$$a^b := \exp(b \ln a).$$

Approach III: The exponential series (a popular approach, can be readily generalized to complex exponents)

We know that the series

$$\sum_{k=0}^{+\infty} \frac{x^k}{k!}$$

converges absolutely for each $x \in \mathbb{R}$. Thus we can make the following definition.

Definition. For each $x \in \mathbb{R}$, we define

$$\exp x := \sum_{k=0}^{+\infty} \frac{x^k}{k!}.$$

Lemma

For each $x, y \in \mathbb{R}$, we have

$$\exp(x + y) = \exp x \, \exp y.$$

Sloppy proof. We have

$$\begin{aligned}\exp x \, \exp y &= \sum_{k=0}^{+\infty} \sum_{j=0}^{+\infty} \frac{x^k}{k!} \frac{y^j}{j!} \overset{(?)}{=} \sum_{m=0}^{+\infty} \sum_{j+k=m} \frac{x^k}{k!} \frac{y^j}{j!} = \sum_{m=0}^{+\infty} \sum_{k=0}^{m} \frac{x^k y^{m-k}}{k!(m-k)!} \\ &= \sum_{m=0}^{+\infty} \frac{1}{m!} \sum_{k=0}^{m} \binom{m}{k} x^k y^{m-k} = \sum_{m=0}^{+\infty} \frac{(x+y)^m}{m!} = \exp(x+y),\end{aligned}$$

where the second last step follows from applying binomial theorem to the inner sum. However, the rearrangement of the double summation at the second step is sloppy. Much more effort is needed to rigorously justify this rearrangement. ■

We then need the idea of "uniform convergence" in mathematical analysis when proving the continuity of exp.

Lemma

The function $\exp : \mathbb{R} \to (0, +\infty)$ is continuous and one-to-one.

Now we can conclude that exp has a continuous inverse, which is then called ln : $(0, +\infty) \to \mathbb{R}$. Finally, we define a^b using this function ln.

Definition. For each $a > 0$ and $b \in \mathbb{R}$, we define

$$a^b := \exp(b \ln a).$$

Approach IV: The integral definition of ln **(the most elementary approach to rigorously define real exponents)**

The three approaches above all sound elementary at the beginning, but in each of them we cannot justify the continuity and injectivity of the exponential function in an elementary way. The following fourth approach sounds a bit indirect at the beginning, but it turns out that we can already justify the story in full rigor with what we have learnt in MATH1014.

We know that the function

$$f(t) = \frac{1}{t}$$

is continuous on $(0, +\infty)$, so it is integrable on each bounded closed subinterval of $(0, +\infty)$. Hence we can make the following definition.

Definition (natural logarithm). For each $x \in (0, +\infty)$, we define

$$\ln x := \int_1^x \frac{1}{t}\, dt.$$

We then show that this definition indeed gives us the natural logarithm as intended, by verifying that the law of logarithm is valid.

Lemma

For each $x, y \in (0, +\infty)$, we have

$$\ln(xy) = \ln x + \ln y.$$

Proof. For each $x, y \in (0, +\infty)$, we have

$$\ln(xy) = \int_1^{xy} \frac{1}{t}\, dt = \int_1^x \frac{1}{t}\, dt + \int_x^{xy} \frac{1}{t}\, dt = \int_1^x \frac{1}{t}\, dt + \int_1^y \frac{1}{u}\, du = \ln x + \ln y,$$

where in the second integral we have made use of the substitution $u = \dfrac{t}{x}$. ■

It is also easy to see that the function ln defined as above is continuous and one-to-one.

Lemma

The function ln : $(0, +\infty) \to \mathbb{R}$ is continuous and one-to-one.

Proof. Since $f(x) = \dfrac{1}{x}$ is continuous on $(0, +\infty)$, the function ln is differentiable on $(0, +\infty)$ by the first version of the Fundamental Theorem of Calculus. Since $f(x) = \dfrac{1}{x} > 0$ on $(0, +\infty)$, it follows that ln is strictly increasing on $(0, +\infty)$ also. ■

Next, we have learnt that the improper integrals

$$\int_1^{+\infty} \frac{1}{t}\, dt \qquad \text{and} \qquad \int_0^1 \frac{1}{t}\, dt$$

both diverge to $+\infty$. As we have just proved that ln is continuous, these imply that the range of ln is the whole $\mathbb{R}$. We conclude that ln has a continuous inverse defined on $\mathbb{R}$,

which is then denoted as $\exp : \mathbb{R} \to (0, +\infty)$. This function exp satisfies our usual law of exponents because ln satisfies the law of logarithms.

Lemma

For each $x, y \in \mathbb{R}$, we have

$$\exp(x + y) = \exp x \, \exp y.$$

Proof. For each $x, y \in \mathbb{R}$, we have

$$\ln(\exp(x + y)) = x + y = \ln(\exp(x)) + \ln(\exp(y)) = \ln(\exp(x) \exp(y)).$$

Since ln is one-to-one, we have $\exp(x + y) = \exp x \, \exp y$. ■

Finally, we can define a^b using this function exp.

Definition. For each $a > 0$ and $b \in \mathbb{R}$, we define

$$a^b := \exp(b \ln a).$$

9 Taylor Series and Fourier Series

In chapter 8, we used a power series to construct a function defined on the interval of convergence, and we saw that the sum of a power series can sometimes be written in a "closed form", e.g.

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1 - x} \qquad \text{for every } x \in (-1, 1),$$

$$x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots = \ln(1 + x) \qquad \text{for every } x \in (-1, 1].$$

In this chapter we study the opposite problem: given a function,

- can this function be "expressed as" a power series? Or equivalently,
- is this function the sum of a certain power series?

We then study an analogous representation problem for periodic functions using trigonometric series (Fourier series).

9.1 Polynomial Approximation of a Function

We begin by approximating a function by a polynomial (a "finite" version of a power series). Recall the linear approximation from chapter 3.

Definition. Let a be a real number and f be a function differentiable at a. The linear polynomial L defined by

$$L(x) = f(a) + f'(a)(x - a)$$

is called the linear approximation (or linearization) of f at a. (Figure: the tangent line $y = L(x)$ touching the curve $y = f(x)$ at $x = a$.)

The linear approximation is a "good" approximation of $f(x)$ when x is close to a, because the difference $|f(x) - L(x)|$ is small compared with the distance $|x - a|$:

$$\lim_{x \to a} \frac{|f(x) - L(x)|}{|x - a|} = \lim_{x \to a} \left| \frac{f(x) - f(a)}{x - a} - \frac{L(x) - f(a)}{x - a} \right| = |f'(a) - f'(a)| = 0.$$

Definition. Let a be a real number, let f be a function defined near a, and let n be a non-negative integer. An nth order approximation of f at a is a polynomial P_n of degree

at most n such that

$$\lim_{x\to a}\frac{f(x)-P_n(x)}{(x-a)^n}=0,$$

which is equivalent to $\lim_{x\to a}\dfrac{|f(x)-P_n(x)|}{|x-a|^n}=0$.

Definition. Let a be a real number and let f, g be functions defined near a. We write the little-o notation

$$f(x)=o(g(x)) \qquad \text{as } x\to a$$

if

$$\lim_{x\to a}\frac{f(x)}{g(x)}=0.$$

Thus if P_n is an nth order approximation of f at a, we may write

$$f(x)=P_n(x)+o((x-a)^n) \qquad \text{as } x\to a.$$

Trap — The "equal sign" in little-o notation

The equal sign in the little-o notation is not really an equal sign. When

$$f_1(x)=o(g(x)) \qquad \text{and} \qquad f_2(x)=o(g(x)) \qquad \text{as } x\to a,$$

we do NOT have $f_1(x)=f_2(x)$ in general. The notation $f(x)=o(g(x))$ as $x\to a$ should be understood as "set belonging": f belongs to the set of all functions satisfying $\lim_{x\to a}\dfrac{f(x)}{g(x)}=0$, i.e. the set of all functions that "decay faster" than g as $x\to a$.

Example — A polynomial approximates itself

Let $f:\mathbb{R}\to\mathbb{R}$ be a polynomial of degree m. Then for every $n\geq m$, f is an nth order approximation of itself at every $a\in\mathbb{R}$, because

$$\lim_{x\to a}\frac{f(x)-f(x)}{(x-a)^n}=0.$$

Example — Quadratic approximation of $\cos x$ at 0

Let $f(x)=\cos x$. Then $P_2(x)=1-\dfrac{1}{2}x^2$ is a second order approximation of f at 0, because

$$\lim_{x\to 0}\frac{f(x)-P_2(x)}{x^2}=\lim_{x\to 0}\frac{\cos x-(1-\frac{1}{2}x^2)}{x^2}=0$$

by l'Hôpital's rule. We may write

$$\cos x=1-\frac{1}{2}x^2+o(x^2) \qquad \text{as } x\to 0.$$

(Figure: the parabola $y=1-\dfrac{1}{2}x^2$ hugging the curve $y=\cos x$ near $x=0$.)

Example — Approximations of $\sin x$ at 0

Let $f(x)=\sin x$.

- Since $\lim_{x\to 0}\dfrac{\sin x}{x}=1$, we have $\lim_{x\to 0}\dfrac{\sin x-x}{x}=0$, so $P_1(x)=x$ is a first order approximation of f at 0.

- By l'Hôpital's rule, $\lim_{x\to 0}\dfrac{\sin x - x}{x^3} = -\dfrac{1}{6}$, so
$$\lim_{x\to 0}\frac{\sin x - (x - \frac{1}{6}x^3)}{x^3} = 0;$$
hence $P_3(x) = x - \dfrac{1}{6}x^3$ is a third order approximation of f at 0. We may write
$$\sin x = x - \frac{1}{6}x^3 + o(x^3) \qquad \text{as } x \to 0.$$

(Figure: the cubic $y = x - \dfrac{1}{6}x^3$ hugging the curve $y = \sin x$ near $x = 0$.)

Lemma 9.7 — Uniqueness of the nth order approximation

If a function f has an nth order approximation at a, then it is unique.

Proof. Suppose P_n and Q_n are both nth order approximations of f at a. Then
$$\lim_{x\to a}\frac{f(x) - P_n(x)}{(x-a)^n} = \lim_{x\to a}\frac{f(x) - Q_n(x)}{(x-a)^n} = 0,$$
which implies
$$\lim_{x\to a}\frac{P_n(x) - Q_n(x)}{(x-a)^n} = \lim_{x\to a}\frac{f(x) - Q_n(x)}{(x-a)^n} - \lim_{x\to a}\frac{f(x) - P_n(x)}{(x-a)^n} = 0.$$
Since $P_n - Q_n$ is a polynomial, this limit forces it to be a multiple of $(x-a)^{n+1}$. But $P_n - Q_n$ has degree at most n, so it must be the zero polynomial. ∎

Corollary 9.8 — Truncation

Let n be a non-negative integer. If
$$P_n(x) = c_0 + c_1(x-a) + \cdots + c_n(x-a)^n$$
is the nth order approximation of a function f at a, then for each non-negative integer $m \le n$,
$$P_m(x) = c_0 + c_1(x-a) + \cdots + c_m(x-a)^m$$
is the mth order approximation of f at a.

Proof. For each $0 \le m \le n$,
$$\lim_{x\to a}\frac{P_n(x) - P_m(x)}{(x-a)^m} = \lim_{x\to a}\big(c_{m+1}(x-a) + \cdots + c_n(x-a)^{n-m}\big) = 0.$$
So
$$\lim_{x\to a}\frac{f(x) - P_m(x)}{(x-a)^m} = \lim_{x\to a}\frac{f(x) - P_n(x)}{(x-a)^m} = \lim_{x\to a}\left(\frac{f(x) - P_n(x)}{(x-a)^n}\cdot(x-a)^{n-m}\right) = 0.$$
This shows that P_m is an mth order approximation of f at a; by Lemma 9.7 it is the only one. ∎

Example — Approximations of x^3 at 0 and at 1

Let $f(x) = x^3$. The nth order approximations of f at 0 are

$$P_0(x) = 0, \quad P_1(x) = 0, \quad P_2(x) = 0, \quad P_3(x) = x^3, \quad P_4(x) = x^3, \quad P_5(x) = x^3, \ \ldots$$

For approximations at 1, note that

$$x^3 = (1 + (x-1))^3 = 1 + 3(x-1) + 3(x-1)^2 + (x-1)^3,$$

so the nth order approximations of f at 1 are

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= 1 + 3(x-1), \\ P_2(x) &= 1 + 3(x-1) + 3(x-1)^2, \\ P_3(x) &= 1 + 3(x-1) + 3(x-1)^2 + (x-1)^3, \\ P_4(x) &= P_5(x) = \cdots = P_3(x). \end{aligned}$$

Lemma 9.10 — Parity of approximations

Suppose $P_n(x) = c_0 + c_1 x + \cdots + c_n x^n$ is the nth order approximation of a function f at 0.

(i) If f is even, then P_n consists of even powers only, i.e. $c_1 = c_3 = c_5 = \cdots = 0$.

(ii) If f is odd, then P_n consists of odd powers only, i.e. $c_0 = c_2 = c_4 = \cdots = 0$.

Proof. We have $\lim\limits_{x\to 0} \dfrac{f(x) - P_n(x)}{x^n} = 0$. Replacing x by $-x$ gives

$$\lim_{x\to 0} \frac{f(-x) - P_n(-x)}{x^n} = 0.$$

If f is even, i.e. $f(-x) = f(x)$ for all x, then $\lim\limits_{x\to 0} \dfrac{f(x) - P_n(-x)}{x^n} = 0$, which shows $P_n(-x) = P_n(x)$ since the nth order approximation is unique. The odd case is similar. ■

Theorem 9.11 — Taylor's Theorem

Let a be a real number, n a non-negative integer, and f a function which is n-times differentiable at a. Then the polynomial

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

called the nth degree Taylor polynomial of f at a, is the nth order approximation of f at a; i.e.

$$f(x) = \underbrace{f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n}_{P_n(x)} + o((x-a)^n)$$

as $x \to a$.

Proof. Q5 of Problem Set 8 in MATH1013. ■

Example — Taylor polynomials of e^x

Let $f(x) = e^x$. For every non-negative integer n, $f^{(n)}(x) = e^x$, so $f^{(n)}(0) = 1$. Hence the nth degree Taylor polynomial of f at 0 is

$$P_n(x) = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n.$$

By Taylor's Theorem,

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n + o(x^n) \qquad \text{as } x \to 0.$$

Example — Taylor polynomials of $\ln(1+x)$

Let $f(x) = \ln(1+x)$. Then $f'(x) = \dfrac{1}{1+x}$, and by induction

$$f^{(n)}(x) = (-1)^{n-1}\frac{(n-1)!}{(1+x)^n} \qquad \text{for each } n \in \mathbb{N},$$

so $f^{(n)}(0) = (-1)^{n-1}(n-1)!$. Thus the nth degree Taylor polynomial of f at 0 is

$$P_n(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots + \frac{(-1)^{n-1}}{n}x^n.$$

By Taylor's Theorem,

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots + \frac{(-1)^{n-1}}{n}x^n + o(x^n) \qquad \text{as } x \to 0.$$

Example — Taylor polynomials of $(1+x)^p$

Let $p \in \mathbb{R}$ and $f(x) = (1+x)^p$. For each non-negative integer n,

$$f^{(n)}(x) = p(p-1)(p-2)\cdots(p-n+1)(1+x)^{p-n},$$

so $f^{(n)}(0) = p(p-1)(p-2)\cdots(p-n+1)$. Hence the nth degree Taylor polynomial of f at 0 is

$$P_n(x) = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots + \frac{p(p-1)\cdots(p-n+1)}{n!}x^n.$$

In particular:

- For $p = -1$, Taylor's Theorem gives, for each non-negative integer n,

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + o(x^n) \qquad \text{as } x \to 0.$$

- For $p = \dfrac{1}{2}$,

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + o(x^3) \qquad \text{as } x \to 0.$$

We can in turn extract information about $f^{(n)}$ from the Taylor polynomials of f.

Example — Computing $f^{(n)}(0)$ for $f(x) = \dfrac{1}{1+x^2}$

Let $f(x) = \dfrac{1}{1+x^2}$. The previous example gives

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + o(x^n) \qquad \text{as } x \to 0,$$

so

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n} + o(x^{2n}) \qquad \text{as } x \to 0.$$

By Taylor's Theorem, the polynomial $1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$ is identical to

$$P_{2n}(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(2n)}(0)}{(2n)!}x^{2n}.$$

Comparing coefficients, for every non-negative integer m,

$$f^{(2m)}(0) = (2m)!(-1)^m \qquad \text{and} \qquad f^{(2m+1)}(0) = 0.$$

Note how tedious it would be to compute $f^{(n)}(0)$ by directly differentiating f for n times.

Solution. (As above; the key step is the uniqueness of the Taylor polynomial, which lets us read off the derivatives from the coefficients.)

Remark 9.16 — Composition with little-o

In the example above we applied the following principle: if $f(x) = o(x^m)$ and $g(x) = c_n x^n + o(x^n)$ as $x \to 0$, then $f(g(x)) = o(x^{mn})$ as $x \to 0$. Indeed, with the change of variable $t = g(x)$,

$$\lim_{x\to 0} \frac{f(g(x))}{x^{mn}} = \lim_{x\to 0} \frac{f(g(x))}{g(x)^m} \cdot \lim_{x\to 0} \frac{g(x)^m}{x^{mn}} = \left(\lim_{t\to 0} \frac{f(t)}{t^m}\right)\left(\lim_{x\to 0} \frac{g(x)}{x^n}\right)^m = 0 \cdot c_n^m = 0.$$

Example — Maclaurin polynomial of $\arctan x$

Let $f(x) = \arctan x$. Then $f'(x) = \dfrac{1}{1+x^2}$, so from the previous example we conclude that for every non-negative integer m,

$$f^{(2m+1)}(0) = (2m)!(-1)^m \qquad \text{and} \qquad f^{(2m+2)}(0) = 0.$$

Also $f(0) = 0$. Therefore by Taylor's Theorem,

$$\arctan x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots + \frac{(-1)^n}{2n+1}x^{2n+1} + o(x^{2n+1}) \qquad \text{as } x \to 0.$$

Corollary 9.18 — Differentiating Taylor polynomials

Let $a \in \mathbb{R}$, $n \in \mathbb{N}$, let f be n-times differentiable at a, and let P_n be a polynomial with $P_n(a) = f(a)$. Then P_n is the nth degree Taylor polynomial of f at a if and only if P_n' is the $(n-1)$th degree Taylor polynomial of f' at a; i.e.

$$f(x) = P_n(x) + o((x-a)^n) \quad \Longleftrightarrow \quad f'(x) = P_n'(x) + o((x-a)^{n-1}) \qquad \text{as } x \to a.$$

Example — Maclaurin polynomials of $\sin x$ and $\cos x$

Let $f(x) = \sin x$. The successive derivatives cycle as

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$	$f^{(5)}(x)$
at x	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$
at 0	0	1	0	-1	0	1

and so on. By Taylor's Theorem, for each non-negative integer n,

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + o(x^{2n+2}) \qquad \text{as } x \to 0.$$

There are no even powers because sin is odd. By Corollary 9.18,

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + o(x^{2n+1}) \qquad \text{as } x \to 0.$$

Evaluating limits using Taylor's Theorem sometimes helps avoid the repeated use of l'Hôpital's rule.

Example — A limit via Taylor's Theorem

Evaluate

$$\lim_{x\to 0} \frac{\cos(\sin x) - \cos x}{x^4}.$$

Solution. Instead of applying l'Hôpital's rule four times, use Taylor's Theorem. The fourth order approximation of the numerator at 0 is

$$\begin{aligned}
\cos(\sin x) - \cos x &= \left[1 - \frac{(\sin x)^2}{2!} + \frac{(\sin x)^4}{4!} + o((\sin x)^4)\right] \\
&\quad - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} + o(x^4)\right] \\
&= \left[1 - \frac{1}{2}\left(x - \frac{x^3}{3!} + o(x^3)\right)^2 + \frac{1}{24}\left(x - \frac{x^3}{3!} + o(x^3)\right)^4\right] \\
&\quad - \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + o(x^4) \\
&= \left(1 - \frac{1}{2}\left(x^2 - \frac{x^4}{3}\right) + \frac{1}{24}x^4\right) \\
&\quad - \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + o(x^4) = \frac{1}{6}x^4 + o(x^4)
\end{aligned}$$

as $x \to 0$, where we ignore all terms in x^5 or higher. Therefore

$$\lim_{x\to 0} \frac{\cos(\sin x) - \cos x}{x^4} = \lim_{x\to 0} \frac{\frac{1}{6}x^4 + o(x^4)}{x^4} = \frac{1}{6}.$$

Example — 5th degree Taylor polynomial of $\tan x$

Let $f(x) = \tan x$. Find the 5th degree Taylor polynomial of f at 0.
Solution. Writing $\sin x$ and $\cos x$ as their 5th degree Taylor polynomials at 0,

$$\tan x = \frac{\sin x}{\cos x} = \frac{x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + o(x^5)}{1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + o(x^5)} \qquad \text{as } x \to 0.$$

For the denominator, recall $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + o(x^3)$ as $x \to 0$. Replacing x by $\dfrac{1}{2}x^2 - \dfrac{1}{24}x^4 + o(x^5)$ and ignoring all terms in x^6 or higher,

$$\begin{aligned}\frac{1}{1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + o(x^5)} &= 1 + \left(\frac{1}{2}x^2 - \frac{1}{24}x^4 + o(x^5)\right) + \left(\frac{1}{2}x^2 - \frac{1}{24}x^4 + o(x^5)\right)^2 \\ &\quad + \left(\frac{1}{2}x^2 - \frac{1}{24}x^4 + o(x^5)\right)^3 + o(x^6) \\ &= 1 + \left(\frac{1}{2}x^2 - \frac{1}{24}x^4\right) + \frac{1}{4}x^4 + o(x^5) \\ &= 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + o(x^5).\end{aligned}$$

Therefore

$$\begin{aligned}\tan x &= \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + o(x^5)\right)\left(1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + o(x^5)\right) \\ &= x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + o(x^5)\end{aligned}$$

as $x \to 0$. Note that the coefficients of the Taylor polynomials of $\tan x$ have no general formula as simple as those of $\cos x$ and $\sin x$, because a division of two polynomial approximations is involved.

9.2 Taylor Series of a Function

We now associate a special power series to each given nice function.

Definition. Let a be a real number and f a function infinitely differentiable at a. The power series

$$\sum_{k=0}^{+\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

is called the Taylor series of f at a. The Taylor series of f at 0 is also called the Maclaurin series of f.

Remark 9.23 — The remainder term

For each non-negative integer n, the difference between f and its nth degree Taylor polynomial at a is called the remainder term,

$$R_n(x) := f(x) - P_n(x).$$

Taylor's Theorem says that for each fixed n, $\lim\limits_{x\to a} R_n(x) = 0$. In this section we investigate whether for each fixed x we have $\lim\limits_{n\to+\infty} R_n(x) = 0$.

- This issue only makes sense for those x in the interval of convergence of the Taylor series.
- If the limit holds for a certain x, then we may write

$$f(x) = \sum_{k=0}^{+\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

i.e. we have successfully "expressed f as a power series".

Example — Remainder of $\dfrac{1}{1-x}$

Let $f(x) = \dfrac{1}{1-x}$. We have found the Taylor polynomial $P_n(x) = 1 + x + x^2 + \cdots + x^n$, so the remainder term is explicitly

$$R_n(x) = f(x) - P_n(x) = \frac{1}{1-x} - \frac{1-x^{n+1}}{1-x} = \frac{x^{n+1}}{1-x}.$$

Hence $\lim\limits_{n\to+\infty} R_n(x) = 0$ if and only if $x \in (-1, 1)$. Therefore f can be expressed as a power series on the interval $(-1, 1)$:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots \qquad \text{for every } x \in (-1, 1),$$

agreeing with chapter 8.

It is in general difficult to find a closed-form expression for $R_n(x)$. Since we only need to determine for which x we have $\lim\limits_{n\to+\infty} R_n(x) = 0$, it is enough to estimate $R_n(x)$.

Theorem 9.25 — Lagrange's remainder formula

Let a be a real number, n a non-negative integer, and f a function $(n+1)$-times differentiable near a. Then for each x near a, there exists a number c between a and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1},$$

i.e.

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

(Note that c depends on n as well.)

Proof. Assume without loss of generality that $a = 0$, and that $x > a$ is fixed and near a. Let F be the function

$$F(t) = \frac{x^{n+1}}{(n+1)!}\left(f(t) - \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}t^k\right) - \frac{t^{n+1}}{(n+1)!}\left(f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k\right).$$

Then $F^{(n)}$ is continuous on $[0, x]$ and differentiable on $(0, x)$. One checks that

$$F(0) = F'(0) = F''(0) = \cdots = F^{(n)}(0) = 0 \qquad \text{and} \qquad F(x) = 0.$$

Applying Rolle's Theorem to F, there exists $c_1 \in (0, x)$ with $F'(c_1) = 0$. Applying Rolle's Theorem to F', there exists $c_2 \in (0, c_1)$ with $F''(c_2) = 0$. Continuing with $F'', F''', \ldots$, and finally with $F^{(n)}$, there exists $c \in (0, c_n)$ with

$$F^{(n+1)}(c) = 0,$$

which is equivalent to

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k + \frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1},$$

as desired. ■

Remark 9.26 — Lagrange's formula generalizes MVT

Lagrange's remainder formula is a generalization of the Mean Value Theorem: in the case $n = 0$ it reduces to the statement that for each x near a there exists c between a and x such that

$$f(x) = f(a) + f'(c)(x - a).$$

Example — Higher-order Fermat's theorem

Let a be a real number and f a function. If $f^{(2n+1)}$ is continuous near a,

$$f'(a) = f''(a) = \cdots = f^{(2n)}(a) = 0 \qquad \text{and} \qquad f^{(2n+1)}(a) \neq 0,$$

then for every x near a there exists c between a and x such that

$$f(x) = f(a) + \underbrace{\frac{f^{(2n+1)}(c)}{(2n+1)!}}_{\neq 0}(x - a)^{2n+1},$$

so f attains no local extremum at a. The case $n = 0$ is "Fermat's Theorem" in MATH1013.

Example — Higher-order second derivative test

Let a be a real number and f a function. If $f^{(2n)}$ is continuous near a,

$$f'(a) = f''(a) = \cdots = f^{(2n-1)}(a) = 0 \qquad \text{and} \qquad f^{(2n)}(a) > 0,$$

then for every x near a there exists c between a and x such that

$$f(x) = f(a) + \underbrace{\frac{f^{(2n)}(c)}{(2n)!}}_{>0}(x - a)^{2n},$$

so f attains a local minimum at a. The case $n = 1$ is the "second derivative test" in MATH1013.

Example — Maclaurin series of e^x converges to e^x

Let $f(x) = e^x$ and consider its remainder term at 0. For each n, $f^{(n)}(x) = e^x$. By Lagrange's remainder formula, for each fixed $x \in \mathbb{R}$ there exists c between 0 and x such that

$$|R_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \right| = \frac{e^c}{(n+1)!}|x|^{n+1} \leq \frac{e^{|x|}}{(n+1)!}|x|^{n+1}.$$

Since x is fixed, $\lim\limits_{n\to+\infty} \dfrac{|x|^{n+1}}{(n+1)!} = 0$, so by the squeeze theorem $\lim\limits_{n\to+\infty} R_n(x) = 0$. Therefore the Maclaurin series of f converges to $f(x)$ for every $x \in \mathbb{R}$:

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots \qquad \text{for every } x \in \mathbb{R}.$$

Example — Maclaurin series of $\sin x$ converges to $\sin x$

Let $f(x) = \sin x$ and consider its remainder term at 0. For each n, $f^{(n+1)}(x)$ is one of $\pm \sin x, \pm \cos x$. By Lagrange's remainder formula, for each fixed $x \in \mathbb{R}$ there exists

c between 0 and x such that

$$|R_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \right| \leq \frac{1}{(n+1)!} |x|^{n+1}.$$

As in the previous example, $\lim_{n\to+\infty} R_n(x) = 0$ by the squeeze theorem. Therefore

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots \qquad \text{for every } x \in \mathbb{R}.$$

Estimation of $R_n(x)$ using Lagrange's formula is useful only when $|f^{(n+1)}(c)|$ can be estimated easily. When such an estimation is too difficult, we may instead directly compute the sum of the Taylor series of f and check whether it equals $f(x)$.

Example — $\ln(1+x)$: limitations of Lagrange's estimate

Let $f(x) = \ln(1+x)$. For each non-negative integer n,

$$f^{(n+1)}(x) = \frac{(-1)^n n!}{(1+x)^{n+1}}.$$

Lagrange's remainder formula would only give

$$|R_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \right| = \frac{1}{n+1} \left| \frac{x}{1+c} \right|^{n+1},$$

from which we could only show $\lim_{n\to+\infty} R_n(x) = 0$ for $x \in [-\frac{1}{2}, 1]$. Fortunately, using termwise integration and Abel's limit theorem (chapter 8), we already have

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots \qquad \text{for } x \in (-1, 1].$$

Example 9.32 — The binomial series

Let $p \in \mathbb{R}$ and $f(x) = (1+x)^p$, and consider the Maclaurin series of f. For each n,

$$f^{(n)}(x) = p(p-1)(p-2)\cdots(p-n+1)(1+x)^{p-n}.$$

The estimation of $|f^{(n+1)}(c)|$ is too difficult, so instead we directly compute the sum of the Maclaurin series,

$$g(x) := \sum_{k=0}^{+\infty} \frac{p(p-1)(p-2)\cdots(p-k+1)}{k!} x^k.$$

Solution. We find all x for which $g(x)$ converges to $f(x)$. Since

$$\lim_{n\to+\infty} \frac{|c_n|}{|c_{n+1}|} = \lim_{n\to+\infty} \left| \frac{\frac{p(p-1)\cdots(p-n+1)}{n!}}{\frac{p(p-1)\cdots(p-n+1)(p-n)}{(n+1)!}} \right| = \lim_{n\to+\infty} \frac{n+1}{|n-p|} = 1,$$

the radius of convergence is 1, i.e. $g(x)$ converges at least on $(-1, 1)$. Using termwise differentiation one can deduce that

$$(1+x)g'(x) = pg(x) \qquad \text{for every } x \in (-1, 1).$$

Now let $h(x) = (1+x)^{-p} g(x)$. Then

$$h'(x) = -p(1+x)^{-p-1}g(x) + (1+x)^{-p}g'(x) = (1+x)^{-p-1}\big(-pg(x) + (1+x)g'(x)\big) = 0$$

for every $x \in (-1, 1)$, so h is constant on $(-1, 1)$; i.e. there exists $c \in \mathbb{R}$ with $g(x) =$

$c(1+x)^p$ on $(-1,1)$. Since $g(0)=1$, we get $c=1$, so $g(x)=(1+x)^p$, i.e.

$$(1+x)^p = \sum_{k=0}^{+\infty} \frac{p(p-1)(p-2)\cdots(p-k+1)}{k!} x^k \qquad \text{for every } x \in (-1,1).$$

The convergence at the end-points ± 1 is left as an exercise (Q6 of Problem Set 8). It turns out that the identity holds for every x in

$$\begin{cases} \mathbb{R} & \text{if } p \in \mathbb{N} \cup \{0\}, \\ [-1,1] & \text{if } p \in (0,+\infty) \setminus \mathbb{N}, \\ (-1,1] & \text{if } p \in (-1,0), \\ (-1,1) & \text{if } p \in (-\infty,-1]. \end{cases}$$

Remark 9.33 — Binomial theorem for general exponents

If a, b are real numbers with $a > |b|$, then $|\frac{b}{a}| < 1$, so the binomial series gives, for every real p,

$$\begin{aligned} (a+b)^p &= a^p \left(1 + \frac{b}{a}\right)^p \\ &= a^p \sum_{k=0}^{+\infty} \frac{p(p-1)\cdots(p-k+1)}{k!} \left(\frac{b}{a}\right)^k \\ &= \sum_{k=0}^{+\infty} \frac{p(p-1)\cdots(p-k+1)}{k!} a^{p-k} b^k. \end{aligned}$$

This is the binomial theorem for general exponents.

Remark 9.34 — Table of standard Maclaurin series

$$\frac{1}{1-x} = \sum_{k=0}^{+\infty} x^k = 1 + x + x^2 + x^3 + \cdots, \qquad x \in (-1,1),$$

$$e^x = \sum_{k=0}^{+\infty} \frac{1}{k!} x^k = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots, \qquad x \in \mathbb{R},$$

$$\cos x = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots, \qquad x \in \mathbb{R},$$

$$\sin x = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots, \qquad x \in \mathbb{R},$$

$$\arctan x = \sum_{k=0}^{+\infty} \frac{(-1)^k}{2k+1} x^{2k+1} = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots, \qquad x \in [-1,1],$$

$$\ln(1+x) = \sum_{k=1}^{+\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots, \qquad x \in (-1,1],$$

$$(1+x)^p = \sum_{k=0}^{+\infty} \frac{p(p-1)\cdots(p-k+1)}{k!} x^k = 1 + px + \frac{p(p-1)}{2}x^2 + \cdots, \qquad \text{see above.}$$

Using this table and evaluating at special values of x, we can compute the sums of

more series.

Example — Summing series via Maclaurin series

Evaluate the sum of each series, if it converges:

$$\text{(a) } \sum_{k=1}^{+\infty} \frac{(-1)^{k-1}}{2^k k} \qquad \text{(b) } 1+\sum_{k=1}^{+\infty} \frac{1\cdot 3\cdot 5\cdot 7\cdots(2k-1)}{k!\cdot 4^k} \qquad \text{(c) } \sum_{k=0}^{+\infty} \frac{1}{(2k)!}.$$

Solution.

(a) The series $\displaystyle\sum_{k=1}^{+\infty} \frac{(-1)^{k-1}}{2^k k} = \sum_{k=1}^{+\infty} \frac{(-1)^{k-1}}{k}\left(\frac{1}{2}\right)^k$ is the Maclaurin series of $\ln(1+x)$ evaluated at $\dfrac{1}{2}$. Hence its sum is $\ln\dfrac{3}{2}$.

(b) The series

$$1+\sum_{k=1}^{+\infty} \frac{1\cdot 3\cdot 5\cdots(2k-1)}{k!\cdot 4^k} = 1+\sum_{k=1}^{+\infty} \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})\cdots(-\frac{2k-1}{2})}{k!}\left(-\frac{1}{2}\right)^k$$

is the Maclaurin series of $(1+x)^{-\frac{1}{2}}$ evaluated at $-\dfrac{1}{2}$. Hence its sum is $\left(1-\dfrac{1}{2}\right)^{-\frac{1}{2}} = \sqrt{2}$.

(c) Note that $\displaystyle\sum_{k=0}^{+\infty} \frac{1}{k!}$ and $\displaystyle\sum_{k=0}^{+\infty} \frac{(-1)^k}{k!}$ are the Maclaurin series of e^x at 1 and -1. Hence

$$\sum_{k=0}^{+\infty} \frac{1}{(2k)!} = \frac{1}{2}\left(\sum_{k=0}^{+\infty} \frac{1}{k!} + \sum_{k=0}^{+\infty} \frac{(-1)^k}{k!}\right) = \frac{e+e^{-1}}{2}.$$

So far, the Taylor series of every function we have seen converges back to $f(x)$ on its interval of convergence. In general, however, the Taylor series of a function may not converge back to $f(x)$ even if it converges. The next example gives an infinitely differentiable function whose Maclaurin series converges for every $x \in \mathbb{R}$, but not always to $f(x)$.

Example 9.36 — A non-analytic smooth function

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \le 0. \end{cases}$$

Find the Taylor series of f centered at 0.

Solution. First, $f(0) = 0$. Since

$$\lim_{x\to 0^+} \frac{f(x)-f(0)}{x} = \lim_{x\to 0^+} \frac{e^{-\frac{1}{x}}}{x} = \lim_{t\to+\infty} te^{-t} = 0$$

$$\text{and} \qquad \lim_{x\to 0^-} \frac{f(x)-f(0)}{x} = \lim_{x\to 0^-} 0 = 0,$$

we have $f'(0) = 0$, and so

$$f'(x) = \begin{cases} \dfrac{1}{x^2}e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \le 0. \end{cases}$$

Similarly,

$$\lim_{x\to 0^+} \frac{f'(x)-f'(0)}{x} = \lim_{x\to 0^+} \frac{\frac{1}{x^2}e^{-\frac{1}{x}}}{x} = \lim_{t\to+\infty} t^3 e^{-t} = 0$$

$$\text{and} \qquad \lim_{x\to 0^-} \frac{f'(x)-f'(0)}{x} = 0,$$

so $f''(0) = 0$, and

$$f''(x) = \begin{cases} \dfrac{1-2x}{x^4} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \le 0. \end{cases}$$

Continuing similarly, one shows that $f^{(n)}(0) = 0$ for every $n \in \mathbb{N}$; in fact

$$f^{(n)}(x) = \begin{cases} \dfrac{p_{n-1}(x)}{x^{2n}} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \le 0, \end{cases}$$

where p_{n-1} is some polynomial of degree at most $n-1$. Therefore the Maclaurin series of f is the zero power series, which converges to 0 for every $x \in \mathbb{R}$. But it converges to $f(x)$ only for $x \le 0$, not for any $x > 0$. (Figure: the graph of $y = f(x)$ is flat ($y = 0$) up to $x = 0$ and rises smoothly for $x > 0$.)

Remark 9.37 — Analytic functions

Let a be a real number. A function f defined near a is said to be analytic at a if its Taylor series at a converges to $f(x)$ for every x in some interval $(a-\delta, a+\delta)$. All functions listed in Remark 9.34 are analytic at 0, while

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \le 0, \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} e^{-\frac{1}{|x|}} & \text{if } x \ne 0, \\ 0 & \text{if } x = 0, \end{cases}$$

are not analytic at 0.

9.3 Fourier Series of a Function

Some infinitely differentiable functions can be nicely represented by power series. We now study another type of series of functions, trigonometric series, and use them to represent periodic functions.

Definition. A function $f : \mathbb{R} \to \mathbb{R}$ is called a periodic function if there exists $c > 0$ such that

$$f(x+c) = f(x) \qquad \text{for every } x \in \mathbb{R}.$$

The number c is called a period of f.

Apart from famous periodic functions like cos and sin, a periodic function can be constructed by specifying its values on an interval and extending the definition periodically.

Example — Periodic extension of x^2

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = x^2 \qquad \text{for every } x \in [-1, 1),$$

extended periodically by $f(x+2) = f(x)$ for every $x \in \mathbb{R}$. Then f is a continuous periodic function with a period 2. (Figure: repeated copies of the parabola segment $y = x^2$ on each interval $[2k-1, 2k+1)$, forming a continuous sawtooth-like wave of height 1.)

Example — Periodic extension of x on $[0, \pi)$

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = x \qquad \text{for every } x \in [0, \pi),$$

extended periodically by $f(x + \pi) = f(x)$ for every $x \in \mathbb{R}$. Then f is periodic with a period π. (Figure: repeated rising segments of slope 1 and height π, with jump discontinuities at every integer multiple of π.)

Lemma 9.40 — Coefficients of a trigonometric polynomial

Let $a_0, a_1, \ldots, a_n, b_1, \ldots, b_n$ be real numbers and consider the finite sum (called a trigonometric polynomial)

$$f(x) := \frac{a_0}{2} + \sum_{k=1}^{n} \big(a_k \cos kx + b_k \sin kx\big).$$

Then f is periodic with a period 2π, and the coefficients are related to the values of f via

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx \qquad \text{and} \qquad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx$$

for every k.

Proof. Q1 of Problem Set 3. The key is to observe that for all non-negative integers m, n,

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \begin{cases} 2\pi & \text{if } m = n = 0, \\ \pi & \text{if } m = n \neq 0, \\ 0 & \text{if } m \neq n, \end{cases}$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \begin{cases} \pi & \text{if } m = n \neq 0, \\ 0 & \text{otherwise,} \end{cases} \qquad \int_{-\pi}^{\pi} \cos mx \sin nx \, dx = 0.$$

■

Remark 9.41 — Orthogonality

Borrowing the dot product idea from vectors, define the "dot product" of two functions by

$$f \cdot g := \int_{-\pi}^{\pi} f(x) g(x) \, dx.$$

Then the observations in the above proof say that every two different members of

$$\{1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \sin 3x, \ldots\}$$

are orthogonal to each other. More on this idea in linear algebra.

Example — Integrals of $\sin^4 x$ via its trigonometric form

Let $f(x) = \sin^4 x$. Applying the double angle formula twice,

$$\sin^4 x = \frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x,$$

so f is a trigonometric polynomial. By Lemma 9.40 we deduce immediately that

$$\int_{-\pi}^{\pi} \sin^4 x\,dx = \pi a_0 = \frac{3\pi}{4}, \qquad \int_{-\pi}^{\pi} \sin^4 x \cos 2x\,dx = \pi a_2 = -\frac{\pi}{2},$$
$$\int_{-\pi}^{\pi} \sin^4 x \cos 4x\,dx = \pi a_4 = \frac{\pi}{8}.$$

For the rest of this section we only consider periodic functions with a period 2π.

Definition. Let f be a periodic function with a period 2π, integrable on $[-\pi, \pi]$. The series of trigonometric functions

$$\mathcal{F}f(x) := \frac{a_0}{2} + \sum_{k=1}^{+\infty} \big(a_k \cos kx + b_k \sin kx\big),$$

where the coefficients are defined by

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \qquad \text{and} \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$$

is called the Fourier series of f (on the interval $[-\pi, \pi]$).

Example — Fourier series of a trigonometric polynomial

Let f be a trigonometric polynomial, i.e. a finite sum

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{n} \big(a_k \cos kx + b_k \sin kx\big).$$

Then the Fourier series of f is just f itself. For example, $f(x) = \sin^4 x$ equals its own Fourier series.

Trap — f need not equal its Fourier series

In general, a periodic function does not necessarily equal its Fourier series. We usually write

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{+\infty} \big(a_k \cos kx + b_k \sin kx\big),$$

since the sequence of partial sums of the Fourier series may not converge to $f(x)$.

Example — Fourier series of a square wave

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0 & \text{if } x \in [-\pi, 0), \\ 1 & \text{if } x \in [0, \pi), \end{cases}$$

extended periodically by $f(x + 2\pi) = f(x)$. (Figure: a square wave alternating between 0 on $[-\pi, 0)$ and 1 on $[0, \pi)$.)

Solution. The Fourier coefficients of f are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{0}^{\pi} 1\,dx = 1,$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} \cos nx\,dx = 0,$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_0^{\pi}\sin nx\,dx = \frac{1}{n\pi}\big[-\cos nx\big]_0^{\pi} = \begin{cases} 0 & \text{if } n \text{ is even,} \\ \dfrac{2}{n\pi} & \text{if } n \text{ is odd,} \end{cases}$$

for every positive integer n; so the Fourier series of f is

$$f(x) \sim \frac{1}{2} + \sum_{k=1}^{+\infty} \frac{2}{(2k-1)\pi}\sin(2k-1)x = \frac{1}{2} + \frac{2}{\pi}\sin x + \frac{2}{3\pi}\sin 3x + \frac{2}{5\pi}\sin 5x + \cdots.$$

(Figure: partial sums $S_5(x)$ and $S_{25}(x)$ oscillating around the square wave, with overshoots near the jumps.) Although each term of the Fourier series is continuous, the sum is discontinuous: at those x where f is continuous the series converges to $f(x)$, but at those x where f is discontinuous the series does not converge to $f(x)$. (Figure: the graph of $y = \mathcal{F}f(x)$, which passes through $\dfrac{1}{2}$ at each jump.)

Theorem 9.47 — Convergence of Fourier series

Let f be a periodic function with period 2π, and let a be a real number. If the one-sided limits of f and f' at a, i.e.

$$\lim_{x\to a^-} f(x), \qquad \lim_{x\to a^+} f(x), \qquad \lim_{x\to a^-} f'(x) \qquad \text{and} \qquad \lim_{x\to a^+} f'(x),$$

all exist as finite real numbers, then the Fourier series of f converges at a to the value

$$\mathcal{F}f(a) = \frac{\lim_{x\to a^-} f(x) + \lim_{x\to a^+} f(x)}{2}.$$

In particular, if f is continuous at a and $f'_-(a), f'_+(a)$ both exist as real numbers, then

$$\mathcal{F}f(a) = f(a).$$

(Figure: at a jump of $y = f(x)$, the actual value $f(a)$ is irrelevant; the Fourier series converges to the midpoint of the left and right limits, provided both one-sided slopes are finite.)

Corollary 9.48

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous periodic function with period 2π. If there exists $a \in \mathbb{R}$ such that

- f is differentiable on $(a, a+2\pi)$, and
- f' is continuous and bounded on $(a, a+2\pi)$,

then $\mathcal{F}f(x) = f(x)$ for every $x \in \mathbb{R}$.

Example 9.49 — Fourier series of the sawtooth wave

Let $f(x) = x$ for every $x \in (-\pi, \pi)$, extended periodically to a period-2π function. (We allow f to remain undefined at the end-points $\pm\pi$.) (Figure: repeated rising segments of slope 1 between $-\pi$ and π, with jumps at odd multiples of π.)

Solution. Note that f is odd. The Fourier coefficients are

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos nx\,dx = 0$$

for every non-negative integer n, and

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin nx\,dx = \frac{1}{\pi}\left[\frac{1}{n^2}\sin nx - \frac{x}{n}\cos nx\right]_{-\pi}^{\pi} = \frac{2(-1)^{n+1}}{n}$$

for every positive integer n; so the Fourier series of f is

$$f(x) \sim 2\sin x - \frac{2}{2}\sin 2x + \frac{2}{3}\sin 3x - \frac{2}{4}\sin 4x + \cdots .$$

For every $x \in (-\pi, \pi)$, Theorem 9.47 gives the equality

$$x = 2\sin x - \sin 2x + \frac{2}{3}\sin 3x - \frac{1}{2}\sin 4x + \cdots .$$

- When $x = \dfrac{\pi}{2}$, we recover

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right),$$

  as obtained in chapter 8.
- When $x = \pi$, although $f(\pi)$ is undefined, $\lim\limits_{x\to\pi^-} f(x) = \pi$ and $\lim\limits_{x\to\pi^+} f(x) = -\pi$, and $\lim\limits_{x\to\pi^-} f'(x) = \lim\limits_{x\to\pi^+} f'(x) = 1$ all exist as finite real numbers. We observe

$$\mathcal{F}f(\pi) = 0 \qquad \text{and} \qquad \frac{\lim_{x\to\pi^-} f(x) + \lim_{x\to\pi^+} f(x)}{2} = \frac{\pi + (-\pi)}{2} = 0,$$

  verifying the conclusion of Theorem 9.47.

Example 9.50 — The Basel problem

Let $f(x) = x^2$ for every $x \in [-\pi, \pi]$, extended periodically to a continuous period-2π function. (Figure: repeated parabola arcs of height π^2, forming a continuous wave with corners at odd multiples of π.)

Solution. Note that f is even. The Fourier coefficients are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^2}{3},$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos nx\,dx = \frac{1}{\pi}\left[\left(\frac{x^2}{n} - \frac{2}{n^3}\right)\sin nx + \frac{2x}{n^2}\cos nx\right]_{-\pi}^{\pi} = \frac{4(-1)^n}{n^2}$$

for every positive integer n, and

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\sin nx\,dx = 0$$

for every positive integer n; so the Fourier series of f is

$$f(x) \sim \frac{\pi^2}{3} - 4\cos x + \frac{4}{2^2}\cos 2x - \frac{4}{3^2}\cos 3x + \frac{4}{4^2}\cos 4x - \cdots .$$

For every $x \in [-\pi, \pi]$, f is continuous at x with finite one-sided derivatives, so by Theorem 9.47 we have the equality

$$x^2 = \frac{\pi^2}{3} - 4\cos x + \frac{4}{2^2}\cos 2x - \frac{4}{3^2}\cos 3x + \frac{4}{4^2}\cos 4x - \cdots .$$

When $x = \pi$, we obtain

$$\pi^2 = \frac{\pi^2}{3} + 4 + \frac{4}{2^2} + \frac{4}{3^2} + \frac{4}{4^2} + \cdots = \frac{\pi^2}{3} + 4\left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots\right).$$

Rearranging,

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6},$$

which is the solution to the Basel problem.

Remark 9.51

Check out the video `https://www.youtube.com/watch?v=d-o3eB9sfls`, which gives a geometric explanation of why π^2 appears in the sum of inverse squares of positive integers.

In the two examples above, the Fourier series of an even function consists of cosine terms only, while the Fourier series of an odd function consists of sine terms only.

Theorem 9.52 — Fourier series of even and odd functions

Let f be a periodic function with period 2π, with Fourier series

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{+\infty} \big(a_k \cos kx + b_k \sin kx\big).$$

- If f is even, then $b_n = 0$ for all n, i.e. the Fourier series of f consists of cosine terms only:
$$f(x) \sim \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \cdots .$$
- If f is odd, then $a_n = 0$ for all n, i.e. the Fourier series of f consists of sine terms only:
$$f(x) \sim b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \cdots .$$

Summary of Chapter 9

What you need to know in this chapter:

- **Polynomial approximation:** nth order approximation at a number a; little-o notation $f(x) = o(g(x))$ as $x \to a \Leftrightarrow \lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = 0$; nth degree Taylor polynomial at a; evaluating limits using Taylor polynomials instead of l'Hôpital's rule.
- **Taylor series:** Taylor series at a and the remainder term; analyzing convergence of the remainder via Lagrange's remainder formula; computing Taylor series of elementary functions; evaluating numerical series via Taylor series; analytic functions.
- **Fourier series:** periodic functions and periodic extension; trigonometric polynomials; Fourier series of a period-2π function; convergence of Fourier series; computing Fourier series of elementary functions; the Basel problem and evaluating numerical series via Fourier series.